Advanced Algorithms for Discrete Optimization

Models, Paradigms, and Algorithms

Pierre Schaus

ICTEAM Institute
UCLouvain, Belgium



October 12, 2026

© 2026 Pierre Schaus

Advanced Algorithms for Discrete Optimization: Models, Paradigms, and Algorithms

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means without prior written permission of the author.

Written with the assistance of artificial intelligence tools.

Published with LaTeX and biber.

First edition, October 12, 2026

Preface

Discrete optimization lies at the core of computer science, operations research, and applied mathematics. It addresses problems where decision variables take values from discrete domains, such as scheduling tasks, routing vehicles, allocating scarce resources, or configuring communication networks.

While many discrete optimization problems are computationally intractable in the worst case (**NP**-hard), real-world applications demand fast, robust, and often provably optimal solutions. To meet this challenge, several complementary algorithmic paradigms have been developed over the last several decades:

- **Dynamic Programming**: Decomposing problems into overlapping subproblems governed by the principle of optimality.
- **Branch and Bound**: Systematically enumerating the solution space while aggressively pruning non-promising regions through relaxation bounds.
- **Linear Programming & Duality**: Exploiting geometric and algebraic structures of polyhedra to find exact solutions or compute strong dual bounds.
- **Integer Linear Programming**: Formulating discrete problems as mixed-integer programs, exploiting total unimodularity, generating cutting planes, and executing LP-based branch-and-bound.
- **Lagrangian Relaxation**: Dualizing complicating constraints to decompose hard problems into tractable subproblems.
- **Network Flows**: Exploiting flow-cut duality, augmenting paths, scaling, and strongly polynomial combinatorial graph algorithms.
- **Local Search & Metaheuristics**: Exploring neighborhood structures to quickly uncover high-quality heuristic solutions in massive search spaces.
- **Constraint Programming**: Leveraging declarative modeling, domain filtering, and constraint propagation to prune infeasible regions efficiently.
- **Benchmarking Optimization Algorithms**: Rigorous empirical evaluation methodologies, shifted geometric means, and Dolan-Moré performance profiles.

Audience and Prerequisites

This book is designed for advanced graduate students (Master's and Ph.D.), researchers, and practitioners who want to master both the mathematical foundations and the practical implementation of modern discrete optimization techniques. Readers are expected to have a solid background in data structures, design and analysis of algorithms, graph theory, and basic linear algebra.

Pedagogical Approach and Software Framework

A distinguishing philosophy of this textbook is bridging mathematical rigor with high-performance algorithmic engineering. Theoretical formulations, polyhedral combinatorics, and optimality theorems are coupled with clean, idiomatic Java implementations. Rather than treating solvers as black boxes, students and practitioners are encouraged to examine, build, and extend core solver engines—including custom bounding procedures, branch-and-bound search strategies, and constraint propagators.

Structure of the Book

Each chapter is dedicated to one foundational paradigm. Chapters feature formal definitions, theoretical properties, pseudocode for key algorithms, concrete examples, and end-of-chapter exercises suitable for course assignments or self-study. The material can support a one-semester graduate course in discrete optimization or serve as a modular reference for specialized seminars in operations research and artificial intelligence.

Acknowledgments and AI Assistance

This textbook was written with the interactive assistance of artificial intelligence tools (specifically large language models). AI assistance was employed to support drafting, prose refinement, TikZ diagram typesetting, and Java code structuring under the continuous guidance, technical direction, and mathematical verification of the author. All pedagogical choices, theoretical formulations, proofs, and algorithmic designs remain the sole responsibility of the author.

Pierre Schaus
Louvain-la-Neuve, Belgium
October 12, 2026

Contents

Preface **iii**

1 Introduction to Discrete Optimization **1**
1.1 From Polynomial Algorithms to NP-Hardness . . . 1
1.2 Industrial Applications and Impact . . . 2
1.2.1 Vehicle Routing and Fleet Logistics (VRP) . . . 2
1.2.2 Workforce and Industrial Scheduling . . . 2
1.2.3 Cloud Computing and Resource Allocation . . . 3
1.2.4 Network Traffic Engineering . . . 3
1.3 Tackling NP-Hard Problems in Practice . . . 3
1.4 The Diagnostic Compass . . . 4
1.5 The Art of Practical Modeling . . . 4
1.5.1 High-Leverage Decision Variables . . . 5
1.5.2 Optimality with Imperfect Data . . . 5
1.6 Formal Problem Formulation . . . 6
1.7 A Taxonomy of Algorithmic Paradigms . . . 7
1.8 Generic Solvers and Practical Projects . . . 7
1.8.1 Generic Optimization Engines . . . 7
1.8.2 Hands-On Programming Projects . . . 8
1.8.3 Language Selection: Why Java? . . . 9
1.9 Exercises . . . 10

2 Dynamic Programming **11**
2.1 Computational Complexity of Knapsack . . . 11
2.1.1 The Polynomial-Time Reduction Technique . . . 12
2.2 Bellman Recurrence for Knapsack . . . 13
2.3 Naive Recursive Implementation . . . 14
2.3.1 Complexity Analysis of the Naive Recursion . . . 14
2.4 Memoization and Overlapping States . . . 14
2.4.1 Memoization with a Hash Map . . . 15
2.5 Tabulation with a 2D Array . . . 16
2.5.1 Iterative Algorithm (Bottom-Up) . . . 16
2.5.2 Subproblem Dependencies . . . 18
2.5.3 The Tabulation Matrix Illustrated . . . 19
2.5.4 Solution Reconstruction via Backtracking . . . 19
2.5.5 The Tabulation Grid as a DAG . . . 20
2.6 Complexity and Pseudo-Polynomial Time . . . 21
2.6.1 Is this Algorithm Polynomial? . . . 21

2.6.2 Weak vs. Strong NP-Hardness . . . 22
2.7 Case Study: Range Partitioning . . . 22
2.7.1 Dynamic Programming Formulation . . . 22
2.7.2 Prefix Sum Precomputation . . . 23
2.7.3 Java Implementation . . . 23
2.8 DP as Shortest Paths in a DAG . . . 24
2.9 A Generic Java Framework for DP . . . 25
2.9.1 The State Abstraction . . . 25
2.9.2 The Transition Class . . . 26
2.9.3 The Model Abstraction . . . 26
2.9.4 The Generic Solver Engine . . . 26
2.10 Knapsack in the Framework . . . 28
2.10.1 The Knapsack State . . . 28
2.10.2 The Knapsack Model . . . 28
2.11 Exercises . . . 30

3 Branch and Bound **31**
3.1 Foundations of Branch and Bound . . . 31
3.1.1 The Three Core Mechanisms . . . 32
3.1.2 Maximization versus Minimization Duality . . . 33
3.2 Branch and Bound for 0/1 Knapsack . . . 34
3.2.1 Binary Search Tree Representation . . . 34
3.2.2 Bounding Procedure 1: Capacity Relaxation . . . 34
3.2.3 Bounding 2: Continuous LP Relaxation . . . 35
3.2.4 Worked Example and Search Tree . . . 37
3.3 Combinatorial Relaxations for the TSP . . . 40
3.3.1 TSP Relaxations and Formulations . . . 40
3.3.2 Comparison of the TSP Relaxations . . . 48
3.3.3 Branching Schemes and Partial Decisions . . . 48
3.4 Search Strategies and Optimality Gap . . . 54
3.4.1 Depth-First Search (DFS) . . . 54
3.4.2 Best-First Search (BFS) . . . 56
3.4.3 The Optimality Gap . . . 56
3.4.4 Hybrid Search: Diving (DFS-BFS) . . . 57
3.5 A Generic Java Framework for B&B . . . 57
3.5.1 Architecture and Core Interfaces . . . 57
3.5.2 Instantiating Knapsack in the Framework . . . 60
3.6 Practical Implementation: TSP in B&B . . . 62
3.7 Exercises . . . 63

4 Linear Programming **65**
4.1 Foundations of Linear Programming . . . 65
4.2 A Motivating Case: The Brewery Problem . . . 66
4.2.1 Problem Formulation . . . 66
4.2.2 Geometric Perspective in Two Dimensions . . . 67
4.3 Geometry in Higher Dimensions . . . 68
4.3.1 A Three-Dimensional Polytope Example . . . 68
4.3.2 Polyhedral Theory and Convexity . . . 68
4.3.3 Extreme Points and Vertex Optimality . . . 70
4.3.4 Combinatorial Explosion of Vertices . . . 71

4.4 Canonical and Standard Forms . . . 71
4.4.1 Motivation for Generic Formulations . . . 71
4.4.2 Canonical and Standard Equational Forms . . . 71
4.4.3 Form Transformations and Slack Variables . . . 72
4.5 The Simplex Algorithm . . . 73
4.5.1 Simplex by Example: The Brewery Problem . . . 73
4.5.2 Formalization of the Simplex Method . . . 76
4.5.3 Algorithmic Implementation in Java . . . 79
4.6 Initialization: Two-Phase and Big-M . . . 82
4.6.1 The Initial Basis Challenge . . . 82
4.6.2 The Two-Phase Simplex Method . . . 82
4.6.3 The Big-M Penalty Approach . . . 91
4.6.4 Comparison and Numerical Stability . . . 92
4.7 Duality Theory and Certificates . . . 92
4.7.1 The Primal-Dual Pair . . . 92
4.7.2 Weak and Strong Duality . . . 93
4.7.3 Complementary Slackness . . . 94
4.7.4 Dual Recovery and Shadow Prices . . . 94
4.8 Complexity, Cycling, and Solvers . . . 95
4.8.1 Degeneracy and Bland's Rule . . . 95
4.8.2 Worst-Case Exponential Complexity . . . 95
4.8.3 Polynomial-Time Interior Algorithms . . . 95
4.8.4 Warm-Starting and Solver Architecture . . . 96
4.9 Applications of Linear Programming . . . 96
4.9.1 The Nutritional Diet Problem . . . 96
4.9.2 From Continuous to Integer Programming . . . 96
4.10 Chapter Summary and Exercises . . . 97
4.10.1 Key Conceptual Takeaways . . . 97
4.10.2 Exercises . . . 97

5 Integer Linear Programming 101
5.1 Foundations of Integer Linear Programming . . . 101
5.1.1 Mathematical Formulations . . . 101
5.1.2 The Geometry of the Integer Hull . . . 102
5.1.3 The Continuous Relaxation Bound . . . 102
5.2 Total Unimodularity and Polyhedral Integrality . . . 103
5.2.1 Network Flow and Bipartite Matching . . . 104
5.3 LP-Based Branch and Bound . . . 105
5.3.1 The Search Algorithm . . . 105
5.3.2 Branching Decisions: Variable Selection . . . 106
5.3.3 Updating the Simplex Tableau with Branching Constraints . . . 107
5.3.4 Dual Simplex Warm-Starting . . . 108
5.3.5 Dual Simplex Implementation in Java . . . 110
5.4 Cutting Planes and Branch-and-Cut . . . 111
5.4.1 Gomory Fractional Cuts . . . 111
5.4.2 The Branch-and-Cut Paradigm . . . 112
5.5 Mixed-Integer Linear Programming Modeling . . . 112
5.6 Chapter Summary and Exercises . . . 113
5.6.1 Key Conceptual Takeaways . . . 113

5.6.2 Exercises . . . 113

6 Lagrangian Relaxation **117**
6.1 The Lagrangian Relaxation Principle . . . 117
6.1.1 The Lagrangian Dual Problem . . . 118
6.2 Resource-Constrained Shortest Path . . . 119
6.2.1 Mathematical Formulation . . . 119
6.2.2 Lagrangian Dualization . . . 120
6.2.3 A Worked Example . . . 120
6.2.4 Geometry of the Lagrangian Dual . . . 122
6.3 Solving the Dual via Subgradients . . . 123
6.3.1 Subgradients and the Subdifferential . . . 123
6.3.2 The Subgradient Iteration Algorithm . . . 124
6.4 Lagrangian Relaxation for the TSP . . . 124
6.4.1 Structural Constraints of the TSP . . . 124
6.4.2 Dualizing Degree Constraints . . . 125
6.4.3 Modified Edge Costs and Multipliers . . . 126
6.4.4 Multiplier Invariance and Optimality . . . 126
6.4.5 A Worked Example on 4 Vertices . . . 127
6.5 Subgradient Optimization for Held–Karp . . . 129
6.5.1 Subgradients and Zero-Sum Property . . . 129
6.5.2 Adaptive Step Size Control . . . 129
6.5.3 The Held–Karp Subgradient Procedure . . . 130
6.6 Polyhedral Strength and Held–Karp . . . 132
6.6.1 The DFJ Formulation of the TSP . . . 132
6.6.2 Subtour LP and Held–Karp Equivalence . . . 134
6.6.3 Empirical Strength of the Bound . . . 135
6.7 Chapter Summary . . . 135
6.8 Exercises . . . 136

7 Network Flows **139**
7.1 Foundations and Problem Formulations . . . 139
7.1.1 Historical Context: Network Interdiction . . . 139
7.1.2 The Minimum Cut Problem . . . 140
7.1.3 The Maximum Flow Problem . . . 141
7.1.4 Case Study: Seminar Scheduling . . . 141
7.2 Linear Programming and Duality . . . 142
7.2.1 Primal and Dual Formulations . . . 142
7.2.2 Total Unimodularity and Integrality . . . 143
7.3 The Ford-Fulkerson Method . . . 144
7.3.1 Residual Graphs and Augmentation . . . 144
7.3.2 The Max-Flow Min-Cut Theorem . . . 147
7.4 Path Selection and Complexity . . . 149
7.4.1 Arbitrary Paths and Pseudo-Polynomiality . . . 149
7.4.2 The Capacity Scaling Algorithm . . . 150
7.4.3 Shortest Augmenting Path: Edmonds–Karp . . . 152
7.4.4 Maximum Capacity Augmenting Paths . . . 153
7.4.5 Comparison of Algorithmic Variants . . . 153
7.5 Object-Oriented Implementation . . . 153
7.5.1 The FlowEdge Data Structure . . . 153

7.5.2 The FlowNetwork Representation . . . 155
7.5.3 The Ford-Fulkerson Solver . . . 155
7.6 Modeling Reductions and Extensions . . . 157
7.6.1 Multiple Sources and Multiple Sinks . . . 157
7.6.2 Vertex Capacities and Node Splitting . . . 157
7.6.3 Lower Bounds and Edge Demands . . . 158
7.6.4 Bipartite Matching and Domain Filtering . . . 158
7.7 Minimum-Cost Flow . . . 159
7.7.1 Mathematical Formulation . . . 159
7.7.2 Residual Costs and Negative Cycles . . . 159
7.7.3 The Successive Shortest Path Algorithm . . . 161
7.8 Chapter Summary and Exercises . . . 161
7.8.1 Key Conceptual Takeaways . . . 162
7.8.2 Exercises . . . 162

8 Local Search and Metaheuristics 165

9 Constraint Programming 167

10 Benchmarking Optimization Algorithms 169

A Kruskal's Algorithm and Disjoint Sets 171
A.1 The Minimum Spanning Tree Problem . . . 171
A.2 The Cut Property and Greedy Optimality . . . 172
A.3 Kruskal's Algorithm and Invariants . . . 173
A.4 The Disjoint-Set Data Structure . . . 174
A.4.1 Union by Rank and Path Compression . . . 174
A.5 Java Implementation . . . 175
A.6 Complexity and Comparison with Prim . . . 178
A.6.1 Kruskal vs. Prim: Trade-Offs . . . 178

References 179

CHAPTER 1

Introduction to Discrete Optimization

Discrete optimization is the science and engineering of making optimal decisions from a finite or countably infinite set of possibilities. It stands at the intersection of computer science, operations research, and applied mathematics, providing the algorithmic backbone for managing complex modern infrastructure, logistics networks, manufacturing supply chains, and computing systems.

Unlike continuous optimization, where calculus, gradients, and smooth geometric structures guide search toward local or global extrema, discrete optimization operates on combinatorial landscapes. In these landscapes, moving a single discrete decision variable from 0 to 1 can drastically alter the feasibility and quality of an entire system.

1.1 From Polynomial Algorithms to NP-Hardness

In introductory courses on data structures and algorithms, students are introduced to a collection of celebrated polynomial-time algorithms:

- **Dijkstra's and Bellman-Ford's algorithms** for single-source shortest paths on graphs in $\mathcal{O}(|E| + |V| \log |V|)$ time.
- **Kruskal's and Prim's algorithms** for minimum spanning trees in $\mathcal{O}(|E| \log |V|)$ time.
- **Edmonds-Karp and push-relabel algorithms** for maximum network flow and bipartite matching in polynomial time.
- **Divide-and-conquer sorting** operating in $\mathcal{O}(n \log n)$ time.

These classical algorithms are elegant and practical. However, they share a common trait: they solve problems characterized by very specific algebraic or matroid structures that prevent combinatorial explosion.

When graduating from textbook abstractions to real-world industrial systems, a sobering truth emerges: **almost every real-world discrete optimization problem is NP-hard**. Consider what happens when we introduce a single realistic constraint into the classical polynomial problems:

- Find a shortest path between two nodes in a network: *Solvable in polynomial time* by Dijkstra's algorithm.
- Find a shortest path subject to a total transit budget or battery capacity limit (the Constrained Shortest Path Problem): *Immediately* **NP**-*hard* [GJ79].

- Find a minimum spanning tree connecting a set of terminals: *Solvable in polynomial time* by Prim's algorithm.
- Find a minimum spanning tree where the degree of each node cannot exceed 3 (the Degree-Constrained Spanning Tree Problem): *Immediately* **NP**-*hard*.
- Match jobs to identical machines to maximize throughput: *Solvable in polynomial time* via maximum bipartite matching.
- Match jobs to machines where tasks share disjunctive resources and strict deadlines: *Immediately* **NP**-*hard*.

The combinatorial explosion is not a rare mathematical curiosity; it is the default condition of real-world decision-making.

1.2 Industrial Applications and Impact

To appreciate the reach of discrete optimization, consider the critical industrial systems that rely on the algorithms developed in this book:

1.2.1 Vehicle Routing and Fleet Logistics (VRP)

Global e-commerce and logistics networks (such as Amazon, DHL, FedEx, and UPS) dispatch hundreds of thousands of delivery vehicles daily. The canonical *Vehicle Routing Problem (VRP)* requires determining an optimal set of routes for a fleet of vehicles based at one or more depots to service a set of customers with known demands:

- **Time Windows ($VRPTW$)**: Deliveries must arrive within customer-specified intervals (e.g., 10:00 AM – 11:30 AM).
- **Capacity and Compartment Constraints**: Vehicles have weight and volume thresholds, with refrigerated vs. non-refrigerated compartments.
- **Driver Rest and Regulations**: Routes must comply with legal driving hours and mandatory rest periods.

A fleet efficiency improvement of merely 1% in a major carrier translates into tens of millions of liters of fuel saved annually, hundreds of thousands of tons of reduced CO_2 emissions, and substantial cost reductions.

1.2.2 Workforce and Industrial Scheduling

Scheduling problems allocate scarce physical resources and human personnel over time:

- **Airline Crew Pairing and Rostering**: Major airlines (such as Air France-KLM, Delta, and Lufthansa) must assign tens of thousands of pilots and flight attendants to flights over monthly horizons. Feasible schedules must satisfy hundreds of regulatory requirements from civil aviation authorities (FAA, EASA), pilot certifications, rest periods, seniority privileges, and collective bargaining agreements.
- **Semiconductor Manufacturing**: In wafer fabrication plants (fabs), expensive cluster tools and photolithography machines process microchips through hundreds of sequential stages. Optimal job sequencing must account for sequence-dependent setup times, chemical re-entry constraints, and clean-room transport delays.
- **Operating Theater Scheduling**: In healthcare, hospitals must schedule elective surgeries across limited operating rooms, coordinating surgeon availability, post-operative

intensive care unit (ICU) beds, and specialized medical instrumentation under emergency disruption risks.

1.2.3 Cloud Computing and Resource Allocation

Hyper-scale cloud providers (such as Google Cloud, Microsoft Azure, and Amazon Web Services) host millions of virtual machines and containers on physical server racks across globally distributed data centers:

- **VM Placement and Consolidation**: Mapping customer workloads to physical servers to minimize total electricity and cooling power while avoiding CPU, RAM, and disk I/O bottlenecks.
- **High-Availability Constraints**: Preventing correlated failures by ensuring that primary and backup replicas of critical services are never hosted on the same physical server rack, power strip, or availability zone.

1.2.4 Network Traffic Engineering

Modern Internet backbones and Software-Defined Networks (SDN) deploy *Traffic Engineering (TE)* to route petabytes of data traffic across mesh topologies:

- Given a network topology graph $G = (V, E)$ with link capacities c_e and an origin-destination traffic matrix demand, compute packet paths (using Segment Routing or MPLS tunnels) that avoid link congestion, minimize end-to-end latency, and reserve failover capacity for link outages.

1.3 Tackling NP-Hard Problems in Practice

Recognizing that a problem is **NP**-hard is the beginning of the algorithm designer's task, not the end of it. **NP-hardness does not mean that a problem cannot be solved.**

Theoretical complexity analysis is inherently pessimistic: it characterizes the asymptotic computational effort required to solve the *pathological worst-case instance* among all conceivable inputs. However:

1. Real-world instances rarely resemble worst-case adversarial constructions. They exhibit geographic clustering, sparse interactions, metric triangle inequalities, and structured demand distributions.
2. Small- and medium-sized instances of **NP**-hard problems can routinely be solved to **proven mathematical optimality** using exact methods such as Branch-and-Bound, Branch-and-Cut, Dynamic Programming, and Constraint Programming.
3. Massive industrial instances with millions of variables can be solved to within fractions of a percent of optimality within minutes using decompositions, relaxations, and Large Neighborhood Search heuristics.

Example 1.1 *Exact Solvability at Scale: The Concorde TSP Solver*

A striking demonstration of exact optimization in action is the celebrated *Concorde TSP Solver*, developed by David Applegate, Robert Bixby, Vašek Chvátal, and William Cook [App+06].
Although the Traveling Salesperson Problem on n vertices has $(n-1)!/2$ possible tours—an astronomically vast search space even for modest n—Concorde routinely

solves instances of several hundred to a few thousand cities (such as the standard TSPLIB benchmarks with 1,000 to 2,000 cities) to **proven mathematical optimality in seconds or minutes** on an ordinary workstation.
Even more remarkably, Concorde holds the world record for the largest TSP instance solved to proven mathematical optimality: the benchmark instance `pla85900`, comprising **85,900 cities** derived from a programmable logic array circuit layout at Bell Laboratories.
Concorde achieves this through a state-of-the-art *Branch-and-Cut* framework: tightly coupling linear programming relaxations, iterative separation of facet-defining cutting planes (subtour elimination constraints, comb inequalities, and clique trees), primal metaheuristics (the Chained Lin-Kernighan algorithm) to uncover near-optimal upper bounds early, and branch-and-bound search. This achievement demonstrates that **NP**-hardness is not an insurmountable barrier: with deep polyhedral insight and algorithmic engineering, exact optimization routinely conquers instances of immense practical scale.

1.4 The Diagnostic Compass

When confronted with a new discrete optimization problem in academic research or industrial practice, the seasoned practitioner does not immediately open a solver or write code. Instead, they orient their algorithmic strategy by answering two fundamental questions:

The Two Diagnostic Questions of Discrete Optimization

1. **What size is your problem?**
   - How many discrete decision variables are involved? (Tens? Thousands? Millions?)
   - What is the planning horizon and constraint density? (A single warehouse with 10 vehicles, or a continental supply chain with 50,000 stops?)
2. **How much time do you give me to solve it?**
   - *Real-Time Online Operation*: Decisions must be returned in milliseconds or a few seconds (e.g., dynamic dispatch in ride-hailing, real-time packet rerouting).
   - *Operational Short-Turnaround*: Solutions needed within 1 to 15 minutes (e.g., flight gate re-assignment during an airport storm, morning truck dispatch).
   - *Tactical and Strategic Offline Planning*: Overnight or multi-hour computational budget (e.g., monthly pilot rostering, annual warehouse location).

The answers to these two questions determine which algorithmic paradigm—or combination of paradigms—should be deployed:

1.5 The Art of Practical Modeling

Solving a discrete optimization problem is not merely a mechanical application of algorithms; it is an **engineering art**. A sophisticated algorithm running on a poorly conceived model will fail, whereas a clean, insightful formulation can solve in seconds on standard solvers.

Table 1.1: Algorithmic Paradigm Selection Matrix based on Problem Scale and Time Budget.

Problem Scale	Generous Time Budget (Hours)	Tight Time Budget (Seconds to Minutes)
Small	**Exact Solvers**: Mixed-Integer Programming (MIP), Constraint Programming (CP), Branch-and-Bound. *Result: Provably optimal solution.*	**Dynamic Programming / Fast MIP**: Table lookup, prefix evaluation, or fast tree search with early cutoff. *Result: Optimal or near-optimal.*
Medium	**Decompositions**: Lagrangian Relaxation, Column Generation, Branch-and-Cut. *Result: Proven optimality or* $< 0.1\%$ *optimality gap.*	**Constraint Propagation + Heuristics**: CP with custom variable selection, restricted tree search. *Result: High-quality feasible solutions.*
Massive	**Hybrid Matheuristics**: Large Neighborhood Search (LNS) combined with MIP/CP subproblem repair. *Result: Deeply optimized solution with strong dual bounds.*	**Metaheuristics**: Tabu Search, Simulated Annealing, greedy randomized adaptive search (GRASP). *Result: Fast, robust heuristic solution.*

1.5.1 High-Leverage Decision Variables

The foremost task of the modeler is to discern which decisions provide high leverage and business value:

- Should we model the exact microsecond when a truck engine ignites, or simply the sequence of delivery stops and time windows?
- Should a railway dispatching model schedule every individual switch point and signal light, or model blocks of track between mainline stations?

There is an inevitable tension between **model fidelity** (capturing every operational nuance) and **computational tractability** (ensuring the model can be solved within the allotted time). The art of modeling lies in abstracting secondary details while rigorously preserving the constraints that dictate system feasibility.

1.5.2 Optimality with Imperfect Data

In academic theory, an algorithm designer strives to prove mathematical optimality down to an optimality gap of 0.00%. In real-world industrial practice, however, **obsessing over exact optimality on imperfect data is an illusion of precision**.

Consider two illustrative industrial realities:

1. **Industrial Manufacturing and Project Scheduling**: A mathematical model treats task durations as fixed integers (e.g., task A takes exactly 45 minutes). In a physical factory, however, task durations vary due to worker fatigue, micro-stoppages, machine calibration drifts, and raw material variations. A mathematically "optimal" schedule that packs jobs with zero buffer time will collapse the moment a task runs 3 minutes late. A slightly sub-optimal schedule with robust time buffers is practically superior to an optimal brittle schedule.
2. **Traffic Engineering in Computer Networks**: Network routing optimization computes paths assuming that the traffic demand between each origin and destination is known and summarized in a traffic matrix. In practice, a traffic matrix is only an *estimation* extrapolated from historical telemetry counters (such as NetFlow or IPFIX

samples). Actual network traffic fluctuates stochastically with video streaming bursts, software downloads, and unexpected link outages. Spending eight hours to find a routing scheme that is 0.05% better on yesterday's estimated traffic matrix is useless if today's traffic shifts by 10%.

When input data contains 5% uncertainty, spending hours of computational effort to close the final 0.1% optimality gap is an ineffective use of resources. What matters in practice is:

- **Robustness**: Solutions remain feasible and high-performing under modest real-world perturbations.
- **Speed of Response**: Delivering actionable solutions within operational decision cycles.
- **Transparent Constraints**: Providing operators with clear explanations of bottlenecks and trade-offs.

1.6 Formal Problem Formulation

Formally, a discrete optimization problem is defined as follows:

Definition 1.1 *Combinatorial Optimization Problem*

A *combinatorial optimization problem* is a tuple $(\mathcal{S}, \mathcal{X}, f)$, where:

- $\mathcal{S}$ is the configuration space, typically $\mathcal{S} \subseteq \mathbb{Z}^n$ or $\mathcal{S} \subseteq \{0,1\}^n$,
- $\mathcal{X} \subseteq \mathcal{S}$ is the set of *feasible solutions* satisfying a finite collection of constraints $g_j(\mathbf{x}) \leq 0$ for $j \in \{1, \ldots, m\}$,
- $f : \mathcal{S} \to \mathbb{R}$ is the *objective function*.

The goal is to find an optimal solution $\mathbf{x}^* \in \mathcal{X}$ such that:

$$f(\mathbf{x}^*) = \min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \quad \text{(for minimization)}, \quad \text{or} \quad f(\mathbf{x}^*) = \max_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \quad \text{(for maximization)}. \tag{1.1}$$

- If $\mathcal{X} = \emptyset$, the problem is **infeasible** (no configuration satisfies all constraints simultaneously).
- If there exist feasible solutions $\mathbf{x} \in \mathcal{X}$ with $f(\mathbf{x}) \to -\infty$ (for minimization), the problem is **unbounded**.
- Any configuration $\mathbf{x} \in \mathcal{X}$ is a **feasible solution**; its objective value $f(\mathbf{x})$ provides a valid *primal bound* on the global optimum $f(\mathbf{x}^*)$.

Theorem 1.1 *Combinatorial Explosion of Binary Spaces*

Let $\mathcal{S} = \{0,1\}^n$ be the space of binary decisions on n independent choices. The cardinality of the search space is:

$$|\mathcal{S}| = 2^n. \tag{1.2}$$

Adding a single variable doubles the size of the search space. For $n = 100$, $|\mathcal{S}| = 2^{100} \approx 1.27 \times 10^{30}$, a number exceeding the estimated number of grains of sand on Earth.

1.7 A Taxonomy of Algorithmic Paradigms

To navigate this combinatorial explosion, this book covers the foundational pillars of modern discrete optimization:

1. **Dynamic Programming (Chapter 2)**: Recursive decomposition into overlapping subproblems governed by Bellman's principle of optimality, DAG representations, and generic solver architectures.
2. **Branch and Bound (Chapter 3)**: Systematic state-space search over decision trees, employing mathematical relaxations and dominance bounds to prune non-promising subtrees.
3. **Linear Programming (Chapter 4)**: Continuous relaxations, polyhedral combinatorics, extreme point geometry, duality theory, and Simplex pivoting algorithms.
4. **Integer Linear Programming (Chapter 5)**: Polyhedral integrality, total unimodularity, LP-based branch-and-bound with dual simplex warm-starting, and cutting planes.
5. **Lagrangian Relaxation (Chapter 6)**: Decomposition of hard problems into tractable components by pricing out complicating constraints via subgradient optimization.
6. **Network Flows (Chapter 7)**: Maximum flow, minimum cut duality, Ford–Fulkerson, capacity scaling, Edmonds–Karp, and minimum-cost flows.
7. **Local Search & Metaheuristics (Chapter 8)**: Neighborhood structure exploration, trajectory methods (Simulated Annealing, Tabu Search), and Large Neighborhood Search (LNS) for rapid discovery of near-optimal solutions in massive spaces.
8. **Constraint Programming (Chapter 9)**: Declarative reasoning over finite variable domains, consistency notions, filtering algorithms for global constraints, and propagation engines.
9. **Benchmarking Optimization Algorithms (Chapter 10)**: Scientific methodology for empirical performance evaluation, shifted geometric means, Dolan-Moré performance profiles, and standard problem repositories.

1.8 Generic Solvers and Practical Projects

A defining philosophy of this course and textbook is that discrete optimization cannot be truly mastered solely through mathematical equations on paper or through one-off, hardcoded scripts. To develop the algorithmic intuition and problem-solving skills of an expert, one must bridge rigorous theoretical concepts with clean, robust, and extensible software architectures.

1.8.1 Generic Optimization Engines

In introductory computer science, dynamic programming or tree search algorithms are frequently presented as rigid, problem-specific routines—such as a hardcoded 2D array and nested loops for the 0/1 Knapsack problem. While such implementations demonstrate the recurrence relation, they fail to provide a reusable foundation for tackling new, unfamiliar combinatorial challenges.

In industrial operations research and advanced software engineering, practitioners do not reinvent search algorithms from scratch for each new application. Instead, they design and utilize **generic solver frameworks** governed by clean declarative abstractions:

- **Generic Dynamic Programming**: Decoupling the problem topology (states, transitions, rewards) from the search engine (memoization, topological traversal, and solution reconstruction). The same generic engine that solves Knapsack can solve the Traveling Salesperson Problem or shortest path problems without modifying a single line of search logic.
- **Generic Branch-and-Bound Engines**: Providing reusable search tree explorations, customizable node selection strategies, priority queues, and dual bounding interfaces independent of the concrete problem formulation.
- **Constraint Propagation Engines**: Implementing variable domain representations, dependency graphs, and event-driven filtering loops that execute independently of specific global constraints.
- **Metaheuristic and Local Search Architectures**: Designing modular neighborhood operators, accept/reject evaluators, and trajectory managers (e.g., Tabu Search and Large Neighborhood Search).

Throughout this textbook, every major algorithmic paradigm is presented alongside its generic software architecture, demonstrating how mathematical principles translate into elegant, production-grade tools.

1.8.2 Hands-On Programming Projects

To ground these theoretical abstractions in practice, the reader is invited to complete and extend some of these generic optimization tools across **six structured programming projects**:

1. **Project 1: Dynamic Programming**: Completing the core search and solution reconstruction engine (`getValueForState`, `getSolution`, `rebuildSolution`) in the generic parameterized DP framework (`Model<S>`, `State`, `Transition<S>`) in package `dynamicprogramming`. Validating the engine on 0/1 Knapsack, and then modeling and solving the Traveling Salesperson Problem (Held-Karp algorithm) using bitset state encodings (`TSPState`).
2. **Project 2: Branch-and-Bound and Lagrangian Relaxation**: Tackling the TSP with tree search and mathematical relaxations in package `branchandbound`. Implementing a simple 1-tree lower bound (`SimpleOneTree`), constructing a Branch-and-Bound solver (`BranchAndBoundTSP`), and deriving an enhanced dual bound reinforced by Lagrangian relaxation (`HeldKarpOneTree`) using subgradient optimization to penalize vertex degree violations.
3. **Project 3: Linear Programming, Simplex, and Maximum Flows**: Formulating combinatorial problems into standard linear programs ($\max \boldsymbol{cx}$ s.t. $\boldsymbol{Ax} \leq \boldsymbol{b}, \boldsymbol{x} \geq 0$) in package `linearprogramming`. Generating constraint matrices for Maximum Flow (`FlowMatrices`) and Maximum Bipartite Matching (`MatchingMatrices`), and implementing the Big-M Simplex method (`BigMSimplex`) to handle initial infeasibilities via lexicographic single-phase optimization, evaluated on Stigler's Diet problem.
4. **Project 4: Local Search and Invariants**: Developing a local search metaheuristic in `localsearch` for the Pigment Sequencing Problem (PSP), a discrete lot-sizing problem with sequence-dependent changeovers and deadlines. Implementing $\mathcal{O}(1)$ incremental cost invariants (`StockingCostInvariant`, `ChangeoverCostInvariant`), feasibility checks, restarts, and swap neighborhood exploration.
5. **Project 5: Constraint Programming**: Building core engine and filtering com-

ponents for the `TinyCSP` solver in package `constraintprogramming`. Implementing domain reduction operations (`removeAbove`, `removeBelow`) and bound-consistent filtering algorithms for global constraints (`Sum` and `LessOrEqual`), and modeling complex combinatorial puzzles such as the Magic Square and Killer Sudoku.

6. **Project 6: Dynamic Programming with A^* Search**: Unifying dynamic programming state spaces with heuristic search algorithms. Formulating a DP model for the Traveling Salesperson Problem and implementing admissible heuristic evaluation functions $h(s)$ inspired by Branch-and-Bound bounds. Implementing and benchmarking A^* search variants in package `astar`: standard A^*, Weighted A^* ($w_h \in \{1.25, 1.5, 2.0\}$) to trade bounded sub-optimality for search speed, and Anytime Weighted A^* to iteratively deliver and refine solutions under tight computational time limits.

All project starter code, automated unit test suites, benchmark problem instances, and validation harnesses are openly available in the accompanying GitHub repository:

`https://github.com/pschaus/linfo2266-students`

The internal mechanisms of these codebases are explained progressively throughout the chapters, directly accompanying the theoretical development of each topic.

1.8.3 Language Selection: Why Java?

A natural question arises: why is **Java** chosen as the primary programming language for this book and its projects, rather than C++ or Python?

In commercial operations research and high-performance solver development, low-level languages such as **C and C++** have historically been dominant. Commercial solvers (such as Gurobi, CPLEX, or Concorde) are predominantly implemented in C or C++ to exploit direct pointer manipulation, manual memory management, and bespoke cache layouts.

However, for a Master's level curriculum and for building deep algorithmic intuition, C++ presents significant pedagogical drawbacks:

- **Excessive Cognitive Overhead**: Complex template metaprogramming, manual memory management, dangling pointers, segmentation faults, and platform-dependent build configurations (Make, CMake, linker errors) divert the student's valuable cognitive energy away from discrete optimization principles toward systems-level debugging. The bar is unnecessarily high, making it harder to compile, debug, and rapidly experiment with algorithms.
- **Fragile Compilation and Toolchains**: Ensuring consistent compilation across macOS, Linux, and Windows environments when linking external C++ libraries can quickly become an unproductive hurdle.

At the opposite extreme, dynamically typed scripting languages such as **Python** offer rapid prototyping, but suffer from critical shortcomings for solver engineering:

- **Execution Speed**: The interpreted nature of Python incurs a performance penalty of $20\times$ to $100\times$ compared to compiled languages, making it impractical to execute deep tree searches with millions of nodes or high-frequency constraint propagation within acceptable time budgets.
- **Lack of Strict Static Typing**: Discrete optimization engines manipulate intricate, interconnected state graphs. Static typing is indispensable for defining clear contracts (such as `Model<S extends State>`) and catching conceptual and lifecycle bugs at compile time.

Java strikes the ideal balance between performance, safety, and readability:

1. **Strong Static Typing with Generics**: Java's robust type system and generic polymorphism enable clean, expressive abstractions for states, decisions, and models, allowing students to design generic solvers with ironclad compile-time safety.
2. **Sweet Spot Between Execution Speed and Readability**: Modern Java Virtual Machines (JVM) feature sophisticated HotSpot Just-In-Time (JIT) compilers, delivering execution throughput that is frequently within a factor of 1.5× to 2× of optimized C++ for memory-bound combinatorial algorithms, while maintaining clean, legible code.
3. **Frictionless Developer Experience**: Automatic garbage collection eliminates the danger of dangling pointers and memory leaks during complex tree traversals. Furthermore, Java guarantees seamless cross-platform portability, first-class IDE tooling, straightforward compilation, informative stack traces, and standardized unit testing with JUnit.

This pedagogical and engineering rationale mirrors the design philosophy of the seminal textbook *Algorithms* by Robert Sedgewick and Kevin Wayne [SW11], which famously adopted Java to teach foundational data structures and graph algorithms. By using Java, we ensure that students and practitioners spend their time thinking about state spaces, bounding functions, pruning theorems, and filtering logic—while still constructing production-grade tools that run with genuine industrial speed.

1.9 Exercises

Exercise 1.1 *Knapsack Formulation*

Formulate the 0/1 Knapsack problem with items $I = \{0, \ldots, n-1\}$, values $v_i > 0$, weights $w_i > 0$, and capacity $C > 0$ as a discrete optimization problem according to Definition 1.1. Explicitly specify the configuration space $\mathcal{S}$, the constraint defining $\mathcal{X}$, and the objective function f.

Exercise 1.2 *Diagnostic Compass for Industrial Problems*

For each of the following industrial scenarios, answer the two diagnostic questions (problem size and available time budget) and recommend an appropriate algorithmic paradigm from Table 1.1:

1. A ride-hailing platform matching 50 passengers to 50 nearby drivers every 3 seconds.
2. An airline creating a monthly crew schedule for 3,000 pilots across 15,000 flight legs over a weekend run.
3. An e-commerce warehouse routing 12 automated guided vehicles (AGVs) to pick 200 customer orders before a 2:00 PM shipping cutoff.

CHAPTER 2

Dynamic Programming

Dynamic Programming (DP) is one of the foundational algorithmic paradigms in discrete optimization, pioneered by Richard Bellman in the 1950s [Bel57]. It provides a systematic method for solving complex optimization problems by decomposing them into a collection of overlapping subproblems, solving each subproblem once, and storing their solutions in a table or cache to avoid redundant work.

Before diving into the mechanics of dynamic programming, it is crucial to understand the computational complexity of the problems we aim to solve. We begin with a rigorous reminder of the **NP**-completeness of the 0/1 Knapsack problem.

2.1 Computational Complexity of Knapsack

In discrete optimization, the *0/1 Knapsack Problem* is formulated as follows: given a set of n items $I = \{0, \dots, n-1\}$, where each item i has an integer profit $v_i > 0$ and an integer weight $w_i > 0$, and a knapsack with maximum weight capacity $C \in \mathbb{Z}^+$, select a subset of items to maximize the total profit without exceeding the capacity:

$$\begin{aligned} \text{maximize} \quad & \sum_{i \in I} v_i x_i \\ \text{subject to} \quad & \sum_{i \in I} w_i x_i \leq C \\ & x_i \in \{0,1\}, \quad \forall i \in I. \end{aligned} \tag{2.1}$$

To analyze its computational complexity from the perspective of complexity theory [GJ79], we consider its decision version:

Definition 2.1 *0/1 Knapsack Decision Problem (KNAPSACK)*

Given item values $v_0, \dots, v_{n-1} \in \mathbb{Z}^+$, weights $w_0, \dots, w_{n-1} \in \mathbb{Z}^+$, capacity $C \in \mathbb{Z}^+$, and a target profit threshold $K \in \mathbb{Z}^+$, does there exist a binary assignment $\mathbf{x} \in \{0,1\}^n$ such that:

$$\sum_{i=0}^{n-1} w_i x_i \leq C \quad \text{and} \quad \sum_{i=0}^{n-1} v_i x_i \geq K \ ? \tag{2.2}$$

2.1.1 The Polynomial-Time Reduction Technique

To prove that a decision problem B is **NP**-complete, one must follow the standard reduction methodology [Kar72]:

1. **Membership in NP**: Show that $B \in \mathbf{NP}$ by providing a polynomial-time verifier that takes an instance and a candidate certificate (witness) and checks its validity in polynomial time.
2. **Hardness by Reduction**: Choose a known **NP**-complete problem A and construct a function $f : \text{Instances}(A) \to \text{Instances}(B)$ such that:
   - f is computable in polynomial time in the size of the instance of A ($f \in \mathbf{P}$).
   - f preserves the answer: an instance x of A is a YES-instance if and only if $f(x)$ is a YES-instance of B:
   $$x \in A \iff f(x) \in B. \tag{2.3}$$

Definition 2.2 *The Subset-Sum Problem (SUBSET-SUM)*

Given a multiset of positive integers $A = \{a_0, a_1, \ldots, a_{n-1}\}$ and a target integer $T \in \mathbb{Z}^+$, does there exist a subset of indices $S \subseteq \{0, \ldots, n-1\}$ such that:

$$\sum_{i \in S} a_i = T \; ? \tag{2.4}$$

The Subset-Sum problem is one of Karp's 21 classic **NP**-complete problems [Kar72].

Theorem 2.1 *NP-Completeness of 0/1 Knapsack*

The Knapsack decision problem is **NP**-complete.

Proof. First, we show that Knapsack $\in \mathbf{NP}$. Given a candidate solution $\mathbf{x} = (x_0, \ldots, x_{n-1}) \in \{0,1\}^n$, a verifier computes $\sum_{i=0}^{n-1} w_i x_i$ and $\sum_{i=0}^{n-1} v_i x_i$, and checks whether the total weight is $\leq C$ and the total profit is $\geq K$. This requires $\mathcal{O}(n)$ additions and comparisons, which is polynomial in the input size. Hence, Knapsack $\in \mathbf{NP}$.

Second, we reduce Subset-Sum to Knapsack (Subset-Sum $\leq_P$ Knapsack). Let (A, T) be an arbitrary instance of Subset-Sum, where $A = \{a_0, \ldots, a_{n-1}\}$ and $T \in \mathbb{Z}^+$. We construct an instance of Knapsack as follows:

- We create n items, with item i having weight $w_i = a_i$ and value $v_i = a_i$ for all $i \in \{0, \ldots, n-1\}$.
- We set the knapsack capacity $C = T$.
- We set the target value threshold $K = T$.

This transformation requires copying n integers and setting $C = K = T$, which takes $\mathcal{O}(n)$ time.

We now establish correctness ($x \in$ Subset-Sum $\iff f(x) \in$ Knapsack):

- ($\Longrightarrow$) Suppose (A, T) is a YES-instance. Then there exists a subset $S \subseteq \{0, \ldots, n-1\}$ such that $\sum_{i \in S} a_i = T$. Setting $x_i = 1$ for $i \in S$ and $x_i = 0$ otherwise yields:
  $$\sum_{i=0}^{n-1} w_i x_i = \sum_{i \in S} a_i = T \leq C \quad \text{and} \quad \sum_{i=0}^{n-1} v_i x_i = \sum_{i \in S} a_i = T \geq K. \tag{2.5}$$
  Thus, the constructed knapsack instance is a YES-instance.

- ($\Longleftarrow$) Conversely, suppose the knapsack instance is a YES-instance. Then there exists $\mathbf{x} \in \{0,1\}^n$ such that $\sum_{i=0}^{n-1} w_i x_i \leq C = T$ and $\sum_{i=0}^{n-1} v_i x_i \geq K = T$. Because $w_i = v_i = a_i$ for each item:

$$T \leq \sum_{i=0}^{n-1} v_i x_i = \sum_{i=0}^{n-1} a_i x_i = \sum_{i=0}^{n-1} w_i x_i \leq T. \tag{2.6}$$

  Squeezing between T and T forces $\sum_{i=0}^{n-1} a_i x_i = T$. The set $S = \{i \mid x_i = 1\}$ is therefore a valid subset summing to T, proving (A, T) is a YES-instance.

Since the reduction is polynomial and sound, Knapsack is **NP**-complete. □

2.2 Bellman Recurrence for Knapsack

Since Knapsack is **NP**-hard, exhaustive search must evaluate all 2^n possible subsets of items. For $n = 50$, $2^{50} \approx 1.12 \times 10^{15}$ candidate solutions; testing one solution every microsecond would require more than 35,000 years.

Dynamic programming avoids this combinatorial explosion by formulating the problem recursively. Let the items be indexed $0, 1, \ldots, n-1$.

Definition 2.3 *Knapsack Subproblem State*

Let $O(j, k)$ denote the optimal (maximum) profit that can be obtained by selecting a subset from the prefix of items $\{0, 1, \ldots, j\}$ subject to a maximum capacity of k:

$$O(j,k) = \max_{\mathbf{x} \in \{0,1\}^{j+1}} \left\{ \sum_{i=0}^{j} v_i x_i \;\middle|\; \sum_{i=0}^{j} w_i x_i \leq k \right\}. \tag{2.7}$$

Our ultimate goal is to determine $O(n-1, C)$.

To compute $O(j, k)$, we consider the binary decision on item j:

1. **Do not select item** j ($x_j = 0$): The available capacity remains k, and the best achievable profit using the remaining items $\{0, \ldots, j-1\}$ is $O(j-1, k)$.
2. **Select item** j ($x_j = 1$): This decision is feasible only if $w_j \leq k$. If chosen, it consumes w_j units of capacity and provides profit v_j. The remaining capacity is $k - w_j$, and the optimal profit on the remaining items $\{0, \ldots, j-1\}$ is $O(j-1, k-w_j)$, yielding total profit $v_j + O(j-1, k-w_j)$.

Bellman's Principle of Optimality dictates that we simply take the maximum over these two options.

Theorem 2.2 *Bellman Recurrence for 0/1 Knapsack*

For any item index $j \in \{0, \ldots, n-1\}$ and capacity $k \in \{0, \ldots, C\}$:

$$O(j,k) = \begin{cases} 0 & \text{if } j < 0 \text{ or } k \leq 0, \\ O(j-1,k) & \text{if } w_j > k, \\ \max\Big(O(j-1,k),\ v_j + O(j-1,k-w_j)\Big) & \text{if } w_j \leq k. \end{cases} \tag{2.8}$$

2.3 Naive Recursive Implementation

Directly translating the recurrence relation (2.8) into Java yields the recursive procedure shown in Listing 2.1.

```java
public class KnapsackNaive {
    int[] value;
    int[] weight;
    int capa;
    int calls = 0; // counter to track recursive invocations

    public KnapsackNaive(int[] value, int[] weight,
                         int capa) {
        this.value = value;
        this.weight = weight;
        this.capa = capa;
    }

    public int solve() {
        return opt(value.length - 1, capa);
    }

    /**
     * Best value on items [0..j] with residual capacity k.
     */
    private int opt(int j, int k) {
        calls++;
        if (j < 0) {
            return 0;
        } else if (weight[j] > k) {
            return opt(j - 1, k);
        } else {
            int optWithout = opt(j - 1, k);
            int optWith = value[j]
                    + opt(j - 1, k - weight[j]);
            return Math.max(optWithout, optWith);
        }
    }
}
```

Listing 2.1: Naive recursive implementation of 0/1 Knapsack in Java (`KnapsackNaive.java`).

2.3.1 Complexity Analysis of the Naive Recursion

Let $T(j)$ denote the number of calls required to compute $\texttt{opt}(j, k)$ in the worst case (when $w_i \le k$ for all i):

$$T(j) = 2\,T(j-1) + \mathcal{O}(1), \quad T(0) = 1. \tag{2.9}$$

Solving this recurrence reveals that the recursion tree is a full binary tree of depth n:

$$T(n) = \sum_{\ell=0}^{n} 2^{\ell} = 2^{n+1} - 1 = \Theta(2^n). \tag{2.10}$$

The naive recursion re-computes identical subproblems an exponential number of times.

2.4 Memoization and Overlapping States

Why does the naive recursion take exponential time? The reason is *overlapping subproblems*.

Consider the following concrete instance:

$$\begin{aligned} \text{Values } v &= [1, 6, 18, 22, 28] \\ \text{Weights } w &= [2, 3, 5, 6, 7] \\ \text{Capacity } C &= 11. \end{aligned} \tag{2.11}$$

Let us observe two different decision paths from the root state ($j = 4, k = 11$):

- Path A: Do not take item 4, take item 3 ($w_3 = 6$), and do not take item 2 ($w_2 = 5$). The consumed weight is 6, leaving residual capacity $11 - 6 = 5$. The subproblem to solve is now $\texttt{opt}(1, 5)$.
- Path B: Do not take item 4, do not take item 3, but take item 2 ($w_2 = 5$). The consumed weight is again 5, leaving residual capacity $11 - 5 = 6$. Looking deeper, choosing items 0 and 1 ($w_0 + w_1 = 2 + 3 = 5$) also leaves residual capacity $11 - 5 = 6$.

Whenever two execution branches arrive at the same item index j with the **exact same remaining capacity** k, the subtrees rooted at those calls are identical. Solving them more than once is completely redundant.

2.4.1 Memoization with a Hash Map

The core idea of dynamic programming is **memoization**: storing the result of $\texttt{opt}(j, k)$ the first time it is computed, and looking it up in $\mathcal{O}(1)$ time whenever it is encountered again.

Listing 2.2 shows the implementation using a Java `HashMap` where the key is a pair (j, k).

```
1  import java.util.HashMap;
2  import util.Pair;
3
4  public class KnapsackHash {
5      int[] value;
6      int[] weight;
7      int capa;
8      HashMap<Pair<Integer, Integer>, Integer> cache;
9
10     public KnapsackHash(int[] value, int[] weight,
11                         int capa) {
12         this.value = value;
13         this.weight = weight;
14         this.capa = capa;
15         this.cache = new HashMap<>();
16     }
17
18     public int solve() {
19         return optCache(value.length - 1, capa);
20     }
21
22     private int optCache(int j, int k) {
23         Pair<Integer, Integer> entry = new Pair<>(j, k);
24         if (cache.containsKey(entry)) {
25             return cache.get(entry); // cache hit: O(1)
26         }
27         int res = opt(j, k);
28         cache.put(entry, res);       // cache store
29         return res;
30     }
31
32     private int opt(int j, int k) {
33         if (j < 0) return 0;
```

```
34          if (weight[j] > k) return optCache(j - 1, k);
35          int optWithout = optCache(j - 1, k);
36          int optWith = value[j]
37                  + optCache(j - 1, k - weight[j]);
38          return Math.max(optWithout, optWith);
39      }
40  }
```

Listing 2.2: Memoized Knapsack using a Hash Map (`KnapsackHash.java`).

Top-Down Sparsity: Reachable vs. Theoretical States. On our 5-item instance with $C = 11$, the Cartesian product of subproblem coordinates contains $(11 + 1) \times 5 = 60$ possible state pairs (j, k). However, top-down memoization visits and caches **only 30 distinct states**—evaluating exactly half of the theoretical state space by exploring only subproblems reachable along valid item selection paths, while completely bypassing the exponential recursion of the naive algorithm.

2.5 Tabulation with a 2D Array

While a `HashMap` provides an elegant top-down memoization strategy, it incurs object allocation overhead, auto-boxing, and hash collision costs. Because the state space is bounded by integer intervals $j \in \{0, \ldots, n-1\}$ and $k \in \{0, \ldots, C\}$, we can store the subproblem values in a simple **2D primitive integer array**:

$$\texttt{table}[k][i] \quad \text{where } k \in \{0, \ldots, C\},\ i \in \{0, \ldots, n-1\}. \tag{2.12}$$

Here, $\texttt{table}[k][i]$ stores the optimal profit achievable using a subset of items from $\{0, \ldots, i\}$ with maximum capacity k.

2.5.1 Iterative Algorithm (Bottom-Up)

Instead of recursion, we iteratively fill the table in topological order: from smaller item subsets to larger ones, and from smaller capacities to larger ones.

Algorithm 1: 0/1 Knapsack via 2D Array Tabulation

Input: Item values $v[0 \ldots n-1]$, weights $w[0 \ldots n-1]$, knapsack capacity C
Output: Optimal profit

Allocate 2D array $\texttt{table}[0 \ldots C][0 \ldots n-1] \leftarrow 0$;
`// Base case: item 0`
for $k \leftarrow w[0]$ **to** C **do**
  $\texttt{table}[k][0] \leftarrow v[0]$;
`// Fill table item by item`
for $i \leftarrow 1$ **to** $n-1$ **do**
  for $k \leftarrow 0$ **to** $w[i]-1$ **do**
    $\texttt{table}[k][i] \leftarrow \texttt{table}[k][i-1]$;
  for $k \leftarrow w[i]$ **to** C **do**
    $\texttt{optWithout} \leftarrow \texttt{table}[k][i-1]$;
    $\texttt{optWith} \leftarrow v[i] + \texttt{table}[k-w[i]][i-1]$;
    $\texttt{table}[k][i] \leftarrow \max(\texttt{optWithout}, \texttt{optWith})$;
return $\texttt{table}[C][n-1]$;

The Java implementation is given in Listing 2.3.

```java
public class KnapsackTable {
    int[] value;
    int[] weight;
    int capa;
    int[][] table;

    public KnapsackTable(int[] value, int[] weight,
                         int capa) {
        this.value = value;
        this.weight = weight;
        this.capa = capa;
        // table[k][i]: best value on items [0..i], capa k
        this.table = new int[capa + 1][value.length];
    }

    public int solve() {
        // Base case: item 0
        for (int k = weight[0]; k <= capa; k++) {
            table[k][0] = value[0];
        }
        // Fill table for items 1 to n-1
        for (int i = 1; i < value.length; i++) {
            for (int k = 0; k < weight[i]; k++) {
                table[k][i] = table[k][i - 1];
            }
            for (int k = weight[i]; k <= capa; k++) {
                int optWithout = table[k][i - 1];
                int optWith = value[i]
                        + table[k - weight[i]][i - 1];
                table[k][i] = Math.max(optWith, optWithout);
            }
        }
        return table[capa][value.length - 1];
    }

    /**
     * Reconstructs optimal subset by backtracking.
     * @return boolean array; true indicates item selected
     */
    public boolean[] reconstructSolution() {
        boolean[] selected = new boolean[value.length];
        int k = capa;
        for (int i = value.length - 1; i > 0; i--) {
            // Differing value means item i was taken
            if (table[k][i] != table[k][i - 1]) {
                selected[i] = true;
                k -= weight[i];
            }
        }
        // Base case: item 0
        if (k >= weight[0] && table[k][0] > 0) {
            selected[0] = true;
        }
        return selected;
    }
}
```

Listing 2.3: Iterative 2D table implementation of 0/1 Knapsack (`KnapsackTable.java`).

2.5.2 Subproblem Dependencies

A fundamental observation about the 2D dynamic programming grid is that the computation of any cell $\texttt{table}[k][i]$ depends *exclusively* on values in the immediately preceding column $i-1$. Specifically, every entry has **at most two incoming candidate edges**:

1. **The Horizontal Transition (Exclude Item i)**: If item i is not selected ($x_i = 0$), the knapsack capacity remains k and no value is added:

$$\texttt{table}[k][i-1] \xrightarrow{\text{reward } 0} \texttt{table}[k][i]. \tag{2.13}$$

2. **The Diagonal Transition (Include Item i)**: If item i is selected ($x_i = 1$), which is feasible only when $k \geq w_i$, item i consumes w_i units of capacity and contributes profit v_i:

$$\texttt{table}[k-w_i][i-1] \xrightarrow{\text{reward } +v_i} \texttt{table}[k][i]. \tag{2.14}$$

The stored value $\texttt{table}[k][i]$ is simply the maximum between these two incoming transitions. When the capacity is insufficient to accommodate item i ($k < w_i$), the diagonal edge is infeasible, and the cell trivially inherits its value from the horizontal predecessor $\texttt{table}[k][i-1]$.

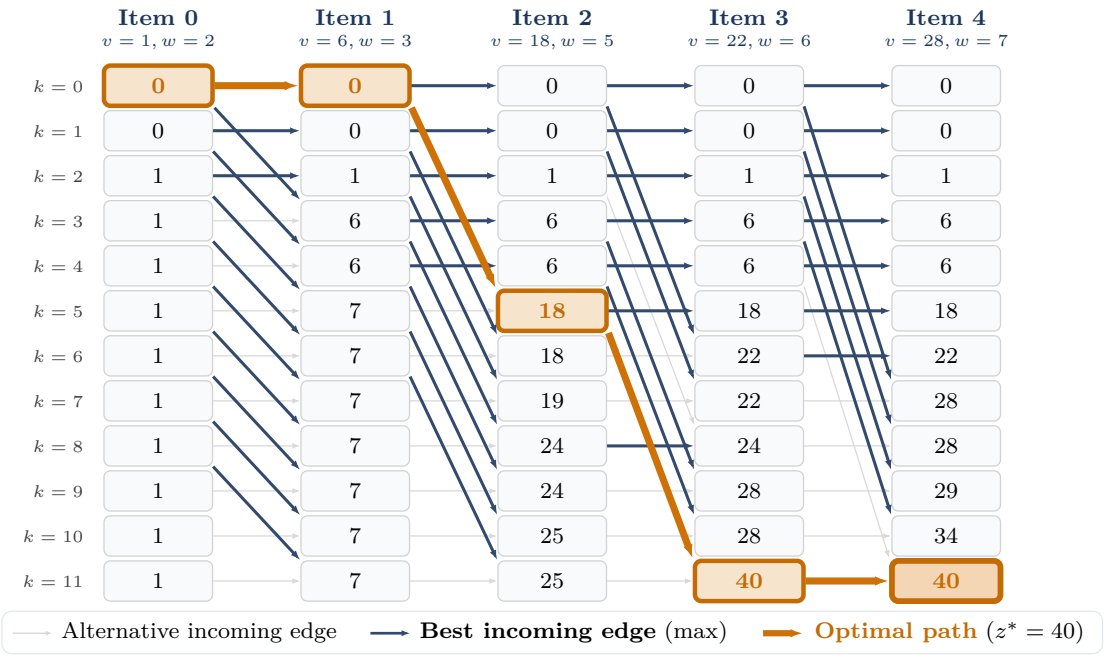


Figure 2.1: Complete 0/1 Knapsack dynamic programming table displaying all subproblem transitions across all 12×5 states. Faint gray arrows represent alternative candidate transitions; bold navy arrows indicate the **best incoming edge** (max) that determines each cell's value; the prominent amber trajectory highlights the **optimal solution path** originating at $(0, 0)$ and culminating at the optimal profit $\texttt{table}[11][4] = \mathbf{40}$.

Figure 2.1 visualizes this complete transition structure:

- **Faint gray arrows**: Transitions evaluated that did not achieve the maximum.
- **Bold navy arrows**: Winning transitions determining cell values (e.g., cell $(11, 3)$ selects the diagonal arrow from $(5, 2)$ since $18 + 22 = 40 > 25$).
- **Amber trajectory**: Tracing the winning arrows backwards from $\texttt{table}[11][4]$ identifies the exact **optimal path** and packed items.

2.5.3 The Tabulation Matrix Illustrated

On our running instance ($v = [1, 6, 18, 22, 28]$, $w = [2, 3, 5, 6, 7]$, $C = 11$), Table 2.1 displays the completed numerical values of the DP table. The optimal profit of **40** is found at the bottom-right corner `table`$[11][4]$.

Table 2.1: Dynamic Programming Table `table`$[k][i]$ for $C = 11$. Bold values with asterisks (*) mark the subproblems visited along the optimal solution path.

Capacity k	**Item 0** $(1, 2)$	**Item 1** $(6, 3)$	**Item 2** $(18, 5)$	**Item 3** $(22, 6)$	**Item 4** $(28, 7)$
0	**0***	**0***	0	0	0
1	0	0	0	0	0
2	1	1	1	1	1
3	1	6	6	6	6
4	1	6	6	6	6
5	1	7	**18***	18	18
6	1	7	18	22	22
7	1	7	19	22	28
8	1	7	24	24	28
9	1	7	24	28	29
10	1	7	25	28	34
11	1	7	25	**40***	**40***

2.5.4 Solution Reconstruction via Backtracking

While the matrix entry `table`$[C][n-1]$ computes the optimal numerical objective value ($z^* = 40$), practical applications require the actual binary decision vector:

$$\boldsymbol{x}^* = [x_0^*, x_1^*, \ldots, x_{n-1}^*] \in \{0, 1\}^n. \tag{2.15}$$

Because the table was filled in forward topological order, the optimal decisions can be recovered in reverse order by **backtracking** from cell $(C, n-1)$ back to stage 0, as formalized in Algorithm 2.

Algorithm 2: Knapsack Solution Reconstruction via Backtracking

Input: Completed DP table `table`$[0 \ldots C][0 \ldots n-1]$, values v, weights w, capacity C
Output: Binary decision vector $\boldsymbol{x} \in \{0, 1\}^n$

Allocate array $x[0 \ldots n-1] \leftarrow 0$;
$k \leftarrow C$;
for $i \leftarrow n-1$ **to** 1 ***step*** -1 **do**
  if *table*$[k][i] \neq$ *table*$[k][i-1]$ **then**
    $x[i] \leftarrow 1$; // Item i was selected
    $k \leftarrow k - w[i]$; // Deduct capacity consumed by item i
  else
    $x[i] \leftarrow 0$; // Item i was skipped
// Base case: evaluate item 0
if $k \geq w[0]$ ***and*** *table*$[k][0] > 0$ **then**
  $x[0] \leftarrow 1$;
return $\boldsymbol{x}$;

Backtracking Walkthrough

Applying Algorithm 2 to our running example reveals the exact sequence of optimal decisions:

1. **Start at** $(k = 11, i = 4)$: We compare $\texttt{table}[11][4] = 40$ with $\texttt{table}[11][3] = 40$. Because the values are identical, item 4 was **not** needed to achieve profit 40. We set $x_4^* = 0$, keep capacity $k = 11$, and step to $(11, 3)$.
2. **At** $(k = 11, i = 3)$: We compare $\texttt{table}[11][3] = 40$ with $\texttt{table}[11][2] = 25$. Because $40 \neq 25$, the value *must* have originated from including item 3! We set $x_3^* = 1$, deduct weight $w_3 = 6$ $(k \leftarrow 11 - 6 = 5)$, and step to $(5, 2)$.
3. **At** $(k = 5, i = 2)$: We compare $\texttt{table}[5][2] = 18$ with $\texttt{table}[5][1] = 7$. Because $18 \neq 7$, item 2 was selected! We set $x_2^* = 1$, deduct weight $w_2 = 5$ $(k \leftarrow 5 - 5 = 0)$, and step to $(0, 1)$.
4. **At** $(k = 0, i = 1)$: Here $k = 0 < w_1 = 3$. Item 1 cannot be packed. We set $x_1^* = 0$ and step to $(0, 0)$.
5. **At** $(k = 0, i = 0)$: Capacity is exhausted $(k = 0 < w_0 = 2)$. We set $x_0^* = 0$.

The reconstructed optimal solution is:

$$\boldsymbol{x}^* = [0, 0, \mathbf{1}, \mathbf{1}, 0], \tag{2.16}$$

selecting items 2 and 3 with total weight $5 + 6 = 11 \leq 11$ and total profit $18 + 22 = \mathbf{40}$.

Complexity of Solution Reconstruction. While computing the full dynamic programming table requires $\Theta(C \cdot n)$ time, reconstructing the optimal solution vector via backtracking requires only n comparisons and subtractions:

$$\text{Reconstruction Time} = \mathcal{O}(n), \qquad \text{Auxiliary Space} = \mathcal{O}(1). \tag{2.17}$$

Solution recovery is computationally negligible compared to the table construction phase.

2.5.5 The Tabulation Grid as a DAG

The observation that each cell in Figure 2.1 has incoming transitions with associated weights leads directly to a unifying theoretical concept: **the dynamic programming grid is a Directed Acyclic Graph (DAG)**, and finding the optimal solution is mathematically equivalent to finding the **Longest Path** in this DAG.

Let us define the knapsack state DAG $G = (V, E)$ explicitly:

- **Vertices** V: Every state (k, i) for $k \in \{0, \ldots, C\}$ and $i \in \{0, \ldots, n-1\}$, augmented with an initial source vertex $(0, \text{Start})$.
- **Edges** E: Directed edges connecting states at stage $i - 1$ to states at stage i:
  - *Skip edge*: $(k, i-1) \to (k, i)$ with edge weight 0 $(x_i = 0)$.
  - *Take edge*: $(k - w_i, i - 1) \to (k, i)$ with edge weight $+v_i$ $(x_i = 1$, valid for $k \geq w_i)$.

Because all directed edges point strictly from stage $i - 1$ to stage i, the graph cannot contain cycles: it is strictly a *layered acyclic graph*.

Notice that the dynamic programming recurrence (2.8) can be rewritten directly in graph-theoretic terms:

$$\texttt{table}[k][i] = \max_{(u,v) \in E,\, v=(k,i)} \left(\texttt{table}[u] + \text{weight}(u, v)\right). \tag{2.18}$$

Notice that this is *precisely* the Bellman-Ford recurrence for computing the **longest path in a Directed Acyclic Graph (DAG)** using topological ordering! Every path from the source $(0, \text{Start})$ to a terminal state represents a feasible item selection, its cumulative edge weight is the total profit, and the dynamic programming solution is simply the maximum-weight trajectory through the grid.

Keep This in Mind for Our Generic DP Solver. We invite the reader to remember this observation for later. In Section 2.8, we will formalize how *every* discrete dynamic program reduces to a shortest or longest path problem on a DAG. Furthermore, when we design our **generic Dynamic Programming solver framework in Java** in Section 2.9, we will capitalize directly on this graph abstraction: by viewing states as graph vertices and decisions as weighted transitions, we will construct an elegant, reusable solver engine that solves DP problems without requiring hand-crafted table dimensions or hardcoded loops.

2.6 Complexity and Pseudo-Polynomial Time

The 2D table contains $(C + 1) \times n$ cells. Computing the value in each cell requires $\mathcal{O}(1)$ operations (one table lookup, one addition, and one comparison). Therefore:

$$\text{Time Complexity} = \Theta(C \cdot n), \qquad \text{Space Complexity} = \Theta(C \cdot n). \tag{2.19}$$

2.6.1 Is this Algorithm Polynomial?

At first glance, $\Theta(C \cdot n)$ might appear polynomial because it is the product of C and n. **However, it is not polynomial in the theoretical sense.**

In computational complexity theory, the running time of an algorithm is measured as a function of the **length of the input**, denoted $|I|$, defined as the number of bits required to encode the instance:

- The number of items n requires $\mathcal{O}(\log n)$ bits.
- The weights and profits require $\mathcal{O}\left(\sum \log w_i + \sum \log v_i\right)$ bits.
- The integer capacity C is encoded in binary using:

$$L = \lceil \log_2(C + 1) \rceil \quad \text{bits.} \tag{2.20}$$

Notice that the numerical value C is exponentially related to its bit length L:

$$C \approx 2^L. \tag{2.21}$$

Expressing the runtime $\Theta(C \cdot n)$ in terms of the input bit-length L reveals:

$$\text{Runtime} = \Theta(n \cdot 2^L). \tag{2.22}$$

The algorithm is **exponential in the length of the input representing** C!

Definition 2.4 *Pseudo-Polynomial Time*

An algorithm for a problem whose inputs include numerical values is said to run in *pseudo-polynomial time* if its execution time is polynomial in the number of input elements n and the **numeric magnitudes** of the numbers involved, but exponential in the number of bits required to represent them.

2.6.2 Weak vs. Strong NP-Hardness

This distinction partitions **NP**-hard problems into two classes:

- **Weakly NP-Hard**: Problems (such as 0/1 Knapsack and Subset-Sum) that admit a pseudo-polynomial time algorithm. When data numbers are bounded by a polynomial in n (or small in practice, e.g., $C \leq 10^5$), dynamic programming solves them efficiently.
- **Strongly NP-Hard**: Problems (such as the Traveling Salesperson Problem, Vertex Cover, and Bin Packing) that remain **NP**-hard even if all numbers in the input are bounded by a polynomial in n. Strongly **NP**-hard problems cannot admit pseudo-polynomial time algorithms unless $\mathbf{P} = \mathbf{NP}$.

Practical Advice for Knapsack. If the capacity C and weights w_i share a greatest common divisor $g = \gcd(C, w_0, \ldots, w_{n-1}) > 1$, dividing all weights and C by g scales down the table size without altering the optimal solution. When C is exceptionally large (e.g., $C = 10^{12}$), dynamic programming on the profit space or fully polynomial-time approximation schemes (FPTAS) can be employed.

2.7 Case Study: Range Partitioning

To reinforce the dynamic programming methodology, we now examine an optimization problem where the objective is to minimize a maximum bottleneck cost: the *Range Partitioning Problem*.

Definition 2.5 *Range Partitioning Problem*

Given:

1. A sequence $S = [s_0, s_1, \ldots, s_{n-1}]$ of n non-negative numbers,
2. An integer $k \geq 1$,

partition S into at most k contiguous sub-ranges without reordering the elements, such that the **maximum sum over all ranges is minimized**.

Example 2.1 *Range Partitioning Example*

Let $S = [1, 2, 3, 4, 5, 6, 7, 8, 9]$ and $k = 3$. A feasible partition into 3 contiguous ranges is:

$$[1, 2, 3, 4, 5], \quad [6, 7], \quad [8, 9]. \tag{2.23}$$

The sums of these ranges are 15, 13, and 17, respectively. The maximum range sum is $\max(15, 13, 17) = \mathbf{17}$, which is the optimal solution.

2.7.1 Dynamic Programming Formulation

Let $O(i, \ell)$ denote the optimal objective value on the prefix sequence $[s_0, \ldots, s_i]$ using at most ℓ partitions (with $i \in \{0, \ldots, n-1\}$ and $\ell \in \{1, \ldots, k\}$).

To determine $O(i, \ell)$, we make a decision: *where to place the split* between the $(\ell - 1)$-th partition and the final (ℓ-th) partition. Suppose the last partition covers elements from index $j + 1$ to i, namely $[s_{j+1}, \ldots, s_i]$:

- The sum of the last partition is $\sum_{m=j+1}^{i} s_m$.
- The preceding prefix $[s_0, \ldots, s_j]$ is partitioned into at most $\ell - 1$ sub-ranges, yielding optimal bottleneck cost $O(j, \ell - 1)$.

- The bottleneck cost for this choice is $\max\left(O(j, \ell-1), \sum_{m=j+1}^{i} s_m\right)$.

Since we wish to minimize the bottleneck, we minimize over all valid split indices $j \in \{0, \ldots, i-1\}$:

$$O(i, \ell) = \min_{0 \le j \le i-1} \max\left(O(j, \ell-1), \sum_{m=j+1}^{i} s_m\right). \tag{2.24}$$

Base Cases:

1. With only $\ell = 1$ partition:

$$O(i, 1) = \sum_{m=0}^{i} s_m, \quad \forall i \in \{0, \ldots, n-1\}. \tag{2.25}$$

2. For a sequence of length 1 ($i = 0$):

$$O(0, \ell) = s_0, \quad \forall \ell \in \{1, \ldots, k\}. \tag{2.26}$$

The optimal global solution is given by $O(n-1, k)$.

2.7.2 Prefix Sum Precomputation

Computing $\sum_{m=j+1}^{i} s_m$ naively takes $\mathcal{O}(n)$ time. By precomputing a *prefix sum array*:

$$\texttt{prefixSum}[i] = \sum_{m=0}^{i} s_m, \quad \text{with } \texttt{prefixSum}[i] = \texttt{prefixSum}[i-1] + s_i, \tag{2.27}$$

any range sum is obtained in $\mathcal{O}(1)$ time:

$$\sum_{m=j+1}^{i} s_m = \texttt{prefixSum}[i] - \texttt{prefixSum}[j]. \tag{2.28}$$

2.7.3 Java Implementation

The complete Java implementation using 2D tabulation is presented in Listing 2.4.

```
1  import java.util.Arrays;
2
3  class RangePartitioningTable {
4      int[] input;
5      int[] prefixSum;
6      int n;
7      int nPartitions;
8      int[][] cache;
9
10     public RangePartitioningTable(int[] input, int k) {
11         this.input = input;
12         this.n = input.length;
13         this.nPartitions = k;
14         // cache[k][i]: best value for k parts on [0..i]
15         this.cache = new int[k + 1][n];
16
17         // Precompute prefix sums for O(1) interval queries
18         prefixSum = new int[n];
19         prefixSum[0] = input[0];
20         for (int i = 1; i < n; i++) {
```

```
            prefixSum[i] = prefixSum[i - 1] + input[i];
        }
    }

    private int sumInterval(int start, int end) {
        return prefixSum[end] - prefixSum[start - 1];
    }

    int solve() {
        // Base cases
        for (int k = 1; k <= nPartitions; k++) {
            // k partitions on sequence of 1 element
            cache[k][0] = input[0];
        }
        for (int i = 1; i < n; i++) {
            // 1 partition up to index i
            cache[1][i] = cache[1][i - 1] + input[i];
        }

        // Fill table: k in 2..nPartitions, i in 1..n-1
        for (int k = 2; k <= nPartitions; k++) {
            for (int i = 1; i < n; i++) {
                int min = Integer.MAX_VALUE;
                for (int j = 0; j <= i - 1; j++) {
                    int res = Math.max(
                        sumInterval(j + 1, i),
                        cache[k - 1][j]);
                    if (res < min) {
                        min = res;
                    }
                }
                cache[k][i] = min;
            }
        }
        return cache[nPartitions][n - 1];
    }
}
```

Listing 2.4: Range Partitioning with 2D Tabulation (`RangePartitioningTable.java`).

Theorem 2.3 *Complexity of Range Partitioning DP*

The dynamic program fills a table of size $(k+1) \times n$. For each cell (k, i), testing all split points $j \in \{0, \ldots, i-1\}$ takes $\mathcal{O}(i) = \mathcal{O}(n)$ time using prefix sums. Thus, the total time complexity is $\mathcal{O}(k \cdot n^2)$ and the space complexity is $\mathcal{O}(k \cdot n)$.

2.8 DP as Shortest Paths in a DAG

A unifying insight in discrete optimization is that **every discrete dynamic program can be reduced to a shortest or longest path problem in a Directed Acyclic Graph (DAG)**, specifically a *layered acyclic graph*.

Consider the state graph $G = (V, E)$ defined by the dynamic program:

- **Vertices** V: Each node corresponds to a reachable state $s \in \mathcal{S}$ (e.g., (j, k) for Knapsack).
- **Edges** E: A directed edge $(u, v) \in E$ corresponds to a feasible transition from state u to successor state v, with edge weight equal to the transition cost or value.

- **Source**: The root state $r \in V$ representing the unassigned problem (e.g., $(n-1, C)$).
- **Sink**: The terminal base case state $t \in V$ (e.g., $j < 0$ or $k = 0$).

Because decisions progress strictly through stages (e.g., item indices decrease monotonically $j \to j-1$), the graph contains no cycles: it is strictly a DAG.

Theorem 2.4 *DP Equivalence to DAG Paths*

- Any **maximization** DP problem is equivalent to finding a **Longest Path** from the root r to a sink in the state DAG.
- Any **minimization** DP problem is equivalent to finding a **Shortest Path** from the root r to a sink in the state DAG.

Why DAGs are Special. Finding the longest simple path in an arbitrary directed graph is a notorious **NP**-hard problem (it contains the Hamiltonian Path problem as a special case). However, on a **Directed Acyclic Graph**, topological sorting guarantees that longest and shortest paths can be solved in **linear time**:

$$\mathcal{O}(|V| + |E|). \tag{2.29}$$

Dynamic programming is precisely this linear-time traversal over the state DAG!

2.9 A Generic Java Framework for DP

Recognizing that every dynamic program is fundamentally a path search over a state DAG opens up an elegant software engineering opportunity: rather than writing ad-hoc loops and arrays for every new problem, we can build a **reusable, generic Dynamic Programming solver framework**.

We now present a generic framework, located in package `dynamicprogramming`. The framework is composed of four clean abstractions:

1. `State`: The nodes of the DAG.
2. `Transition<S>`: The directed edges of the DAG.
3. `Model<S>`: The declarative specification of the problem (root, transitions, base cases).
4. `DynamicProgramming<S>`: The generic solver engine that performs memoized search and solution reconstruction.

2.9.1 The State Abstraction

Every state must be hashable so that the generic solver can store it in a global hash table. In Java, this means providing implementations of `hashCode()` and `equals()`, as shown in Listing 2.5.

```
1  public abstract class State {
2      abstract int hash();
3      abstract boolean isEqual(State state);
4
5      @Override
6      public int hashCode() {
7          return this.hash();
8      }
9
10     @Override
```

```
    public boolean equals(Object o) {
        if (o instanceof State) {
            return isEqual((State) o);
        }
        return false;
    }
}
```

Listing 2.5: The `State` abstract class (`DynamicProgramming.java`).

2.9.2 The Transition Class

A transition represents a directed edge pointing to a successor state, carrying the decision identifier and the transition value (weight):

```
public class Transition<S extends State> {
    private S successor;
    private int decision;
    private double value;

    public Transition(S successor, int decision,
                      double value) {
        this.successor = successor;
        this.decision  = decision;
        this.value     = value;
    }

    public S getSuccessor()   { return successor; }
    public int getDecision()  { return decision; }
    public double getValue()  { return value; }
}
```

Listing 2.6: The `Transition` class (`DynamicProgramming.java`).

2.9.3 The Model Abstraction

The problem model defines the topology of the DAG: what the root state is, how to generate outgoing transitions, when a base case is reached, and whether the problem is a maximization or minimization:

```
import java.util.List;

public abstract class Model<S extends State> {
    abstract S getRootState();
    abstract boolean isBaseCase(S state);
    abstract double getBaseCaseValue(S state);
    abstract List<Transition<S>> getTransitions(S state);
    abstract boolean isMaximization();
}
```

Listing 2.7: The `Model` abstract class (`DynamicProgramming.java`).

2.9.4 The Generic Solver Engine

The generic solver class `DynamicProgramming<S>` implements top-down memoization using a hash map (`HashMap<State, Double>`) and reconstructs the optimal decision sequence:

```java
import java.util.HashMap;
import java.util.LinkedList;
import java.util.List;
import util.Solution;

public class DynamicProgramming<S extends State> {
    Model<S> model;
    HashMap<State, Double> table; // memoization cache

    public DynamicProgramming(Model<S> model) {
        this.model = model;
        this.table = new HashMap<>();
    }

    public Solution getSolution() {
        getValueForState(model.getRootState());
        return rebuildSolution();
    }

    private double getValueForState(S state) {
        if (model.isBaseCase(state)) {
            return model.getBaseCaseValue(state);
        }
        if (table.containsKey(state)) {
            return table.get(state);
        }

        double best = model.isMaximization()
                ? -Double.MAX_VALUE : Double.MAX_VALUE;
        for (Transition<S> transition :
                model.getTransitions(state)) {
            double succVal = getValueForState(
                    transition.getSuccessor());
            double value = succVal + transition.getValue();
            if (model.isMaximization()) {
                if (value > best) best = value;
            } else {
                if (value < best) best = value;
            }
        }
        table.put(state, best);
        return best;
    }

    private Solution rebuildSolution() {
        S currentState = model.getRootState();
        double best = table.get(currentState);
        List<Integer> decisions = new LinkedList<>();
        double currentValue = best;

        while (!model.isBaseCase(currentState)) {
            for (Transition<S> t :
                    model.getTransitions(currentState)) {
                double succVal = getValueForState(
                        t.getSuccessor());
                double value = succVal + t.getValue();
                if (value == currentValue) {
                    currentState = t.getSuccessor();
                    decisions.add(t.getDecision());
                    currentValue -= t.getValue();
```

```
                        break;
                    }
                }
            }
            return new Solution(best, decisions);
        }
    }
```

Listing 2.8: Core evaluation logic of `DynamicProgramming<S>`.

2.10 Knapsack in the Framework

With the generic solver in place, implementing a new dynamic program requires only two concise classes:

1. A problem-specific state class extending `State`.
2. A problem model class extending `Model<S>`.

2.10.1 The Knapsack State

The state encapsulates the current item index and residual capacity. It computes its hash code using `Objects.hash(item, capacity)`:

```
import java.util.Objects;

public class KnapsackState extends State {
    int item, capacity;

    public KnapsackState(int index, int capacity) {
        this.item = index;
        this.capacity = capacity;
    }

    @Override
    int hash() {
        return Objects.hash(item, capacity);
    }

    @Override
    boolean isEqual(State s) {
        if (s instanceof KnapsackState) {
            KnapsackState other = (KnapsackState) s;
            return this.item == other.item
                && this.capacity == other.capacity;
        }
        return false;
    }
}
```

Listing 2.9: Knapsack State implementation (`KnapsackState.java`).

2.10.2 The Knapsack Model

The `Knapsack` model defines the transitions corresponding to leaving or taking each item:

```
import java.util.LinkedList;
import java.util.List;
```

```java
import util.knapsack.KnapsackInstance;

public class Knapsack extends Model<KnapsackState> {
    KnapsackInstance instance;
    KnapsackState root;

    public Knapsack(KnapsackInstance instance) {
        this.instance = instance;
        this.root = new KnapsackState(0, instance.capacity);
    }

    @Override
    boolean isBaseCase(KnapsackState state) {
        return state.item == instance.n
            || state.capacity == 0;
    }

    @Override
    double getBaseCaseValue(KnapsackState state) {
        return 0.0;
    }

    @Override
    List<Transition<KnapsackState>> getTransitions(
            KnapsackState state) {
        List<Transition<KnapsackState>> transitions =
                new LinkedList<>();

        // Decision 0: Do not take the item
        transitions.add(new Transition<>(
            new KnapsackState(state.item + 1,
                              state.capacity),
            0,
            0.0
        ));

        // Decision 1: Take item if capacity allows
        if (instance.weight[state.item] <= state.capacity) {
            int remCapa = state.capacity
                    - instance.weight[state.item];
            transitions.add(new Transition<>(
                new KnapsackState(state.item + 1, remCapa),
                1,
                instance.value[state.item]
            ));
        }

        return transitions;
    }

    @Override
    KnapsackState getRootState() { return root; }

    @Override
    boolean isMaximization()     { return true; }
}
```

Listing 2.10: Knapsack Model implementation (`Knapsack.java`).

To solve the knapsack instance, the user simply runs:

```java
Knapsack model = new Knapsack(instance);
```

```
DynamicProgramming<KnapsackState> solver =
        new DynamicProgramming<>(model);
Solution solution = solver.getSolution();
```

The generic framework executes top-down memoized traversal, handles base cases, and reconstructs the optimal set of selected items seamlessly.

Student Capstone Project: The Traveling Salesperson Problem. The power of this generic DAG formulation extends far beyond knapsack. In the accompanying laboratory project, you will formulate the *Traveling Salesperson Problem* (Held-Karp algorithm) within this very framework by defining a `TSPState` consisting of the current city and a bitset of visited cities.

2.11 Exercises

Exercise 2.1 *Prefix Sum Boundary Conditions*

In Listing 2.4, method `sumInterval(start, end)` reads `prefixSum[start - 1]`. Explain what happens when `start` = 0. How should the method or prefix array be structured to handle `start` = 0 cleanly without throwing an `ArrayIndexOutOfBoundsException`?

Exercise 2.2 *Memory Optimization for Knapsack DP*

Notice that computing column i in `KnapsackTable.java` requires only values from column $i-1$. Show how to reduce the space complexity of the tabulation approach from $\mathcal{O}(C \cdot n)$ to $\mathcal{O}(C)$ using only two 1D arrays (or a single 1D array traversed backwards).

Exercise 2.3 *Partitioning with Non-Contiguous Ranges*

Show how to modify the 0/1 Knapsack recurrence equation to solve the Subset-Sum problem with boolean values `true`/`false` indicating whether a target sum is achievable.

CHAPTER 3

Branch and Bound

In combinatorial optimization, many fundamental problems—including the 0/1 Knapsack problem, the Traveling Salesperson Problem (TSP), and general Mixed-Integer Linear Programs (MILP)—are **NP**-hard. Exhaustive enumeration of all feasible candidate solutions is computationally prohibitive: a knapsack instance with $n = 60$ items entails $2^{60} \approx 1.15 \times 10^{18}$ configurations, while a symmetric TSP on $n = 30$ cities requires evaluating $\frac{(n-1)!}{2} \approx 4.42 \times 10^{30}$ distinct tours. Even at the rate of one billion evaluations per second, brute-force enumeration would span millions of years.

To solve such problems to provable optimality without inspecting every candidate, we turn to *Branch and Bound* (B&B), an implicit enumeration paradigm pioneered in the foundational work of Ailsa Land and Alison Doig Harcourt [LD60], George Dantzig [Dan57], and John Little et al. [Lit+63]. Rather than examining every individual leaf of the combinatorial search tree, Branch and Bound constructs a hierarchical decomposition of the solution space and systematically prunes entire subtrees whose solutions are mathematically guaranteed to be suboptimal or infeasible.

At the core of Branch and Bound lies an indispensable principle that guides all modern integer programming:

"Combinatorial optimization is the art of relaxing."

When an optimization problem is too constrained or combinatorial to solve directly in polynomial time, we relax one or more difficult constraints. The resulting relaxed subproblem can be solved efficiently, yielding an optimistic bound that proves whether exploring a region of the search space is worthwhile.

3.1 Foundations of Branch and Bound

Consider an abstract discrete optimization problem P_0 formulated as:

$$z^* = \max_{\mathbf{x} \in X} f(\mathbf{x}), \tag{3.1}$$

where $X \subseteq \mathbb{Z}^n \times \mathbb{R}^m$ represents the discrete or mixed-integer feasible region, and $f : X \to \mathbb{R}$ denotes the objective function to be maximized. Branch and Bound solves P_0 by coordinating three algorithmic mechanisms: *branching*, *bounding*, and *fathoming* (pruning).

3.1.1 The Three Core Mechanisms

Definition 3.1 *Branching (Space Partitioning)*

Let $X_s \subseteq X$ represent the feasible domain of a subproblem P_s. *Branching* partitions X_s into $k \geq 2$ non-empty, mutually exclusive (or covering) subsets $X_{s,1}, X_{s,2}, \ldots, X_{s,k}$ such that:

$$X_s = \bigcup_{i=1}^{k} X_{s,i}. \tag{3.2}$$

Consequently, the optimal objective over X_s satisfies:

$$\max_{\mathbf{x} \in X_s} f(\mathbf{x}) = \max_{i \in \{1,\ldots,k\}} \left(\max_{\mathbf{x} \in X_{s,i}} f(\mathbf{x}) \right). \tag{3.3}$$

In binary optimization, branching typically splits a subproblem by selecting an undecided binary variable $x_j \in \{0, 1\}$ and generating two children: one setting $x_j = 1$ and the other setting $x_j = 0$.

Definition 3.2 *Bounding Procedure (Relaxation)*

For any subproblem P_s with feasible domain X_s, a *bounding procedure* computes an optimistic estimate $UB(X_s)$ on the optimal value achievable within X_s:

$$UB(X_s) \geq \max_{\mathbf{x} \in X_s} f(\mathbf{x}). \tag{3.4}$$

In a maximization context, $UB(X_s)$ is an *upper bound*. Symmetrically, for minimization problems, the procedure computes a *lower bound* $LB(X_s) \leq \min_{\mathbf{x} \in X_s} f(\mathbf{x})$.

An optimistic bound is obtained by solving a *relaxation* $\mathcal{R}(P_s)$ of P_s, formed by expanding the feasible space ($X_s \subseteq X_s^{\text{rel}}$) or by replacing the objective function with an over-estimator. For the bound to be useful in practice, $\mathcal{R}(P_s)$ must satisfy two conflicting criteria:

1. **Tractability**: The relaxation must be solvable in polynomial (or pseudo-polynomial) time, typically via linear programming, greedy heuristics, or polynomial-time graph algorithms.
2. **Tightness**: The bound $UB(X_s)$ must be as close as possible to $\max_{\mathbf{x} \in X_s} f(\mathbf{x})$. A loose bound fails to prune unpromising branches, degenerating the search into exhaustive enumeration.

Throughout the search, the algorithm maintains an *incumbent solution* $\mathbf{x}^* \in X$, representing the best feasible solution discovered so far, with objective value:

$$LB^* = f(\mathbf{x}^*). \tag{3.5}$$

Because $\mathbf{x}^*$ is feasible for the original problem P_0, LB^* serves as a global lower bound on the true optimum ($LB^* \leq z^*$). If no feasible solution is known initially, we set $LB^* = -\infty$.

Theorem 3.1 *The Three Fathoming (Pruning) Rules*

During the exploration of the search tree, a subproblem P_s with domain X_s can be immediately *fathomed* (pruned without further branching) if and only if at least one of

the following three conditions holds:

1. **Pruning by Bound (Domination)**:

$$UB(X_s) \leq LB^*. \tag{3.6}$$

Since no candidate solution in X_s can yield an objective value strictly exceeding $UB(X_s)$, X_s cannot contain a solution strictly better than the current incumbent $\mathbf{x}^*$.

2. **Pruning by Infeasibility**:

$$X_s^{\text{rel}} = \emptyset \implies X_s = \emptyset. \tag{3.7}$$

If the relaxed problem admits no feasible solution, the restricted original subproblem is guaranteed to be empty.

3. **Pruning by Optimality (Integrality)**: Let $\mathbf{x}_s^*$ be an optimal solution to the relaxation $\mathcal{R}(P_s)$. If $\mathbf{x}_s^*$ satisfies all constraints of the original problem ($\mathbf{x}_s^* \in X_s$), then $\mathbf{x}_s^*$ is optimal for P_s, and $f(\mathbf{x}_s^*) = UB(X_s)$. If $f(\mathbf{x}_s^*) > LB^*$, we update the incumbent:

$$\mathbf{x}^* \leftarrow \mathbf{x}_s^*, \quad LB^* \leftarrow f(\mathbf{x}_s^*). \tag{3.8}$$

Since the optimum of P_s is fully determined, no further branching on X_s is required.

3.1.2 Maximization versus Minimization Duality

Optimization problems can be formulated interchangeably as maximization or minimization through the fundamental sign-reversal identity:

$$\min_{\mathbf{x}\in X} c(\mathbf{x}) \iff \max_{\mathbf{x}\in X} \big(-c(\mathbf{x})\big). \tag{3.9}$$

While theoretical expositions often adopt a maximization perspective, general-purpose optimization solvers and algorithmic frameworks (including the generic Java solver developed in this chapter) conventionally standardize on *minimization*. The duality between the two paradigms is summarized in Table 3.1. In minimization, relaxations provide *lower bounds* $LB(X_s) \leq c(\mathbf{x})$ for all $\mathbf{x} \in X_s$, while the incumbent provides an *upper bound* $UB^* = c(\mathbf{x}^*)$. A node is pruned by domination whenever $LB(X_s) \geq UB^*$.

Table 3.1: Duality between Maximization and Minimization in Branch and Bound.

Role / Property	**Maximization** ($\max f(\mathbf{x})$)	**Minimization** ($\min c(\mathbf{x})$)
Relaxation bound	$UB(X_s) \geq \max_{\mathbf{x}\in X_s} f(\mathbf{x})$	$LB(X_s) \leq \min_{\mathbf{x}\in X_s} c(\mathbf{x})$
Incumbent value	$LB^* = f(\mathbf{x}^*)$	$UB^* = c(\mathbf{x}^*)$
Initial incumbent	$LB^* = -\infty$	$UB^* = +\infty$
Incumbent update	$LB^* \leftarrow f(\mathbf{x})$ (if $f(\mathbf{x}) > LB^*$)	$UB^* \leftarrow c(\mathbf{x})$ (if $c(\mathbf{x}) < UB^*$)
Fathoming by bound	$UB(X_s) \leq LB^*$	$LB(X_s) \geq UB^*$
Fathoming by optimality	$LB^* \leftarrow f(\mathbf{x}_s^*)$ (if $\mathbf{x}_s^* \in X$)	$UB^* \leftarrow c(\mathbf{x}_s^*)$ (if $\mathbf{x}_s^* \in X$)

3.2 Branch and Bound for 0/1 Knapsack

We now illustrate the mechanics of Branch and Bound on the 0/1 Knapsack problem. Recall from Chapter 2 that given n items with values $v_0, \dots, v_{n-1} \in \mathbb{Z}^+$, weights $w_0, \dots, w_{n-1} \in \mathbb{Z}^+$, and a total knapsack capacity $C \in \mathbb{Z}^+$, the problem is formulated as:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=0}^{n-1} v_i x_i \\ \text{subject to} \quad & \sum_{i=0}^{n-1} w_i x_i \leq C, \\ & x_i \in \{0,1\}, \quad \forall i \in \{0,\dots,n-1\}. \end{aligned} \tag{3.10}$$

3.2.1 Binary Search Tree Representation

We construct a binary search tree by deciding the inclusion of items sequentially. At depth $k \in \{0,\dots,n-1\}$, the search branches on variable x_k:

- **Left Branch** ($x_k = 1$): The item is included in the knapsack. This decision is feasible only if the remaining knapsack capacity satisfies $C_{\text{residual}} \geq w_k$. If included, accumulated value increases by v_k and residual capacity becomes $C_{\text{residual}} - w_k$.
- **Right Branch** ($x_k = 0$): The item is excluded. The accumulated value and residual capacity remain unchanged.

A node in this tree is defined by the decision prefix $(x_0, \dots, x_{k-1})$, the accumulated value $V_{\text{fixed}} = \sum_{i=0}^{k-1} v_i x_i$, and the residual capacity $C_{\text{residual}} = C - \sum_{i=0}^{k-1} w_i x_i$. Unconstrained brute-force search would expand $2^{n+1} - 1$ nodes. To prune this tree effectively, we examine two candidate bounding procedures.

3.2.2 Bounding Procedure 1: Capacity Relaxation

The simplest relaxation consists in dropping the knapsack capacity constraint entirely for all undecided items $\{k,\dots,n-1\}$:

$$\begin{aligned} \text{maximize} \quad & V_{\text{fixed}} + \sum_{i=k}^{n-1} v_i x_i \\ \text{subject to} \quad & x_i \in \{0,1\}, \quad \forall i \in \{k,\dots,n-1\}. \end{aligned} \tag{3.11}$$

Since capacity is unconstrained, the optimal assignment trivially sets $x_i = 1$ for all remaining items:

$$UB_{\text{capa}} = V_{\text{fixed}} + \sum_{i=k}^{n-1} v_i. \tag{3.12}$$

While UB_{capa} can be computed in $\mathcal{O}(1)$ time after an $\mathcal{O}(n)$ prefix-sum precomputation, it is extremely loose because it completely ignores weights. It only prunes branches when the sum of *all* remaining item values cannot beat the incumbent.

3.2.3 Bounding 2: Continuous LP Relaxation

A vastly superior bound is obtained by preserving the capacity constraint while relaxing the integrality restriction $x_i \in \{0,1\}$ to the continuous interval $x_i \in [0,1]$:

$$\begin{aligned} \text{maximize} \quad & V_{\text{fixed}} + \sum_{i=k}^{n-1} v_i x_i \\ \text{subject to} \quad & \sum_{i=k}^{n-1} w_i x_i \le C_{\text{residual}}, \\ & 0 \le x_i \le 1, \quad \forall i \in \{k, \dots, n-1\}. \end{aligned} \tag{3.13}$$

This continuous knapsack problem was solved by George Dantzig [Dan57] via a greedy efficiency rule.

Definition 3.3 *Value-to-Weight Ratio (Item Efficiency)*

For each item i, its *efficiency* r_i is defined as the ratio of its value to its weight:

$$r_i = \frac{v_i}{w_i}. \tag{3.14}$$

Without loss of generality, we assume the items are sorted in non-increasing order of efficiency:

$$\frac{v_0}{w_0} \ge \frac{v_1}{w_1} \ge \cdots \ge \frac{v_{n-1}}{w_{n-1}}. \tag{3.15}$$

Definition 3.4 *Critical (Split) Item*

Given items sorted by non-increasing efficiency and an available capacity C_{residual}, let j denote the index of the *critical item* (or split item), defined as the first item that cannot be added in its entirety without violating capacity:

$$j = \min \left\{ i \in \{k, \dots, n-1\} \;\middle|\; \sum_{\ell=k}^{i} w_\ell > C_{\text{residual}} \right\}. \tag{3.16}$$

If $\sum_{\ell=k}^{n-1} w_\ell \le C_{\text{residual}}$, all remaining items fit, and no critical item exists (all $x_i = 1$).

Theorem 3.2 *Optimality of the Greedy Linear Relaxation*

Suppose the undecided items $\{k, \dots, n-1\}$ are sorted by non-increasing efficiency $r_i = v_i/w_i$. Let j be the critical item defined in Definition 3.4. The optimal solution $\mathbf{x}^*$ to the continuous knapsack problem (3.13) is given by:

$$x_i^* = \begin{cases} 1 & \text{for } k \le i < j, \\ \dfrac{C_{\text{residual}} - \sum_{\ell=k}^{j-1} w_\ell}{w_j} & \text{for } i = j, \\ 0 & \text{for } j < i \le n-1. \end{cases} \tag{3.17}$$

The corresponding continuous upper bound is:

$$UB_{\text{LP}} = V_{\text{fixed}} + \sum_{i=k}^{j-1} v_i + x_j^* \cdot v_j. \tag{3.18}$$

Proof. Suppose for contradiction that another feasible solution $\mathbf{x}'$ exists with strictly higher objective value: $\sum_{i=k}^{n-1} v_i x_i' > \sum_{i=k}^{n-1} v_i x_i^*$. Since $\mathbf{x}' \neq \mathbf{x}^*$, there must exist at least one index $a < j$ where $x_a' < x_a^* = 1$ and at least one index $b > j$ where $x_b' > x_b^* = 0$ (or a fractional imbalance at j).

Consider the difference vector $\boldsymbol{\delta} = \mathbf{x}^* - \mathbf{x}'$. We can decompose the difference into indices where $\mathbf{x}^*$ takes more than $\mathbf{x}'$ ($i \leq j$, where $\delta_i > 0$) and indices where $\mathbf{x}^*$ takes less than $\mathbf{x}'$ ($i \geq j$, where $\delta_i < 0$). Because both solutions are feasible and $\mathbf{x}^*$ exhausts the capacity:

$$\sum_{i \leq j} w_i \delta_i \geq \sum_{i \geq j} w_i(-\delta_i) > 0. \tag{3.19}$$

Now consider the net difference in objective value:

$$\sum_{i=k}^{n-1} v_i x_i^* - \sum_{i=k}^{n-1} v_i x_i' = \sum_{i=k}^{n-1} v_i \delta_i = \sum_{i \leq j} \frac{v_i}{w_i} w_i \delta_i - \sum_{i \geq j} \frac{v_i}{w_i} w_i(-\delta_i). \tag{3.20}$$

Because the items are sorted in non-increasing ratio order, $r_i \geq r_j$ for all $i \leq j$, and $r_i \leq r_j$ for all $i \geq j$. Therefore:

$$\sum_{i \leq j} r_i(w_i \delta_i) \geq r_j \sum_{i \leq j} w_i \delta_i \geq r_j \sum_{i \geq j} w_i(-\delta_i) \geq \sum_{i \geq j} r_i(w_i(-\delta_i)). \tag{3.21}$$

Hence, $\sum_{i=k}^{n-1} v_i x_i^* - \sum_{i=k}^{n-1} v_i x_i' \geq 0$, which contradicts the assumption that $\mathbf{x}'$ achieves a strictly higher profit. Thus, $\mathbf{x}^*$ is optimal. □

Example 3.1 *Continuous Relaxation Computation*

Consider a knapsack instance with capacity $C = 14$ and 5 items already sorted by non-increasing efficiency:

Item Index i	0	1	2	3	4
Value v_i	28	22	18	6	1
Weight w_i	7	6	5	2	1
Ratio $r_i = v_i/w_i$	4.00	3.67	3.60	3.00	1.00
Cumulative Weight $\sum_{\ell=0}^{i} w_\ell$	7	13	18	20	21

At the root node ($k = 0$, $C_{\text{residual}} = 14$):

- Item 0 is taken completely ($x_0^* = 1$), consuming $w_0 = 7$; residual capacity becomes $14 - 7 = 7$.
- Item 1 is taken completely ($x_1^* = 1$), consuming $w_1 = 6$; residual capacity becomes $7 - 6 = 1$.
- Item 2 has weight $w_2 = 5 > 1$, making $j = 2$ the critical item. We take fraction $x_2^* = \frac{1}{5}$.
- Items 3 and 4 are excluded: $x_3^* = x_4^* = 0$.

The continuous upper bound is:

$$UB_{\text{LP}} = v_0 + v_1 + \frac{1}{5}v_2 = 28 + 22 + 0.2 \times 18 = 50 + 3.6 = 53.6. \tag{3.22}$$

Since all item values are integers, any valid integer solution must satisfy $\sum v_i x_i \leq \lfloor 53.6 \rfloor = 53$.

3.2.4 Worked Example and Search Tree

To see how Branch and Bound operates in practice, consider the following 0/1 knapsack instance:

$$\begin{aligned} \text{maximize} \quad & 28x_0 + 30x_1 + 20x_2 \\ \text{subject to} \quad & 4x_0 + 6x_1 + 4x_2 \leq 9, \\ & x_0, x_1, x_2 \in \{0, 1\}. \end{aligned} \tag{3.23}$$

The instance parameters and item efficiencies are:

- Item 0: $v_0 = 28$, $w_0 = 4 \implies r_0 = \frac{28}{4} = 7.0$.
- Item 1: $v_1 = 30$, $w_1 = 6 \implies r_1 = \frac{30}{6} = 5.0$.
- Item 2: $v_2 = 20$, $w_2 = 4 \implies r_2 = \frac{20}{4} = 5.0$.

The items are already sorted in non-increasing efficiency order ($7.0 \geq 5.0 \geq 5.0$). We explore the tree using Depth-First Search (DFS), branching on x_0, then x_1, then x_2, prioritizing the inclusion branch ($x_i = 1$) before the exclusion branch ($x_i = 0$). At each node, we track the tuple:

$$(\text{Accumulated Value},\ \text{Residual Capacity},\ \text{Upper Bound } UB,\ \text{Incumbent } LB^*). \tag{3.24}$$

Trace Under LP Relaxation

We now walk through every step of the search tree under the continuous LP relaxation, as depicted in Figure 3.1:

1. **Node 1 (Root)**: No decisions made ($V_{\text{fixed}} = 0, C_{\text{residual}} = 9, LB^* = -\infty$). The LP relaxation takes Item 0 fully ($w_0 = 4, v_0 = 28$), leaving $9 - 4 = 5$ units of capacity. Item 1 has weight $w_1 = 6 > 5$, so it becomes the critical item ($j = 1$). We take fraction $\frac{5}{6}$ of Item 1, yielding fractional profit $\frac{5}{6} \times 30 = 25$. Thus:

   $$UB = 28 + 25 = 53. \tag{3.25}$$

   Since $53 > LB^* = -\infty$, the node is active. We branch on $x_0 = 1$.

2. **Node 2 ($x_0 = 1$)**: Item 0 is included ($V_{\text{fixed}} = 28, C_{\text{residual}} = 9 - 4 = 5, LB^* = -\infty$). The remaining capacity is 5. For the remaining items $\{1, 2\}$, Item 1 is the critical item with fraction $\frac{5}{6}$, giving bound $UB = 28 + \frac{5}{6} \times 30 = 53$. Since $53 > -\infty$, we branch on $x_1 = 1$.

3. **Node 3 ($x_0 = 1, x_1 = 1$)**: Including Item 1 requires $w_1 = 6$, but $C_{\text{residual}} = 5$. The required total weight is $4 + 6 = 10 > 9$. This node violates capacity:

   $$C_{\text{residual}} = 5 - 6 = -1 < 0 \implies \textbf{Pruned by Infeasibility}. \tag{3.26}$$

   The algorithm backtracks to Node 2 and explores the right branch $x_1 = 0$.

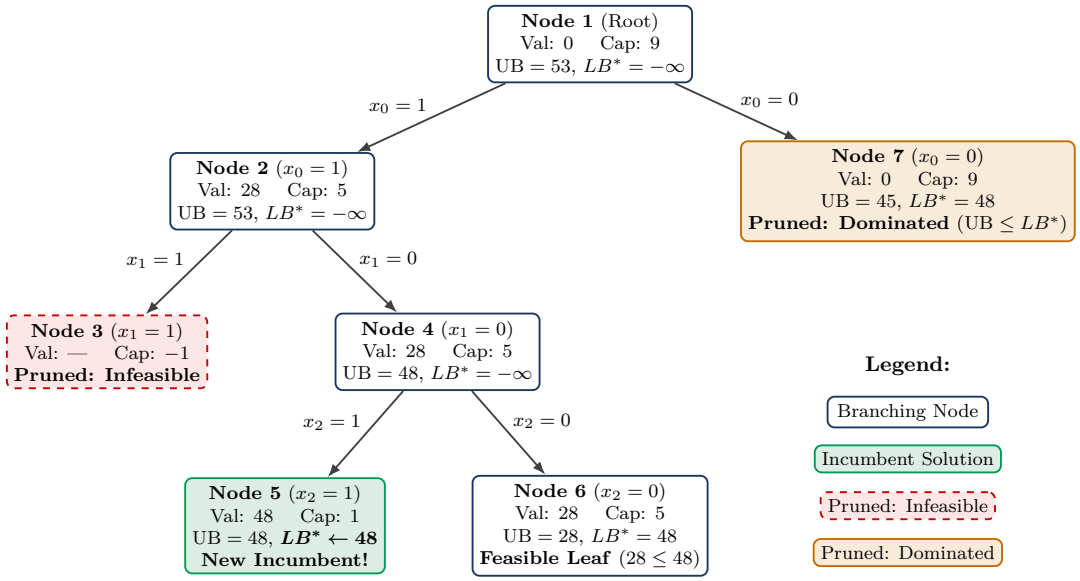


Figure 3.1: Branch and Bound search tree for the 0/1 knapsack instance (3.23) using the continuous LP relaxation. The search explores left branches first ($x_i = 1$). When Node 5 establishes the incumbent $LB^* = 48$, Node 7 on the right branch is evaluated. Its upper bound $UB = 45 \leq 48$ allows pruning the entire right subtree without exploring x_1 and x_2.

4. **Node 4 ($x_0 = 1, x_1 = 0$)**: Item 1 is excluded ($V_{\text{fixed}} = 28, C_{\text{residual}} = 5, LB^* = -\infty$). The only remaining item is Item 2 ($w_2 = 4, v_2 = 20$). Since $w_2 \leq C_{\text{residual}}$, it fits completely. The upper bound is $UB = 28 + 20 = 48$. We branch on $x_2 = 1$.
5. **Node 5 ($x_0 = 1, x_1 = 0, x_2 = 1$)**: Item 2 is included ($V_{\text{fixed}} = 28 + 20 = 48, C_{\text{residual}} = 5 - 4 = 1$). All items have been assigned, so this is a complete, feasible integer solution. We update the incumbent:

$$\mathbf{x}^* = (1, 0, 1), \quad LB^* \leftarrow 48. \tag{3.27}$$

   Being a leaf, the node terminates. The algorithm backtracks to Node 4 and tests $x_2 = 0$.
6. **Node 6 ($x_0 = 1, x_1 = 0, x_2 = 0$)**: Item 2 is excluded ($V_{\text{fixed}} = 28, C_{\text{residual}} = 5$). This leaf represents candidate solution $(1, 0, 0)$ with value 28. Since $28 \leq LB^* = 48$, it does not improve the incumbent. The search backtracks all the way to Root (Node 1).
7. **Node 7 ($x_0 = 0$)**: Item 0 is excluded ($V_{\text{fixed}} = 0, C_{\text{residual}} = 9, LB^* = 48$). We solve the continuous relaxation on undecided items $\{1, 2\}$ with available capacity 9:
   - Item 1 is taken fully ($w_1 = 6, v_1 = 30$), leaving $9 - 6 = 3$ units of capacity.
   - Item 2 has weight $w_2 = 4 > 3$, so it is the critical item. We take fraction $\frac{3}{4}$ of Item 2, contributing $\frac{3}{4} \times 20 = 15$.

   The resulting continuous upper bound is:

$$UB = 30 + 15 = 45. \tag{3.28}$$

   We now evaluate the pruning condition against the incumbent $LB^* = 48$:

$$UB = 45 \leq LB^* = 48 \implies \textbf{Pruned by Bound (Dominated)}! \tag{3.29}$$

   The entire subtree beneath $x_0 = 0$ (which contains candidate solutions $(0, 1, 1)$, $(0, 1, 0)$, $(0, 0, 1)$, and $(0, 0, 0)$) is discarded in a single bounding test.

The search terminates. The optimal solution is $\mathbf{x}^* = (1, 0, 1)$ with optimal profit $z^* = 48$. Exactly **7 nodes** were visited, out of 15 in the full binary tree.

Comparison with Capacity Relaxation

To appreciate the decisive role of bound tightness, let us examine the execution of Branch and Bound on the identical instance using the naive *Capacity Relaxation* (3.12), illustrated in Figure 3.2:

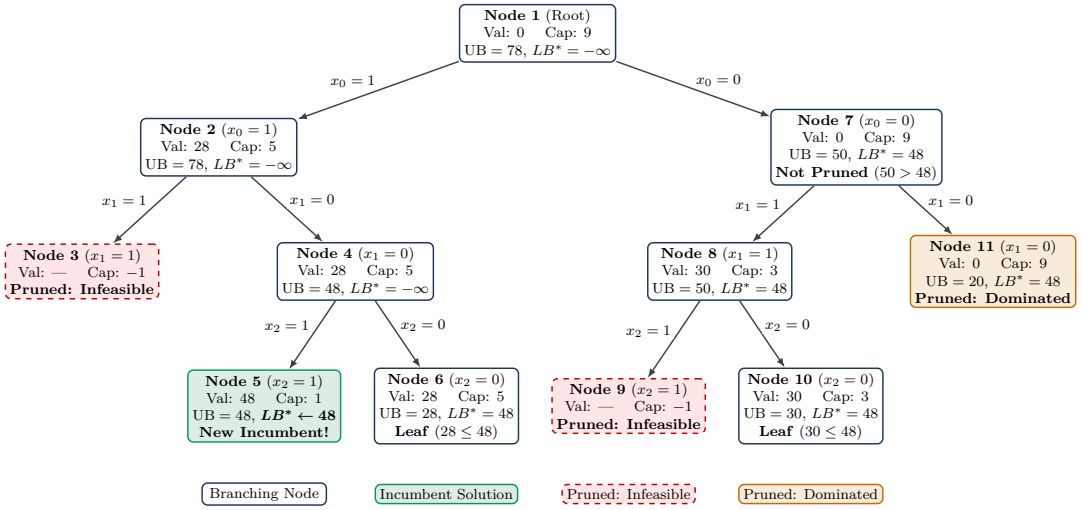


Figure 3.2: Branch and Bound search tree for instance (3.23) using the weak Capacity Relaxation. Because UB_{capa} ignores item weights, Node 7 produces an upper bound of $50 > 48$. Consequently, the branch cannot be pruned, forcing the algorithm to explore 4 additional nodes (Nodes 8, 9, 10, 11).

Under capacity relaxation:

- At Root (Node 1), $UB = 28 + 30 + 20 = 78$.
- At Node 2 ($x_0 = 1$), $UB = 28 + 30 + 20 = 78$.
- At Node 4 ($x_0 = 1, x_1 = 0$), $UB = 28 + 20 = 48$.
- At Node 5, the incumbent is updated to $LB^* = 48$.
- Now consider **Node 7 ($x_0 = 0$)**: The accumulated value is 0, and remaining items are $\{1, 2\}$. The capacity relaxation bound is:

$$UB_{\text{capa}} = 0 + v_1 + v_2 = 30 + 20 = 50. \tag{3.30}$$

  Because $50 > LB^* = 48$, **Node 7 cannot be pruned!**
- The search is forced to expand Node 8 ($x_0 = 0, x_1 = 1$, $UB = 50$), evaluate Node 9 ($x_0 = 0, x_1 = 1, x_2 = 1$, infeasible), evaluate Node 10 ($x_0 = 0, x_1 = 1, x_2 = 0$, leaf with value 30), and expand Node 11 ($x_0 = 0, x_1 = 0$, $UB = 20 \leq 48$, pruned).

A quantitative comparison between the two bounding procedures is summarized in Table 3.2.

The Cost-Benefit Tradeoff of Bounding. Computing the LP relaxation requires sorting items once in $\mathcal{O}(n \log n)$ time and evaluating fractional allocations in $\mathcal{O}(n)$ time per node (or $\mathcal{O}(1)$ incrementally). While slightly more expensive per node than the $\mathcal{O}(1)$ capacity sum, the dramatic reduction in tree size saves exponential amounts of work as n increases. Tight bounds are the lifeblood of efficient Branch and Bound.

Table 3.2: Quantitative comparison of Branch and Bound under Capacity and LP Relaxations on instance (3.23).

Metric	Capacity Relaxation	Continuous LP Relaxation
Root Upper Bound (UB_0)	78	53
Nodes Visited	11	7
Infeasible Nodes Detected	2 (Nodes 3, 9)	1 (Node 3)
Subtrees Pruned by Bound	1 (Node 11)	1 (Node 7)
Search Space Eliminated by Bound	2 leaves (25%)	4 leaves (50%)
Optimal Solution Found	$(1, 0, 1)$, Value 48	$(1, 0, 1)$, Value 48

3.3 Combinatorial Relaxations for the TSP

We now turn from the 0/1 Knapsack problem to one of the most celebrated and historically significant problems in combinatorial optimization: the *Traveling Salesperson Problem* (TSP). The TSP is **NP**-hard and exhibits factorial growth in its naive search space: on a complete graph with n vertices, there are $\frac{(n-1)!}{2}$ distinct undirected tours. Solving the TSP to provable optimality for non-trivial instances requires designing bounding procedures that compute strong, admissible lower bounds by relaxing hard combinatorial constraints.

3.3.1 TSP Relaxations and Formulations

Let $G = (V, E)$ be a complete undirected weighted graph with vertex set $V = \{0, 1, \ldots, n-1\}$, edge set $E = \{\{i, j\} : i, j \in V, i < j\}$, and non-negative symmetric edge costs $c_e \geq 0$ for each $e \in E$. A *tour* (or Hamiltonian cycle) is a simple closed cycle that visits every vertex in V exactly once and returns to the starting vertex. The objective is to find a tour of minimum total cost.

To solve the TSP via Branch and Bound, we require strong, admissible lower bounds that can be evaluated quickly at each search node. Rather than relying on classical algebraic integer linear programming models with an exponential family of subtour elimination constraints (such as the Dantzig–Fulkerson–Johnson formulation [DFJ54], which requires dynamic separation algorithms in branch-and-cut solvers), we characterize a Hamiltonian cycle through three complementary graph-theoretic formulations. Each formulation isolates different topological properties of a tour and immediately suggests natural, polynomial-time combinatorial relaxations to derive lower bounds:

1. **Formulation 1**: Unique connected component and degree-2 regularity, yielding the Assignment (2-factor) and Cheapest Incident Edges relaxations.
2. **Formulation 2**: Spanning tree on V plus one edge, yielding the Minimum Spanning Tree plus Cheapest Edge relaxation.
3. **Formulation 3**: 1-tree rooted at vertex 0 with degree-2 regularity, yielding the celebrated Held–Karp 1-tree relaxation.

Formulation 1: Connected 2-Factor

Rather than specifying an exponential family of subset inequalities, we can characterize a Hamiltonian cycle purely in terms of graph topology. Let $G(x) = (V, E(x))$ denote the spanning subgraph induced by the edge selection vector $\boldsymbol{x} \in \{0, 1\}^{|E|}$, where $E(x) = \{e \in E : x_e = 1\}$.

We state the optimization problem directly on the structural properties of $G(x)$:

$$\begin{aligned} \text{minimize} \quad & \sum_{e \in E} c_e x_e \\ \text{subject to} \quad & \sum_{e \in \delta(i)} x_e = 2, \quad \forall i \in V \quad \text{(degree constraints)}, \\ & \kappa(G(x)) = 1 \quad \text{(unique connected component)}, \\ & x_e \in \{0,1\}, \quad \forall e \in E, \end{aligned} \tag{3.31}$$

where $\kappa(G(x))$ denotes the number of connected components in the subgraph $G(x)$.

Proposition 3.3 *Equivalence of the Connected 2-Regular Subgraph Formulation*

A spanning subgraph $G' = (V, E')$ is a valid Hamiltonian cycle on n vertices if and only if it satisfies:

1. Every vertex has degree exactly 2: $\deg_{G'}(i) = 2$ for all $i \in V$.
2. The subgraph consists of a unique connected component: $\kappa(G') = 1$.

Proof. ($\Rightarrow$) If G' is a Hamiltonian cycle, it is a single simple cycle visiting every vertex in V exactly once. By definition, every vertex has degree 2, and the cycle is path-connected across all n vertices; hence, $\kappa(G') = 1$.

($\Leftarrow$) Suppose G' satisfies both conditions. In any graph where every vertex has degree exactly 2, every connected component must itself be a 2-regular connected graph. The only connected 2-regular graphs are simple cycles. Therefore, G' is a disjoint union of simple cycles (a 2-factor). Because $\kappa(G') = 1$, there exists exactly one connected component, which means G' consists of a single simple cycle. Since G' spans all n vertices of V, this unique cycle has length n, which is by definition a Hamiltonian cycle. □

Formulation (3.31) isolates the two orthogonal dimensions of the TSP:

- **A local condition**: each vertex must satisfy 2-regularity ($\deg(i) = 2$).
- **A global topological condition**: the subgraph must form a single connected component ($\kappa = 1$).

Deriving Relaxations from Formulation 1. By relaxing one or both of these two conditions, Formulation (3.31) directly yields two combinatorial lower bounds:

Relaxation 1A: Minimum Weight Assignment (Bipartite Matching / 2-Factor) Relaxation. What if we drop the global connectivity condition ($\kappa = 1$) while strictly preserving degree-2 regularity across all vertices? In directed terms, every vertex in a tour must have in-degree 1 and out-degree 1. Dropping the unique connected component constraint

yields the classical *Linear Assignment Problem* (AP):

$$\begin{aligned} \text{minimize} \quad & \sum_{i=0}^{n-1} \sum_{j=0, j\neq i}^{n-1} c_{ij} x_{ij} \\ \text{subject to} \quad & \sum_{j=0, j\neq i}^{n-1} x_{ij} = 1, \quad \forall i \in \{0, \ldots, n-1\}, \\ & \sum_{i=0, i\neq j}^{n-1} x_{ij} = 1, \quad \forall j \in \{0, \ldots, n-1\}, \\ & x_{ij} \in \{0, 1\}, \quad \forall i \neq j. \end{aligned} \tag{3.32}$$

The first constraint ensures each city has exactly one successor (out-degree 1), while the second ensures each city has exactly one predecessor (in-degree 1). Setting $c_{ii} = \infty$ explicitly prevents self-loops ($x_{ii} = 0$).

Definition 3.5 *2-Factor (Cycle Cover)*

A *2-factor* (or cycle cover) of a graph G is a spanning subgraph in which every vertex has degree exactly 2. In a directed graph, a 2-factor is a set of vertex-disjoint directed cycles that collectively visit every vertex in V exactly once.

Theorem 3.4 *Assignment Relaxation Bound*

Every valid TSP tour is a 2-factor consisting of a single cycle of length n. Consequently, the optimal solution to the Assignment Problem (3.32) provides an admissible lower bound on the TSP:

$$LB_{\text{AP}} \leq z^*_{\text{TSP}}. \tag{3.33}$$

If the optimal assignment solution forms a single connected cycle, it is a feasible TSP tour and solves the problem to global optimality.

Proof. Any valid TSP tour corresponds to an assignment matrix x_{ij} where each city has exactly one successor and one predecessor, with no subtours. Since the Assignment Problem minimizes total cost over all 2-factors (with or without subtours), its feasible domain strictly contains all TSP tours. Thus, $LB_{\text{AP}} \leq z^*_{\text{TSP}}$. If the solution contains only one cycle, it satisfies global connectivity, making it feasible and optimal for the original TSP. □

- **Constraints Relaxed**: Global connectivity ($\kappa = 1$) is completely dropped. The solution may decompose into multiple disconnected cycles (subtours).
- **Constraints Preserved**: 2-regularity across all vertices is strictly enforced ($\deg(i) = 2$ for all $i \in V$).
- **Algorithmic Complexity**: The Assignment Problem can be solved in $\mathcal{O}(n^3)$ time using the Hungarian algorithm (Kuhn-Munkres [Kuh55]) or the Jonker-Volgenant shortest augmenting path algorithm [JV87].
- **Bound Quality**: Strong. Because degree 2 is enforced at every vertex, LB_{AP} is often close to z^*_{TSP}. However, solving an $\mathcal{O}(n^3)$ matching problem at every search node is computationally intensive.

Example 3.2 *Computing the Assignment (2-Factor) Bound*

Solving the Linear Assignment Problem on our canonical 7-vertex instance (Figure 3.3(c)) enforces in-degree 1 and out-degree 1 at every vertex. The optimal 2-factor decomposes into two disconnected directed subtours:

- Subtour 1 (5 vertices): $0 \to 5 \to 4 \to 6 \to 1 \to 0$ with directed edge costs $1 + 2 + 2 + 3 + 4 = 12$.
- Subtour 2 (2 vertices): $2 \leftrightarrow 3$ (directed 2-cycle $2 \to 3 \to 2$) with directed edge costs $3 + 3 = 6$.

The resulting assignment lower bound is:

$$LB_{\text{AP}} = 12 + 6 = 18. \tag{3.34}$$

Although every vertex strictly satisfies 2-regularity ($\deg(i) = 2$), the solution is not a feasible tour because it violates the unique connected component constraint by forming two disjoint cycles.

Relaxation 1B: Cheapest Incident Edges Relaxation. What if we relax Formulation (3.31) even further by dropping global connectivity AND uncoupling the coordinated degree choices? In an undirected graph, every tour vertex must have degree 2. A very fast lower bound is obtained by independently selecting the two cheapest available incident edges for each vertex $i \in V$ and summing their weights. Because every edge $e = \{u, v\}$ is incident to two endpoints, this sum counts each edge twice:

$$LB_{\text{incident}} = \frac{1}{2} \sum_{i \in V} \min_{\substack{j_1, j_2 \in V \setminus \{i\} \\ j_1 \neq j_2}} (c_{i,j_1} + c_{i,j_2}). \tag{3.35}$$

In directed or asymmetric models where each city must have in-degree 1, the bound simplifies to selecting the single cheapest incoming edge into each vertex:

$$LB_{\text{in}} = \sum_{i \in V} \min_{j \in V \setminus \{i\}} c_{ji}. \tag{3.36}$$

- **Constraints Relaxed**: Global connectivity ($\kappa = 1$) is entirely dropped. Degree-2 regularity is relaxed to independent local edge choices; the union of edges need not form valid cycles.
- **Constraints Preserved**: Local incident coverage (each vertex has at least two incident edge choices, or in-degree at least 1).
- **Algorithmic Complexity**: Scanning each row of the distance matrix requires $\mathcal{O}(n^2)$ time total (or $\mathcal{O}(n)$ incrementally).
- **Bound Quality**: Very weak, but computationally inexpensive.

Formulation 2: Spanning Tree Plus Edge

A second, profoundly influential graph-theoretic characterization models a Hamiltonian cycle through tree connectivity:

$$\begin{aligned} \text{minimize} \quad & \sum_{e \in E} c_e x_e \\ \text{subject to} \quad & \{e \in E : x_e = 1\} \text{ forms a spanning tree on } V \text{ plus exactly 1 edge}, \\ & \sum_{e \in \delta(i)} x_e = 2, \quad \forall i \in V \quad \text{(degree constraints)}, \\ & x_e \in \{0,1\}, \quad \forall e \in E. \end{aligned} \tag{3.37}$$

Proposition 3.5 *Equivalence of the Spanning Tree Formulation*

Every valid TSP tour on n vertices consists of a spanning tree on V plus exactly one additional edge, with every vertex having degree exactly 2.

Proof. Let T be a valid Hamiltonian cycle on n vertices. By definition, T contains exactly n vertices and n edges, and every vertex has degree 2. If we remove any single edge $e^* = \{u, v\}$ from T, the remaining subgraph $T \setminus \{e^*\}$ is a simple path visiting all n vertices. A path on n vertices has $n-1$ edges, is connected, and contains no cycles; hence, $T \setminus \{e^*\}$ is a spanning tree on V. Reinserting e^* yields a spanning tree plus one edge that closes the cycle, where every vertex maintains degree 2. □

Deriving Relaxation from Formulation 2: Minimum Spanning Tree plus Cheapest Edge. What happens if we drop the degree-2 constraint on all vertices in Formulation (3.37), while strictly preserving tree connectivity on V plus one edge?

Definition 3.6 *Spanning Tree plus Cheapest Edge Bound*

The *Minimum Spanning Tree plus Cheapest Edge* relaxation drops the requirement that $\sum_{e \in \delta(i)} x_e = 2$ for all $i \in V$, while retaining the constraint that $\{e : x_e = 1\}$ forms a spanning tree on V plus one additional edge:

$$LB_{\text{MST+1}} = w(\text{MST}(V)) + \min_{e \in E \setminus \text{MST}(V)} c_e, \tag{3.38}$$

where $\text{MST}(V)$ denotes a Minimum Spanning Tree on the complete graph $G = (V, E)$, and $w(\text{MST}(V)) = \sum_{e \in \text{MST}(V)} c_e$.

Theorem 3.6 *Validity of the MST plus Cheapest Edge Bound*

For any TSP instance with non-negative edge costs, $LB_{\text{MST+1}} \leq z^*_{\text{TSP}}$.

Proof. Let T^* be an optimal TSP tour with cost z^*_{TSP}. By Theorem 3.5, T^* can be decomposed into a spanning tree $T_{\text{span}} \subseteq T^*$ on V ($n-1$ edges) and a single remaining edge $e^* \in T^* \setminus T_{\text{span}}$. By definition of the Minimum Spanning Tree, $w(\text{MST}(V)) \leq w(T_{\text{span}})$. Furthermore, e^* is an edge in $E \setminus T_{\text{span}}$, so $\min_{e \notin \text{MST}(V)} c_e \leq c_{e^*}$. Summing these two inequalities yields:

$$LB_{\text{MST+1}} = w(\text{MST}(V)) + \min_{e \notin \text{MST}(V)} c_e \leq w(T_{\text{span}}) + c_{e^*} = z^*_{\text{TSP}}. \tag{3.39}$$

□

- **Constraints Relaxed**: Degree-2 regularity on all vertices is completely relaxed. Vertices in MST(V) can be leaves ($\deg(i) = 1$) or branching hubs ($\deg(i) \geq 3$).
- **Constraints Preserved**: Global connectivity is strictly enforced (the spanning tree connects all n vertices in a single component), and exactly n edges are chosen.
- **Algorithmic Complexity**: An MST on n vertices can be computed in $\mathcal{O}(n^2)$ time via Prim's algorithm. Identifying the cheapest non-tree edge takes $\mathcal{O}(n^2)$ time (or $\mathcal{O}(|E|)$). Total time is strictly $\mathcal{O}(n^2)$.
- **Bound Quality**: Noticeably tighter than cheapest incident edges because global connectivity is preserved. However, relaxing degree 2 everywhere leaves many leaves and hubs.

Example 3.3 *Computing the MST plus Cheapest Edge Bound*

Consider the canonical 7-vertex weighted graph depicted in Figure 3.3 with vertex set $V = \{0, 1, 2, 3, 4, 5, 6\}$ and 12 symmetric edge costs:

$$\begin{aligned} &c_{0,1} = 4, \quad c_{0,5} = 1, \quad c_{0,2} = 5, \quad c_{1,6} = 3, \quad c_{1,5} = 2, \quad c_{6,5} = 6, \\ &c_{6,4} = 2, \quad c_{5,4} = 2, \quad c_{5,2} = 9, \quad c_{4,2} = 1, \quad c_{4,3} = 2, \quad c_{2,3} = 3. \end{aligned} \tag{3.40}$$

To evaluate $LB_{\text{MST+1}}$:

1. The Minimum Spanning Tree on all 7 vertices (MST(V)) selects 6 edges: $\{0, 5\}$ (cost 1), $\{4, 2\}$ (cost 1), $\{1, 5\}$ (cost 2), $\{5, 4\}$ (cost 2), $\{6, 4\}$ (cost 2), and $\{4, 3\}$ (cost 2). The total tree cost is:

$$w(\text{MST}(V)) = 1 + 1 + 2 + 2 + 2 + 2 = 10. \tag{3.41}$$

2. The remaining non-tree edges are $\{0, 1\}$ (cost 4), $\{0, 2\}$ (cost 5), $\{1, 6\}$ (cost 3), $\{6, 5\}$ (cost 6), $\{5, 2\}$ (cost 9), and $\{2, 3\}$ (cost 3). The cheapest non-tree edge is $\{2, 3\}$ with cost 3 (tied with $\{1, 6\}$).

Adding this cheapest edge yields the lower bound:

$$LB_{\text{MST+1}} = 10 + 3 = 13. \tag{3.42}$$

As illustrated in Figure 3.3(a), this relaxed solution leaves leaf vertices 0, 1, and 6 with degree 1, while vertex 4 has degree 4 and vertex 5 has degree 3.

Formulation 3: Rooted 1-Tree

Notice the primary weakness of Formulation 2's relaxation: it relaxes the degree constraint on *every* vertex. In their seminal work, Held and Karp [HK70] observed that we can anchor the tree connectivity by designating a specific root vertex, say vertex $0 \in V$. This yields the *1-tree* formulation:

$$\begin{aligned} \text{minimize} \quad & \sum_{e \in E} c_e x_e \\ \text{subject to} \quad & \{e \in E : x_e = 1\} \text{ forms a 1-tree rooted at vertex 0,} \\ & \sum_{e \in \delta(i)} x_e = 2, \quad \forall i \in V \quad \text{(degree constraints),} \\ & x_e \in \{0, 1\}, \quad \forall e \in E. \end{aligned} \tag{3.43}$$

Definition 3.7 *1-Tree*

Let $G = (V, E)$ be a connected graph and let $0 \in V$ be a designated root vertex. A *1-tree rooted at vertex* 0 consists of:

1. A spanning tree on the subgraph induced by the remaining $n-1$ vertices $V \setminus \{0\} = \{1, 2, \ldots, n-1\}$.
2. Exactly two distinct edges connecting root vertex 0 to the induced subgraph $V \setminus \{0\}$.

Notice that a 1-tree on n vertices contains exactly $(n-2)+2 = n$ edges, is connected, and contains *exactly one simple cycle*, which necessarily passes through root vertex 0.

Proposition 3.7 *Equivalence of the 1-Tree Formulation*

A spanning subgraph $T = (V, E')$ is a valid Hamiltonian cycle on n vertices if and only if T is a 1-tree rooted at vertex 0 and every vertex in V has degree exactly 2.

Proof. ($\Rightarrow$) Let T be a valid Hamiltonian cycle on n vertices. By definition, root vertex 0 has degree exactly 2 in T. Removing vertex 0 and its two incident edges leaves a simple path spanning all remaining $n-1$ vertices $V \setminus \{0\}$. A path on $n-1$ vertices has $n-2$ edges, is connected, and contains no cycles; hence, it is a spanning tree on $V \setminus \{0\}$. Thus, T is by definition a 1-tree rooted at vertex 0, where $\deg(i) = 2$ for all $i \in V$.

($\Leftarrow$) Suppose T is a 1-tree rooted at vertex 0 and satisfies $\deg_T(i) = 2$ for all $i \in V$. A 1-tree on n vertices contains $(n-2)+2 = n$ edges and is connected, so it has a unique connected component ($\kappa(T) = 1$). Since every vertex has degree exactly 2, by Theorem 3.3, T must be a single simple cycle spanning all n vertices—a valid Hamiltonian cycle. □

Deriving Relaxations from Formulation 3. Formulation (3.43) is structurally superior to Formulation (3.37) because the degree-2 constraint at root vertex 0 ($\deg(0) = 2$) is *built directly into the topological definition of the 1-tree*!

Relaxation 3: The Minimum Weight 1-Tree (Held–Karp) Relaxation. What happens if we strictly enforce the 1-tree structure (guaranteeing tree connectivity on $V \setminus \{0\}$ and $\deg(0) = 2$) while relaxing the degree-2 constraints on the remaining $n-1$ vertices $V \setminus \{0\}$?

Theorem 3.8 *The 1-Tree Bound on the TSP*

Every valid TSP tour is a 1-tree in which every vertex has degree exactly 2. Consequently, the minimum weight 1-tree provides an admissible lower bound on the optimal TSP tour cost:

$$LB_{\text{1-tree}} = w(\text{MST}(V \setminus \{0\})) + c_{0,j_1} + c_{0,j_2} \leq z^*_{\text{TSP}}, \tag{3.44}$$

where $j_1, j_2 \in V \setminus \{0\}$ are the two distinct vertices with the smallest edge costs to root vertex 0.

Proof. Let T be any valid TSP tour. By Theorem 3.7, T is a 1-tree rooted at vertex 0 with $\deg(i) = 2$ for all $i \in V$. Since the set of all valid TSP tours is a strict subset of all possible 1-trees on G, the minimum weight 1-tree cannot exceed the cost of the optimal tour. □

Proposition 3.9 *Dominance of the 1-Tree over MST plus Cheapest Edge*

For any choice of root vertex 0, the 1-tree bound is at least as tight as the MST plus cheapest edge bound:

$$LB_{\text{1-tree}} \geq LB_{\text{MST+1}}. \tag{3.45}$$

Proof. The MST plus cheapest edge relaxation allows vertex 0 to have degree 1, 3, or more in the selected n edges. The 1-tree explicitly restricts the feasible set of relaxed subgraphs by enforcing $\sum_{e \in \delta(0)} x_e = 2$. Restricting the feasible region of a minimization problem can never decrease the optimal objective value. Hence, $LB_{\text{1-tree}} \geq LB_{\text{MST+1}}$. □

- **Constraints Relaxed**: Degree-2 regularity on the $n-1$ vertices $V \setminus \{0\}$ is relaxed.
- **Constraints Preserved**: Degree-2 regularity at root vertex 0 is strictly enforced ($\deg(0) = 2$), global connectivity is preserved, and the graph contains exactly one cycle passing through vertex 0.
- **Algorithmic Complexity**: Computing the 1-tree requires finding an MST on $n-1$ vertices via Prim's algorithm in $\mathcal{O}(n^2)$ time, followed by an $\mathcal{O}(n)$ scan to locate the two cheapest edges incident to root 0. Total time is strictly $\mathcal{O}(n^2)$.
- **Bound Quality**: Excellent balance of speed and tightness. It forms the backbone of exact TSP solvers.

Example 3.4 *Computing the Minimum 1-Tree Bound*

We evaluate the 1-tree relaxation on the same 7-vertex instance, choosing vertex 0 as the designated root (Figure 3.3(b)):

1. An MST on the remaining 6 vertices $V \setminus \{0\} = \{1, 2, 3, 4, 5, 6\}$ selects 5 edges: $\{4, 2\}$ (cost 1), $\{1, 5\}$ (cost 2), $\{5, 4\}$ (cost 2), $\{6, 4\}$ (cost 2), and $\{4, 3\}$ (cost 2). The total weight of this tree is:

$$w(\text{MST}(V \setminus \{0\})) = 1 + 2 + 2 + 2 + 2 = 9. \tag{3.46}$$

2. The three edges incident to root vertex 0 in the original graph are $\{0, 5\}$ (cost 1), $\{0, 1\}$ (cost 4), and $\{0, 2\}$ (cost 5). The two cheapest incident edges are $\{0, 5\}$ and $\{0, 1\}$, with cost:

$$c_{0,5} + c_{0,1} = 1 + 4 = 5. \tag{3.47}$$

Summing these two components yields the 1-tree lower bound:

$$LB_{\text{1-tree}} = 9 + 5 = 14. \tag{3.48}$$

Notice that $LB_{\text{1-tree}} = 14 > LB_{\text{MST+1}} = 13$, confirming that enforcing degree 2 at vertex 0 tightens the relaxation (Theorem 3.9).

Connecting to Lagrangian Relaxation: Strengthening the 1-Tree Bound. Formulation (3.43) provides the foundation for Lagrangian relaxation. Since the 1-tree relaxation already enforces connectivity and $\deg(0) = 2$, the degree constraints $\deg(i) = 2$ for the remaining $n-1$ vertices $i \in V \setminus \{0\}$ can be dualized using Lagrange multipliers $\boldsymbol{\pi} \in \mathbb{R}^n$. In Chapter 6, we will explain how subgradient optimization iteratively adjusts these multipliers

to penalize degree violations, strengthening the 1-tree bound to obtain significantly tighter lower bounds (the celebrated Held–Karp bound).

3.3.2 Comparison of the TSP Relaxations

The four relaxations represent complementary projections of the Traveling Salesperson polytope. Their theoretical properties and computational trade-offs are summarized in Table 3.3.

Table 3.3: Comparison of Combinatorial Relaxations for the Traveling Salesperson Problem.

Relaxation	Parent Formulation	Degree Constraints	Connectivity	Complexity	Relative Tightness
Cheapest Incident	Formulation 1	Relaxed ($\deg \geq 1$)	Dropped	$\mathcal{O}(n^2)$	Very Weak
Assignment (2-Factor)	Formulation 1	$\deg(i) = 2 \quad \forall i \in V$	Dropped	$\mathcal{O}(n^3)$	Strong
MST + Cheapest Edge	Formulation 2	Relaxed ($\forall i \in V$)	Preserved	$\mathcal{O}(n^2)$	Moderate
Minimum 1-Tree	Formulation 3	$\deg(0) = 2$, others relaxed	Preserved	$\mathcal{O}(n^2)$	Strong

Example 3.5 *Optimal Tour and Bounding Hierarchy*

Figure 3.3(d) displays the unique optimal Hamiltonian tour on this 7-vertex graph:

$$0 - 5 - 1 - 6 - 4 - 3 - 2 - 0, \tag{3.49}$$

with total optimal cost:

$$z^*_{\text{TSP}} = 1 + 2 + 3 + 2 + 2 + 3 + 5 = 18. \tag{3.50}$$

Comparing all bounds across Figure 3.3 illustrates the bounding hierarchy:

$$LB_{\text{MST+1}}\ (13) \quad \leq \quad LB_{\text{1-tree}}\ (14) \quad \leq \quad LB_{\text{AP}}\ (18) \quad = \quad z^*_{\text{TSP}}\ (18). \tag{3.51}$$

Remarkably, while the assignment bound attains the optimal value ($LB_{\text{AP}} = 18 = z^*_{\text{TSP}}$), it cannot certify optimality on its own because its relaxed solution is infeasible for the TSP (disconnected into two subtours). Within Branch and Bound, branching would still be required to eliminate the subtour, or tree connectivity must be imposed.

Orthogonal Relaxations. Notice the striking structural orthogonality between the *1-Tree* and the *Assignment* relaxations:

- The 1-Tree relaxation (derived from Formulation 3 (3.43)) **preserves connectivity** but **relaxes degrees** (except at vertex 0).
- The Assignment relaxation (derived from Formulation 1 (3.31)) **preserves degrees** everywhere but **relaxes connectivity**.

In Chapter 6, we will explain how to strengthen the 1-tree bound using Lagrangian relaxation to obtain significantly tighter lower bounds.

3.3.3 Branching Schemes and Partial Decisions

When Branch and Bound explores the search space of the TSP, every subproblem P_s corresponds to a node in the search tree where a *partial set of decisions* has already been committed along the path from the root node to P_s. Formally, the decision variables $\boldsymbol{x} \in \{0,1\}^{|E|}$ are partitioned into three subsets:

- **Mandatory (Fixed) Edges** $E_{\text{fixed}} = \{e \in E : x_e = 1\}$: edges that *must* be included in any tour completed from this node.

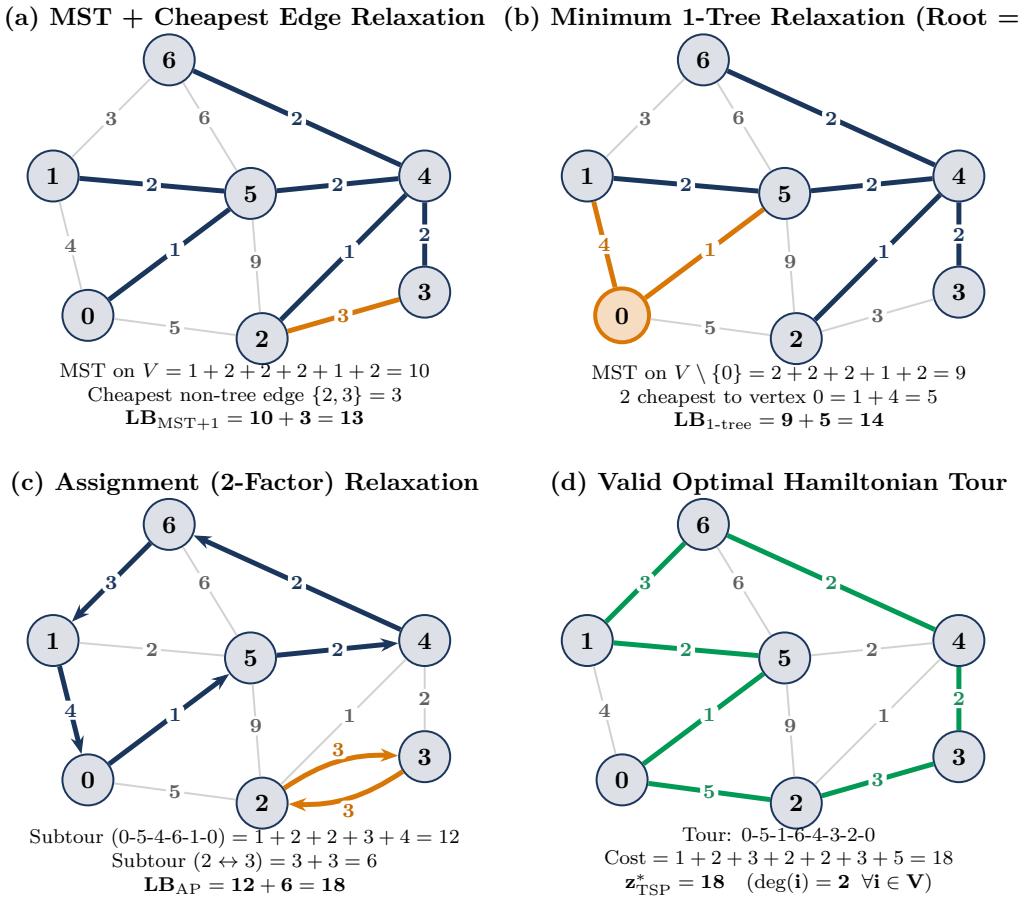


Figure 3.3: Comparison of combinatorial relaxations and the optimal Hamiltonian tour on the canonical 7-vertex graph. Inactive graph edges are shown in light gray with their costs; active edges are highlighted. (a) Minimum Spanning Tree on V (cost 10) plus cheapest non-tree edge $\{2,3\}$ (cost 3) yields $LB_{\text{MST+1}} = 13$ (from Formulation 2). (b) Minimum 1-Tree with root 0: MST on $V \setminus \{0\}$ (cost 9) plus two cheapest edges to vertex 0 (cost 5) yields $LB_{\text{1-tree}} = 14$ (from Formulation 3). (c) Assignment (2-factor) relaxation enforces in-degree 1 and out-degree 1 at every vertex, decomposing into two subtours of cost $12+6 = 18$ (from Formulation 1). (d) Optimal Hamiltonian tour of cost $z^*_{\text{TSP}} = 18$. Across all relaxations, the bounding hierarchy $13 \leq 14 \leq 18 = z^*_{\text{TSP}}$ holds.

- **Excluded (Forbidden) Edges** $E_{\text{excluded}} = \{e \in E : x_e = 0\}$: edges that *cannot* be selected in any completion.
- **Free (Undecided) Edges** $E_{\text{free}} = E \setminus (E_{\text{fixed}} \cup E_{\text{excluded}})$: candidate edges available for selection.

Conditioning Lower Bounds on Ancestor Decisions. To enable effective pruning during search, the lower-bounding procedure at any node P_s must strictly take into account all branching decisions committed along the path from the root node down to P_s. Conditioning the relaxation on E_{fixed} and E_{excluded} restricts the feasible space of completions, forcing the relaxed solution to incorporate all mandatory edges and avoid all forbidden ones. Consequently, the computed lower bound is monotonically non-decreasing along every branch of the search tree:

$$LB(P_0) \leq LB(P_1) \leq \cdots \leq LB(P_s). \tag{3.52}$$

As search descends deeper into the tree, the accumulated restrictions force the relaxation to adopt increasingly constrained, higher-cost substructures. As soon as $LB(P_s) \geq UB^*$, the entire subtree rooted at P_s is fathomed and pruned immediately, preventing the exploration of exponential numbers of suboptimal completions.

Partitioning the Search Space

Before examining how relaxations incorporate committed edges, we must address the branching operator itself: how do we divide an active subproblem into mutually exclusive, smaller subproblems? Two primary branching paradigms exist for the Traveling Salesperson Problem:

Scheme 1: Sequential Path Extension (Tour Construction). In this intuitive, constructive scheme, the search tree builds a Hamiltonian cycle sequentially as a growing path. Fixing origin vertex 0 as the starting point, each search node at depth k represents an ordered partial path:

$$P = [0, v_1, v_2, \ldots, v_k], \tag{3.53}$$

together with the set of unvisited candidate vertices:

$$V_{\text{remaining}} = V \setminus \{0, v_1, \ldots, v_k\}. \tag{3.54}$$

The branching step generates $|V_{\text{remaining}}|$ child subproblems, each corresponding to choosing a distinct candidate vertex $v_{k+1} \in V_{\text{remaining}}$ to append to the path, committing edge $\{v_k, v_{k+1}\}$ to E_{fixed}.

The sequential path-extension scheme possesses clear operational advantages: the maximum search depth is strictly bounded by n, and detecting a leaf solution is straightforward (when $k = n - 1$, the path visits all cities and is closed by appending edge $\{v_{n-1}, 0\}$). However, it imposes an arbitrary chronological ordering on tour construction: early decisions must choose between local paths even when the global topology of the remaining tour is still completely unconstrained.

Scheme 2: Conflict-Driven Edge Branching (Relaxation-Guided on the 1-Tree). Rather than forcing edges to form a single continuous path from the start, a more flexible and mathematically elegant paradigm allows edges to be fixed anywhere across the graph, letting the violations of the relaxation directly dictate the branching choices.

Suppose we evaluate the minimum 1-tree relaxation at search node P_s, respecting all currently committed edges E_{fixed} and E_{excluded}:

1. **Exact Optimality Detection**: If every vertex in the resulting 1-tree T happens to have degree exactly 2 ($\deg_T(i) = 2$ for all $i \in V$), then T is a connected 2-regular spanning graph. By Theorem 3.3, T is a valid Hamiltonian cycle! The relaxation has solved the subproblem to global optimality without further branching: we update the incumbent upper bound $UB^* \leftarrow \min(UB^*, w(T))$ and fathom the node immediately.
2. **Degree-Conflict Identification**: If T is not a valid tour, it cannot satisfy 2-regularity everywhere. Because the total degree sum in a 1-tree is $\sum_{i \in V} \deg_T(i) = 2n$ and root vertex 0 strictly satisfies $\deg_T(0) = 2$, there must exist at least one vertex $v \in V \setminus \{0\}$ that has degree strictly greater than 2:

$$\deg_T(v) \geq 3. \tag{3.55}$$

   We refer to such a vertex as a *conflict hub*. In any valid tour, vertex v can have at most 2 incident edges. Therefore, among the $d = \deg_T(v) \geq 3$ incident edges chosen by the 1-tree, at least one edge *cannot* belong to any feasible tour completion!
3. **Resolving the Conflict by Branching**: We can directly resolve this conflict by branching on the incident edges of vertex v:
   - *Binary Edge Branching*: Select an edge $e \in \delta_T(v) \cap E_{\text{free}}$ incident to conflict vertex v in the 1-tree (for example, the edge with the highest cost or the greatest degree violation). Partition into two child subproblems:
     - **Exclude Edge** e: enforce $x_e = 0$ ($e \in E_{\text{excluded}}$). This branch directly eliminates the conflict edge, forcing the 1-tree in this subtree to find an alternative, more costly connection. This almost always produces an immediate increase in the lower bound ($LB_{\text{child}} > LB_{\text{parent}}$).
     - **Include Edge** e: enforce $x_e = 1$ ($e \in E_{\text{fixed}}$). This branch commits edge e to the tour, which consumes one unit of the degree quota at vertex v and triggers constraint propagation to exclude other edges.
   - *Multi-Way Exclusion Branching*: Alternatively, let $\{e_1, e_2, \ldots, e_d\}$ be the $d \geq 3$ edges incident to vertex v in the 1-tree. Since any valid tour can contain at most 2 of these edges, at least one must be excluded. One can create mutually disjoint child branches by excluding e_1 in the first branch, fixing e_1 and excluding e_2 in the second branch, and so forth.

Conflict-driven branching is widely considered the gold standard for relaxation-based Branch and Bound because every branch is purposeful: it directly targets and eliminates the structural flaw (the degree violation) that prevented the relaxation from being a feasible solution.

Conditioning Bounds with Fixed Edges

Regardless of whether path-extension or conflict-driven branching is used, the lower-bounding engine must correctly enforce the sets E_{excluded} and E_{fixed} inherited from all ancestors up to the current node.

1. Handling Excluded Edges ($x_e = 0$). An edge $e = \{u, v\} \in E_{\text{excluded}}$ cannot appear in any completion of the current subproblem:

- **Cost Masking**: In the distance matrix D, excluded edges are effectively eliminated by setting $c_{uv} \leftarrow \infty$ (or tracked via a fast boolean mask `excluded`$[u][v] =$ `true`). Any graph algorithm (Prim, Kruskal, Dijkstra, Hungarian) simply skips masked edges.

- **Logical Deductions (Constraint Propagation)**: Committing edges to E_{fixed} logically forces additional edges into E_{excluded} through constraint propagation:

  1. *Degree-2 Exhaustion*: As soon as any vertex $v_i \in V$ has two incident edges fixed in E_{fixed}, its degree quota is completely exhausted. To maintain 2-regularity, every other incident edge must be excluded:

  $$\forall u \notin \mathcal{N}_{\text{fixed}}(v_i), \quad \text{mark edge } \{v_i, u\} \text{ as } \textbf{excluded}. \tag{3.56}$$

  In sequential path extension $P = [0, v_1, \ldots, v_k]$, every internal vertex v_i $(1 \leq i < k)$ already has its incoming edge (v_{i-1}, v_i) and outgoing edge (v_i, v_{i+1}) fixed; hence, all other edges incident to v_i are immediately excluded.

  2. *Premature Subtour Prevention*: If a set of mandatory edges forms a path connecting vertices u and w of length $\ell < n$, adding edge $\{u, w\}$ would close an isolated cycle of length $\ell < n$. Therefore:

  $$\text{if } |E_{\text{fixed}}| < n - 1, \quad \text{mark edge } \{u, w\} \text{ as } \textbf{excluded}. \tag{3.57}$$

  In sequential path extension $P = [0, v_1, \ldots, v_k]$, this rule dictates that return edge $\{v_k, 0\}$ is strictly forbidden as long as unvisited cities remain $(k < n - 1)$.

  3. *Anti-Parallel Exclusion*: In directed models, selecting directed arc (u, v) immediately excludes reverse arc (v, u) to prevent forming an invalid 2-cycle.

2. Handling Mandatory (Fixed) Edges ($x_e = 1$). When edges E_{fixed} are committed to the tour, the relaxation must be constrained to include every edge in E_{fixed}:

- **In Spanning Tree and 1-Tree Relaxations**: The most efficient way to incorporate fixed edges into an MST or 1-tree is via Kruskal's algorithm equipped with a Disjoint-Set (Union-Find) data structure (see Appendix A for a comprehensive review of Kruskal's algorithm, the Cut Property, and the Union-Find data structure):

  1. Initialize a Union-Find partition where each vertex is in its own singleton set.
  2. *Pre-insert all mandatory edges* $e = \{u, v\} \in E_{\text{fixed}}$:
     - If any vertex has more than 2 incident fixed edges ($\deg_{E_{\text{fixed}}}(v) > 2$), the subproblem violates 2-regularity and is **pruned immediately by infeasibility** ($LB \leftarrow \infty$).
     - If find(u) == find(v), edge e connects two vertices already in the same connected component, forming a cycle! If $|E_{\text{fixed}}| < n$, this constitutes a premature subtour, and the subproblem is **pruned immediately by infeasibility** ($LB \leftarrow \infty$).
     - Otherwise, call union(u, v) and accumulate the cost c_e into the baseline bound.
  3. *Greedy Completion on Free Edges*: Once all mandatory edges are merged, iterate through the remaining non-excluded free edges E_{free} in non-decreasing order of cost, adding each edge whose endpoints belong to different sets until the spanning tree on $V \setminus \{0\}$ is complete.
  4. *Root Connections for the 1-Tree*: For root vertex 0, if $m \in \{0, 1, 2\}$ edges incident to vertex 0 are already in E_{fixed}, they are retained; the remaining $2 - m$ connections are selected greedily as the cheapest available non-excluded edges incident to 0.

- **In Assignment Relaxations**: If directed arc (u, v) is fixed $(x_{uv} = 1)$:

  - Remove row u and column v from the cost matrix, reducing the assignment problem from dimension m to $m-1$.
  - Add the constant cost c_{uv} directly to the accumulated baseline lower bound.
  - Alternatively, set $c_{uv} \leftarrow -M$ (or 0) while setting all other entries in row u and column v to ∞.
- **In Cheapest Incident Edges**: If an edge incident to vertex v is fixed to u, the first incident cost for v is locked to c_{uv}, and only the second incident edge is chosen greedily among non-excluded neighbors.

Example 3.6 *Branching Decisions and Bound Conditioning on a 7-Vertex Graph*

We illustrate both branching paradigms and the bound-conditioning mechanics on the canonical 7-vertex TSP instance from Figure 3.3, whose edge weights are given in (3.40).

1. Conflict-Driven 1-Tree Branching. Recall from Example 3.4 that the unconstrained 1-tree at root node P_0 (with root vertex 0) selects the 7 edges highlighted in Figure 3.3(b), achieving lower bound $LB(P_0) = 14$:

- Vertex degrees in the 1-tree: $\deg(0) = 2$, $\deg(1) = 2$, leaves $\deg(2) = 1$, $\deg(3) = 1$, $\deg(6) = 1$, and *conflict hubs* $\deg(5) = 3$ and $\deg(4) = 4$.

Because vertex 4 has degree 4 (incident to $\{4,2\}$, $\{4,3\}$, $\{6,4\}$, and $\{5,4\}$), it violates the degree-2 requirement of a valid tour. We choose incident edge $e = \{5,4\}$ (cost 2) and perform binary edge branching:

1. **Left Branch (Exclude $\{5,4\}$)**: We commit $x_{5,4} = 0$, so $E_{\text{excluded}} = \{\{5,4\}\}$ and $E_{\text{fixed}} = \emptyset$. When Kruskal's algorithm computes the MST on $V \setminus \{0\} = \{1,2,3,4,5,6\}$, edge $\{5,4\}$ is masked with $c_{5,4} \leftarrow \infty$. The algorithm selects:
   - $\{4,2\}$ (cost 1), $\{1,5\}$ (cost 2), $\{6,4\}$ (cost 2), $\{4,3\}$ (cost 2).
   - At this stage, the forest consists of two disconnected components: $\{1,5\}$ and $\{2,3,4,6\}$. Since edge $\{5,4\}$ is excluded, the cheapest available edge bridging these components is $\{1,6\}$ with cost 3 (cheaper than $\{6,5\}$ of cost 6 or $\{5,2\}$ of cost 9).

   The resulting MST on $V \setminus \{0\}$ has weight $1+2+2+2+3 = 10$ (increased by 1). Adding the two cheapest incident edges to root 0 ($\{0,5\}$ of cost 1 and $\{0,1\}$ of cost 4, sum 5) yields the updated lower bound:

$$LB(P_{\text{left}}) = 10 + 5 = 15. \tag{3.58}$$

   Notice that conditioning on the single exclusion $x_{5,4} = 0$ has **strictly strengthened the lower bound** from 14 to 15 ($LB(P_{\text{left}}) > LB(P_0)$) while reducing the degree of conflict vertex 4 from 4 to 3!
2. **Right Branch (Include $\{5,4\}$)**: We commit $x_{5,4} = 1$, so $E_{\text{fixed}} = \{\{5,4\}\}$. Edge $\{5,4\}$ is pre-inserted into the Union-Find structure, and Kruskal completes the tree using the remaining cheapest free edges. The lower bound remains $LB(P_{\text{right}}) = 14$, but vertex 4 has now consumed 1 of its 2 available degrees, allowing subsequent branching on its remaining edges to rapidly resolve the conflict or trigger infeasibility pruning.

2. Sequential Path Extension Branching. In the path-extension scheme, search starts at root $P_0 = [0]$:

1. Branching from $[0]$ chooses the next city. Suppose we branch to child $P_1 = [0, 5]$ (edge $\{0, 5\}$ fixed, cost 1). Return edge $\{5, 0\}$ is marked excluded to prevent premature 2-cycle $0 - 5 - 0$.
2. From $[0, 5]$, branching on candidate city 4 yields node $P_2 = [0, 5, 4]$ with fixed path edges $\{0, 5\}$ and $\{5, 4\}$ (accumulated path cost $1 + 2 = 3$):
   - *Degree-2 Exhaustion*: Vertex 5 is now an internal path vertex with its two tour edges committed (incoming from 0, outgoing to 4). Constraint propagation immediately marks all other edges incident to 5 as **excluded**:
   $$E_{\text{excluded}} \leftarrow E_{\text{excluded}} \cup \{\{5, 1\}, \{5, 6\}, \{5, 2\}\}. \tag{3.59}$$
   - *Subtour Prevention*: Because unvisited cities $\{1, 2, 3, 6\}$ remain, edge $\{4, 0\}$ is marked **excluded**.

   When computing the completion lower bound on remaining vertices $\{1, 2, 3, 6\}$ (which must connect to path endpoints 0 and 4), the excluded edges $\{5, 1\}$, $\{5, 6\}$, $\{5, 2\}$, and $\{4, 0\}$ are completely inaccessible. This forces the relaxation to use more expensive detours (such as $\{1, 6\}$ of cost 3), directly driving the conditioned lower bound upward and accelerating bound pruning.

3.4 Search Strategies and Optimality Gap

Regardless of the problem under consideration, every Branch and Bound solver can be conceptualized as an exploration engine operating over a dynamic collection of active, unexpanded subproblems called the *search frontier*, *fringe*, or *open nodes* $\mathcal{F}$.

Algorithm 3 outlines the abstract Branch and Bound algorithm using this open-nodes formulation (standardized on minimization). The algorithm maintains the frontier $\mathcal{F}$ alongside the incumbent solution $\mathbf{x}^*$ and its objective upper bound UB^*. At each iteration, a node n is extracted from $\mathcal{F}$, tested against the current bound, and either fathomed or expanded into child subproblems.

The critical observation is that the exploration order of the search tree is entirely determined by line 6:

$$n \leftarrow \mathcal{F}.\text{remove}(). \tag{3.60}$$

By substituting different data structure implementations for the open-nodes collection $\mathcal{F}$, we obtain fundamentally distinct exploration paradigms with dramatically different performance characteristics.

3.4.1 Depth-First Search (DFS)

Depth-First Search implements $\mathcal{F}$ as a Last-In, First-Out (LIFO) stack. The algorithm immediately expands the child of the most recently visited node, diving down a single branch until a leaf is reached or fathomed.

- **Memory Efficiency**: The maximum number of nodes residing simultaneously on the stack is bounded by $\mathcal{O}(b \cdot d)$, where b is the branching factor ($b = 2$ for binary trees)

Algorithm 3: Generic Branch and Bound Framework with Open Nodes.

Input: Root subproblem P_0; lower-bounding procedure $LB(\cdot)$; branching operator children$(\cdot)$
Output: Globally optimal solution $\mathbf{x}^*$ with optimal cost UB^*

$\mathcal{F} \leftarrow$ OpenNodes();
`// Initialize search frontier with root`
$\mathcal{F}$.add(P_0);
$UB^* \leftarrow \infty$;
`// Incumbent upper bound`
$\mathbf{x}^* \leftarrow$ null;
while $\mathcal{F}$ *is not empty* **do**
  $n \leftarrow \mathcal{F}$.remove();
  `// Node selection dictated by queue policy`
  if n.*lowerBound*() $\geq UB^*$ **then**
    continue;
    `// Pruned by bound (incumbent improved since insertion)`
  if n.*isSolutionCandidate*() **then**
    if n.*objectiveFunction*() $< UB^*$ **then**
      $UB^* \leftarrow n$.objectiveFunction();
      `// Update incumbent`
      $\mathbf{x}^* \leftarrow n$.getSolution();
  else
    for ***each*** *child* $\in n$.*children*() **do**
      if *child*.*lowerBound*() $< UB^*$ **then**
        $\mathcal{F}$.add(child);
        `// Filter and enqueue promising subproblems`
return $\mathbf{x}^*$

and d is the maximum search depth ($d \leq n$). Memory consumption is strictly linear in problem size.

- **Rapid Incumbent Discovery**: By diving directly to leaves, DFS uncovers full feasible solutions early in the search. This quickly raises LB^*, enabling aggressive pruning of subsequent branches.
- **Susceptibility to Poor Branching Decisions**: If the search takes a suboptimal branch near the root of the tree, DFS can spend billions of iterations exhaustively exploring an unproductive subtree before backtracking.

3.4.2 Best-First Search (BFS)

Best-First Search (also called Best-Bound Search) implements $\mathcal{F}$ as a priority queue ordered by the optimistic bound of each node:

$$n^* = \arg\max_{n \in \mathcal{F}} UB(n) \quad (\text{or } \arg\min_{n \in \mathcal{F}} LB(n) \text{ in minimization}). \tag{3.61}$$

The algorithm always expands the globally most promising node in the entire search frontier.

- **Minimal Node Exploration**: Best-First Search is provably optimal in the number of nodes expanded: it never expands any node whose optimistic bound is worse than the optimal objective value z^* [Wol20].
- **Exponential Memory Consumption**: In the worst case, the frontier $\mathcal{F}$ can grow exponentially with tree depth. For large problems, BFS frequently exhausts available RAM long before reaching an optimal leaf, terminating with an out-of-memory failure.
- **Delayed Incumbent Updates**: BFS tends to expand nodes in breadth-like layers across the top of the tree, often discovering its first feasible integer solution very late in the search.

3.4.3 The Optimality Gap

In practical industrial optimization, one rarely has the computational budget to run Branch and Bound to absolute completion. An essential feature of Branch and Bound is that it operates as an *anytime algorithm*: it provides both a valid feasible solution and a mathematical guarantee on how far that solution could possibly be from the global optimum.

Definition 3.8 *Global Dual Bound and Optimality Gap*

Let $\mathcal{F}$ be the set of open nodes in the search frontier at any instant. For a minimization problem with incumbent upper bound $UB^* = c(\mathbf{x}^*)$, the *global lower bound* across the entire unexplored search space is:

$$LB_{\text{global}} = \min_{n \in \mathcal{F}} LB(n). \tag{3.62}$$

The *absolute optimality gap* and *relative optimality gap* are defined respectively as:

$$\text{Gap}_{\text{abs}} = UB^* - LB_{\text{global}}, \qquad \text{Gap}_{\text{rel}} = \frac{UB^* - LB_{\text{global}}}{UB^*}. \tag{3.63}$$

Symmetrically, for a maximization problem with incumbent LB^* and open nodes providing upper bounds:

$$UB_{\text{global}} = \max_{n \in \mathcal{F}} UB(n), \qquad \text{Gap}_{\text{rel}} = \frac{UB_{\text{global}} - LB^*}{LB^*}. \tag{3.64}$$

When $\text{Gap}_{\text{rel}} = 0$, the incumbent is proven to be mathematically optimal. In commercial practice, solvers typically terminate when $\text{Gap}_{\text{rel}} \leq \varepsilon$ (for example, $\varepsilon = 10^{-4}$ or 0.01%), guaranteeing that the discovered solution is within a fraction of a percent of the theoretical optimum.

3.4.4 Hybrid Search: Diving (DFS-BFS)

To combine the lean memory footprint and rapid incumbent discovery of DFS with the global guidance of BFS, state-of-the-art solvers employ *diving heuristics* (hybrid BFS-DFS).

As shown in Algorithm 4, the algorithm extracts the globally best node n from a Best-First priority queue. Instead of returning immediately to the queue after expanding n, the solver performs a greedy depth-first *dive*: at each step, it selects the child with the best local lower bound, pushes all other sibling children into the global priority queue, and continues diving down that single trajectory until reaching a leaf or encountering a node pruned by bound. This discovers high-quality feasible solutions rapidly while maintaining a globally bounded frontier.

Algorithm 4: Hybrid Best-First / Depth-First Diving Strategy.

Input: Initial open nodes priority queue $\mathcal{F}$ containing the root node n_0; incumbent $UB^* \leftarrow \infty$
Output: Optimal or proven ε-optimal solution $\mathbf{x}^*$

```
while F is not empty do
    n ← F.extractMin();
    // Greedy dive from node n until a leaf or pruned state
    while not n.isSolutionCandidate() and n.lowerBound() < UB* do
        children ← n.children();
        if children is empty then
            break;
        Sort children in ascending order of lowerBound();
        n ← children[0];
        // Follow the best child
        for i ← 1 to |children| − 1 do
            F.add(children[i]);
            // Store siblings in global frontier
    if n.isSolutionCandidate() then
        if n.objectiveFunction() < UB* then
            UB* ← n.objectiveFunction();
            x* ← n.getSolution();
return x*
```

3.5 A Generic Java Framework for B&B

We now present a clean, generic, and extensible Java framework for Branch and Bound. The framework cleanly decouples the core search engine from problem-specific state representations and frontier queuing strategies.

3.5.1 Architecture and Core Interfaces

The architecture is structured around three core abstractions:

1. `Node<T>`: Represents an active node in the search tree, encapsulating problem state T, depth, bound computation, and branching logic.
2. `OpenNodes<T>`: Represents the search frontier (fringe) ADT.
3. `BranchAndBound`: The generic solver engine executing the exploration loop.

To standardize across all combinatorial problems, the framework formulates every problem as **minimization**. For maximization problems (such as Knapsack), the implementation maps $\max f(\mathbf{x}) \iff \min(-f(\mathbf{x}))$, returning negated upper bounds: lowerBound() $= -UB$.

Listing 3.1 presents the generic `Node<T>` interface.

```java
import java.util.List;

/**
 * Node interface for Branch and Bound.
 * Formulated for minimization problems.
 * @param <T> Domain state associated with this node.
 */
public interface Node<T> {
    /** True if node is complete, feasible solution. */
    boolean isSolutionCandidate();

    /** Objective value (valid if isSolutionCandidate). */
    double objectiveFunction();

    /** Lower bound on completions of this node. */
    double lowerBound();

    /** Depth in search tree (root has depth 0). */
    int depth();

    /** Generates children of this node via branching. */
    List<Node<T>> children();

    /** Returns domain state associated with this node. */
    T getState();
}
```

Listing 3.1: The generic `Node<T>` interface (`Node.java`).

Listing 3.2 defines the `OpenNodes<T>` interface alongside its standard concrete implementations: Best-First (priority queue), Depth-First (stack), and Depth-First with Best-First tie-breaking.

```java
import java.util.Comparator;
import java.util.PriorityQueue;
import java.util.Stack;

public interface OpenNodes<T> {
    void add(Node<T> n);
    Node<T> remove();
    boolean isEmpty();
    int size();
}

/** Best-First Search: extracts node with minimal bound. */
public class BestFirstOpenNodes<T> implements OpenNodes<T> {
    private final PriorityQueue<Node<T>> queue;

    public BestFirstOpenNodes() {
        queue = new PriorityQueue<>(
```

```
            Comparator.comparingDouble(Node::lowerBound)
        );
    }
    public void add(Node<T> n) { queue.add(n); }
    public Node<T> remove()    { return queue.remove(); }
    public boolean isEmpty()   { return queue.isEmpty(); }
    public int size()          { return queue.size(); }
}

/** Depth-First Search: LIFO exploration using stack. */
public class DepthFirstOpenNodes<T>
        implements OpenNodes<T> {
    private final Stack<Node<T>> stack = new Stack<>();

    public void add(Node<T> n) { stack.push(n); }
    public Node<T> remove()    { return stack.pop(); }
    public boolean isEmpty()   { return stack.isEmpty(); }
    public int size()          { return stack.size(); }
}

/** DFS with Best-First tie-breaking on equal depth. */
public class DepthFirstBestFirstOpenNodes<T>
        implements OpenNodes<T> {
    private final PriorityQueue<Node<T>> queue;

    public DepthFirstBestFirstOpenNodes() {
        queue = new PriorityQueue<>((o1, o2) -> {
            if (o1.depth() != o2.depth()) {
                // deeper first
                return Integer.compare(
                        o2.depth(), o1.depth());
            }
            return Double.compare(
                    o1.lowerBound(), o2.lowerBound());
        });
    }
    public void add(Node<T> n) { queue.add(n); }
    public Node<T> remove()    { return queue.remove(); }
    public boolean isEmpty()   { return queue.isEmpty(); }
    public int size()          { return queue.size(); }
}
```

Listing 3.2: Open nodes interface and concrete search strategies.

Listing 3.3 shows the core `BranchAndBound.minimize` solver method. Notice the direct correspondence with the abstract algorithm presented in Algorithm 3: the polymorphic `OpenNodes<T>` collection instantiates the search frontier $\mathcal{F}$, while the `while` loop executes node selection, candidate evaluation, and child expansion identically.

```
import java.util.function.Consumer;

public class BranchAndBound {
    /**
     * Explores search space using Branch and Bound.
     * @param openNodes Collection containing root node.
     * @param onSolution Callback on improving solution.
     */
    public static <T> void minimize(
            OpenNodes<T> openNodes,
            Consumer<Node<T>> onSolution) {
        double upperBound = Double.MAX_VALUE;
```

```
13
14          while (!openNodes.isEmpty()) {
15              Node<T> n = openNodes.remove();
16
17              if (n.isSolutionCandidate()) {
18                  double objective = n.objectiveFunction();
19                  if (objective < upperBound) {
20                      upperBound = objective;
21                      onSolution.accept(n);
22                  }
23              } else if (n.lowerBound() < upperBound) {
24                  for (Node<T> child : n.children()) {
25                      openNodes.add(child);
26                  }
27              }
28          }
29      }
30  }
```

Listing 3.3: The generic Branch and Bound exploration loop (`BranchAndBound.java`).

3.5.2 Instantiating Knapsack in the Framework

To instantiate the 0/1 Knapsack problem in our framework, we define the `NodeKnapsack` class presented in Listing 3.4.

```
1  import java.util.ArrayList;
2  import java.util.List;
3
4  public class NodeKnapsack implements Node<NodeKnapsack> {
5      final int[] value;
6      final int[] weight;
7      final int selectedValue;
8      final int capaLeft;
9      final int index;
10     final boolean selected;
11     final NodeKnapsack parent;
12     final double ub;
13     final int depth;
14
15     public NodeKnapsack(NodeKnapsack parent,
16                         int[] value, int[] weight,
17                         int selectedValue, int capaLeft,
18                         int index, boolean selected) {
19         this.parent = parent;
20         this.value = value;
21         this.weight = weight;
22         this.selectedValue = selectedValue;
23         this.capaLeft = capaLeft;
24         this.index = index;
25         this.selected = selected;
26         this.depth = (parent == null) ? 0
27                 : parent.depth + 1;
28         this.ub = lpRelaxUBound();
29     }
30
31     /** Continuous LP relaxation on undecided items. */
32     private double lpRelaxUBound() {
33         double valueUb = selectedValue;
34         int c = capaLeft;
```

```
35          for (int i = index + 1; i < value.length; i++) {
36              if (weight[i] <= c) {
37                  valueUb += value[i];
38                  c -= weight[i];
39              } else {
40                  double frac = (double) c / weight[i];
41                  valueUb += frac * value[i];
42                  break;
43              }
44          }
45          return valueUb;
46      }
47
48      @Override
49      public boolean isSolutionCandidate() {
50          return index == value.length - 1;
51      }
52
53      @Override
54      public double objectiveFunction() {
55          return -selectedValue; // negated for min
56      }
57
58      @Override
59      public double lowerBound() {
60          return -ub; // negated upper bound is lower bound
61      }
62
63      @Override
64      public int depth() {
65          return depth;
66      }
67
68      @Override
69      public List<Node<NodeKnapsack>> children() {
70          List<Node<NodeKnapsack>> children =
71                  new ArrayList<>(2);
72          int nextIndex = index + 1;
73
74          // Right child: do not select item at nextIndex
75          children.add(new NodeKnapsack(
76              this, value, weight, selectedValue,
77              capaLeft, nextIndex, false));
78
79          // Left child: select item at nextIndex
80          if (capaLeft >= weight[nextIndex]) {
81              children.add(new NodeKnapsack(
82                  this, value, weight,
83                  selectedValue + value[nextIndex],
84                  capaLeft - weight[nextIndex],
85                  nextIndex, true));
86          }
87          return children;
88      }
89
90      @Override
91      public NodeKnapsack getState() { return this; }
92  }
```

Listing 3.4: Knapsack node implementation (`BranchAndBoundKnapsack.java`).

Executing the solver requires only pre-sorting items by ratio and instantiating the root node:

```
// Pre-sort items in non-increasing ratio order
sortValueOverWeight(value, weight);

OpenNodes<NodeKnapsack> openNodes =
        new BestFirstOpenNodes<>();
NodeKnapsack root = new NodeKnapsack(
        null, value, weight, 0, capacity, -1, false);
openNodes.add(root);

BranchAndBound.minimize(openNodes, solNode -> {
    System.out.println("New best knapsack profit: "
            + (-solNode.lowerBound()));
});
```

3.6 Practical Implementation: TSP in B&B

To synthesize the theoretical and algorithmic foundations developed in this chapter, readers are invited to implement an exact Branch and Bound solver for the Traveling Salesperson Problem by bridging the combinatorial relaxations from Section 3.3 with the generic search engine established in Section 3.5.

The implementation involves three core components:

1. **Search Node State and Branching**: Design a search node class (`NodeTSP`) implementing the generic `Node<NodeTSP>` interface. The node encapsulates the partial decisions accumulated along the branch from the root node (such as the growing partial path in sequential path extension or the sets E_{fixed} and E_{excluded} in edge-based branching; see Section 3.3.3). When generating child nodes, perform the necessary constraint propagation:
    - Exclude all non-tour edges incident to vertices whose degree-2 quota has been reached.
    - Forbid edges that would prematurely close a subtour before all n cities have been visited.
2. **Conditioned Lower-Bounding Procedures**: Implement the lower-bounding procedures developed in Section 3.3:
    - *Cheapest Incident Edges*: A fast baseline bound scanning non-excluded incident edges.
    - *Minimum 1-Tree Relaxation*: Utilizing Kruskal's algorithm and the Union-Find data structure (reviewed in Appendix A) to compute the conditioned 1-tree on $V \setminus \{0\}$ respecting all mandatory and excluded edges.
3. **Search Execution and Benchmarking**: Integrate the search node and bounding routines into the generic `BranchAndBound.minimize` exploration engine. Compare the impact of search strategies (Depth-First Search vs. Best-First Search) and bound quality (cheapest incident vs. 1-tree) on the number of expanded nodes and total CPU time across representative TSP instances.

3.7 Exercises

Exercise 3.1 *Branch and Bound Tree under Variable Reordering*

Consider the following 0/1 knapsack instance:

$$\begin{aligned} \text{maximize} \quad & 25x_0 + 18x_1 + 36x_2 \\ \text{subject to} \quad & 5x_0 + 6x_1 + 4x_2 \leq 11, \\ & x_0, x_1, x_2 \in \{0, 1\}. \end{aligned} \tag{3.65}$$

1. Compute the value-to-weight efficiency ratios r_0, r_1, r_2. Is the given ordering x_0, x_1, x_2 sorted by non-increasing efficiency?
2. Draw the complete Branch and Bound search tree under Depth-First Search with the continuous LP relaxation using the original variable ordering x_0, x_1, x_2.
3. Now sort the variables by non-increasing efficiency: x_2, x_0, x_1. Recompute the Branch and Bound search tree.
4. Compare the number of nodes evaluated and branches pruned under both orderings. Explain why ordering variables by efficiency improves pruning performance.

Exercise 3.2 *Optimality of the Split Item in Continuous Knapsack*

Let $\mathbf{x}^*$ be the optimal solution to the continuous knapsack relaxation with critical item j. Prove that if the capacity C is an integer and all item weights w_i are integers, the fractional variable x_j^* satisfies $x_j^* = 0$ if and only if the optimal solution to the continuous knapsack relaxation is strictly integer feasible.

Exercise 3.3 *Proof of the 1-Tree Cycle Property*

Let $G = (V, E)$ be a connected undirected graph with $|V| = n$. Prove graph-theoretically that any 1-tree on G contains:

1. Exactly n edges.
2. Exactly one simple cycle, and that this cycle necessarily contains the designated vertex 0.

Exercise 3.4 *Edge-Based Branching for the Traveling Salesperson Problem*

In Section 3.6, branching is performed by extending a partial tour vertex by vertex. An alternative branching scheme branches on edges: selecting an undecided edge $e = \{u, v\}$ and generating two children:

- Child 1: Force edge e to be included in the tour ($x_e = 1$).
- Child 2: Forbid edge e from being included in the tour ($x_e = 0$).

1. How should edge e be chosen to maximize pruning efficiency?
2. What logical deductions can be made immediately when $x_e = 1$ is posted?
3. Compare the depth and branching factor of this edge-branching scheme with the vertex-extension scheme.

Exercise 3.5 *Memory Complexity of Best-First Search*

Consider a binary Branch and Bound search tree of uniform depth d. Suppose that at depth $k \leq d$, all nodes have identical upper bounds, forcing Best-First Search to explore the tree in breadth-first layers. Derive the exact number of nodes residing in the priority queue when the search reaches depth k. What does this imply about the scalability of pure Best-First Search on large-scale combinatorial problems?

CHAPTER 4

Linear Programming

Linear programming (LP) is the cornerstone of continuous and discrete optimization. Developed by George Dantzig in 1947 [Dan63] alongside the revolutionary Simplex algorithm, linear programming addresses the problem of optimizing a linear objective function subject to linear equality and inequality constraints. Beyond its immense direct utility in industrial operations, logistics, and telecommunications, linear programming serves as the algorithmic engine of discrete optimization. As established in Chapter 3, relaxing integrality constraints in mixed-integer linear programs produces continuous linear relaxations that yield rigorous dual bounds and prune combinatorial search trees.

In this chapter, we develop the geometric, algebraic, and computational foundations of linear programming. We begin with a concrete motivating case study—the Brewery Problem—and examine its geometric perspective in two and three dimensions. We formalize polyhedral theory, prove the fundamental theorem of vertex optimality, and analyze why naive vertex enumeration fails. We then introduce canonical and standard forms, derive the Simplex tableau mechanics, and provide clean Java implementations. Finally, we explore initialization techniques (Two-Phase and Big-M), duality theory, complexity, total unimodularity, and network flow modeling.

4.1 Foundations of Linear Programming

A linear program optimizes a linear objective function over a continuous domain restricted by linear equations or linear inequalities.

Definition 4.1 *Linear Program*

Let $n \in \mathbb{Z}^+$ denote the number of continuous decision variables, and let $m \in \mathbb{Z}^+$ denote the number of constraints. A *linear program* optimizes a linear function of

$\mathbf{x} = (x_0, x_1, \ldots, x_{n-1})^\top \in \mathbb{R}^n$ subject to linear constraints:

$$\begin{aligned} \text{maximize} \quad & \mathbf{c}^\top \mathbf{x} = \sum_{j=0}^{n-1} c_j x_j \\ \text{subject to} \quad & \sum_{j=0}^{n-1} a_{i,j} x_j \leq b_i, \quad \forall i \in M_{\leq}, \\ & \sum_{j=0}^{n-1} a_{i,j} x_j = b_i, \quad \forall i \in M_{=}, \\ & \sum_{j=0}^{n-1} a_{i,j} x_j \geq b_i, \quad \forall i \in M_{\geq}, \\ & x_j \geq 0, \quad \forall j \in N_{\geq}, \qquad x_j \in \mathbb{R} \text{ (free)}, \quad \forall j \in N_{\text{free}}, \end{aligned} \tag{4.1}$$

where $M_{\leq} \cup M_{=} \cup M_{\geq} = \{0, 1, \ldots, m-1\}$ partitions the constraints and $N_{\geq} \cup N_{\text{free}} = \{0, 1, \ldots, n-1\}$ partitions the decision variables.

In vector-matrix notation, when all constraints are less-than-or-equal-to inequalities and all variables are non-negative, the problem is concisely stated as:

$$\max \left\{ \mathbf{c}^\top \mathbf{x} \;\middle|\; \mathbf{A}\mathbf{x} \leq \mathbf{b},\ \mathbf{x} \geq \mathbf{0} \right\}, \tag{4.2}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the constraint matrix, $\mathbf{b} \in \mathbb{R}^m$ is the right-hand-side vector, and $\mathbf{c} \in \mathbb{R}^n$ is the objective cost vector.

4.2 A Motivating Case: The Brewery Problem

4.2.1 Problem Formulation

To build geometric and algorithmic intuition, we examine a classic manufacturing optimization model: *The Brewery Problem* [SW11].

A small brewery produces two handcrafted beverages: ale and beer. Production is strictly limited by scarce agricultural resources: corn, hops, and malt. The production recipes and economic yields are defined as follows:

- One barrel of ale requires 5 units of corn, 4 units of hops, and 35 units of malt, generating a profit of 13 monetary units.
- One barrel of beer requires 15 units of corn, 4 units of hops, and 20 units of malt, generating a profit of 23 monetary units.
- Available ingredient supplies are capped at 480 units of corn, 160 units of hops, and 1190 units of malt.

Adhering to our zero-based indexing convention, let $x_0 \geq 0$ denote the number of barrels of ale produced, and let $x_1 \geq 0$ denote the number of barrels of beer produced. The linear

programming formulation is:

$$\begin{aligned} \text{maximize} \quad & Z = 13x_0 + 23x_1 & \\ \text{subject to} \quad & 5x_0 + 15x_1 \leq 480 & \text{(Corn inventory)} \\ & 4x_0 + 4x_1 \leq 160 & \text{(Hops inventory)} \\ & 35x_0 + 20x_1 \leq 1190 & \text{(Malt inventory)} \\ & x_0 \geq 0,\ x_1 \geq 0. & \end{aligned} \tag{4.3}$$

4.2.2 Geometric Perspective in Two Dimensions

Because Equation (4.3) involves only two decision variables (x_0 and x_1), we can visualize its entire mathematical structure in the Cartesian plane, as illustrated in Figure 4.1.

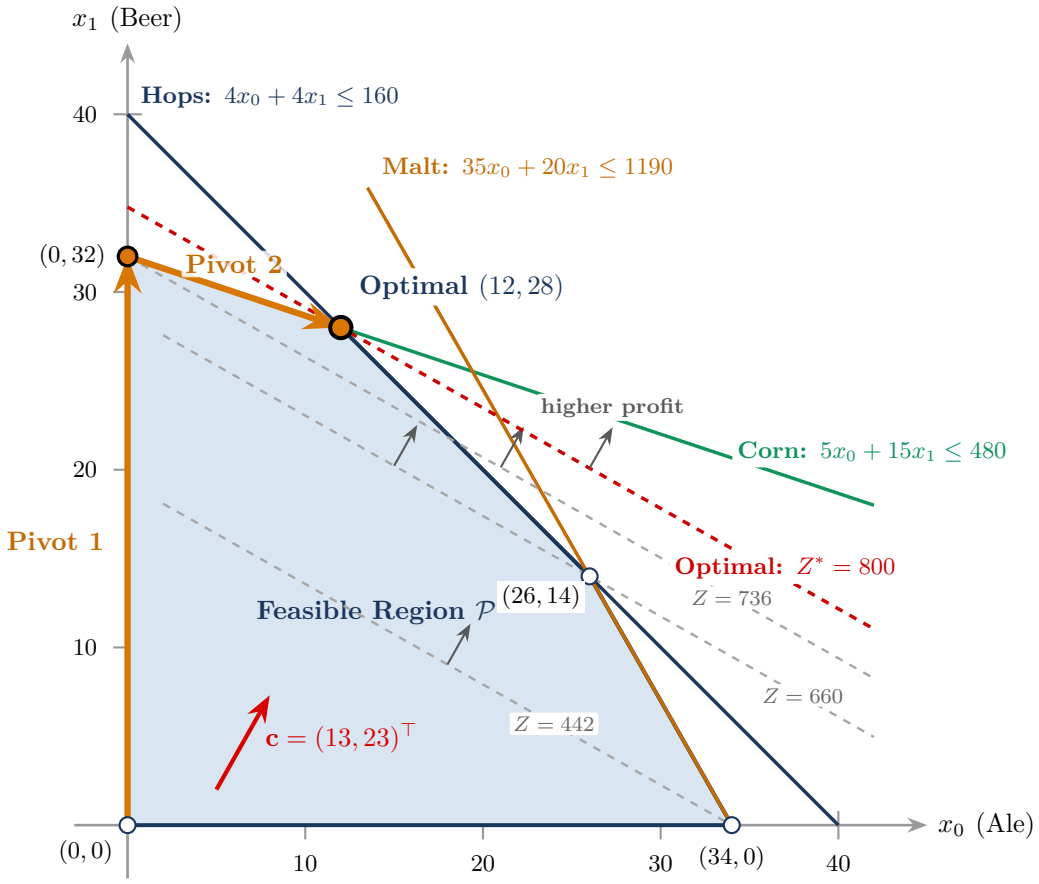


Figure 4.1: Feasible region of the Brewery Problem (Equation (4.3)). The shaded polygon $\mathcal{P}$ is the intersection of five closed halfspaces. Dashed lines depict iso-profit contours at the extreme points ($Z = 442, 660, 736$) up to the optimal contour ($Z^* = 800$), with orthogonal arrows indicating the gradient direction of increasing profit. The Simplex algorithm moves from the origin $(0, 0)$ to $(0, 32)$ along Pivot 1, and then to the optimal vertex $(12, 28)$ along Pivot 2.

Each linear inequality defines a closed *halfspace*:

$$H_i = \left\{ \mathbf{x} \in \mathbb{R}^2 \;\middle|\; \mathbf{a}_i^\top \mathbf{x} \leq b_i \right\}. \tag{4.4}$$

The set of feasible production plans is the intersection of these five halfspaces:

$$\mathcal{P} = \left\{ \mathbf{x} \in \mathbb{R}^2_{\geq 0} \;\middle|\; 5x_0 + 15x_1 \leq 480,\ 4x_0 + 4x_1 \leq 160,\ 35x_0 + 20x_1 \leq 1190 \right\}. \tag{4.5}$$

The boundary of $\mathcal{P}$ is a closed, convex polygon with five corners, called *vertices* or *extreme points*:

- $V_0 = (0, 0)$ with profit $Z = 13(0) + 23(0) = 0$.
- $V_1 = (0, 32)$ with profit $Z = 13(0) + 23(32) = 736$.
- $V_2 = (12, 28)$ with profit $Z = 13(12) + 23(28) = 156 + 644 = 800$.
- $V_3 = (26, 14)$ with profit $Z = 13(26) + 23(14) = 338 + 322 = 660$.
- $V_4 = (34, 0)$ with profit $Z = 13(34) + 23(0) = 442$.

The objective function $Z = 13x_0 + 23x_1$ defines a family of parallel lines with normal vector $\mathbf{c} = (13, 23)^\top$, called *iso-profit contour lines*. As profit increases, the line moves in the direction of the gradient $\mathbf{c}$. The last point of contact between the expanding contour line and the feasible region $\mathcal{P}$ is the vertex $V_2 = (12, 28)$, establishing that the optimal production plan yields $Z^* = 800$.

4.3 Geometry in Higher Dimensions

4.3.1 A Three-Dimensional Polytope Example

The geometric principles observed in two dimensions generalize naturally to three and higher dimensions. Consider the following 3-dimensional linear program with variables $x_0, x_1, x_2 \geq 0$:

$$\begin{aligned} \text{maximize} \quad & Z = x_0 + x_1 \\ \text{subject to} \quad & x_0 + x_1 + x_2 \leq 4 \\ & x_0 \leq 2 \\ & x_2 \leq 3 \\ & 3x_1 + x_2 \leq 6 \\ & x_0, x_1, x_2 \geq 0. \end{aligned} \tag{4.6}$$

As shown in Figure 4.2, each linear inequality defines a halfspace bounded by a 2-dimensional plane in $\mathbb{R}^3$. The intersection of these halfspaces forms a 3-dimensional bounded polytope whose extreme points are:

$$(0, 0, 0),\ (2, 0, 0),\ (2, 2, 0),\ (0, 2, 0),\ (0, 0, 3),\ (1, 0, 3),\ (0, 1, 3),\ (2, 0, 2).$$

For the objective function $\max(x_0 + x_1)$, evaluating the objective across all vertices reveals that the maximum value $Z^* = 4$ is attained at the vertex $\mathbf{x}^* = (2, 2, 0)$. For an alternative objective such as $\max(x_0 + x_1 + x_2)$, the maximum value $Z^* = 4$ is simultaneously attained at multiple vertices: $(2, 2, 0)$, $(1, 0, 3)$, $(0, 1, 3)$, and $(2, 0, 2)$, meaning the entire face connecting these vertices is optimal.

4.3.2 Polyhedral Theory and Convexity

We now formalize the mathematics governing polyhedral geometry in $\mathbb{R}^n$ [BT97].

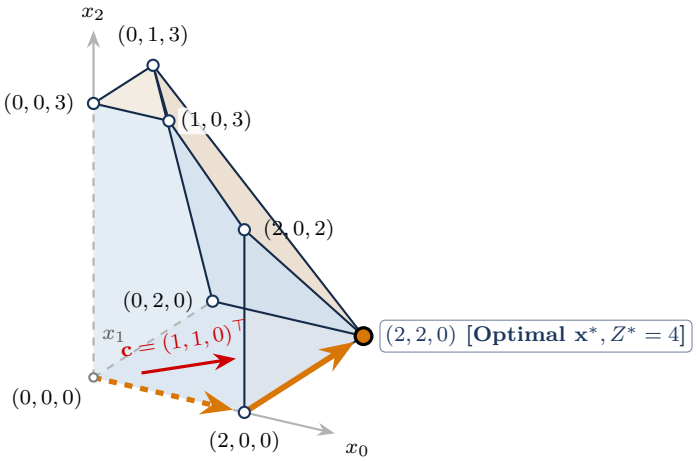


Figure 4.2: Feasible 3D polytope for Equation (4.6). The feasible region is bounded by planar facets. Vertices correspond to basic solutions where three independent planes intersect. Edges connect adjacent vertices, and Simplex walks along improving edges to reach the optimal vertex $\mathbf{x}^* = (2, 2, 0)$.

Definition 4.2 *Convex Set and Convex Combination*

A set $S \subseteq \mathbb{R}^n$ is *convex* if, for any two points $\mathbf{x}, \mathbf{y} \in S$, the entire line segment connecting them belongs to S:

$$\alpha\mathbf{x} + (1 - \alpha)\mathbf{y} \in S, \quad \forall \alpha \in [0, 1]. \tag{4.7}$$

A point $\mathbf{x} \in \mathbb{R}^n$ is a *convex combination* of points $\mathbf{v}^{(0)}, \mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(k-1)}$ if there exist non-negative weights $\alpha_0, \ldots, \alpha_{k-1} \geq 0$ such that $\sum_{i=0}^{k-1} \alpha_i = 1$ and:

$$\mathbf{x} = \sum_{i=0}^{k-1} \alpha_i \mathbf{v}^{(i)}. \tag{4.8}$$

Definition 4.3 *Polyhedron and Polytope*

A *polyhedron* $\mathcal{P} \subseteq \mathbb{R}^n$ is the intersection of a finite number of closed halfspaces:

$$\mathcal{P} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{A}\mathbf{x} \leq \mathbf{b}\}, \tag{4.9}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. A bounded polyhedron is called a *polytope*.

Proposition 4.1 *Convexity of Polyhedra*

Every polyhedron $\mathcal{P} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{A}\mathbf{x} \leq \mathbf{b}\}$ is convex.

Proof. Let $\mathbf{x}, \mathbf{y} \in \mathcal{P}$ and let $\alpha \in [0, 1]$. By definition, $\mathbf{A}\mathbf{x} \leq \mathbf{b}$ and $\mathbf{A}\mathbf{y} \leq \mathbf{b}$. Consider $\mathbf{z} = \alpha\mathbf{x} + (1 - \alpha)\mathbf{y}$. By linearity of matrix multiplication:

$$\mathbf{A}\mathbf{z} = \mathbf{A}(\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}) = \alpha(\mathbf{A}\mathbf{x}) + (1 - \alpha)(\mathbf{A}\mathbf{y}) \leq \alpha\mathbf{b} + (1 - \alpha)\mathbf{b} = \mathbf{b}. \tag{4.10}$$

Hence, $\mathbf{z} \in \mathcal{P}$, proving that $\mathcal{P}$ is convex. □

4.3.3 Extreme Points and Vertex Optimality

Definition 4.4 *Extreme Point and Vertex*

Let $\mathcal{P} \subseteq \mathbb{R}^n$ be a polyhedron.

1. A point $\mathbf{x} \in \mathcal{P}$ is an *extreme point* if it cannot be expressed as a strict convex combination of two distinct points in $\mathcal{P}$:

$$\mathbf{x} = \alpha\mathbf{y} + (1-\alpha)\mathbf{z}, \quad \mathbf{y}, \mathbf{z} \in \mathcal{P},\ \alpha \in (0,1) \implies \mathbf{x} = \mathbf{y} = \mathbf{z}. \tag{4.11}$$

2. A point $\mathbf{x} \in \mathcal{P}$ is a *vertex* if there exists a cost vector $\mathbf{c} \in \mathbb{R}^n$ such that $\mathbf{x}$ is the unique maximizer of $\mathbf{c}^\top\mathbf{y}$ over all $\mathbf{y} \in \mathcal{P}$:

$$\mathbf{c}^\top\mathbf{x} > \mathbf{c}^\top\mathbf{y}, \quad \forall \mathbf{y} \in \mathcal{P} \setminus \{\mathbf{x}\}. \tag{4.12}$$

Theorem 4.2 *Minkowski-Weyl Representation Theorem*

Every point in a bounded polytope $\mathcal{P} \subset \mathbb{R}^n$ is a convex combination of its finite set of extreme points (vertices) $\{\mathbf{v}^{(0)}, \mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(k-1)}\}$:

$$\forall \mathbf{x} \in \mathcal{P}, \quad \exists \boldsymbol{\alpha} \in \mathbb{R}^k_{\geq 0} \text{ with } \sum_{i=0}^{k-1} \alpha_i = 1 \quad \text{such that} \quad \mathbf{x} = \sum_{i=0}^{k-1} \alpha_i \mathbf{v}^{(i)}. \tag{4.13}$$

Theorem 4.3 *Fundamental Theorem of Linear Programming*

Let $\mathcal{P} \subset \mathbb{R}^n$ be a non-empty bounded polytope. For any linear objective function $\mathbf{c}^\top\mathbf{x}$, at least one optimal solution $\mathbf{x}^*$ is an extreme point (vertex) of $\mathcal{P}$.

Proof. Let $\{\mathbf{v}^{(0)}, \mathbf{v}^{(1)}, \ldots, \mathbf{v}^{(k-1)}\}$ denote the finite collection of extreme points of $\mathcal{P}$. Because $\mathcal{P}$ is compact and the objective function $\mathbf{c}^\top\mathbf{x}$ is continuous, the maximum of $\mathbf{c}^\top\mathbf{x}$ over $\mathcal{P}$ is attained at some point $\mathbf{x}^* \in \mathcal{P}$.

By the Minkowski-Weyl Representation Theorem (Theorem 4.2), $\mathbf{x}^*$ can be represented as a convex combination of the vertices:

$$\mathbf{x}^* = \sum_{i=0}^{k-1} \alpha_i \mathbf{v}^{(i)}, \quad \text{where } \alpha_i \geq 0, \quad \sum_{i=0}^{k-1} \alpha_i = 1. \tag{4.14}$$

Applying the linear objective function $\mathbf{c}^\top$ yields:

$$\mathbf{c}^\top\mathbf{x}^* = \mathbf{c}^\top \left(\sum_{i=0}^{k-1} \alpha_i \mathbf{v}^{(i)} \right) = \sum_{i=0}^{k-1} \alpha_i \left(\mathbf{c}^\top\mathbf{v}^{(i)} \right). \tag{4.15}$$

Let $\mathbf{v}^{(j)}$ be a vertex that achieves the maximum objective value among all vertices: $\mathbf{c}^\top\mathbf{v}^{(j)} = \max_{0 \leq i < k} \mathbf{c}^\top\mathbf{v}^{(i)}$. Then:

$$\mathbf{c}^\top\mathbf{x}^* = \sum_{i=0}^{k-1} \alpha_i \left(\mathbf{c}^\top\mathbf{v}^{(i)} \right) \leq \sum_{i=0}^{k-1} \alpha_i \left(\mathbf{c}^\top\mathbf{v}^{(j)} \right) = \left(\mathbf{c}^\top\mathbf{v}^{(j)} \right) \sum_{i=0}^{k-1} \alpha_i = \mathbf{c}^\top\mathbf{v}^{(j)}. \tag{4.16}$$

Since $\mathbf{v}^{(j)} \in \mathcal{P}$ and $\mathbf{x}^*$ is optimal over all points in $\mathcal{P}$, we also have $\mathbf{c}^\top\mathbf{v}^{(j)} \leq \mathbf{c}^\top\mathbf{x}^*$. Therefore:

$$\mathbf{c}^\top\mathbf{x}^* = \mathbf{c}^\top\mathbf{v}^{(j)}. \tag{4.17}$$

Thus, the vertex $\mathbf{v}^{(j)}$ attains the optimal objective value, proving the theorem. □

4.3.4 Combinatorial Explosion of Vertices

Theorem 4.3 implies that solving a continuous optimization problem over infinitely many feasible points reduces to searching over a finite set of vertices. However, exhaustive vertex enumeration is computationally intractable because the number of vertices grows exponentially with the dimension:

- In 1 dimension, the unit interval $[0, 1]$ has $2 = 2^1$ vertices (defined by 2 inequalities).
- In 2 dimensions, the unit square $[0, 1]^2$ has $4 = 2^2$ vertices (defined by 4 inequalities).
- In 3 dimensions, the unit cube $[0, 1]^3$ has $8 = 2^3$ vertices (defined by 6 inequalities).
- In n dimensions, the unit hypercube $[0, 1]^n$ has 2^n vertices (defined by $2n$ linear inequalities).

For $n = 50$, $2^{50} \approx 1.12 \times 10^{15}$ vertices; evaluating one vertex per microsecond would require more than 35,000 years. For a general system with m constraints and $n + m$ variables, the number of potential vertices can reach $\binom{n+m}{m}$.

This combinatorial explosion demonstrates that brute-force enumeration is impractical. We require an algorithm that navigates through the vertices intelligently: the Simplex algorithm.

4.4 Canonical and Standard Forms

4.4.1 Motivation for Generic Formulations

In two and three dimensions, geometry allows us to visualize halfspaces, draw contour lines, and identify optimal vertices visually. In real-world discrete optimization, however, models routinely involve tens of thousands of variables and constraints. To solve arbitrary linear programs on a computer, we cannot rely on geometric visualization; we must design a generic, purely algebraic algorithm.

Just as the Gauss-Jordan elimination algorithm operates on arbitrary systems of linear equations $\mathbf{Ax} = \mathbf{b}$, an algebraic linear programming solver requires a standardized mathematical formulation. Standardizing the problem allows a single solver engine to process any optimization instance regardless of its original domain-specific structure.

4.4.2 Canonical and Standard Equational Forms

Two canonical representations dominate the optimization literature:

Definition 4.5 *Canonical Form*

A linear program is in *canonical form* if it is a maximization problem where all functional constraints are less-than-or-equal-to inequalities ($\leq$) and all variables are non-negative:

$$\begin{aligned} \text{maximize} \quad & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} \quad & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}, \end{aligned} \tag{4.18}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, and $\mathbf{c} \in \mathbb{R}^n$.

Definition 4.6 *Standard Form*

A linear program is in *standard form* (or equational form) if the objective is maximization, all functional constraints are linear equations (=), and all variables are non-negative:

$$\begin{aligned} \text{maximize} \quad & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}, \end{aligned} \tag{4.19}$$

where $\mathbf{b} \geq \mathbf{0}$. Standard form is the direct operational format for the Simplex algorithm.

4.4.3 Form Transformations and Slack Variables

Every linear program can be converted into standard form through the following transformations:

1. **Minimization to Maximization**: Negate the cost coefficients:

$$\min \mathbf{d}^\top \mathbf{x} \iff -\max(-\mathbf{d}^\top \mathbf{x}). \tag{4.20}$$

2. **Inequalities to Equalities (Slack Variables)**: For each less-than-or-equal-to constraint $\sum_{j=0}^{n-1} a_{i,j} x_j \leq b_i$, introduce a non-negative *slack variable* $x_{n+i} \geq 0$:

$$\sum_{j=0}^{n-1} a_{i,j} x_j + x_{n+i} = b_i, \quad x_{n+i} \geq 0. \tag{4.21}$$

   If $x_{n+i} = 0$, the constraint holds with strict equality and is said to be *tight* or *active*.

3. **Surplus Variables**: For a greater-than-or-equal-to constraint $\sum_{j=0}^{n-1} a_{i,j} x_j \geq b_i$, subtract a non-negative *surplus variable* $s_i \geq 0$:

$$\sum_{j=0}^{n-1} a_{i,j} x_j - s_i = b_i, \quad s_i \geq 0. \tag{4.22}$$

4. **Equalities to Inequalities**: An equality $\mathbf{a}_i^\top \mathbf{x} = b_i$ can be converted into two inequalities: $\mathbf{a}_i^\top \mathbf{x} \leq b_i$ and $-\mathbf{a}_i^\top \mathbf{x} \leq -b_i$.
5. **Unconstrained (Free) Variables**: A decision variable $x_j \in \mathbb{R}$ with no sign restriction is split into two non-negative variables:

$$x_j = x_j^+ - x_j^-, \quad \text{with } x_j^+ \geq 0,\ x_j^- \geq 0. \tag{4.23}$$

Example 4.1 *Converting the Brewery Problem to Standard Form*

The Brewery Problem (Equation (4.3)) has $n = 2$ original decision variables (x_0, x_1) and $m = 3$ resource constraints. We introduce three slack variables: $x_2 \geq 0$ (unused

corn), $x_3 \geq 0$ (unused hops), and $x_4 \geq 0$ (unused malt):

$$\begin{aligned} \text{maximize} \quad & Z = 13x_0 + 23x_1 + 0x_2 + 0x_3 + 0x_4 \\ \text{subject to} \quad & 5x_0 + 15x_1 + x_2 = 480 \\ & 4x_0 + 4x_1 + x_3 = 160 \\ & 35x_0 + 20x_1 + x_4 = 1190 \\ & x_0, x_1, x_2, x_3, x_4 \geq 0. \end{aligned} \tag{4.24}$$

The resulting system has $n' = n + m = 5$ variables and $m = 3$ equations.

4.5 The Simplex Algorithm

4.5.1 Simplex by Example: The Brewery Problem

To understand how Dantzig's Simplex algorithm systematically navigates the boundary of a polytope, we return to the standard-form Brewery Problem established in Equation (4.24):

$$\begin{aligned} \text{maximize} \quad & Z = 13x_0 + 23x_1 + 0x_2 + 0x_3 + 0x_4 & & \\ \text{subject to} \quad & 5x_0 + 15x_1 + x_2 = 480 & \text{(Corn)} & \\ & 4x_0 + 4x_1 + x_3 = 160 & \text{(Hops)} & \\ & 35x_0 + 20x_1 + x_4 = 1190 & \text{(Malt)} & \\ & x_0, x_1, x_2, x_3, x_4 \geq 0. & & \end{aligned} \tag{4.25}$$

This system comprises $m = 3$ equations and $n' = n + m = 5$ non-negative variables, leaving $5 - 3 = 2$ degrees of freedom. Because there are more variables than equations, the system cannot be uniquely inverted without fixing two variables.

The core insight of the Simplex method is to partition the variables into two complementary sets:

- **Non-basic variables** ($n = 2$ variables): set temporarily to zero.
- **Basic variables** ($m = 3$ variables): uniquely solved in terms of the non-basic variables.

Expressing the basic variables and the objective function Z exclusively in terms of the non-basic variables produces an algebraic representation known as a *dictionary*.

Step 0: Initial Basic Feasible Solution (BFS) We designate the original decision variables as non-basic: $x_0 = 0$ (ale) and $x_1 = 0$ (beer). Isolating the three slack variables $\{x_2, x_3, x_4\}$ as basic variables yields the initial dictionary:

$$x_2 = 480 - 5x_0 - 15x_1, \tag{4.26}$$
$$x_3 = 160 - 4x_0 - 4x_1, \tag{4.27}$$
$$x_4 = 1190 - 35x_0 - 20x_1, \tag{4.28}$$
$$Z = 0 + 13x_0 + 23x_1. \tag{4.29}$$

Setting $x_0 = 0$ and $x_1 = 0$ immediately determines the basic solution:

$$x_2 = 480, \quad x_3 = 160, \quad x_4 = 1190, \quad Z = 0. \tag{4.30}$$

Because all variables are non-negative ($x \geq \mathbf{0}$), this basic solution is feasible—a *basic feasible solution* (BFS). Geometrically, this initial point corresponds to the origin vertex $V_0 = (0, 0)$ in Figure 4.1.

Step 1: First Pivot (Moving from V_0 to V_1) To increase the profit $Z = 13x_0 + 23x_1$, we observe that both non-basic variables have positive coefficients. Increasing either variable improves the objective:

- **Entering Variable Selection**: Producing beer (x_1) increases profit at a rate of \$23 per barrel, while ale (x_0) yields \$13 per barrel. Following Dantzig's largest-coefficient rule, we choose x_1 as the *entering variable* to enter the basis. We hold the other non-basic variable at $x_0 = 0$.
- **Leaving Variable Selection (Minimum Ratio Rule)**: As x_1 increases, the basic slack variables decrease according to Equations (4.26) to (4.28):

$$\begin{aligned} x_2 = 480 - 15x_1 \geq 0 &\implies x_1 \leq \frac{480}{15} = 32, \\ x_3 = 160 - 4x_1 \geq 0 &\implies x_1 \leq \frac{160}{4} = 40, \\ x_4 = 1190 - 20x_1 \geq 0 &\implies x_1 \leq \frac{1190}{20} = 59.5. \end{aligned}$$

  The strictest bound is imposed by the corn constraint ($x_2 \geq 0$): x_1 cannot exceed 32. At $x_1 = 32$, corn is fully exhausted ($x_2 = 0$). Increasing x_1 further would make $x_2 < 0$, violating feasibility. Hence, x_2 is the *leaving variable*: it leaves the basis and becomes non-basic.
- **Dictionary Substitution**: We invert the corn equation to express the new basic variable x_1 in terms of the new non-basic variables (x_0, x_2):

$$15x_1 = 480 - 5x_0 - x_2 \implies x_1 = 32 - \frac{1}{3}x_0 - \frac{1}{15}x_2. \tag{4.31}$$

  Substituting Equation (4.31) into the remaining dictionary equations yields:

$$x_1 = 32 - \frac{1}{3}x_0 - \frac{1}{15}x_2, \tag{4.32}$$

$$x_3 = 160 - 4x_0 - 4\left(32 - \frac{1}{3}x_0 - \frac{1}{15}x_2\right) = 32 - \frac{8}{3}x_0 + \frac{4}{15}x_2, \tag{4.33}$$

$$x_4 = 1190 - 35x_0 - 20\left(32 - \frac{1}{3}x_0 - \frac{1}{15}x_2\right) = 550 - \frac{85}{3}x_0 + \frac{4}{3}x_2, \tag{4.34}$$

$$Z = 13x_0 + 23\left(32 - \frac{1}{3}x_0 - \frac{1}{15}x_2\right) = 736 + \frac{16}{3}x_0 - \frac{23}{15}x_2. \tag{4.35}$$

Setting the non-basic variables $x_0 = 0$ and $x_2 = 0$ produces the new BFS:

$$x_1 = 32, \quad x_3 = 32, \quad x_4 = 550, \quad x_0 = 0, \quad x_2 = 0, \quad Z = 736. \tag{4.36}$$

Geometrically, this pivot traversed the edge connecting $V_0(0, 0)$ to $V_1(0, 32)$ in Figure 4.1, boosting profit from \$0 to \$736.

Step 2: Second Pivot (Moving from V_1 to V_2) We inspect the updated objective formula Equation (4.35):

$$Z = 736 + \frac{16}{3}x_0 - \frac{23}{15}x_2. \tag{4.37}$$

- **Entering Variable Selection**: The coefficient of x_2 is $-23/15 < 0$, confirming that leaving corn idle reduces profit. However, the coefficient of x_0 is $+16/3 > 0$. Even though ale initially appeared less profitable than beer (\$13 vs. \$23), producing ale is

now beneficial because it consumes fewer resources per barrel than beer (only 5 kg of corn vs. 15 kg), allowing resource reallocation. We select x_0 to enter the basis (holding $x_2 = 0$).

- **Leaving Variable Selection**: With $x_2 = 0$, increasing x_0 impacts basic variables according to:

$$x_1 = 32 - \frac{1}{3}x_0 \geq 0 \implies x_0 \leq \frac{32}{1/3} = 96,$$
$$x_3 = 32 - \frac{8}{3}x_0 \geq 0 \implies x_0 \leq \frac{32}{8/3} = 12,$$
$$x_4 = 550 - \frac{85}{3}x_0 \geq 0 \implies x_0 \leq \frac{550}{85/3} \approx 19.41.$$

  The minimum ratio is $\theta^* = 12$, reached by x_3 (hops). At $x_0 = 12$, available hops are fully exhausted ($x_3 = 0$). Thus, x_3 leaves the basis and x_0 enters.

- **Dictionary Substitution**: Isolating x_0 from the hops equation Equation (4.33):

$$\frac{8}{3}x_0 = 32 + \frac{4}{15}x_2 - x_3 \implies x_0 = 12 + \frac{1}{10}x_2 - \frac{3}{8}x_3. \tag{4.38}$$

  Substituting Equation (4.38) into Equations (4.32), (4.34) and (4.35):

$$x_1 = 32 - \frac{1}{3}\left(12 + \frac{1}{10}x_2 - \frac{3}{8}x_3\right) - \frac{1}{15}x_2 = 28 - \frac{1}{10}x_2 + \frac{1}{8}x_3, \tag{4.39}$$
$$x_0 = 12 + \frac{1}{10}x_2 - \frac{3}{8}x_3, \tag{4.40}$$
$$x_4 = 550 - \frac{85}{3}\left(12 + \frac{1}{10}x_2 - \frac{3}{8}x_3\right) + \frac{4}{3}x_2 = 210 - \frac{3}{2}x_2 + \frac{85}{8}x_3, \tag{4.41}$$
$$Z = 736 + \frac{16}{3}\left(12 + \frac{1}{10}x_2 - \frac{3}{8}x_3\right) - \frac{23}{15}x_2 = 800 - x_2 - 2x_3. \tag{4.42}$$

Setting the non-basic variables $x_2 = 0$ and $x_3 = 0$ produces the new BFS:

$$x_0 = 12, \quad x_1 = 28, \quad x_4 = 210, \quad x_2 = 0, \quad x_3 = 0, \quad Z = 800. \tag{4.43}$$

Geometrically, this pivot traversed the edge from $V_1(0, 32)$ to $V_2(12, 28)$ in Figure 4.1.

Step 3: Stopping Criterion and Optimality Certificate Consider the final objective expression Equation (4.42):

$$Z = 800 - x_2 - 2x_3. \tag{4.44}$$

Every feasible production plan must satisfy non-negativity: $x_2 \geq 0$ and $x_3 \geq 0$. Because the coefficients of both non-basic variables are strictly negative (-1 and -2), any feasible point satisfies:

$$-x_2 \leq 0 \quad \text{and} \quad -2x_3 \leq 0 \implies Z = 800 - x_2 - 2x_3 \leq 800. \tag{4.45}$$

This inequality holds for *all* feasible points in the entire polytope $\mathcal{P}$. Because our current basic solution achieves $Z = 800$, no feasible plan can possibly surpass it. The dictionary thus provides an irrefutable, self-verifying **certificate of global optimality**. Pivoting terminates with the optimal solution:

$$x_0^* = 12 \text{ (ale)}, \quad x_1^* = 28 \text{ (beer)}, \quad x_4^* = 210 \text{ (unused malt)}, \quad x_2^* = x_3^* = 0, \tag{4.46}$$

yielding maximal profit $Z^* = 800$.

Step 4: Simplex Tableaux Representation Carrying out symbolic variable substitutions by hand is tedious. The numerical coefficients can be recorded compactly in a 2D matrix called a *simplex tableau*. By rewriting the objective equation as:

$$Z - 13x_0 - 23x_1 + 0x_2 + 0x_3 + 0x_4 = 0, \tag{4.47}$$

the pivots correspond directly to elementary Gauss-Jordan row operations.

Initial Tableau (T_0): Basis $\{x_2, x_3, x_4\}$, $Z = 0$:

$$T_0 = \left[\begin{array}{ccccc|c} x_0 & x_1 & x_2 & x_3 & x_4 & \text{RHS} \\ \hline 5 & \mathbf{15} & 1 & 0 & 0 & 480 \\ 4 & 4 & 0 & 1 & 0 & 160 \\ 35 & 20 & 0 & 0 & 1 & 1190 \\ \hline 13 & 23 & 0 & 0 & 0 & 0 \end{array}\right] \tag{4.48}$$

- Entering column: $q = 1$ (x_1, beer), since $a[3][1] = 23 > 0$.
- Minimum ratio test: Row 0: $480/15 = 32$; Row 1: $160/4 = 40$; Row 2: $1190/20 = 59.5$. Minimum ratio is $\theta^* = 32$ at row $p = 0$. Pivot element is $a[0][1] = \mathbf{15}$.

Tableau 1 (T_1): Basis $\{x_1, x_3, x_4\}$, $Z = 736$:

$$T_1 = \left[\begin{array}{ccccc|c} x_0 & x_1 & x_2 & x_3 & x_4 & \text{RHS} \\ \hline 1/3 & 1 & 1/15 & 0 & 0 & 32 \\ \mathbf{8/3} & 0 & -4/15 & 1 & 0 & 32 \\ 85/3 & 0 & -4/3 & 0 & 1 & 550 \\ \hline 16/3 & 0 & -23/15 & 0 & 0 & -736 \end{array}\right] \tag{4.49}$$

- Entering column: $q = 0$ (x_0, ale), since $a[3][0] = 16/3 > 0$.
- Minimum ratio test: Row 0: $32/(1/3) = 96$; Row 1: $32/(8/3) = 12$; Row 2: $550/(85/3) \approx 19.41$. Minimum ratio is $\theta^* = 12$ at row $p = 1$. Pivot element is $a[1][0] = \mathbf{8/3}$.

Tableau 2 (T_2): Basis $\{x_1, x_0, x_4\}$, $Z = 800$:

$$T_2 = \left[\begin{array}{ccccc|c} x_0 & x_1 & x_2 & x_3 & x_4 & \text{RHS} \\ \hline 0 & 1 & 1/10 & -1/8 & 0 & 28 \\ 1 & 0 & -1/10 & 3/8 & 0 & 12 \\ 0 & 0 & 3/2 & -85/8 & 1 & 210 \\ \hline 0 & 0 & -1 & -2 & 0 & -800 \end{array}\right] \tag{4.50}$$

All reduced costs in row 3 are non-positive ($\bar{\mathbf{c}} = (0, 0, -1, -2, 0) \leq \mathbf{0}$). Optimality is reached at $Z^* = 800$. Notice how this algebraic sequence exactly mirrors the vertex trajectory $V_0(0, 0) \to V_1(0, 32) \to V_2(12, 28)$ identified geometrically in Figure 4.1.

4.5.2 Formalization of the Simplex Method

Having witnessed the mechanics of the Simplex algorithm on the Brewery Problem, we now establish its formal mathematical theory for an arbitrary linear program.

Bases and Matrix Partitions

Consider a standard-form linear program with m equality constraints and $n' = n + m$ non-negative variables:

$$\text{maximize} \quad \mathbf{c}^\top \mathbf{x} \quad \text{s.t.} \quad \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}, \tag{4.51}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n'}$ has full row rank $\operatorname{rank}(\mathbf{A}) = m \leq n'$. We partition the columns of $\mathbf{A}$ into an $m \times m$ invertible basis matrix $\mathbf{A}_B$ and an $m \times (n' - m)$ non-basis matrix $\mathbf{A}_N$:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_B & \mathbf{A}_N \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} \mathbf{x}_B \\ \mathbf{x}_N \end{bmatrix}, \qquad \mathbf{c} = \begin{bmatrix} \mathbf{c}_B \\ \mathbf{c}_N \end{bmatrix}. \tag{4.52}$$

The constraint equation $\mathbf{A}_B\mathbf{x}_B + \mathbf{A}_N\mathbf{x}_N = \mathbf{b}$ determines the basic variables $\mathbf{x}_B$ as a linear function of the non-basic variables $\mathbf{x}_N$:

$$\mathbf{x}_B = \mathbf{A}_B^{-1}\mathbf{b} - \mathbf{A}_B^{-1}\mathbf{A}_N\mathbf{x}_N. \tag{4.53}$$

A *basic solution* sets all non-basic variables to zero: $\mathbf{x}_N = \mathbf{0}$, yielding $\mathbf{x}_B = \mathbf{A}_B^{-1}\mathbf{b}$. If $\mathbf{x}_B \geq \mathbf{0}$, the solution is a *basic feasible solution* (BFS), which corresponds algebraically to an extreme point of the polytope $\mathcal{P} = \{\mathbf{x} \geq \mathbf{0} \mid \mathbf{A}\mathbf{x} = \mathbf{b}\}$.

Substituting Equation (4.53) into the objective function yields:

$$\begin{aligned} Z &= \mathbf{c}_B^\top \mathbf{x}_B + \mathbf{c}_N^\top \mathbf{x}_N \\ &= \mathbf{c}_B^\top \left(\mathbf{A}_B^{-1}\mathbf{b} - \mathbf{A}_B^{-1}\mathbf{A}_N\mathbf{x}_N \right) + \mathbf{c}_N^\top \mathbf{x}_N \\ &= \underbrace{\mathbf{c}_B^\top \mathbf{A}_B^{-1}\mathbf{b}}_{Z_0} + \underbrace{\left(\mathbf{c}_N^\top - \mathbf{c}_B^\top \mathbf{A}_B^{-1}\mathbf{A}_N \right)}_{\bar{\mathbf{c}}_N^\top} \mathbf{x}_N. \end{aligned} \tag{4.54}$$

The scalar Z_0 represents the objective value at the current basic solution, and $\bar{\mathbf{c}}_N^\top$ is the vector of *reduced costs*. The reduced cost $\bar{c}_j$ quantifies the net rate of increase in Z per unit increase in the non-basic variable x_j.

The complete equational system is stored in a 2D tableau `a[i][j]` of dimensions $(m + 1) \times (n + m + 1)$:

Basis	$\mathbf{x}_N$	$\mathbf{x}_B$	RHS
$\mathbf{x}_B$	$\mathbf{A}_B^{-1}\mathbf{A}_N$	$\mathbf{I}_m$	$\mathbf{A}_B^{-1}\mathbf{b}$
Obj	$-(\mathbf{c}_N^\top - \mathbf{c}_B^\top \mathbf{A}_B^{-1}\mathbf{A}_N)$	$\mathbf{0}^\top$	$-Z_0$

(4.55)

In array coordinates:

- Rows $0 \leq i < m$ store the constraint equations transformed by $\mathbf{A}_B^{-1}$.
- Row m stores the objective row (reduced costs in columns $0 \leq j < n + m$, and $-Z_0$ in column $n + m$).
- Column $n + m$ holds the current right-hand-side values $\mathbf{A}_B^{-1}\mathbf{b}$.

Pivoting and the Minimum Ratio Rule

A single iteration of the Simplex method executes a *pivot*, which swaps one non-basic variable into the basis (the *entering variable*) and one basic variable out (the *leaving variable*).

Step 1: Selecting the Entering Variable To increase the objective function in a maximization problem, we must pick a non-basic column q with a positive reduced cost: $a[m][q] > 0$.

- **Dantzig's Rule**: Choose $q = \arg\max_j a[m][j]$. This maximizes the immediate per-unit rate of ascent.
- **Bland's Smallest-Subscript Rule**: Choose $q = \min\{j \mid a[m][j] > 0\}$. This rule provably prevents cycling (Section 4.8).

If $a[m][j] \leq 0$ for all columns $j \in \{0, \ldots, n + m - 1\}$, the current basic feasible solution is optimal.

Step 2: Selecting the Leaving Variable (Minimum Ratio Rule) Increasing x_q causes the basic variables to adjust according to $x_{B(i)} = \bar{b}_i - \bar{a}_{i,q}x_q$. To maintain non-negativity $x_{B(i)} \geq 0$, x_q can increase only until the first basic variable drops to zero:

$$\theta^* = \min_{0 \leq i < m \,:\, a[i][q] > 0} \frac{a[i][n+m]}{a[i][q]}. \tag{4.56}$$

The row p that attains this minimum ratio is selected as the pivot row:

$$p = \arg\min_{0 \leq i < m \,:\, a[i][q] > 0} \frac{a[i][n+m]}{a[i][q]}. \tag{4.57}$$

If $a[i][q] \leq 0$ for all rows $i \in \{0, \ldots, m-1\}$, x_q can be increased indefinitely without violating any constraint. The linear program is *unbounded* ($Z^* = +\infty$).

Step 3: Gauss-Jordan Elimination The tableau is updated using elementary row operations:

$$\begin{aligned} a'[i][j] &= a[i][j] - a[p][j] \cdot \frac{a[i][q]}{a[p][q]}, && \forall i \neq p,\ \forall j, \\ a'[p][j] &= \frac{a[p][j]}{a[p][q]}, && \forall j. \end{aligned} \tag{4.58}$$

Optimality Certificate in the Tableau

Theorem 4.4 *Simplex Optimality Condition*

If every reduced cost in row m of the simplex tableau is non-positive:

$$\bar{c}_j = a[m][j] \leq 0, \quad \forall j \in \{0, \ldots, n+m-1\}, \tag{4.59}$$

then the current basic feasible solution is globally optimal.

Proof. For any feasible solution $\mathbf{x} \geq \mathbf{0}$, Equation (4.54) states:

$$Z(\mathbf{x}) = Z_0 + \sum_{j \in N} \bar{c}_j x_j. \tag{4.60}$$

Because $x_j \geq 0$ and $\bar{c}_j \leq 0$ for every $j \in N$, each term $\bar{c}_j x_j \leq 0$. Hence:

$$Z(\mathbf{x}) = Z_0 + \sum_{j \in N} \underbrace{\bar{c}_j}_{\leq 0} \underbrace{x_j}_{\geq 0} \leq Z_0. \tag{4.61}$$

Since the current basic solution achieves $Z = Z_0$, no feasible solution can yield a larger objective value. □

4.5.3 Algorithmic Implementation in Java

Tableau Representation and Memory Layout As illustrated in Figure 4.3, the tableau is stored as a single two-dimensional array `a` of dimensions $(m+1) \times (n+m+1)$. This layout directly mirrors the partitioned block dictionary of Equation (4.55):

- **Constraint Matrix A** $(m \times n)$: Rows $0 \leq i < m$ and columns $0 \leq j < n$ hold the original constraint coefficients $A_{i,j}$.
- **Slack Variables** $\mathbf{I}_m$ $(m \times m)$: Rows $0 \leq i < m$ and columns $n \leq j < n+m$ contain an identity matrix. Setting the diagonal entries $\texttt{a}[i][n+i] = 1.0$ introduces m non-negative slack variables $x_n, \ldots, x_{n+m-1}$.
- **Right-Hand Side b** $(m \times 1)$: Column $n+m$ stores the resource limits b_i for each constraint row $0 \leq i < m$.
- **Objective Row** $\mathbf{c}^\top$ $(1 \times (n+m+1))$: Row m stores the objective function coefficients. Columns $0 \leq j < n$ hold the costs c_j (the initial reduced costs). Columns $n \leq j < n+m$ represent zero costs for slack variables, and column $n+m$ stores the negated current objective value $-Z_0 = 0.0$.
- **Basis Tracker `basis`**: An integer array of size m tracking which variable is currently basic in each constraint row i. Because all $b_i \geq 0$, the slacks form the initial basic feasible solution: $\texttt{basis}[i] = n+i$.

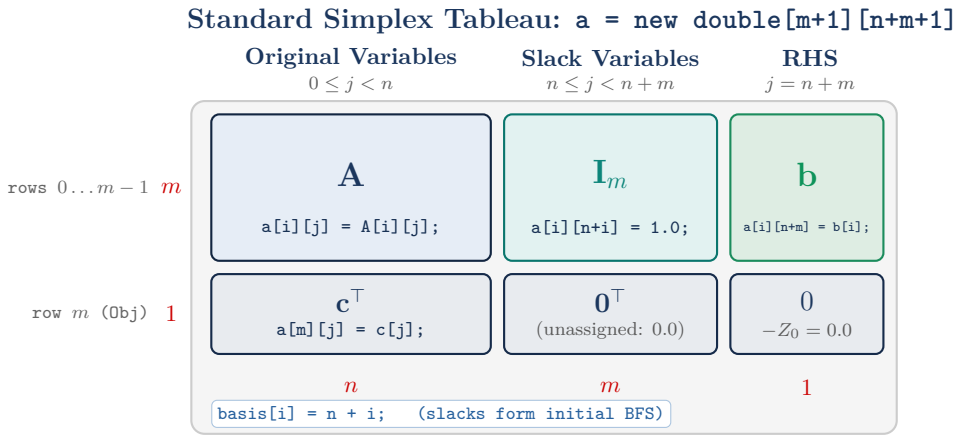


Figure 4.3: Standard Simplex tableau 2D array memory layout and Java initialization (`LinearProgramming.java`) storing m constraints and 1 objective row across n original variables, m slacks, and 1 RHS column in an array of size $(m+1) \times (n+m+1)$.

Listing 4.1 presents the complete Java implementation of the Simplex solver.

```
public class LinearProgramming {
    private static final double EPSILON = 1.0E-10;
    private double[][] a;   // (m+1) x (n+m+1) tableau
    private int m, n;
    private int[] basis;    // basis[i] = basic var for row i

    public LinearProgramming(double[][] A, double[] b,
                             double[] c) {
        m = b.length; n = c.length;
        for (int i = 0; i < m; i++)
            if (b[i] < 0) throw new IllegalArgumentException("RHS < 0");

        a = new double[m + 1][n + m + 1];
        for (int i = 0; i < m; i++)
            for (int j = 0; j < n; j++) a[i][j] = A[i][j];

        basis = new int[m];
        for (int i = 0; i < m; i++) {
            a[i][n + i] = 1.0;
            a[i][m + n] = b[i];
            basis[i] = n + i;
        }
        for (int j = 0; j < n; j++) a[m][j] = c[j];

        solve();
    }

    private void solve() {
        while (true) {
            int q = bland();
            if (q == -1) break; // optimal
            int p = minRatioRule(q);
            if (p == -1) throw new ArithmeticException("LP is unbounded");
            pivot(p, q);
            basis[p] = q;
        }
    }

    // Bland: smallest index with positive reduced cost
    private int bland() {
        for (int j = 0; j < m + n; j++)
            if (a[m][j] > EPSILON) return j;
        return -1;
    }

    // Dantzig: largest positive reduced cost
    private int dantzig() {
        int q = -1;
        double maxCost = EPSILON;
        for (int j = 0; j < m + n; j++)
            if (a[m][j] > maxCost) { maxCost = a[m][j]; q = j; }
        return q;
    }
```

```
54
55      private int minRatioRule(int q) {
56          int p = -1;
57          for (int i = 0; i < m; i++) {
58              if (a[i][q] <= EPSILON) continue;
59              if (p == -1 ||
60                  a[i][m+n] / a[i][q] < a[p][m+n] / a[p][q]) p = i;
61          }
62          return p;
63      }
64
65      private void pivot(int p, int q) {
66          for (int i = 0; i <= m; i++)
67              for (int j = 0; j <= m + n; j++)
68                  if (i != p && j != q)
69                      a[i][j] -= a[p][j] * (a[i][q] / a[p][q]);
70          for (int i = 0; i <= m; i++)
71              if (i != p) a[i][q] = 0.0;
72          for (int j = 0; j <= m + n; j++)
73              if (j != q) a[p][j] /= a[p][q];
74          a[p][q] = 1.0;
75      }
76
77      public double value() { return -a[m][m + n]; }
78
79      public double[] primal() {
80          double[] x = new double[n];
81          for (int i = 0; i < m; i++)
82              if (basis[i] < n) x[basis[i]] = a[i][m + n];
83          return x;
84      }
85
86      public double[] dual() {
87          double[] y = new double[m];
88          for (int i = 0; i < m; i++) y[i] = -a[m][n + i];
89          return y;
90      }
91  }
```

Listing 4.1: Self-contained Simplex solver using Bland's rule.

Pivot Selection Strategies Method `bland()` implements Bland's smallest-subscript rule, scanning row m from left to right and returning the first index j satisfying `a[m][j] > EPSILON`. Alternatively, method `dantzig()` selects the column with the maximum positive reduced cost. The threshold `EPSILON` $= 10^{-10}$ protects against numerical inaccuracies: a tiny residual such as $+10^{-16}$ caused by roundoff error will not trigger an extraneous pivot.

Minimum Ratio Test and Unboundedness Method `minRatioRule(q)` determines the leaving row p by computing the ratio `a[i][m+n]`/`a[i][q]` across all rows where `a[i][q]` > `EPSILON`. Non-positive entries are strictly ignored because they do not constrain an increase in x_q. If all entries in column q are non-positive, no row constrains the growth of x_q; `p` remains -1, signaling that the linear program is unbounded and triggering an exception.

In-Place Gauss-Jordan Pivoting Method `pivot(p, q)` updates the entire tableau in $\mathcal{O}(m(n+m))$ operations:

1. For all rows $i \neq p$ and columns $j \neq q$, the entry is updated by subtracting the scaled pivot row: $a[i][j] \leftarrow a[i][j] - a[p][j] \cdot \frac{a[i][q]}{a[p][q]}$.
2. Column q is zeroed out for all rows $i \neq p$.
3. The pivot row p is normalized by dividing each entry by $a[p][q]$, setting $a[p][q] = 1$.
4. Finally, `basis[p]` is updated to record that variable q is now basic for row p.

Extracting Primal and Dual Solutions Upon termination:

- `value()` returns the optimal objective value $-a[m][n+m]$.
- `primal()` extracts the primal decision variables $x_0, \ldots, x_{n-1}$. For each row i, if `basis[i]` $< n$, the corresponding variable is assigned the RHS value $a[i][n+m]$; all non-basic variables remain zero.
- `dual()` extracts the optimal dual variables $y_i = -a[m][n+i]$ directly from the reduced costs of the slack variables, a property formalized by Linear Programming Duality in Section 4.7.

4.6 Initialization: Two-Phase and Big-M

4.6.1 The Initial Basis Challenge

In the Brewery Problem, finding an initial basic feasible solution was immediate because $\mathbf{b} \geq \mathbf{0}$, allowing slack variables to serve as an initial identity basis.

However, when constraints have negative right-hand sides ($b_i < 0$), setting original variables to zero violates non-negativity. Recall that canonical form (Definition 4.5) expresses all functional constraints as less-than-or-equal-to inequalities ($\leq$). For instance, if an application requires $2x_0 - x_1 \geq 2$, multiplying by -1 transforms it into the canonical $\leq$ inequality $-2x_0 + x_1 \leq -2$. Consider:

$$\begin{aligned} \text{maximize} \quad & Z = 2x_0 - x_1 && (4.62)\\ \text{subject to} \quad & -2x_0 + x_1 \leq -2, && (4.63)\\ & x_0 - 5x_1 \leq -4, && (4.64)\\ & x_0, x_1 \geq 0. && (4.65) \end{aligned}$$

In canonical form $\mathbf{Ax} \leq \mathbf{b}$, the right-hand side vector $\mathbf{b} = (-2, -4)^\top$ has negative entries for both constraint (4.63) ($b_0 = -2 < 0$) and constraint (4.64) ($b_1 = -4 < 0$). At the origin $(x_0, x_1) = (0, 0)$, both constraints are violated: $0 \not\leq -2$ and $0 \not\leq -4$. Introducing standard slack variables $s_0, s_1 \geq 0$ yields $-2x_0 + x_1 + s_0 = -2$ and $x_0 - 5x_1 + s_1 = -4$; setting $x_0 = x_1 = 0$ requires $s_0 = -2 < 0$ and $s_1 = -4 < 0$, violating non-negativity.

4.6.2 The Two-Phase Simplex Method

Geometric Intuition and Principles

The Two-Phase Simplex method resolves the initial basis challenge through a two-stage geometric strategy, illustrated conceptually in Figure 4.4:

1. **Phase I (Feasibility via Embedding)**: When the origin $\mathbf{x} = \mathbf{0}$ lies outside the feasible polytope $\mathcal{P}$, the standard identity basis formed by slack variables is infeasible because it violates non-negativity. Phase I overcomes this by embedding the n-dimensional space into an augmented $(n + m)$-dimensional space. For each constraint i, we introduce a non-negative *artificial variable* $x_{a,i} \geq 0$, which acts as an extra coordinate measuring the degree of constraint violation at the origin.

   Setting each artificial variable equal to the non-negative right-hand side ($x_{a,i} = b_i \geq 0$) provides a trivial, valid basic feasible solution in this higher-dimensional artificial polytope. We then optimize an auxiliary Phase I objective that penalizes the sum of artificial violations:

$$\text{maximize} \quad W = -\sum_{i=0}^{m-1} x_{a,i} \quad \Longleftrightarrow \quad \text{minimize} \quad \sum_{i=0}^{m-1} x_{a,i}. \tag{4.66}$$

   As depicted by the dashed green trajectory in Figure 4.4, the Simplex algorithm pivots along the edges of the artificial polytope, systematically contracting the artificial dimensions toward zero.

   - If $\max W = 0$, every artificial variable has been driven to zero ($x_{a,i} = 0$ for all i). At this exact instant, the algorithm has eliminated the artificial violation and "landed" directly on an extreme point $\mathbf{x}^{(0)}$ on the boundary of the original feasible polytope $\mathcal{P}$. The active basic variables constitute a certified basic feasible solution for the original problem.
   - If $\max W < 0$, at least one artificial variable remains strictly positive at optimality ($\sum x_{a,i} > 0$). This provides an irrefutable certificate that no combination of original variables can satisfy all constraints simultaneously: the original linear program is **infeasible**.

2. **Phase II (Optimality on the Original Polytope)**: Once Phase I delivers a valid vertex $\mathbf{x}^{(0)} \in \mathcal{P}$, the artificial scaffolding is discarded. We drop the auxiliary objective row W and the artificial variable columns, reinstate the true objective function $\mathbf{c}^\top \mathbf{x}$, and express it in terms of the current non-basic variables. Phase II then resumes standard Simplex pivoting, traversing adjacent vertices along improving edges of $\mathcal{P}$ (illustrated by the solid orange trajectory $\mathbf{x}^{(0)} \to \mathbf{x}^{(1)} \to \cdots \to \mathbf{x}^*$ in Figure 4.4) until an optimal vertex is certified or an unbounded ray is identified.

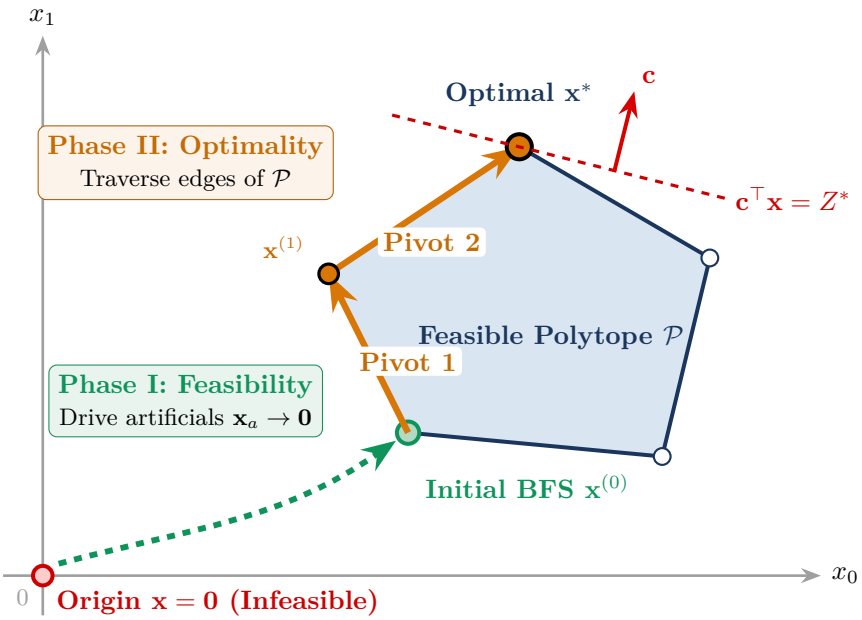


Figure 4.4: Two-Phase Simplex geometry. Phase I introduces artificial variables to create an artificial feasible space, driving them to zero along the dashed green trajectory to reach an extreme point $\mathbf{x}^{(0)}$ of the original polytope $\mathcal{P}$. Phase II then traverses the edges of $\mathcal{P}$ (solid orange path) to the optimal vertex $\mathbf{x}^*$.

Step-by-Step Worked Example

To illustrate both the algebraic transformations and tableau mechanics of the Two-Phase method, we trace the solution of the canonical-form linear program from Equation (4.62):

$$\begin{aligned} \text{maximize} \quad & Z = 2x_0 - x_1 && (4.67)\\ \text{subject to} \quad & -2x_0 + x_1 \leq -2, && (4.68)\\ & x_0 - 5x_1 \leq -4, && (4.69)\\ & x_0, x_1 \geq 0. && (4.70) \end{aligned}$$

Step 1: Adding Slacks and Row Sign Normalization The problem is in canonical form $\mathbf{Ax} \leq \mathbf{b}$ with constraint matrix $\mathbf{A} = \begin{pmatrix} -2 & 1 \\ 1 & -5 \end{pmatrix}$ and right-hand side vector $\mathbf{b} = \begin{pmatrix} -2 \\ -4 \end{pmatrix}$.

Converting constraints (4.68) and (4.69) into standard equality form by adding non-negative slack variables $s_0, s_1 \geq 0$ yields:

$$\begin{aligned} -2x_0 + x_1 + s_0 &= -2, && (4.71)\\ x_0 - 5x_1 + s_1 &= -4. && (4.72) \end{aligned}$$

Setting $x_0 = 0$ and $x_1 = 0$ assigns basic values $s_0 = -2$ and $s_1 = -4$. Both violate the non-negativity requirement ($s_0 \geq 0, s_1 \geq 0$); thus, the standard slack variables cannot serve as an initial basic feasible solution.

To restore non-negative right-hand sides, we multiply any constraint row where $b_i < 0$ by -1. Because both $b_0 = -2 < 0$ in Equation (4.71) and $b_1 = -4 < 0$ in Equation (4.72) are strictly negative, multiplying both equations by -1 normalizes the constants to $+2$ and $+4$:

$$\begin{aligned} 2x_0 - x_1 - s_0 &= 2, && (4.73)\\ -x_0 + 5x_1 - s_1 &= 4. && (4.74) \end{aligned}$$

This row-sign normalization matches the logic of `TwoPhaseSimplex.java`, where row i is negated whenever $b_i < 0$. In Equations (4.73) and (4.74), the slack variables now appear with coefficient -1.0 (functioning as surplus variables).

Step 2: Injecting Artificial Variables and Phase I Objective Because the surplus variables in Equations (4.73) and (4.74) have coefficients -1.0, they cannot provide an initial identity basis (setting non-basic variables to 0 would yield $s_0 = -2 < 0, s_1 = -4 < 0$). We therefore introduce non-negative artificial variables $a_0, a_1 \geq 0$ with coefficients $+1.0$:

$$\begin{aligned} 2x_0 - x_1 - s_0 + a_0 &= 2, && (4.75)\\ -x_0 + 5x_1 - s_1 + a_1 &= 4. && (4.76) \end{aligned}$$

Setting $x_0 = x_1 = s_0 = s_1 = 0$ gives $a_0 = 2 \geq 0$ and $a_1 = 4 \geq 0$, establishing an initial basic feasible solution in the augmented space with basis $\{a_0, a_1\}$.

The Phase I objective maximizes $W = -a_0 - a_1$. Expressing a_0 and a_1 from Equations (4.75) and (4.76):

$$a_0 = 2 - 2x_0 + x_1 + s_0, \qquad a_1 = 4 + x_0 - 5x_1 + s_1,$$

which yields the Phase I objective in non-basic variables:

$$W = -(2 - 2x_0 + x_1 + s_0) - (4 + x_0 - 5x_1 + s_1) = -6 + x_0 + 4x_1 - s_0 - s_1. \quad (4.77)$$

In dictionary form, Equation (4.77) directly provides the reduced costs for row Ph1 ($\bar{w}_{x_0} = 1$, $\bar{w}_{x_1} = 4$, $\bar{w}_{s_0} = -1$, $\bar{w}_{s_1} = -1$), with initial baseline value $W_0 = -6$.

Step 3: Initial Canonicalized Tableau Following the memory layout of Figure 4.5, the tableau has $m + 2 = 4$ rows and $n + 2m + 1 = 7$ columns. Row $m = 2$ stores the original objective ($Z = 2x_0 - x_1$), while row $m + 1 = 3$ stores the Phase I objective (Equation (4.77)):

$$T_0 = \begin{array}{c|cccccc|c} \text{Basis} & x_0 & x_1 & s_0 & s_1 & a_0 & a_1 & \text{RHS} \\ \hline a_0 & 2 & -1 & -1 & 0 & 1 & 0 & 2 \\ a_1 & -1 & 5 & 0 & -1 & 0 & 1 & 4 \\ \hline \text{Obj } (Z) & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ \text{Ph1 } (W) & 1 & 4 & -1 & -1 & 0 & 0 & 6 \end{array} \tag{4.78}$$

The right-hand side of row Ph1 stores the negated value $-W = 6$, corresponding to $W = -6$.

Step 4: Phase I Pivoting

- **Pivot 1**: Scanning row Ph1 using Bland's rule, the first column with a positive reduced cost is column $q = 0$ (x_0, with $\bar{w}_0 = 1 > 0$). We apply the minimum ratio rule among rows where $a[i][0] > 0$:

$$\text{Row 0 } (a_0) : \frac{2}{2} = 1, \qquad \text{Row 1 } (a_1) : a[1][0] = -1 \leq 0 \text{ (ineligible)}.$$

  The minimum ratio is 1, so row $p = 0$ is selected: x_0 enters the basis and artificial variable a_0 leaves. Pivoting on entry $a[0][0] = 2$ yields:

$$T_1 = \begin{array}{c|cccccc|c} \text{Basis} & x_0 & x_1 & s_0 & s_1 & a_0 & a_1 & \text{RHS} \\ \hline x_0 & 1 & -\frac{1}{2} & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & 1 \\ a_1 & 0 & \frac{9}{2} & -\frac{1}{2} & -1 & \frac{1}{2} & 1 & 5 \\ \hline \text{Obj } (Z) & 0 & 0 & 1 & 0 & -1 & 0 & -2 \\ \text{Ph1 } (W) & 0 & \frac{9}{2} & -\frac{1}{2} & -1 & -\frac{1}{2} & 0 & 5 \end{array} \tag{4.79}$$

  The current Phase I objective is $W = -5$, reflecting that artificial variable $a_1 = 5$ remains.

- **Pivot 2**: In row Ph1, column $q = 1$ (x_1) has positive reduced cost $\bar{w}_1 = 9/2 > 0$. Testing the ratio rule:

$$\text{Row 0 } (x_0) : a[0][1] = -\frac{1}{2} \leq 0 \text{ (ineligible)}, \qquad \text{Row 1 } (a_1) : \frac{5}{9/2} = \frac{10}{9}.$$

  Row $p = 1$ is the unique candidate: x_1 enters and the remaining artificial variable a_1 leaves. Pivoting on entry $a[1][1] = 9/2$ yields:

$$T_2 = \begin{array}{c|cccccc|c} \text{Basis} & x_0 & x_1 & s_0 & s_1 & a_0 & a_1 & \text{RHS} \\ \hline x_0 & 1 & 0 & -\frac{5}{9} & -\frac{1}{9} & \frac{5}{9} & \frac{1}{9} & \frac{14}{9} \\ x_1 & 0 & 1 & -\frac{1}{9} & -\frac{2}{9} & \frac{1}{9} & \frac{2}{9} & \frac{10}{9} \\ \hline \text{Obj } (Z) & 0 & 0 & 1 & 0 & -1 & 0 & -2 \\ \text{Ph1 } (W) & 0 & 0 & 0 & 0 & -1 & -1 & 0 \end{array} \tag{4.80}$$

Step 5: Phase I Conclusion and Feasibility Certificate In tableau T_2, all reduced costs in row Ph1 for the non-artificial variables $\{x_0, x_1, s_0, s_1\}$ are $0 \leq 0$. The Phase I objective value is:

$$W^* = -a[3][6] = -0 = 0.$$

Because $\max W = 0$, both artificial variables have been driven to zero ($a_0 = 0, a_1 = 0$). The basic variables are:

$$x_0 = \frac{14}{9}, \qquad x_1 = \frac{10}{9}.$$

Because $x_0 \geq 0$ and $x_1 \geq 0$, $\mathbf{x}^{(0)} = (14/9, 10/9)^\top$ is a certified basic feasible solution (extreme point) of the original polyhedron $\mathcal{P}$. Phase I has successfully concluded.

Step 6: Phase II Transition and Tableau We discard the Phase I objective row and the artificial columns a_0, a_1. Because elementary row operations were simultaneously applied to the original objective row Z during the Phase I pivots (as implemented in `TwoPhaseSimplex.java`), row Z is already in canonical form with respect to the feasible basis $\{x_0, x_1\}$, expressing Z purely in terms of the non-basic surplus variables:

$$T_{\text{II}} = \begin{array}{c|cccc|c} \text{Basis} & x_0 & x_1 & s_0 & s_1 & \text{RHS} \\ \hline x_0 & 1 & 0 & -\frac{5}{9} & -\frac{1}{9} & \frac{14}{9} \\ x_1 & 0 & 1 & -\frac{1}{9} & -\frac{2}{9} & \frac{10}{9} \\ \hline \text{Obj } (Z) & 0 & 0 & 1 & 0 & -2 \end{array} \tag{4.81}$$

The current objective value is $Z = -(-2) = 2$, which coincides with $2x_0 - x_1 = 2(14/9) - (10/9) = 18/9 = 2$.

Step 7: Phase II Execution and Unboundedness Certificate In the objective row of T_{II}, surplus variable s_0 has a positive reduced cost:

$$\bar{c}_{s_0} = 1 > 0.$$

To increase Z, we select s_0 as the entering column ($q = 2$). We inspect the constraint entries in column s_0:

$$a[0][2] = -\frac{5}{9} \leq 0, \qquad a[1][2] = -\frac{1}{9} \leq 0.$$

Every coefficient in the entering column is non-positive ($a[i][2] \leq 0$ for all i). Expressing the basic variables as functions of s_0 (holding $s_1 = 0$):

$$x_0 = \frac{14}{9} + \frac{5}{9}s_0,$$
$$x_1 = \frac{10}{9} + \frac{1}{9}s_0.$$

Because the coefficients $+5/9$ and $+1/9$ are strictly positive, increasing s_0 causes both x_0 and x_1 to increase. Neither basic variable decreases, so neither variable will ever reach zero. The surplus s_0 can be increased toward $+\infty$ without violating feasibility:

$$Z = 2 + 1 \cdot s_0 \xrightarrow{s_0 \to \infty} +\infty.$$

The minimum ratio rule returns $p = -1$, certifying that the linear program is **unbounded**.

Pedagogical Significance of the Example. This trace demonstrates the dual power of the Simplex method: Phase I resolves the challenge of an infeasible origin by navigating through an augmented artificial space to locate a valid initial vertex $\mathbf{x}^{(0)} = (14/9, 10/9)^\top$ on $\mathcal{P}$. Phase II then detects that the feasible region extends infinitely along the ray direction $(5/9, 1/9)^\top$ with an increasing objective, providing an exact algorithmic certificate of unboundedness. If the objective had instead been bounded—for instance, maximizing $-2x_0 + x_1$ (the negation of the original objective)—then the reduced costs in T_{II} would all be non-positive ($\bar{c}_{s_0} = -1 \leq 0, \bar{c}_{s_1} = 0 \leq 0$), certifying that $\mathbf{x}^{(0)}$ is immediately the unique global optimum.

Two-Phase Implementation in Java

Tableau Setup and Phase I Canonicalization As depicted in Figure 4.5, the constructor of `TwoPhaseSimplex` configures the 2D array `a` of dimensions $(m+2) \times (n+2m+1)$, adding m artificial variable columns and an auxiliary Phase I objective row:

- **Row Sign Normalization**: If any right-hand side constant is negative ($b_i < 0$), the entire constraint row i is multiplied by -1, ensuring that all entries in the RHS column $n+2m$ are non-negative ($b_i \geq 0$). The slack variable coefficient is correspondingly set to -1.0.
- **Artificial Variables $\mathbf{I}_m$** (columns $n+m \leq j < n+2m$): An $m \times m$ identity block introduces artificial variables $x_{a,0}, \ldots, x_{a,m-1}$ to supply an immediate starting basis.
- **Phase I Auxiliary Objective** (row $m+1$): Stores the auxiliary objective $W = -\sum_{i=0}^{m-1} x_{a,i}$, initialized with coefficients -1.0 in the artificial columns.
- **Canonicalization via Pivoting**: Because the artificial variables are declared basic (`basis`$[i] = n+m+i$), their reduced costs in the Phase I objective row $m+1$ must be eliminated before Phase I begins. The constructor accomplishes this by executing `pivot(i, n + m + i)` for each row i. This Gauss-Jordan elimination clears the -1.0 entries in row $m+1$ and computes the initial value $W = -\sum b_i$, placing the Phase I tableau into canonical form.

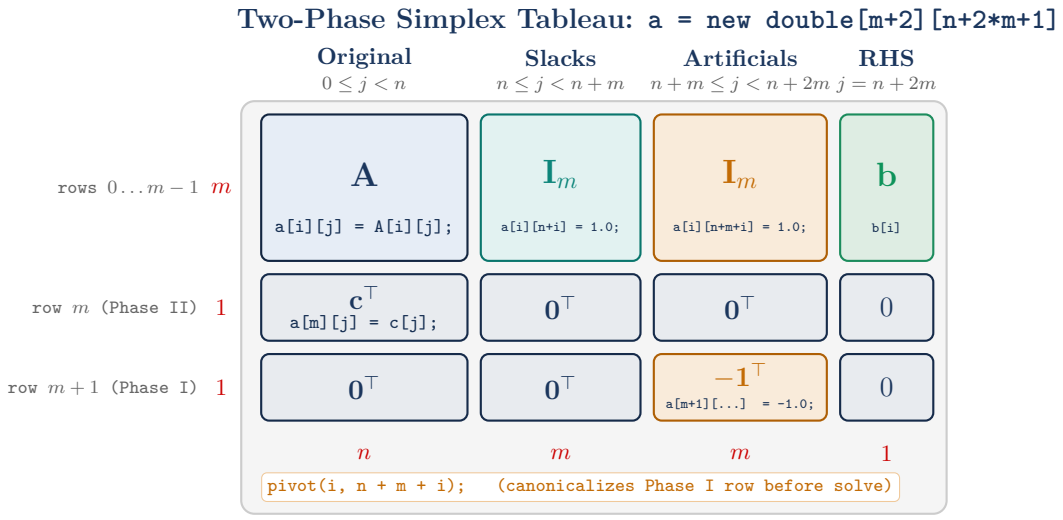


Figure 4.5: Two-Phase Simplex tableau 2D array memory layout and Java initialization (`TwoPhaseSimplex.java`) augmented to size $(m+2) \times (n+2m+1)$ with m artificial variable columns and an auxiliary Phase I objective row.

Listing 4.2 displays the complete implementation of `TwoPhaseSimplex`.

```java
public class TwoPhaseSimplex {
    private static final double EPSILON = 1.0E-8;
    private double[][] a; // tableaux: (m+2) x (n+2*m+1)
    private int m, n;
    private int[] basis;

    public TwoPhaseSimplex(double[][] A, double[] b, double[] c) {
        m = b.length;
        n = c.length;
        a = new double[m + 2][n + m + m + 1];

        for (int i = 0; i < m; i++)
            for (int j = 0; j < n; j++) a[i][j] = A[i][j];

        for (int i = 0; i < m; i++) a[i][n + i] = 1.0;
        for (int i = 0; i < m; i++) a[i][n + m + m] = b[i];
        for (int j = 0; j < n; j++) a[m][j] = c[j];

        // Ensure non-negative RHS
        for (int i = 0; i < m; i++) {
            if (b[i] < 0) {
                a[i][n + m + m] = -b[i];
                for (int j = 0; j < n; j++) a[i][j] = -a[i][j];
                a[i][n + i] = -1.0;
            }
        }

        // Artificial variables form initial basis
        for (int i = 0; i < m; i++) {
            a[i][n + m + i] = 1.0;
            a[m + 1][n + m + i] = -1.0; // minimize sum(x_a)
        }
        for (int i = 0; i < m; i++) pivot(i, n + m + i);

        basis = new int[m];
        for (int i = 0; i < m; i++) basis[i] = n + m + i;

        phase1();
        phase2();
    }

    private void phase1() {
        while (true) {
            int q = -1;
            for (int j = 0; j < n + m; j++)
                if (a[m + 1][j] > EPSILON) { q = j; break; }
            if (q == -1) break; // Phase I optimal

            int p = minRatioRule(q);
            pivot(p, q);
            basis[p] = q;
        }
        if (a[m + 1][n + m + m] > EPSILON)
            throw new ArithmeticException("LP is infeasible");
    }
```

```
56
57      private void phase2() {
58          while (true) {
59              int q = -1;
60              for (int j = 0; j < n + m; j++)
61                  if (a[m][j] > EPSILON) { q = j; break; }
62              if (q == -1) break; // Phase II optimal
63
64              int p = minRatioRule(q);
65              if (p == -1)
66                  throw new ArithmeticException("LP is unbounded");
67              pivot(p, q);
68              basis[p] = q;
69          }
70      }
71
72      private int minRatioRule(int q) {
73          int p = -1;
74          for (int i = 0; i < m; i++) {
75              if (a[i][q] <= EPSILON) continue;
76              if (p == -1 ||
77                  a[i][n+m+m] / a[i][q] < a[p][n+m+m] / a[p][q]) p = i;
78          }
79          return p;
80      }
81
82      private void pivot(int p, int q) {
83          for (int i = 0; i <= m + 1; i++)
84              for (int j = 0; j <= n + m + m; j++)
85                  if (i != p && j != q)
86                      a[i][j] -= a[p][j] * (a[i][q] / a[p][q]);
87          for (int i = 0; i <= m + 1; i++)
88              if (i != p) a[i][q] = 0.0;
89          for (int j = 0; j <= n + m + m; j++)
90              if (j != q) a[p][j] /= a[p][q];
91          a[p][q] = 1.0;
92      }
93
94      public double value() { return -a[m][n + m + m]; }
95
96      public double[] primal() {
97          double[] x = new double[n];
98          for (int i = 0; i < m; i++)
99              if (basis[i] < n) x[basis[i]] = a[i][n + m + m];
100         return x;
101     }
102 }
```

Listing 4.2: Two-Phase Simplex implementation in Java.

4.6.3 The Big-M Penalty Approach

An alternative initialization is the *Big-M method*. Instead of an auxiliary objective, artificial variables $\mathbf{x}_a$ are penalized directly in the original objective function using a large positive constant $M \gg 0$:

$$\begin{aligned} \text{maximize} \quad & \mathbf{c}^\top \mathbf{x} - M \sum_{i=0}^{m-1} x_{a,i} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} + \mathbf{s} + \mathbf{x}_a = \mathbf{b} \\ & \mathbf{x}, \mathbf{s}, \mathbf{x}_a \geq \mathbf{0}. \end{aligned} \tag{4.82}$$

Because the problem is a maximization, any positive value of an artificial variable heavily degrades the objective. If a feasible solution exists, the algorithm automatically drives all artificial variables to zero during standard iterations.

Listing 4.3 shows how the tableau is initialized in `BigMSimplex`.

```
public class BigMSimplex {
    private static final double EPSILON = 1.0E-8;
    private double[][] a;
    private int m, n;
    private int[] basis;

    public BigMSimplex(double[][] A, double[] b,
                       double[] c, double BigM) {
        m = b.length; n = c.length;
        a = new double[m + 1][n + m + m + 1];

        for (int i = 0; i < m; i++)
            for (int j = 0; j < n; j++) a[i][j] = A[i][j];

        for (int i = 0; i < m; i++) a[i][n + i] = 1.0;
        for (int i = 0; i < m; i++) a[i][n + m + m] = b[i];
        for (int j = 0; j < n; j++) a[m][j] = c[j];
        for (int j = n + m; j < n + m + m; j++)
            a[m][j] = -BigM; // penalize artificials

        // Multiply rows with b[i] < 0 by -1
        for (int i = 0; i < m; i++) {
            if (b[i] < 0) {
                a[i][n + m + m] = -b[i];
                for (int j = 0; j < n; j++) a[i][j] = -a[i][j];
                a[i][n + i] = -1.0;
            }
        }

        for (int i = 0; i < m; i++) a[i][n + m + i] = 1.0;
        for (int i = 0; i < m; i++) pivot(i, n + m + i);

        basis = new int[m];
        for (int i = 0; i < m; i++) basis[i] = n + m + i;

        solve();
    }
}
```

Listing 4.3: Big-M Simplex constructor and initialization.

4.6.4 Comparison and Numerical Stability

Numerical Instability of Big-M. While conceptually straightforward, the Big-M method introduces significant numerical instability in floating-point computation:

- **Catastrophic Cancellation**: When M is very large (e.g., $M = 10^9$), arithmetic operations like $c_j - M \cdot a_{i,j}$ cause significant roundoff loss in IEEE 754 floating-point representations. Small genuine cost differences are erased by truncation errors.
- **Ill-Conditioning**: Large disparate magnitudes degrade the condition number of the basis matrix.
- **False Conclusions**: If M is too small, an artificial variable may remain in the basis at optimality, leading to incorrect claims of feasibility.

Consequently, commercial solvers (such as Gurobi, CPLEX, and HiGHS) almost universally implement the Two-Phase Simplex method or dual simplex initialization rather than Big-M.

Table 4.1 summarizes the trade-offs between the two initialization schemes.

Criterion	Two-Phase Simplex	Big-M Method
Tableau Dimension	$(m+2) \times (n+2m+1)$	$(m+1) \times (n+2m+1)$
Parameter Sensitivity	None (parameter-free)	High (sensitive to choice of M)
Numerical Stability	High (avoids ill-conditioned rows)	Low (prone to catastrophic cancellation)
Infeasibility Detection	Exact ($W^* < 0$)	Requires testing if $x_{a,i} > \epsilon$
Implementation Complexity	Requires two objective loops	Single objective loop
Industrial Use	Universal standard	Primarily pedagogical

Table 4.1: Comparison of Two-Phase Simplex versus the Big-M method.

4.7 Duality Theory and Certificates

Every linear program has an associated linear program called its *dual*. Duality is among the most profound principles in mathematical optimization, providing certificates of optimality and economic sensitivities.

4.7.1 The Primal-Dual Pair

Definition 4.7 *Primal and Dual Linear Programs*

Given a primal linear program in canonical form:

$$\begin{aligned} \text{maximize} \quad & \mathbf{c}^\top \mathbf{x} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}, \end{aligned} \tag{4.83}$$

its *dual linear program* is defined as:

$$\begin{aligned} \text{minimize} \quad & \mathbf{b}^\top \mathbf{y} \\ \text{subject to} \quad & \mathbf{A}^\top \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq \mathbf{0}, \end{aligned} \tag{4.84}$$

where $\mathbf{y} = (y_0, y_1, \ldots, y_{m-1})^\top \in \mathbb{R}^m$ is the vector of dual decision variables.

Each dual variable y_i corresponds to the i-th primal constraint, and each dual constraint corresponds to the j-th primal variable.

4.7.2 Weak and Strong Duality

Theorem 4.5 *Weak Duality Theorem*

If $\mathbf{x}$ is a primal feasible solution and $\mathbf{y}$ is a dual feasible solution, then:

$$\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}. \tag{4.85}$$

Proof. Because $\mathbf{x} \geq \mathbf{0}$ and $\mathbf{A}^\top \mathbf{y} \geq \mathbf{c}$, taking the inner product with $\mathbf{x}$ preserves the inequality:

$$\mathbf{c}^\top \mathbf{x} \leq (\mathbf{A}^\top \mathbf{y})^\top \mathbf{x} = \mathbf{y}^\top \mathbf{A}\mathbf{x}. \tag{4.86}$$

Because $\mathbf{y} \geq \mathbf{0}$ and $\mathbf{A}\mathbf{x} \leq \mathbf{b}$, we similarly have:

$$\mathbf{y}^\top (\mathbf{A}\mathbf{x}) \leq \mathbf{y}^\top \mathbf{b} = \mathbf{b}^\top \mathbf{y}. \tag{4.87}$$

Combining both inequalities yields $\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$. □

Weak duality establishes that any dual feasible solution provides an upper bound on the optimal value of the primal. Conversely, any primal feasible solution provides a lower bound on the dual.

Theorem 4.6 *Strong Duality Theorem*

If a linear program has an optimal solution $\mathbf{x}^*$, then its dual linear program also has an optimal solution $\mathbf{y}^*$, and their optimal objective values are identical:

$$\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*. \tag{4.88}$$

4.7.3 Complementary Slackness

Theorem 4.7 *Complementary Slackness Theorem*

Let $\mathbf{x}$ be primal feasible and $\mathbf{y}$ be dual feasible. Then $\mathbf{x}$ and $\mathbf{y}$ are simultaneously optimal for their respective problems if and only if:

$$x_j \left(\sum_{i=0}^{m-1} a_{i,j} y_i - c_j \right) = 0, \quad \forall j \in \{0, \dots, n-1\}, \tag{4.89}$$

and

$$y_i \left(b_i - \sum_{j=0}^{n-1} a_{i,j} x_j \right) = 0, \quad \forall i \in \{0, \dots, m-1\}. \tag{4.90}$$

Proof. Subtracting the primal objective from the dual objective:

$$\mathbf{b}^\top \mathbf{y} - \mathbf{c}^\top \mathbf{x} = \mathbf{b}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{A}\mathbf{x} + \mathbf{y}^\top \mathbf{A}\mathbf{x} - \mathbf{c}^\top \mathbf{x} = \mathbf{y}^\top (\mathbf{b} - \mathbf{A}\mathbf{x}) + (\mathbf{A}^\top \mathbf{y} - \mathbf{c})^\top \mathbf{x}. \tag{4.91}$$

Since $\mathbf{x}$ and $\mathbf{y}$ are feasible, $\mathbf{y} \geq \mathbf{0}$, $\mathbf{b} - \mathbf{A}\mathbf{x} \geq \mathbf{0}$, $\mathbf{A}^\top \mathbf{y} - \mathbf{c} \geq \mathbf{0}$, and $\mathbf{x} \geq \mathbf{0}$. Every term in the sum is non-negative:

$$\sum_{i=0}^{m-1} \underbrace{y_i}_{\geq 0} \underbrace{(b_i - \mathbf{a}_{i,\cdot}^\top \mathbf{x})}_{\geq 0} + \sum_{j=0}^{n-1} \underbrace{x_j}_{\geq 0} \underbrace{(\mathbf{a}_{\cdot,j}^\top \mathbf{y} - c_j)}_{\geq 0} = \mathbf{b}^\top \mathbf{y} - \mathbf{c}^\top \mathbf{x} \geq 0. \tag{4.92}$$

By Strong Duality (Theorem 4.6), $\mathbf{x}$ and $\mathbf{y}$ are optimal if and only if $\mathbf{b}^\top \mathbf{y} - \mathbf{c}^\top \mathbf{x} = 0$. Since a sum of non-negative terms equals zero if and only if each term is zero, Equations (4.89) and (4.90) must hold. □

4.7.4 Dual Recovery and Shadow Prices

Proposition 4.8 *Dual Extraction from Tableau*

At optimality, the reduced costs of the slack variables $s_i = x_{n+i}$ in row m of the simplex tableau satisfy:

$$y_i^* = -a[m][n+i], \quad \forall i \in \{0, \dots, m-1\}. \tag{4.93}$$

In the Brewery Problem (Section 4.5.1), the final row of Tableau 2 has slack reduced costs $a[3][2] = -1$ (corn slack), $a[3][3] = -2$ (hops slack), and $a[3][4] = 0$ (malt slack). The optimal dual variables are therefore:

$$y_0^* = 1, \quad y_1^* = 2, \quad y_2^* = 0. \tag{4.94}$$

Evaluating the dual objective confirms strong duality:

$$\mathbf{b}^\top \mathbf{y}^* = 480(1) + 160(2) + 1190(0) = 480 + 320 = 800 = Z^*. \tag{4.95}$$

Economic Interpretation (Shadow Prices) The dual variable y_i^* represents the *shadow price* or marginal value of resource i:

$$y_i^* = \frac{\partial Z^*}{\partial b_i}. \tag{4.96}$$

- $y_0^* = 1$: Having one additional unit of corn increases maximum profit by 1 unit.
- $y_1^* = 2$: Having one additional unit of hops increases maximum profit by 2 units.
- $y_2^* = 0$: Malt is in excess (slack $x_4^* = 210 > 0$). Adding more malt has zero marginal value. By complementary slackness, since the malt constraint is not tight, its shadow price is strictly zero ($y_2^* = 0$).

4.8 Complexity, Cycling, and Solvers

4.8.1 Degeneracy and Bland's Rule

A basic feasible solution is *degenerate* if at least one basic variable has value zero: $x_{B(i)} = 0$. Geometrically, degeneracy occurs when more than n hyperplanes intersect at a single vertex in $\mathbb{R}^n$.

In the presence of degeneracy, a pivot may yield a minimum ratio of $\theta^* = 0$. In this case, the entering variable enters at value zero, the basis changes, but the geometric point $\mathbf{x}$ and the objective value Z remain unchanged. This is known as *stalling*.

If a sequence of degenerate pivots cycles through a recurring set of bases without altering the solution, the Simplex algorithm can enter an infinite loop. This phenomenon is called *cycling*.

Theorem 4.9 *Bland's Anticycling Theorem*

Under Bland's smallest-subscript rule (always selecting the entering column $q = \min\{j \mid \bar{c}_j > 0\}$ and tie-breaking the leaving row by choosing the smallest index p), the Simplex algorithm never cycles and terminates in a finite number of iterations [Bla77].

4.8.2 Worst-Case Exponential Complexity

Despite its extraordinary practical speed, the Simplex algorithm with Dantzig's rule is not a polynomial-time algorithm in the worst case. In 1972, Victor Klee and George Minty constructed a celebrated family of perturbed hypercubes [KM72]:

$$\begin{aligned} \text{maximize} \quad & \sum_{j=0}^{n-1} 10^{n-1-j} x_j \\ \text{subject to} \quad & 2\sum_{k=0}^{j-1} 10^{j-k} x_k + x_j \le 100^j, \quad \forall j \in \{0, \dots, n-1\}, \\ & x_j \ge 0, \quad \forall j \in \{0, \dots, n-1\}. \end{aligned} \tag{4.97}$$

The feasible polytope of the *Klee-Minty cube* has 2^n vertices. Dantzig's pivot rule visits every single vertex before reaching the optimum, requiring $2^n - 1$ pivots! For $n = 50$, the algorithm would require $2^{50} \approx 10^{15}$ pivots.

4.8.3 Polynomial-Time Interior Algorithms

The theoretical question of whether linear programming could be solved in polynomial time was resolved positively by Leonid Khachiyan in 1979 using the *Ellipsoid Method* [Kha79]. In 1984, Narendra Karmarkar introduced the first practical polynomial-time *Interior Point Method* [Kar84].

Interior point methods traverse the interior of the feasible polytope following the central path, rather than walking along boundary edges. They solve linear programs with n variables in $\mathcal{O}(n^{3.5}L)$ operations, where L is the bit length of the input data.

4.8.4 Warm-Starting and Solver Architecture

If interior point methods possess polynomial worst-case guarantees, why does the Simplex algorithm remain dominant in discrete optimization?

The answer lies in *warm-starting* (re-optimization). In branch-and-bound and cutting-plane algorithms, solvers evaluate thousands of linear programming relaxations that differ by only a single bounding constraint ($x_j \leq \lfloor v \rfloor$ or $x_j \geq \lceil v \rceil$). The dual simplex method can restart directly from the optimal basis of the parent node and restore feasibility in just 2–5 pivots! In contrast, interior point methods cannot be warm-started effectively and must solve each relaxation almost from scratch.

Modern commercial solvers (such as Gurobi and CPLEX) and high-performance open-source solvers (such as HiGHS and SCIP) combine primal simplex, dual simplex, and barrier solvers, employing presolve transformations, sparse LU factorizations, and steepest-edge pricing.

4.9 Applications of Linear Programming

4.9.1 The Nutritional Diet Problem

Pioneered by Nobel laureate George Stigler in 1945 [Sti45], the diet problem seeks the least-cost combination of foods satisfying daily nutritional requirements:

$$\begin{aligned} \text{minimize} \quad & \sum_{j=0}^{n-1} \text{cost}_j \cdot x_j \\ \text{subject to} \quad & \sum_{j=0}^{n-1} \text{nutrient}_{i,j} \cdot x_j \geq \text{minReq}_i, \quad \forall i \in \{0, \ldots, m-1\}, \\ & \sum_{j=0}^{n-1} \text{nutrient}_{i,j} \cdot x_j \leq \text{maxReq}_i, \quad \forall i \in \{0, \ldots, m-1\}, \\ & x_j \geq 0, \quad \forall j \in \{0, \ldots, n-1\}. \end{aligned} \tag{4.98}$$

Because the constraints have positive lower bounds ($\text{minReq}_i > 0$), setting all food quantities to zero ($x_j = 0$) violates nutritional adequacy. As implemented in `DietProblem.java`, this formulation requires Two-Phase Simplex or Big-M to establish an initial feasible diet before optimizing cost.

Furthermore, Duality Theory (Section 4.7) provides profound economic insights into this model: the optimal dual variables y_i^* represent the *marginal shadow price* of nutrient i. If a health authority increases the minimum daily requirement minReq_i by 1 unit, the total grocery expenditure rises by exactly y_i^* monetary units. Conversely, non-binding nutritional constraints exhibit zero dual prices by complementary slackness ($\bar{s}_i > 0 \implies y_i = 0$).

4.9.2 From Continuous to Integer Programming

Throughout this chapter, decisions have been assumed to be continuously divisible ($\mathbf{x} \geq \mathbf{0}$, $\mathbf{x} \in \mathbb{R}^n$). This continuous hypothesis underpins all classical polyhedral results:

- The feasible region is convex, guaranteeing that local optimality implies global optimality.
- Extreme points can be systematically reached and traversed along polyhedral edges in polynomial average time via the Simplex algorithm.
- Linear programming duality provides exact certificates of optimality and unboundedness.

However, in combinatorial optimization, decisions are rarely divisible. Decisions typically represent indivisible entities: selecting an item ($x_j \in \{0, 1\}$), dispatching an integral number of transport vehicles ($x_j \in \mathbb{Z}_{\geq 0}$), or ordering tasks. Restricting variables to integer values breaks convexity and moves the problem into the realm of **NP**-hard discrete optimization.

In Chapter 5, we build directly upon the foundations established here. We explore when continuous linear programming solves integer problems directly (via total unimodularity), and how modern solvers seamlessly combine the Simplex algorithm with the tree search of Chapter 3 through LP-based Branch-and-Bound, dual simplex warm-starting, and cutting planes.

4.10 Chapter Summary and Exercises

4.10.1 Key Conceptual Takeaways

Table 4.2 summarizes the theoretical and algorithmic principles established in this chapter.

Concept	Key Result and Algorithmic Significance
Polyhedral Convexity	Feasible sets are convex polyhedra; local optimality implies global optimality.
Extreme Points	An optimal solution always exists at an extreme point (vertex) of the polytope.
Simplex Algorithm	Traverses adjacent 1-dimensional edges with non-decreasing objective values.
Pivoting	Minimum ratio rule maintains primal feasibility; Bland's rule provably prevents cycling.
Initialization	Two-Phase Simplex and Big-M establish feasibility when the origin is infeasible.
Duality	Primal and dual optimal values coincide ($\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*$); dual variables yield shadow prices.
Sensitivity	Shadow prices and allowable ranges quantify marginal resource valuations.

Table 4.2: Core foundations of linear programming and polyhedral theory.

4.10.2 Exercises

Exercise 4.1 *Tableau Pivot and Optimality Verification*

Consider the following simplex tableau for a maximization problem with non-negative variables $x_0, x_1, x_2, x_3, x_4, x_5 \geq 0$:

x_0	x_1	x_2	x_3	x_4	x_5	RHS
-1	1	-3	1	0	0	25
2	3	4	0	1	0	12
3	-1	4	0	0	1	15
2	-1	-1	0	0	0	-20

1. Identify the current basic feasible solution, its basic variable indices, and its objective value.

2. Identify the entering column using Bland's rule.
3. Compute the minimum ratio for each eligible row and identify the leaving row.
4. Perform the pivot operation and state the updated tableau.
5. Is the resulting basic feasible solution optimal? Justify your answer.

Exercise 4.2 *Form Transformation and Two-Phase Initialization*

Consider the linear program:

$$\begin{aligned} \text{maximize} \quad & 3x_0 - x_1 + 2x_2 \\ \text{subject to} \quad & x_0 + 2x_1 - x_2 \geq 4 \\ & 2x_0 - x_1 + x_2 \leq 2 \\ & x_0 - 3x_1 \leq -3 \\ & x_0, x_1, x_2 \geq 0. \end{aligned}$$

1. Transform the problem into standard form with non-negative right-hand sides.
2. Formulate the Phase I linear program by introducing artificial variables.
3. Set up the initial Phase I simplex tableau and perform the algebraic operations needed to canonicalize the artificial objective row.
4. Determine whether the original problem possesses a basic feasible solution.

Exercise 4.3 *Duality and Shadow Price Analysis*

A manufacturing facility produces three components x_0, x_1, x_2 using machining, assembly, and testing resources with capacities 100, 80, and 60 hours, respectively. The optimal simplex tableau yields slack variable reduced costs $\bar{c}_{s_0} = -4$, $\bar{c}_{s_1} = 0$, and $\bar{c}_{s_2} = -2.5$, with optimal profit $Z^* = 1450$.

1. State the optimal dual vector $\mathbf{y}^* = (y_0^*, y_1^*, y_2^*)^\top$.
2. An external contractor offers to lease 10 additional hours of machining capacity for \$35. Should the plant manager accept the offer?
3. What is the marginal value of expanding testing capacity?
4. Verify complementary slackness for the assembly constraint.

Exercise 4.4 *Total Unimodularity of Bipartite Graphs*

Let $G = (U \cup V, E)$ be an undirected bipartite graph. Let $\mathbf{A} \in \{0,1\}^{(|U|+|V|)\times|E|}$ denote its vertex-edge incidence matrix.

1. Show that every column of $\mathbf{A}$ contains exactly two non-zero entries, each equal to 1.
2. Prove by induction on the submatrix size that $\det(\mathbf{B}) \in \{-1, 0, 1\}$ for every square submatrix $\mathbf{B}$ of $\mathbf{A}$.
3. Conclude that the maximum cardinality matching problem on bipartite graphs can be solved via continuous linear programming.

Exercise 4.5 *The Klee-Minty Cube*

Consider the 3-dimensional Klee-Minty cube (Equation (4.97)) with $n = 3$:

$$\begin{aligned} \text{maximize} \quad & 100x_0 + 10x_1 + x_2 \\ \text{subject to} \quad & x_0 \leq 1 \\ & 20x_0 + x_1 \leq 100 \\ & 200x_0 + 20x_1 + x_2 \leq 10000 \\ & x_0, x_1, x_2 \geq 0. \end{aligned}$$

1. Trace the sequence of basic feasible solutions visited by the Simplex algorithm using Dantzig's largest-coefficient pricing rule, starting from $\mathbf{x} = (0, 0, 0)$.
2. Count the total number of pivots executed before reaching the optimal vertex.
3. Now solve the same problem using Bland's smallest-subscript rule. Does Bland's rule visit fewer vertices on this instance? Explain why.

Exercise 4.6 *Network Flow Modeling and Dual Interpretation*

Consider a directed graph $G = (V, E)$ with source $s = 0$ and sink $t = |V| - 1$.

1. Formulate the maximum (s, t)-flow problem as a linear program in canonical form.
2. Formulate its dual linear program.
3. Show that any dual feasible solution $\mathbf{y}$ with binary variables $p_v \in \{0, 1\}$ corresponds to an (s, t)-cut in G.
4. Using Strong Duality, deduce the Max-Flow Min-Cut Theorem.

CHAPTER 5

Integer Linear Programming

Linear programming provides a polynomial-time computational framework for continuous decisions. However, in combinatorial optimization, decisions are frequently discrete and indivisible: items cannot be partially selected, facilities cannot be fractionally opened, and sequencing decisions are strictly binary. Restricting decision variables to integer values transforms tractable continuous polyhedra into computationally hard (**NP**-hard) discrete search spaces.

This chapter bridges continuous linear programming (Chapter 4) with the systematic tree search of branch and bound (Chapter 3). We investigate the geometric structure of integer polyhedra, identify structural conditions (total unimodularity) under which continuous linear programming naturally yields integer solutions, and develop the modern *LP-based Branch-and-Bound* engine, showing how the Dual Simplex algorithm enables rapid warm-starting when adding branching constraints and cutting planes.

5.1 Foundations of Integer Linear Programming

5.1.1 Mathematical Formulations

An *Integer Linear Program* (ILP) restricts all decision variables to integer values:

$$\begin{aligned} \text{maximize} \quad & Z = \mathbf{c}^\top \mathbf{x} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \in \mathbb{Z}^n_{\geq 0}, \end{aligned} \tag{5.1}$$

where $\mathbf{A} \in \mathbb{Z}^{m \times n}$, $\mathbf{b} \in \mathbb{Z}^m$, and $\mathbf{c} \in \mathbb{R}^n$. When variables are further restricted to binary choices $\mathbf{x} \in \{0,1\}^n$, the formulation is a *0-1 Integer Linear Program* (or Binary Integer Program).

In real-world applications, discrete decisions frequently interact with continuous physical quantities (e.g., binary facility opening decisions paired with continuous commodity flows). Such problems are formulated as *Mixed-Integer Linear Programs* (MILP):

$$\begin{aligned} \text{maximize} \quad & Z = \mathbf{c}^\top \mathbf{x} + \mathbf{d}^\top \mathbf{y} \\ \text{subject to} \quad & \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} \leq \mathbf{b} \\ & \mathbf{x} \in \mathbb{Z}^n_{\geq 0}, \quad \mathbf{y} \in \mathbb{R}^p_{\geq 0}. \end{aligned} \tag{5.2}$$

5.1.2 The Geometry of the Integer Hull

To analyze an integer linear program geometrically, we define two fundamental geometric objects:

Definition 5.1 *Continuous Relaxation and Integer Hull*

Let $S = \{\mathbf{x} \in \mathbb{Z}^n_{\geq 0} \mid \mathbf{Ax} \leq \mathbf{b}\}$ be the set of integer feasible solutions.

1. The *continuous LP relaxation polyhedron* is:
$$\mathcal{P} = \{\mathbf{x} \in \mathbb{R}^n_{\geq 0} \mid \mathbf{Ax} \leq \mathbf{b}\}. \tag{5.3}$$
2. The *integer hull* $\mathcal{P}_I$ is the convex hull of all integer feasible points:
$$\mathcal{P}_I = \text{conv}(S) = \text{conv}(\mathcal{P} \cap \mathbb{Z}^n). \tag{5.4}$$

By construction, the integer hull is a convex polyhedron whose vertices are guaranteed to be integral, and it satisfies the inclusion:

$$\mathcal{P}_I \subseteq \mathcal{P}. \tag{5.5}$$

Theorem 5.1 *Fundamental Theorem of Integer Programming*

Let $\mathcal{P} = \{\mathbf{x} \in \mathbb{R}^n_{\geq 0} \mid \mathbf{Ax} \leq \mathbf{b}\}$ be a bounded polyhedron (polytope). Then optimizing a linear objective over the discrete set $S = \mathcal{P} \cap \mathbb{Z}^n$ is equivalent to optimizing over its integer hull $\mathcal{P}_I$:
$$\max_{\mathbf{x} \in \mathcal{P} \cap \mathbb{Z}^n} \mathbf{c}^\top \mathbf{x} = \max_{\mathbf{x} \in \mathcal{P}_I} \mathbf{c}^\top \mathbf{x}. \tag{5.6}$$

Proof. Because $\mathcal{P}$ is bounded, the set of integer points $S = \mathcal{P} \cap \mathbb{Z}^n$ is finite. Every point $\mathbf{x} \in \mathcal{P}_I = \text{conv}(S)$ can be expressed as a convex combination of points in S: $\mathbf{x} = \sum_k \lambda_k \mathbf{x}^{(k)}$ with $\lambda_k \geq 0$, $\sum_k \lambda_k = 1$, and $\mathbf{x}^{(k)} \in S$. By linearity:
$$\mathbf{c}^\top \mathbf{x} = \sum_k \lambda_k (\mathbf{c}^\top \mathbf{x}^{(k)}) \leq \max_k (\mathbf{c}^\top \mathbf{x}^{(k)}). \tag{5.7}$$
Hence, the maximum over $\mathcal{P}_I$ is attained at an extreme point of $\mathcal{P}_I$, which is an element of S. □

The Intractability of the Integer Hull. Theorem 5.1 reveals a profound theoretical insight: if an explicit linear inequality description $\mathcal{P}_I = \{\mathbf{x} \mid \tilde{\mathbf{A}}\mathbf{x} \leq \tilde{\mathbf{b}}\}$ were readily available, solving an **NP**-hard integer program would reduce to a polynomial-time continuous linear program! However, describing $\mathcal{P}_I$ explicitly requires an exponential number of facet-defining linear inequalities in general. The algorithmic goal of cutting plane methods and branch-and-cut is to approximate $\mathcal{P}_I$ *locally* near the optimal relaxation point.

5.1.3 The Continuous Relaxation Bound

Because $\mathcal{P}_I \subseteq \mathcal{P}$, optimizing over the continuous relaxation $\mathcal{P}$ expands the feasible region. For a maximization problem, this relaxation provides an upper bound on the integer optimum:

$$Z^*_{\text{ILP}} = \max_{\mathbf{x} \in \mathcal{P} \cap \mathbb{Z}^n} \mathbf{c}^\top \mathbf{x} \leq \max_{\mathbf{x} \in \mathcal{P}} \mathbf{c}^\top \mathbf{x} = Z^*_{\text{LP}}. \tag{5.8}$$

The non-negative difference:

$$\Delta_{\text{gap}} = Z^*_{\text{LP}} - Z^*_{\text{ILP}} \geq 0 \tag{5.9}$$

is the *integrality gap*. If the continuous relaxation optimum $\mathbf{x}^*$ happens to satisfy $\mathbf{x}^* \in \mathbb{Z}^n$, then $\mathbf{x}^*$ is immediately certified as an optimal integer solution, and $\Delta_{\text{gap}} = 0$.

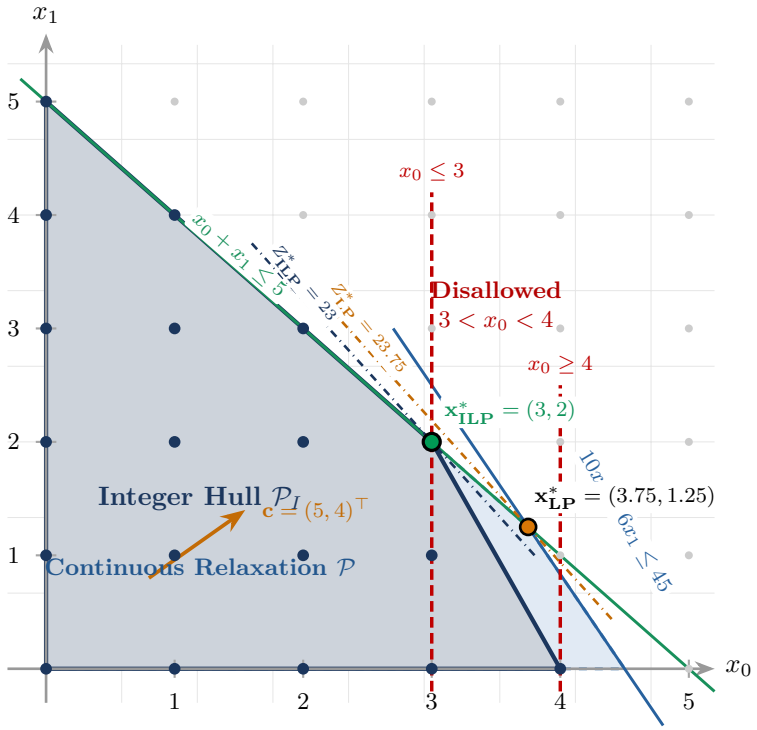


Figure 5.1: Geometric relationship between the continuous LP relaxation $\mathcal{P}$, the integer hull $\mathcal{P}_I = \text{conv}(\mathcal{P} \cap \mathbb{Z}^2)$, and LP-based branching. Discrete integer feasible solutions (navy dots) lie inside $\mathcal{P}$. The continuous relaxation optimum $\mathbf{x}^*_{\text{LP}} = (3.75, 1.25)$ achieves $Z^*_{\text{LP}} = 23.75$ on an exterior fractional corner. Branching on x_0 creates two subproblems ($x_0 \leq 3$ and $x_0 \geq 4$), carving away the non-integral strip $3 < x_0 < 4$ and isolating the integer optimum $\mathbf{x}^*_{\text{ILP}} = (3, 2)$ with $Z^*_{\text{ILP}} = 23$.

Figure 5.1 illustrates this fundamental geometric relationship for the two-variable problem analyzed in Example 5.1. The continuous relaxation polyhedron $\mathcal{P}$ contains points not in the integer hull $\mathcal{P}_I$. The objective function $\mathbf{c}^\top \mathbf{x}$ drives the continuous Simplex algorithm to the fractional vertex $\mathbf{x}^*_{\text{LP}} = (3.75, 1.25)$. Branching on x_0 introduces the disjunction $x_0 \leq 3 \vee x_0 \geq 4$, eliminating the fractional strip $3 < x_0 < 4$ without removing any integer feasible points, and guiding the search toward $\mathbf{x}^*_{\text{ILP}} = (3, 2)$.

5.2 Total Unimodularity and Polyhedral Integrality

Under specific algebraic matrix structures, the continuous LP relaxation is guaranteed to yield an integer solution without branching.

Definition 5.2 *Totally Unimodular Matrix*

A matrix $\mathbf{A} \in \mathbb{Z}^{m \times n}$ is *totally unimodular* (TUM) if the determinant of every square submatrix of $\mathbf{A}$ belongs to $\{-1, 0, 1\}$.

In particular, every individual entry of a totally unimodular matrix must belong to $\{-1, 0, 1\}$, since each entry is a 1×1 submatrix.

Theorem 5.2 *Hoffman-Kruskal Integrality Theorem*

Let $\mathbf{A} \in \mathbb{Z}^{m \times n}$ be an integral matrix. The polyhedron:

$$\mathcal{P} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{A}\mathbf{x} \leq \mathbf{b},\ \mathbf{x} \geq \mathbf{0}\} \tag{5.10}$$

has only integral extreme points for every integer vector $\mathbf{b} \in \mathbb{Z}^m$ for which $\mathcal{P}$ is non-empty if and only if $\mathbf{A}$ is totally unimodular.

Proof. Let $\mathbf{A}_B$ be any basis matrix corresponding to an extreme point $\mathbf{x}$. In standard equational form with slack variables $[\mathbf{A} \mid \mathbf{I}_m]$, any basis matrix is an $m \times m$ invertible submatrix. The basic variables are given by Cramer's rule:

$$x_{B(i)} = \frac{\det(\mathbf{A}_B^{(i)})}{\det(\mathbf{A}_B)}, \quad \forall i \in \{0, \ldots, m-1\}, \tag{5.11}$$

where $\mathbf{A}_B^{(i)}$ is obtained by replacing column i of $\mathbf{A}_B$ with the integer right-hand side $\mathbf{b}$. Because $\mathbf{A}$ is TUM and $\mathbf{A}_B$ is non-singular, $\det(\mathbf{A}_B) \in \{-1, 1\}$. Expanding $\det(\mathbf{A}_B^{(i)})$ along the integer column $\mathbf{b}$ produces an integer. Dividing an integer by ± 1 yields an integer. Hence, every basic feasible solution $\mathbf{x}$ is entirely integral. □

5.2.1 Network Flow and Bipartite Matching

A cornerstone application of total unimodularity is network optimization:

Proposition 5.3 *Node-Arc Incidence Matrices are TUM*

Let $G = (V, E)$ be a directed graph with $V = \{0, \ldots, |V| - 1\}$. The node-arc incidence matrix $\mathbf{M} \in \{-1, 0, 1\}^{|V| \times |E|}$, defined by:

$$M_{v,e} = \begin{cases} +1, & \text{if arc } e \text{ leaves vertex } v, \\ -1, & \text{if arc } e \text{ enters vertex } v, \\ 0, & \text{otherwise,} \end{cases} \tag{5.12}$$

is totally unimodular.

Consequently, for any minimum-cost network flow problem with integer capacities $u_e \in \mathbb{Z}$ and integer node supplies/demands $b_v \in \mathbb{Z}$:

$$\begin{aligned} \text{minimize} \quad & \sum_{e \in E} c_e f_e \\ \text{subject to} \quad & \sum_{e \in \delta^+(v)} f_e - \sum_{e \in \delta^-(v)} f_e = b_v, \quad \forall v \in V, \\ & 0 \leq f_e \leq u_e, \quad \forall e \in E, \end{aligned} \tag{5.13}$$

solving the continuous linear relaxation with the Simplex algorithm guarantees an integer optimal flow $\boldsymbol{f}^* \in \mathbb{Z}^{|E|}$.

Similarly, in a bipartite graph $G = (U \cup V, E)$, the vertex-edge incidence matrix is totally unimodular. Solving the continuous LP relaxation:

$$\max \sum_{e \in E} x_e \quad \text{s.t.} \quad \sum_{e \in \delta(w)} x_e \leq 1 \; (\forall w \in U \cup V), \quad x_e \geq 0 \tag{5.14}$$

automatically yields optimal binary matching variables $x_e^* \in \{0, 1\}$.

Listing 5.1 shows how a directed network is encoded as a linear program in Java.

```java
public class FlowMatrices {
    final double[][] A;
    final double[] b;
    final double[] c;

    public FlowMatrices(FlowNetwork network) {
        int V = network.V(), E = network.E();
        int nRows = E + 2 * V, nCols = E;
        A = new double[nRows][nCols];
        b = new double[nRows];
        c = new double[nCols];

        int source = network.source();
        for (int e : network.edgesFrom(source)) c[e] = 1.0;

        // Capacity constraints: f_e <= capacity_e
        for (int e = 0; e < E; e++) {
            A[e][e] = 1.0;
            b[e] = network.edge(e).capacity();
        }

        // Flow conservation: sum_in - sum_out = 0
        int row = E;
        for (int v = 0; v < V; v++) {
            if (v == source || v == network.sink()) continue;
            for (int e : network.inEdges(v))  A[row][e] =  1.0;
            for (int e : network.outEdges(v)) A[row][e] = -1.0;
            b[row++] = 0.0;
            // Opposite inequality: -(sum_in - sum_out) <= 0
            for (int e : network.inEdges(v))  A[row][e] = -1.0;
            for (int e : network.outEdges(v)) A[row][e] =  1.0;
            b[row++] = 0.0;
        }
    }
}
```

Listing 5.1: Encoding Network Flow constraint matrices in Java.

5.3 LP-Based Branch and Bound

When the constraint matrix $\mathbf{A}$ is not totally unimodular, continuous LP relaxations generally yield fractional extreme points. To eliminate fractional solutions, modern solvers integrate the Simplex algorithm into the tree search framework of Branch and Bound (Chapter 3).

5.3.1 The Search Algorithm

Consider the integer linear program $\max\{\mathbf{c}^\top \mathbf{x} \mid \mathbf{A}\mathbf{x} \leq \mathbf{b},\ \mathbf{x} \in \mathbb{Z}^n_{\geq 0}\}$. The solver maintains a global lower bound $\underline{Z}$ (the objective value of the best known integer feasible solution, initialized to $-\infty$), and a queue $\mathcal{Q}$ of active subproblems (nodes).

At each node $u \in \mathcal{Q}$:

1. **Relaxation Solve**: Solve the continuous linear programming relaxation $LP(u)$ via the Simplex algorithm, obtaining an optimal basic solution $\mathbf{x}^*$ with objective value $Z_{LP}(u)$.
2. **Pruning Checks**:
   - *Pruning by Infeasibility*: If $LP(u)$ is infeasible, prune node u (the subproblem contains no feasible integer points).
   - *Pruning by Bound*: If $Z_{LP}(u) \leq \underline{Z}$, prune node u (no feasible completion in this subtree can strictly improve upon the incumbent).
   - *Pruning by Integrality*: If all integer-constrained variables are integral ($x_j^* \in \mathbb{Z}$ for all j), then $\mathbf{x}^*$ is integer feasible. Update the incumbent $\underline{Z} \leftarrow \max(\underline{Z}, Z_{LP}(u))$ and prune node u.
3. **Branching**: If $Z_{LP}(u) > \underline{Z}$ and at least one variable x_k^* is fractional ($x_k^* \notin \mathbb{Z}$), choose a fractional variable x_k and partition the feasible space into two disjoint subproblems:

$$\text{Left child (Down-branch): } x_k \leq \lfloor x_k^* \rfloor, \qquad \text{Right child (Up-branch): } x_k \geq \lceil x_k^* \rceil. \tag{5.15}$$

5.3.2 Branching Decisions: Variable Selection

When multiple basic variables are fractional at an optimal node solution $\mathbf{x}^*$, the choice of the branching variable x_k exerts an immense impact on the size of the search tree. Solvers employ several heuristics:

1. Most Fractional Rule Select the variable whose fractional part $f_j = x_j^* - \lfloor x_j^* \rfloor$ is closest to 0.5:

$$k = \arg\min_{j\,:\,f_j > 0} |f_j - 0.5|. \tag{5.16}$$

The intuition is that variables furthest from an integer impose the greatest disruption to the continuous relaxation, maximizing bound degradation in the child nodes. While computationally trivial ($\mathcal{O}(n)$), this myopic rule ignores objective coefficients and constraint geometry.

2. Strong Branching Before committing to a branch, the solver tests a candidate subset of fractional variables. For each candidate x_j, the solver tentatively applies both bounds $x_j \leq \lfloor x_j^* \rfloor$ and $x_j \geq \lceil x_j^* \rceil$, executes a limited number of Dual Simplex pivots (typically 10–50 iterations), and measures the objective degradations:

$$\Delta_j^- = Z^* - Z_{LP}(x_j \leq \lfloor x_j^* \rfloor), \qquad \Delta_j^+ = Z^* - Z_{LP}(x_j \geq \lceil x_j^* \rceil). \tag{5.17}$$

The solver then selects the variable that maximizes a combined score, such as the product score:

$$\text{score}(j) = \max(\Delta_j^-, 10^{-6}) \times \max(\Delta_j^+, 10^{-6}). \tag{5.18}$$

Strong branching produces very small search trees, but the computational cost of testing multiple candidates at every node is high.

3. Pseudo-Cost Branching To obtain the quality of strong branching without its computational overhead, solvers track historical performance. The *pseudo-costs* Ψ_j^- and Ψ_j^+ record the average objective degradation per unit change observed whenever variable x_j was branched upon in ancestor nodes:

$$\Psi_j^- = \frac{1}{|K_j^-|} \sum_{u \in K_j^-} \frac{Z_{\text{parent}} - Z_{\text{child}}}{\lfloor x_j^* \rfloor - x_j^*}, \qquad \Psi_j^+ = \frac{1}{|K_j^+|} \sum_{u \in K_j^+} \frac{Z_{\text{parent}} - Z_{\text{child}}}{x_j^* - \lceil x_j^* \rceil}. \tag{5.19}$$

At subsequent nodes, the degradations are estimated cheaply as $\Delta_j^- \approx \Psi_j^- \cdot f_j$ and $\Delta_j^+ \approx \Psi_j^+ \cdot (1 - f_j)$.

4. Reliability Branching State-of-the-art MIP solvers (such as HiGHS, SCIP, and Gurobi) combine strong branching and pseudo-costs via *reliability branching*. For the first η times a variable is considered (typically $\eta \in [4, 8]$), strong branching evaluates the variable directly to initialize reliable pseudo-costs. Once a variable has been evaluated η times, the solver switches permanently to fast pseudo-cost estimates.

5.3.3 Updating the Simplex Tableau with Branching Constraints

The decisive reason LP-based Branch-and-Bound is computationally viable is that child LP relaxations *do not need to be solved from scratch*. When a branching constraint is added to a child node, the parent's optimal Simplex tableau is updated directly.

Algebraic Tableau Representation Recall from Section 4.5 that an optimal simplex tableau partitions variables into basic indices B and non-basic indices N. The canonical dictionary equations express the basic variables and the objective in terms of non-basic variables:

$$x_{B(i)} = \bar{b}_i - \sum_{j \in N} \bar{a}_{i,j} x_j, \quad \forall i \in \{0, \ldots, m-1\}, \tag{5.20}$$

$$Z = Z_0 + \sum_{j \in N} \bar{c}_j x_j. \tag{5.21}$$

Because the parent tableau is optimal for a maximization problem, every reduced cost is non-positive:

$$\bar{c}_j \leq 0, \quad \forall j \in N. \tag{5.22}$$

Adding a Down-Branch Constraint Suppose basic variable $x_{B(k)}$ is fractional with value $\bar{b}_k = x_{B(k)}^* \notin \mathbb{Z}$. Let $f_k = \bar{b}_k - \lfloor \bar{b}_k \rfloor \in (0, 1)$ denote its fractional part. The down-branch imposes:

$$x_{B(k)} \leq \lfloor \bar{b}_k \rfloor. \tag{5.23}$$

We convert this inequality into standard equality form by introducing a non-negative slack variable $s_{\text{branch}} \geq 0$:

$$x_{B(k)} + s_{\text{branch}} = \lfloor \bar{b}_k \rfloor. \tag{5.24}$$

Crucially, $x_{B(k)}$ is already a basic variable! To maintain the canonical tableau form, basic variables cannot appear with non-zero coefficients in other constraint rows. We must substitute the dictionary expression of $x_{B(k)}$ from Equation (5.20) into Equation (5.24):

$$\left(\bar{b}_k - \sum_{j \in N} \bar{a}_{k,j} x_j \right) + s_{\text{branch}} = \lfloor \bar{b}_k \rfloor. \tag{5.25}$$

Rearranging to express the new basic slack variable s_{branch} yields:

$$s_{\text{branch}} - \sum_{j \in N} \bar{a}_{k,j} x_j = \lfloor \bar{b}_k \rfloor - \bar{b}_k = -f_k. \tag{5.26}$$

In standard tableau row notation with basic variable s_{branch} in row m:

$$s_{\text{branch}} + \sum_{j \in N} (-\bar{a}_{k,j}) x_j = -f_k. \tag{5.27}$$

The Fundamental Duality Asymmetry Inspecting the augmented tableau with row (5.27) reveals two fundamental properties:

1. **Primal Infeasibility**: Setting all non-basic variables $x_j = 0$ ($j \in N$) assigns $s_{\text{branch}} = -f_k < 0$. The current basic solution violates non-negativity and is *primal infeasible*.
2. **Dual Feasibility (Optimality)**: The objective row (5.21) is completely unchanged! The new slack variable s_{branch} has an initial cost of 0, so its reduced cost is zero. All existing non-basic variables retain their reduced costs $\bar{c}_j \leq 0$. The tableau is **100% dual feasible**!

5.3.4 Dual Simplex Warm-Starting

Because the augmented tableau is dual feasible but primal infeasible, applying the Primal Simplex algorithm would require discarding the tableau and executing Two-Phase Simplex from scratch.

Instead, the **Dual Simplex Algorithm** (Lemke, 1954 [Lem54]) is applied directly! The Dual Simplex algorithm maintains dual feasibility ($\bar{c}_j \leq 0$) while systematically eliminating negative right-hand sides until primal feasibility is restored.

Step 1: Dual Leaving Row Selection (Primal Feasibility Rule) Select a row p with a negative right-hand side constant:

$$p = \operatorname*{arg\,min}_{i \,:\, a[i][n+m] < 0} a[i][n+m]. \tag{5.28}$$

For a newly added branching row, $p = m$ (the branching row itself, where $a[m][n+m] = -f_k < 0$).

Step 2: Dual Ratio Test (Dual Optimality Rule) To ensure that all reduced costs in row $m+1$ remain non-positive ($\bar{c}'_j \leq 0$) after Gauss-Jordan elimination, the entering column $q \in N$ must be chosen among columns where the pivot row entry is strictly negative:

$$q = \operatorname*{arg\,min}_{j \in N \,:\, a[p][j] < 0} \frac{a[\text{obj}][j]}{a[p][j]} = \operatorname*{arg\,min}_{j \in N \,:\, a[p][j] < 0} \left| \frac{\bar{c}_j}{a[p][j]} \right|. \tag{5.29}$$

- If no entry in row p is negative ($a[p][j] \geq 0$ for all $j \in N$), then no non-basic variable can increase the negative right-hand side. The subproblem is certified to be **infeasible**, and the child node is immediately pruned!
- Otherwise, column q enters the basis, and row p leaves.

Step 3: Gauss-Jordan Pivot Perform the standard row operations on pivot element $a[p][q] < 0$. After typically only **1 to 3 dual pivots**, all basic right-hand sides become non-negative, reaching the optimal solution of the child node!

Example 5.1 *Updating the Tableau via Dual Simplex*

Consider an integer linear program whose continuous relaxation has two non-negative variables $x_0, x_1 \geq 0$:

$$\begin{aligned} \text{maximize} \quad & Z = 5x_0 + 4x_1 \\ \text{subject to} \quad & x_0 + x_1 \leq 5 \\ & 10x_0 + 6x_1 \leq 45 \\ & x_0, x_1 \in \mathbb{Z}_{\geq 0}. \end{aligned}$$

Introducing slack variables $s_0, s_1 \geq 0$, the optimal continuous simplex tableau is:

$T_0 =$

Basis	x_0	x_1	s_0	s_1	RHS
x_1	0	1	$\frac{5}{2}$	$-\frac{1}{4}$	$\frac{5}{4}$
x_0	1	0	$-\frac{3}{2}$	$\frac{1}{4}$	$\frac{15}{4}$
Obj (Z)	0	0	$-\frac{5}{2}$	$-\frac{1}{4}$	$-\frac{95}{4}$

The continuous optimal solution is fractional: $x_0^* = 15/4 = 3.75$, $x_1^* = 5/4 = 1.25$, with $Z^* = 23.75$.

We branch on x_0: its fractional part is $f_0 = 0.75$. The down-branch imposes $x_0 \leq 3$, or $x_0 + s_2 = 3$. Substituting $x_0 = \frac{15}{4} - \left(-\frac{3}{2}s_0 + \frac{1}{4}s_1\right)$ into $x_0 + s_2 = 3$ gives:

$$s_2 = 3 - \frac{15}{4} + \left(-\frac{3}{2}s_0 + \frac{1}{4}s_1\right) \implies s_2 + \frac{3}{2}s_0 - \frac{1}{4}s_1 = -\frac{3}{4}.$$

Appending this row with basic variable s_2 yields the augmented tableau:

$T_{\text{branch}} =$

Basis	x_0	x_1	s_0	s_1	s_2	RHS
x_1	0	1	$\frac{5}{2}$	$-\frac{1}{4}$	0	$\frac{5}{4}$
x_0	1	0	$-\frac{3}{2}$	$\frac{1}{4}$	0	$\frac{15}{4}$
s_2	0	0	$\frac{3}{2}$	$-\frac{1}{4}$	1	$-\frac{3}{4}$
Obj (Z)	0	0	$-\frac{5}{2}$	$-\frac{1}{4}$	0	$-\frac{95}{4}$

All reduced costs are non-positive ($\bar{c}_{s_0} = -5/2$, $\bar{c}_{s_1} = -1/4$), confirming dual feasibility. However, the RHS in row s_2 is $-3/4 < 0$. We perform a Dual Simplex step:

- **Leaving row**: Row $p = s_2$ (the only row with negative RHS).
- **Entering column**: Inspect negative entries in row s_2: the only negative entry is column s_1 with $a[s_2][s_1] = -1/4$. Column $q = s_1$ must enter.
- **Pivot on** (s_2, s_1): Multiplying row s_2 by -4 normalizes the pivot entry to 1, and eliminating s_1 from the other rows produces:

$T_{\text{dual}} =$

Basis	x_0	x_1	s_0	s_1	s_2	RHS
x_1	0	1	1	0	-1	2
x_0	1	0	0	0	1	3
s_1	0	0	-6	1	-4	3
Obj (Z)	0	0	-4	0	-1	-23

In exactly **one dual pivot**, all RHS entries are non-negative! The new optimal basic solution is:

$$x_0 = 3, \quad x_1 = 2, \quad Z = 23.$$

Because both x_0 and x_1 are integers, this child node immediately yields an integer feasible solution with value $Z = 23$, establishing a powerful global lower bound!

5.3.5 Dual Simplex Implementation in Java

Listing 5.2 implements the complete Dual Simplex re-optimization algorithm in Java. When a branching constraint or cutting plane is appended to the tableau, the constructor receives the augmented dictionary matrix. Method `solve()` drives out negative right-hand sides via dual pivots: `selectLeavingRow()` identifies the leaving row ($a[p][\text{RHS}] < 0$), and `dualRatioTest(p)` selects the entering column q minimizing $|\bar{c}_j / a[p][j]|$. If no negative pivot exists in row p, `solve()` returns `false`, immediately certifying child node infeasibility.

```
public class DualSimplex {
    private static final double EPS = 1e-10;
    private final double[][] a;
    private final int[] basis;
    private final int m, nCols;

    public DualSimplex(double[][] a, int[] basis,
                       int m, int nCols) {
        this.a = a;
        this.basis = basis;
        this.m = m;
        this.nCols = nCols;
    }

    /** Dual Simplex loop: returns false if infeasible. */
    public boolean solve() {
        while (true) {
            int p = selectLeavingRow();
            if (p == -1) return true; // Primal feasible!
            int q = dualRatioTest(p);
            if (q == -1) return false; // Infeasible node
            pivot(p, q);
            basis[p] = q;
        }
    }

    /** Select row with the most negative right-hand side. */
    private int selectLeavingRow() {
        int p = -1;
        double minRHS = -EPS;
        for (int i = 0; i < m; i++) {
            if (a[i][nCols] < minRHS) {
                minRHS = a[i][nCols];
                p = i;
            }
        }
        return p;
    }
```

```
39
40      /** Dual ratio test: argmin |c_j / a[p][j]| for a[p][j] < -EPS. */
41      private int dualRatioTest(int p) {
42          int q = -1;
43          double minRatio = Double.POSITIVE_INFINITY;
44          for (int j = 0; j < nCols; j++) {
45              if (a[p][j] < -EPS) {
46                  double ratio = Math.abs(a[m][j] / a[p][j]);
47                  if (ratio < minRatio) {
48                      minRatio = ratio;
49                      q = j;
50                  }
51              }
52          }
53          return q;
54      }
55
56      private void pivot(int p, int q) {
57          for (int i = 0; i <= m; i++)
58              for (int j = 0; j <= nCols; j++)
59                  if (i != p && j != q)
60                      a[i][j] -= a[p][j] * (a[i][q] / a[p][q]);
61          for (int i = 0; i <= m; i++)
62              if (i != p) a[i][q] = 0.0;
63          for (int j = 0; j <= nCols; j++)
64              if (j != q) a[p][j] /= a[p][q];
65          a[p][q] = 1.0;
66      }
67  }
```

Listing 5.2: Compact Dual Simplex warm-starting in Java.

5.4 Cutting Planes and Branch-and-Cut

Branch and bound partitions the search space geometrically. An orthogonal approach, pioneered by Ralph Gomory in 1958 [Gom58], is to *tighten* the continuous relaxation by adding valid inequalities that cut off fractional vertices without removing any integer feasible points.

5.4.1 Gomory Fractional Cuts

Gomory cuts possess a remarkable property: they can be derived directly from the optimal simplex tableau of the continuous relaxation without requiring problem-specific structural knowledge.

Theorem 5.4 *Gomory Fractional Cut Validity*

Let $x_i + \sum_{j \in N} \bar{a}_{i,j} x_j = \bar{b}_i$ be a row of an optimal simplex tableau where $\bar{b}_i \notin \mathbb{Z}$. Decompose each coefficient into its integer floor and fractional part:

$$\bar{a}_{i,j} = \lfloor \bar{a}_{i,j} \rfloor + f_{i,j}, \qquad \bar{b}_i = \lfloor \bar{b}_i \rfloor + f_i, \tag{5.30}$$

where $0 \le f_{i,j} < 1$ and $0 < f_i < 1$. Then the *Gomory fractional cut*:

$$\sum_{j \in N} f_{i,j} x_j \ge f_i \tag{5.31}$$

is satisfied by every integer feasible solution $\mathbf{x} \in \mathcal{P} \cap \mathbb{Z}^n$, but strictly violates the current fractional LP solution $\mathbf{x}^*$.

Proof. Rewriting the tableau row equation:

$$x_i + \sum_{j \in N} \lfloor \bar{a}_{i,j} \rfloor x_j - \lfloor \bar{b}_i \rfloor = f_i - \sum_{j \in N} f_{i,j} x_j. \tag{5.32}$$

For any integer feasible solution $\mathbf{x} \in \mathbb{Z}^n_{\geq 0}$, the left-hand side is an integer because all variables and floor terms are integers. Hence, the right-hand side $f_i - \sum_{j \in N} f_{i,j} x_j$ must also be an integer. Because $f_i < 1$ and $f_{i,j} \geq 0, x_j \geq 0$, the right-hand side is strictly less than 1:

$$f_i - \sum_{j \in N} f_{i,j} x_j < 1. \tag{5.33}$$

An integer strictly less than 1 cannot exceed 0:

$$f_i - \sum_{j \in N} f_{i,j} x_j \leq 0 \iff \sum_{j \in N} f_{i,j} x_j \geq f_i. \tag{5.34}$$

Finally, at the current LP relaxation solution $\mathbf{x}^*$, every non-basic variable is zero ($x^*_j = 0$ for all $j \in N$). Substituting $\mathbf{x}^*$ yields $0 \geq f_i > 0$, a contradiction. Thus, $\mathbf{x}^*$ is cut off! □

Adding a slack variable $s_{\text{cut}} \geq 0$ transforms Equation (5.31) into:

$$s_{\text{cut}} - \sum_{j \in N} f_{i,j} x_j = -f_i. \tag{5.35}$$

Notice that, exactly like a branching constraint, the cut row has a negative right-hand side ($-f_i < 0$) while preserving dual feasibility ($\bar{c}_j \leq 0$). Thus, Gomory cuts are resolved using the exact same **Dual Simplex** engine!

5.4.2 The Branch-and-Cut Paradigm

Pure cutting plane methods historically suffered from slow numerical convergence due to ill-conditioning. Modern commercial and open-source solvers combine cutting planes with tree search in the **Branch-and-Cut** framework:

1. At node u, solve the continuous LP relaxation.
2. **Cut Generation Loop**: Generate valid cutting planes (Gomory cuts, knapsack cover cuts, clique cuts, flow covers). Re-optimize using Dual Simplex. Repeat until bound improvement diminishes (the "tailing-off" effect).
3. **Branch**: If the solution remains fractional, branch on a fractional variable using reliability branching.

5.5 Mixed-Integer Linear Programming Modeling

Binary variables ($x \in \{0, 1\}$) provide expressive modeling power for discrete logic:

1. Logical Implications and Indicator Constraints To enforce the logical implication "if $x = 1$, then $y = 1$", where $x, y \in \{0, 1\}$:

$$x \leq y. \tag{5.36}$$

To enforce "at most k out of n options can be selected":

$$\sum_{j=0}^{n-1} x_j \leq k. \tag{5.37}$$

2. Big-M Formulations and Numerical Pitfalls Consider linking a continuous activity level $y \in [0, U]$ to a binary decision $x \in \{0, 1\}$ (e.g., if a machine is turned off, $x = 0$, its output must be zero, $y = 0$):

$$y \leq Mx, \tag{5.38}$$

where M is a sufficiently large upper bound ($M \geq U$).

The Hazard of Large M. Choosing an unnecessarily large constant M (e.g., $M = 10^9$) causes two severe deficiencies:

1. *Weak LP Relaxation*: At fractional nodes where $x = 10^{-6}$, the constraint allows $y = 10^3$, giving a very loose upper bound and causing massive search trees.
2. *Catastrophic Numerical Cancellation*: When M exceeds solver tolerance ratios ($1/\epsilon \approx 10^8$), roundoff errors trigger false feasibility claims. Always set M to the tightest valid physical upper bound ($M = U$).

3. Strong vs. Weak Formulations: Facility Location In the Uncapacitated Facility Location Problem (UFLP), client $i \in \{0, \ldots, m-1\}$ can be served by facility $j \in \{0, \ldots, n-1\}$ ($x_{i,j} \in [0, 1]$) only if facility j is open ($y_j \in \{0, 1\}$).

- **Weak Formulation**: $\sum_{i=0}^{m-1} x_{i,j} \leq m \cdot y_j$.
- **Strong Formulation**: $x_{i,j} \leq y_j$ for all $i \in \{0, \ldots, m-1\}$.

Although the weak formulation has fewer constraints, the strong formulation defines a much tighter polyhedron: its continuous LP relaxation bound is significantly closer to the integer optimum, reducing branch-and-bound runtimes by orders of magnitude.

5.6 Chapter Summary and Exercises

5.6.1 Key Conceptual Takeaways

Table 5.1 summarizes the core principles of Integer Linear Programming.

5.6.2 Exercises

Exercise 5.1 *Total Unimodularity Proof*

Prove that the constraint matrix of the Maximum Bipartite Matching problem is totally unimodular. Use the criterion that every column contains at most two non-zero entries (± 1), and that vertices can be partitioned into sets U and V.

Concept	Key Result and Algorithmic Significance
Integer Hull $\mathcal{P}_I$	The convex hull of integer feasible points; optimizing over $\mathcal{P}_I$ is linear but **NP**-hard.
LP Relaxation	Provides an upper bound $Z^*_{\text{LP}} \geq Z^*_{\text{ILP}}$; bounding quality determines tree size.
Total Unimodularity	If $\mathbf{A}$ is TUM and $\mathbf{b} \in \mathbb{Z}^m$, all extreme points of the continuous LP are integer.
Branching Split	Partitions fractional space via $x_k \leq \lfloor x^*_k \rfloor$ and $x_k \geq \lceil x^*_k \rceil$.
Tableau Augmentation	Adding a branch constraint leaves the tableau dual feasible but primal infeasible.
Dual Simplex	Restores feasibility in 1–3 pivots without restarting from scratch (warm-starting).
Gomory Cuts	Valid inequalities derived directly from fractional tableau rows; resolved via Dual Simplex.
Branch-and-Cut	State-of-the-art MIP engine combining cut generation at search nodes with tree branching.

Table 5.1: Foundations of Integer Linear Programming and Branch-and-Cut.

Exercise 5.2 *Tableau Augmentation and Dual Simplex Pivot*

An optimal continuous simplex tableau for maximizing $Z = 3x_0 + 4x_1$ with slacks s_0, s_1 has row:

$$x_0 + \frac{2}{3}s_0 - \frac{1}{3}s_1 = \frac{7}{3}.$$

1. Formulate the down-branch constraint $x_0 \leq \lfloor 7/3 \rfloor = 2$ using a non-negative slack $s_2 \geq 0$.
2. Substitute the dictionary expression of x_0 to express the new row with basic variable s_2.
3. Identify the leaving row and apply the dual ratio test to determine the entering column.
4. Perform the dual pivot and verify that primal feasibility is restored.

Exercise 5.3 *Derivation of a Gomory Fractional Cut*

For the tableau row in Exercise 5.2:

$$x_0 + \frac{2}{3}s_0 - \frac{1}{3}s_1 = \frac{7}{3} :$$

1. Write the coefficients of $-1/3$ and $7/3$ in terms of their integer floors and fractional parts ($a = \lfloor a \rfloor + f$ with $0 \leq f < 1$).
2. Derive the corresponding Gomory fractional cut $\sum_j f_{i,j} s_j \geq f_i$.
3. Introduce slack s_{cut} and express the cut as an augmented tableau row.

Exercise 5.4 *Formulation Tightness: Facility Location*

Consider a facility location problem with 3 clients and 2 candidate facilities. Write out the constraints for both the weak formulation ($\sum_{i=0}^{2} x_{i,j} \leq 3y_j$) and the strong

formulation ($x_{i,j} \leq y_j$). Show a fractional point (x, y) that is feasible for the weak formulation relaxation but strictly infeasible for the strong formulation relaxation.

CHAPTER 6

Lagrangian Relaxation

In discrete optimization, NP-hard combinatorial problems often possess an underlying dual structure: they can be viewed as an *easy* core subproblem (such as a shortest path or a minimum spanning tree, solvable in polynomial time) entangled with a collection of *complicating* or *hard* side constraints (such as budget limits, resource capacities, or degree balance).

In Chapter 3, we observed that simply ignoring complicating constraints yields valid lower bounds for minimization problems. However, outright omission frequently produces bounds that are too loose to prune search trees effectively. *Lagrangian relaxation* provides an elegant and mathematically profound bridge between computational tractability and bound tightness. Instead of enforcing complicating constraints strictly (which induces NP-hardness) or discarding them entirely (which degrades bound quality), Lagrangian relaxation *dualizes* them into the objective function, penalizing their violation via a vector of *Lagrange multipliers*.

Originally conceived in continuous calculus by Joseph-Louis Lagrange and extended to discrete and integer programming by Everett [Eve63], Geoffrion [Geo74], and Fisher [Fis81], Lagrangian relaxation represents one of the most powerful paradigms in operational research. In this chapter, we develop the theoretical foundations of Lagrangian duality and subgradient optimization. We first illustrate the mechanics of the approach on the *Resource-Constrained Shortest Path Problem* (RCSPP). We then study its application to the Traveling Salesperson Problem (TSP), leading to the celebrated *Held–Karp lower bound* [HK70; HK71].

6.1 The Lagrangian Relaxation Principle

Consider a generic discrete minimization problem formulated as:

$$z^* = \min_{\mathbf{x}} \quad \mathbf{c}^\top \mathbf{x} \tag{6.1}$$

$$\text{subject to} \quad \mathbf{A}\mathbf{x} \leq \mathbf{b}, \tag{6.2}$$

$$\mathbf{x} \in X, \tag{6.3}$$

where $X \subseteq \mathbb{Z}^n$ represents a discrete set with special combinatorial structure (for example, the set of incidence vectors of all directed s-t paths or all spanning trees in a graph). We assume that optimizing a linear objective over X alone (without constraint (6.2)) can be performed in polynomial time by a specialized combinatorial algorithm. Conversely, the system of linear inequalities $\mathbf{A}\mathbf{x} \leq \mathbf{b}$ (with $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$) represents the *complicating constraints* that render the joint problem (6.1)–(6.3) NP-hard.

Definition 6.1 *Lagrangian Relaxation and Dual Function*

Let $\boldsymbol{\lambda} \in \mathbb{R}^m_+$ be a non-negative vector of *Lagrange multipliers* associated with the complicating constraints $\mathbf{Ax} \leq \mathbf{b}$. The *Lagrangian relaxation* of problem (6.1)–(6.3) is defined as:

$$\mathcal{L}(\boldsymbol{\lambda}) = \min_{\mathbf{x} \in X} \left\{ \mathbf{c}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (\mathbf{Ax} - \mathbf{b}) \right\}. \tag{6.4}$$

The function $\mathcal{L}(\boldsymbol{\lambda})\colon \mathbb{R}^m_+ \to \mathbb{R}$ is called the *Lagrangian dual function*.

Notice that when $\boldsymbol{\lambda} \geq \mathbf{0}$, any candidate solution $\mathbf{x} \in X$ that violates the i-th complicating constraint ($(\mathbf{Ax})_i > b_i$) incurs a positive penalty $\lambda_i((\mathbf{Ax})_i - b_i) > 0$, inflating the objective value. Conversely, solutions that satisfy the constraint with strict slack receive a non-positive adjustment.

Theorem 6.1 *Weak Duality of Lagrangian Relaxation*

For any vector of non-negative multipliers $\boldsymbol{\lambda} \geq \mathbf{0}$, the value of the Lagrangian dual function provides a valid lower bound on the optimal primal objective value:

$$\mathcal{L}(\boldsymbol{\lambda}) \leq z^*. \tag{6.5}$$

Proof. Let $\mathbf{x}^*$ be an optimal solution to the primal problem (6.1)–(6.3), so that $z^* = \mathbf{c}^\top \mathbf{x}^*$. Because $\mathbf{x}^*$ is feasible for the primal problem, it satisfies $\mathbf{x}^* \in X$ and $\mathbf{Ax}^* \leq \mathbf{b}$, which implies $\mathbf{Ax}^* - \mathbf{b} \leq \mathbf{0}$. Since $\boldsymbol{\lambda} \geq \mathbf{0}$, the scalar product satisfies:

$$\boldsymbol{\lambda}^\top (\mathbf{Ax}^* - \mathbf{b}) = \sum_{i=0}^{m-1} \lambda_i ((\mathbf{Ax}^*)_i - b_i) \leq 0. \tag{6.6}$$

Therefore:

$$\mathbf{c}^\top \mathbf{x}^* + \boldsymbol{\lambda}^\top (\mathbf{Ax}^* - \mathbf{b}) \leq \mathbf{c}^\top \mathbf{x}^* = z^*. \tag{6.7}$$

By definition, $\mathcal{L}(\boldsymbol{\lambda})$ minimizes $\mathbf{c}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (\mathbf{Ax} - \mathbf{b})$ over all $\mathbf{x} \in X$. Since $\mathbf{x}^* \in X$, it immediately follows that:

$$\mathcal{L}(\boldsymbol{\lambda}) = \min_{\mathbf{x} \in X} \left\{ \mathbf{c}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (\mathbf{Ax} - \mathbf{b}) \right\} \leq \mathbf{c}^\top \mathbf{x}^* + \boldsymbol{\lambda}^\top (\mathbf{Ax}^* - \mathbf{b}) \leq z^*. \tag{6.8}$$

□

Handling Equality Constraints. If the complicating constraints are formulated as equalities $\mathbf{Ax} = \mathbf{b}$, the corresponding multipliers $\boldsymbol{\lambda} \in \mathbb{R}^m$ are unconstrained in sign ($\boldsymbol{\lambda} \in \mathbb{R}^m$). For any feasible solution $\mathbf{x}^*$, $\mathbf{Ax}^* - \mathbf{b} = \mathbf{0}$, so $\boldsymbol{\lambda}^\top (\mathbf{Ax}^* - \mathbf{b}) = 0$ holds regardless of the sign of $\boldsymbol{\lambda}$. Thus, $\mathcal{L}(\boldsymbol{\lambda}) \leq z^*$ holds unconditionally for all $\boldsymbol{\lambda} \in \mathbb{R}^m$.

6.1.1 The Lagrangian Dual Problem

Since $\mathcal{L}(\boldsymbol{\lambda}) \leq z^*$ for every $\boldsymbol{\lambda} \geq \mathbf{0}$, our natural goal is to identify the multiplier vector that delivers the tightest possible lower bound. This leads to the *Lagrangian dual problem*:

$$\mathcal{L}^* = \max_{\boldsymbol{\lambda} \geq \mathbf{0}} \mathcal{L}(\boldsymbol{\lambda}) = \max_{\boldsymbol{\lambda} \geq \mathbf{0}} \left\{ \min_{\mathbf{x} \in X} \left\{ \mathbf{c}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (\mathbf{Ax} - \mathbf{b}) \right\} \right\}. \tag{6.9}$$

The difference $z^* - \mathcal{L}^* \geq 0$ is known as the *duality gap*. In discrete optimization, because X is non-convex, the duality gap is generally non-zero ($z^* > \mathcal{L}^*$), unlike in continuous linear programming where strong duality holds.

Proposition 6.2 *Concavity and Non-Smoothness of the Dual Function*

The Lagrangian dual function $\mathcal{L}(\boldsymbol{\lambda})$ is concave over its domain $\mathbb{R}^m_+$. Furthermore, if X is a finite discrete set, $\mathcal{L}(\boldsymbol{\lambda})$ is piecewise linear and non-differentiable at points where multiple solutions $\mathbf{x} \in X$ tie for the minimum.

Proof. Since X is finite, let $X = \{\mathbf{x}^{(0)}, \mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(K-1)}\}$. For each fixed $\mathbf{x}^{(k)} \in X$, the function $f_k(\boldsymbol{\lambda}) = \mathbf{c}^\top \mathbf{x}^{(k)} + \boldsymbol{\lambda}^\top (\mathbf{A}\mathbf{x}^{(k)} - \mathbf{b})$ is affine (linear plus a constant) in $\boldsymbol{\lambda}$. The Lagrangian dual function is the pointwise minimum of this finite collection of affine functions:

$$\mathcal{L}(\boldsymbol{\lambda}) = \min_{k \in \{0,\ldots,K-1\}} f_k(\boldsymbol{\lambda}). \tag{6.10}$$

The pointwise minimum (or infimum) of any collection of affine (concave) functions is concave. At any point $\boldsymbol{\lambda}$ where two or more distinct solutions $\mathbf{x}^{(j)}, \mathbf{x}^{(k)} \in X$ achieve the minimum with different constraint residual vectors $\mathbf{A}\mathbf{x}^{(j)} \neq \mathbf{A}\mathbf{x}^{(k)}$, the function $\mathcal{L}(\boldsymbol{\lambda})$ exhibits a slope discontinuity (a "kink") and is therefore non-differentiable. □

Because $\mathcal{L}(\boldsymbol{\lambda})$ is non-differentiable, traditional gradient ascent cannot be employed to solve the dual problem (6.9). Before presenting subgradient optimization to resolve this challenge, we examine how Lagrangian relaxation operates on a concrete network optimization problem.

6.2 Resource-Constrained Shortest Path

To develop concrete intuition, consider the *Resource-Constrained Shortest Path Problem* (RCSPP). Let $G = (V, A)$ be a directed graph with vertex set $V = \{0, 1, \ldots, n-1\}$, origin node $s = 0$, and destination node $t = n - 1$. Each directed arc $(i, j) \in A$ has:

- a traversal monetary cost $c_{ij} \geq 0$, and
- a resource consumption or traversal time $t_{ij} \geq 0$.

The traveler is allotted a maximum resource budget $T > 0$. The objective is to find a directed path from origin 0 to destination $n - 1$ that minimizes total traversal cost without exceeding the budget T.

6.2.1 Mathematical Formulation

Let $x_{ij} \in \{0, 1\}$ be binary decision variables indicating whether arc $(i, j) \in A$ is traversed. The problem is formulated as an integer linear program:

$$z^* = \min_{\mathbf{x}} \quad \sum_{(i,j)\in A} c_{ij} x_{ij} \tag{6.11}$$

$$\text{subject to} \quad \sum_{j:\,(i,j)\in A} x_{ij} - \sum_{j:\,(j,i)\in A} x_{ji} = \begin{cases} 1 & \text{if } i = 0, \\ -1 & \text{if } i = n-1, \\ 0 & \text{if } i \in V \setminus \{0, n-1\}, \end{cases} \tag{6.12}$$

$$\sum_{(i,j)\in A} t_{ij} x_{ij} \leq T, \tag{6.13}$$

$$x_{ij} \in \{0, 1\} \quad \forall (i, j) \in A. \tag{6.14}$$

Constraints (6.12) are standard flow conservation equations, defining the set $\mathcal{P}$ of all directed $0 \to (n-1)$ paths. Without constraint (6.13), the problem reduces to the classical

single-source Shortest Path Problem, solvable in $\mathcal{O}(|A| + |V| \log |V|)$ time via Dijkstra's algorithm.

However, the addition of the scalar budget constraint (6.13) renders RCSPP **NP**-hard (specifically, weakly **NP**-hard by a direct reduction from the 0–1 Knapsack or Subset-Sum problem; see Ahuja et al. [AMO93]). Constraint (6.13) is therefore the complicating constraint.

6.2.2 Lagrangian Dualization

Following the Lagrangian relaxation recipe, we associate a non-negative multiplier $\lambda \geq 0$ with the budget constraint (6.13) and dualize it into the objective function:

$$\begin{aligned} \mathcal{L}(\lambda) &= \min_{\mathbf{x} \in \mathcal{P}} \left\{ \sum_{(i,j) \in A} c_{ij} x_{ij} + \lambda \left(\sum_{(i,j) \in A} t_{ij} x_{ij} - T \right) \right\} \\ &= \min_{\mathbf{x} \in \mathcal{P}} \left\{ \sum_{(i,j) \in A} (c_{ij} + \lambda t_{ij}) x_{ij} \right\} - \lambda T. \end{aligned} \tag{6.15}$$

The Penalized Shortest Path Subproblem. For any fixed multiplier value $\lambda \geq 0$, the quantity $-\lambda T$ is a constant offset. We can define modified arc costs:

$$c'_{ij}(\lambda) = c_{ij} + \lambda t_{ij} \quad \forall (i,j) \in A. \tag{6.16}$$

Because $c_{ij} \geq 0$, $t_{ij} \geq 0$, and $\lambda \geq 0$, the modified costs are strictly non-negative: $c'_{ij}(\lambda) \geq 0$. Consequently, evaluating $\mathcal{L}(\lambda)$ for any given λ requires solving a standard unconstrained Shortest Path Problem on G with arc weights $c'_{ij}(\lambda)$, which Dijkstra's algorithm solves in polynomial time!

The multiplier λ has a transparent economic interpretation: it represents the *internal shadow price* (or toll) per unit of resource consumed. When $\lambda = 0$, the algorithm ignores resource usage and finds the cheapest path, which may severely exceed the budget T. As λ increases, resource-heavy arcs become penalizingly expensive, steering the shortest path toward faster, resource-frugal trajectories.

6.2.3 A Worked Example

To trace the mechanics of Lagrangian relaxation numerically, consider the 6-vertex directed network illustrated in Figure 6.1(a). The vertices are $V = \{0, 1, 2, 3, 4, 5\}$, with origin $s = 0$, destination $t = 5$, and a resource budget $T = 14$.

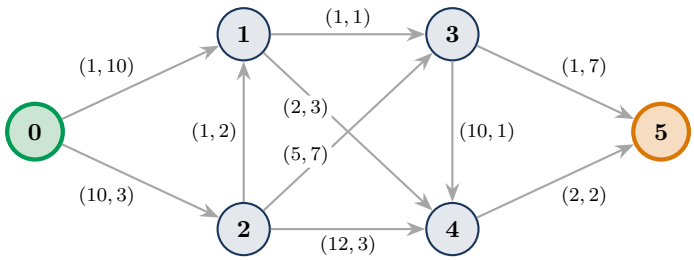
The network contains 9 directed paths connecting vertex 0 to vertex 5. Table 6.1 enumerates each path P, its original traversal cost $c(P) = \sum_{e \in P} c_e$, its resource consumption $t(P) = \sum_{e \in P} t_e$, and its Lagrangian objective value:

$$f_P(\lambda) = c(P) + \lambda(t(P) - T) = c(P) + \lambda(t(P) - 14). \tag{6.17}$$

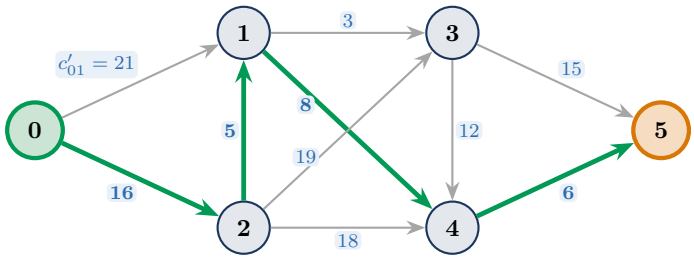
Let us analyze the behavior of $\mathcal{L}(\lambda)$ across different multiplier values:

- **At** $\lambda = 0$: The relaxation completely disregards time constraints. Dijkstra's algorithm selects path $P_0 = (0 \to 1 \to 3 \to 5)$ with unpenalized cost $\mathcal{L}(0) = 3$. However, $t(P_0) = 18 > 14$, so P_0 is severely infeasible. The lower bound 3 is valid, but very loose compared to the optimal feasible solution ($z^* = 13$, achieved by P_3).
- **At** $\lambda = 1$: Traversal time is penalized at a rate of 1 per unit. The modified cost of P_0 rises to $3 + 1 \times (18 - 14) = 7$. Path $P_1 = (0 \to 1 \to 4 \to 5)$ now wins with modified cost $5 + 1 \times (15 - 14) = 6$. The lower bound tightens from 3 to 6.

(a) Original Network: Arc Weights (c_{ij}, t_{ij}), Budget $T = 14$



(b) Lagrangian Subproblem: Modified Costs $c'_{ij} = c_{ij} + 2t_{ij}$



Shortest Path at $\lambda = 2$: $P^* = (0 \to 2 \to 1 \to 4 \to 5)$, Cost $c'(P^*) = 16 + 5 + 8 + 6 = 35$
Lagrangian Lower Bound: $\mathcal{L}(\lambda = 2) = c'(P^*) - \lambda T = 35 - 2 \times 14 = \mathbf{7}$
Original Metrics: $c(P^*) = 10 + 1 + 2 + 2 = 15$, $t(P^*) = 3 + 2 + 3 + 2 = 10 \leq 14$ (Primal Feasible!)

Figure 6.1: The Resource-Constrained Shortest Path network. Panel (a) shows the original arc parameters (c_{ij}, t_{ij}) with resource budget $T = 14$. Panel (b) shows the modified arc costs $c'_{ij} = c_{ij} + 2t_{ij}$ for multiplier $\lambda = 2$. The optimal shortest path under modified weights (highlighted in green) yields the optimal dual lower bound $\mathcal{L}(2) = 7$.

Table 6.1: All directed paths from origin 0 to destination 5 with traversal metrics and Lagrangian evaluation across multipliers $\lambda \in \{0, 1, 2, 3\}$. Feasible paths satisfying $t(P) \leq 14$ are highlighted in bold.

Path P	**Cost** $c(P)$	**Time** $t(P)$	$\lambda = 0$	$\lambda = 1$	$\lambda = 2$	$\lambda = 3$
$P_0 : 0 \to 1 \to 3 \to 5$	3	18	**3**	7	11	15
$P_1 : 0 \to 1 \to 4 \to 5$	5	15	5	**6**	**7**	8
$\mathbf{P_2 : 0 \to 1 \to 3 \to 4 \to 5}$	**14**	**14**	14	14	14	14
$\mathbf{P_3 : 0 \to 2 \to 1 \to 3 \to 5}$	**13**	**13**	13	12	11	10
$\mathbf{P_4 : 0 \to 2 \to 1 \to 4 \to 5}$	**15**	**10**	15	11	**7**	**3**
$\mathbf{P_5 : 0 \to 2 \to 1 \to 3 \to 4 \to 5}$	**24**	**9**	24	19	14	9
$P_6 : 0 \to 2 \to 3 \to 5$	16	17	16	19	22	25
$\mathbf{P_7 : 0 \to 2 \to 3 \to 4 \to 5}$	**27**	**13**	27	26	25	24
$\mathbf{P_8 : 0 \to 2 \to 4 \to 5}$	**24**	**8**	24	18	12	6
Dual Lower Bound $\mathcal{L}(\lambda)$			**3**	**6**	**7**	**3**

- **At $\lambda = 2$:** As shown in Figure 6.1(b), the modified costs on arcs change significantly: $c'_{01} = 21$, while $c'_{02} = 16$. Paths P_1 and P_4 tie for the minimum:

$$f_{P_1}(2) = 5 + 2 \times (15 - 14) = 7, \tag{6.18}$$
$$f_{P_4}(2) = 15 + 2 \times (10 - 14) = 7. \tag{6.19}$$

  The lower bound reaches $\mathcal{L}(2) = 7$. Furthermore, path P_4 consumes $t(P_4) = 10 \leq 14$ time units, providing our first primal feasible path with actual cost $c(P_4) = 15$!

- **At $\lambda = 3$:** Time consumption is penalized so aggressively that the fast, high-cost path P_4 ($t = 10$) dominates, but its dual value drops to $15+3\times(10-14) = 3$. Over-penalizing resource consumption causes the dual bound to decrease!

6.2.4 Geometry of the Lagrangian Dual

Plotting $\mathcal{L}(\lambda)$ as a function of $\lambda \geq 0$ reveals the characteristic structure of Lagrangian duality, as shown in Figure 6.2. Each path P_k corresponds to a straight line of slope $t(P_k) - T$. The Lagrangian dual function $\mathcal{L}(\lambda)$ is the lower envelope (pointwise minimum) of these lines:

$$\mathcal{L}(\lambda) = \min_{k \in \{0,\ldots,8\}} \{c(P_k) + \lambda(t(P_k) - 14)\}. \tag{6.20}$$

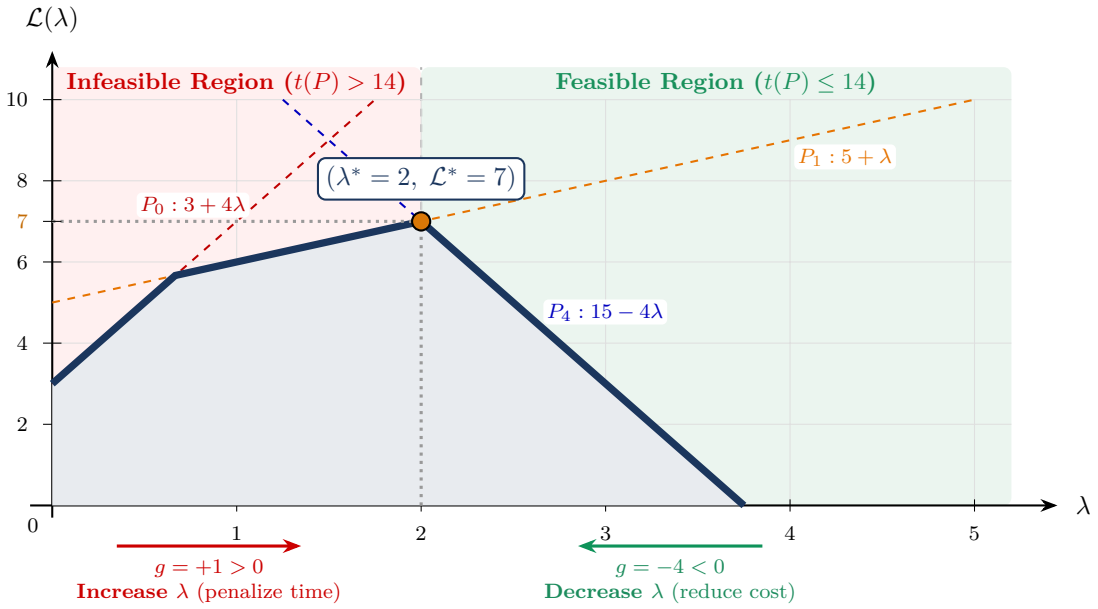


Figure 6.2: The concave piecewise-linear Lagrangian dual function $\mathcal{L}(\lambda)$ for the RCSPP example. The optimal multiplier $\lambda^* = 2$ achieves the peak lower bound $\mathcal{L}^* = 7$. For $\lambda < 2$, the optimal path violates the budget ($t(P) > T$), yielding a positive subgradient that increases λ. For $\lambda > 2$, the subgradient is negative, steering λ back toward the optimum.

The lower envelope is concave and non-smooth. Its maximum occurs at the intersection of the line for P_1 (slope $15 - 14 = +1$) and the line for P_4 (slope $10 - 14 = -4$):

$$5 + 1 \cdot (\lambda - 0) = 15 - 4\lambda \implies 5\lambda = 10 \implies \lambda^* = 2. \tag{6.21}$$

The optimal Lagrangian bound is $\mathcal{L}^* = \mathcal{L}(2) = 7$.

6.3 Solving the Dual via Subgradients

To find the optimal multiplier $\boldsymbol{\lambda}^*$ that maximizes the concave dual function $\mathcal{L}(\boldsymbol{\lambda})$, we cannot compute standard gradients because $\mathcal{L}(\boldsymbol{\lambda})$ has non-differentiable vertices ("kinks") at every point where multiple primal solutions tie for the minimum.

6.3.1 Subgradients and the Subdifferential

Non-smooth convex and concave optimization relies on the concept of a *subgradient* (see Shor [Sho85]).

Definition 6.2 *Subgradient of a Concave Function*

Let $f\colon \mathbb{R}^m \to \mathbb{R}$ be a concave function. A vector $\mathbf{g} \in \mathbb{R}^m$ is called a *subgradient* of f at point $\boldsymbol{\lambda}$ if for all $\boldsymbol{\lambda}' \in \mathbb{R}^m$:

$$f(\boldsymbol{\lambda}') \leq f(\boldsymbol{\lambda}) + \mathbf{g}^\top(\boldsymbol{\lambda}' - \boldsymbol{\lambda}). \tag{6.22}$$

The set of all subgradients of f at $\boldsymbol{\lambda}$ is called the *subdifferential*, denoted by $\partial f(\boldsymbol{\lambda})$.

Condition (6.22) states that the affine hyper-plane $\boldsymbol{\lambda}' \mapsto f(\boldsymbol{\lambda}) + \mathbf{g}^\top(\boldsymbol{\lambda}' - \boldsymbol{\lambda})$ lies everywhere above (or tangent to) the concave function f. A fundamental theorem of non-smooth optimization states that $\boldsymbol{\lambda}^*$ is a global maximizer of a concave function f if and only if $\mathbf{0} \in \partial f(\boldsymbol{\lambda}^*)$.

Theorem 6.3 *Subgradient of the Lagrangian Dual Function*

Let $\boldsymbol{\lambda} \geq \mathbf{0}$, and let $\mathbf{x}(\boldsymbol{\lambda}) \in X$ be any optimal primal solution to the Lagrangian subproblem (6.4), so that $\mathcal{L}(\boldsymbol{\lambda}) = \mathbf{c}^\top\mathbf{x}(\boldsymbol{\lambda}) + \boldsymbol{\lambda}^\top(\mathbf{A}\mathbf{x}(\boldsymbol{\lambda}) - \mathbf{b})$. Then the constraint residual vector:

$$\mathbf{g}(\boldsymbol{\lambda}) = \mathbf{A}\mathbf{x}(\boldsymbol{\lambda}) - \mathbf{b} \tag{6.23}$$

is a valid subgradient of $\mathcal{L}$ at $\boldsymbol{\lambda}$, that is, $\mathbf{g}(\boldsymbol{\lambda}) \in \partial\mathcal{L}(\boldsymbol{\lambda})$.

Proof. Let $\boldsymbol{\lambda}' \geq \mathbf{0}$ be any arbitrary multiplier vector. By definition of the dual function:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\lambda}') &= \min_{\mathbf{x}\in X}\left\{\mathbf{c}^\top\mathbf{x} + (\boldsymbol{\lambda}')^\top(\mathbf{A}\mathbf{x} - \mathbf{b})\right\} \\ &\leq \mathbf{c}^\top\mathbf{x}(\boldsymbol{\lambda}) + (\boldsymbol{\lambda}')^\top(\mathbf{A}\mathbf{x}(\boldsymbol{\lambda}) - \mathbf{b}),\end{aligned} \tag{6.24}$$

where the inequality holds because $\mathbf{x}(\boldsymbol{\lambda}) \in X$ is a feasible (though not necessarily optimal) candidate for the subproblem under $\boldsymbol{\lambda}'$. Subtracting $\mathcal{L}(\boldsymbol{\lambda}) = \mathbf{c}^\top\mathbf{x}(\boldsymbol{\lambda}) + \boldsymbol{\lambda}^\top(\mathbf{A}\mathbf{x}(\boldsymbol{\lambda}) - \mathbf{b})$ from both sides yields:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\lambda}') - \mathcal{L}(\boldsymbol{\lambda}) &\leq (\boldsymbol{\lambda}')^\top(\mathbf{A}\mathbf{x}(\boldsymbol{\lambda}) - \mathbf{b}) - \boldsymbol{\lambda}^\top(\mathbf{A}\mathbf{x}(\boldsymbol{\lambda}) - \mathbf{b}) \\ &= (\boldsymbol{\lambda}' - \boldsymbol{\lambda})^\top(\mathbf{A}\mathbf{x}(\boldsymbol{\lambda}) - \mathbf{b}) = \mathbf{g}(\boldsymbol{\lambda})^\top(\boldsymbol{\lambda}' - \boldsymbol{\lambda}).\end{aligned} \tag{6.25}$$

Rearranging gives $\mathcal{L}(\boldsymbol{\lambda}') \leq \mathcal{L}(\boldsymbol{\lambda}) + \mathbf{g}(\boldsymbol{\lambda})^\top(\boldsymbol{\lambda}' - \boldsymbol{\lambda})$, confirming that $\mathbf{g}(\boldsymbol{\lambda}) \in \partial\mathcal{L}(\boldsymbol{\lambda})$. □

Theorem 6.3 delivers an extraordinarily practical result: *we obtain a valid subgradient for free whenever we solve the Lagrangian subproblem!* For the RCSPP, if P_k is the shortest path computed under multiplier λ_k, the subgradient is simply the scalar constraint violation:

$$g_k = t(P_k) - T. \tag{6.26}$$

- If $t(P_k) > T$, the path exceeds the budget, so $g_k > 0$. The subgradient indicates that λ should be *increased* to penalize time more heavily.
- If $t(P_k) < T$, the path consumes less than the allotted budget, so $g_k < 0$. The subgradient indicates that λ should be *decreased* to seek lower monetary costs.

6.3.2 The Subgradient Iteration Algorithm

The subgradient optimization algorithm iteratively updates the multiplier vector in the direction of the subgradient, projecting the result back onto the non-negative orthant $\mathbb{R}^m_+$:

$$\boldsymbol{\lambda}^{(k+1)} = \max\left\{\mathbf{0},\ \boldsymbol{\lambda}^{(k)} + \mu_k \mathbf{g}^{(k)}\right\}, \tag{6.27}$$

where $\mu_k > 0$ is a scalar *step size* at iteration k.

Unlike smooth gradient ascent, the sequence of dual values $\mathcal{L}(\boldsymbol{\lambda}^{(k)})$ is *not guaranteed to increase monotonically* at every iteration (see Figure 6.2). However, if the step size sequence satisfies the classical convergence conditions:

$$\lim_{k\to\infty} \mu_k = 0 \quad \text{and} \quad \sum_{k=0}^{\infty} \mu_k = \infty, \tag{6.28}$$

then the sequence $\boldsymbol{\lambda}^{(k)}$ is guaranteed to converge to an optimal multiplier $\boldsymbol{\lambda}^*$ (Shor [Sho85]). A common step size rule meeting these criteria is the harmonic series $\mu_k = \mu_0/(k+1)$.

Algorithm 5 outlines the complete subgradient optimization framework for RCSPP. In addition to computing the dual bound $\underline{L}$, the algorithm tracks the best primal feasible path $\bar{P}$ discovered during execution, maintaining an optimality gap $(\mathbf{c}(\bar{P}) - \underline{L})/\mathbf{c}(\bar{P})$.

Binary Search for Single Complicating Constraints. When an optimization problem has only a *single* scalar multiplier $\lambda \in \mathbb{R}_+$ (as in RCSPP), $\mathcal{L}(\lambda)$ is a one-dimensional concave function. In that special case, one can alternatively apply binary search, golden section search, or ternary search on the interval $[0, \lambda_{\max}]$ to find λ^* rapidly without tuning step sizes. However, subgradient optimization remains indispensable for combinatorial problems involving hundreds or thousands of multipliers, such as the Traveling Salesperson Problem.

6.4 Lagrangian Relaxation for the TSP

We now apply Lagrangian relaxation to the Traveling Salesperson Problem on an undirected complete graph $G = (V, E)$ with edge costs $w_e \geq 0$ and $|V| = n$, where vertices are indexed from 0 to $n - 1$.

6.4.1 Structural Constraints of the TSP

In Chapter 3, we recognized that any valid Hamiltonian cycle corresponds to an edge subset $T \subseteq E$ of cardinality $|T| = n$ that simultaneously satisfies two distinct structural conditions:

1. **Subtour-Free Connectivity**: The selected edges must connect all vertices into a single cycle without isolated sub-circuits. As established in Theorem 3.8, this is equivalent to requiring the edge set to form a *1-tree* (a spanning tree on $V \setminus \{0\}$ combined with two distinct edges incident to vertex 0).
2. **Degree Regularity**: Every vertex $i \in V$ must have degree exactly 2:

$$\deg(i) = \sum_{e \in \delta(i)} x_e = 2 \quad \forall i \in V. \tag{6.29}$$

Algorithm 5: Subgradient Optimization for the RCSPP

Input: Directed graph $G = (V, A)$, costs c_{ij}, traversal times t_{ij}, budget T, tolerance ϵ, max iterations K

Output: Best lower bound $\underline{L}$, best feasible path $\bar{P}$

$\underline{L} \leftarrow -\infty$
$\bar{P} \leftarrow$ null, $\quad \bar{z} \leftarrow +\infty$
$\lambda_0 \leftarrow 0, \quad k \leftarrow 0$
while $k < K$ ***and*** *step size* $\mu_k > \epsilon$ **do**
  Define modified costs $c'_{ij} = c_{ij} + \lambda_k t_{ij}$ for all $(i, j) \in A$
  $P_k \leftarrow \text{ShortestPath}(G, c', 0, n-1)$ `// Dijkstra's algorithm`
  $\mathcal{L}_k \leftarrow c(P_k) + \lambda_k (t(P_k) - T)$
  if $\mathcal{L}_k > \underline{L}$ **then**
    $\underline{L} \leftarrow \mathcal{L}_k$ `// Update best dual lower bound`
  if $t(P_k) \leq T$ ***and*** $c(P_k) < \bar{z}$ **then**
    $\bar{P} \leftarrow P_k, \quad \bar{z} \leftarrow c(P_k)$ `// Update best primal feasible solution`
  if $\bar{z} - \underline{L} \leq \epsilon$ **then**
    break `// Certified optimality within tolerance`
  $g_k \leftarrow t(P_k) - T$ `// Compute subgradient`
  $\mu_k \leftarrow 1.0/(k+1)$ `// Divergent step size`
  $\lambda_{k+1} \leftarrow \max(0,\ \lambda_k + \mu_k g_k)$
  $k \leftarrow k + 1$
return $\underline{L}, \bar{P}$

The minimum 1-tree problem satisfies the connectivity requirement and can be solved in polynomial time $\mathcal{O}(|E| \log |V|)$ using Kruskal's or Prim's algorithm (Appendix A). However, an unconstrained minimum 1-tree frequently violates the degree regularity constraints: some vertices may end up as leaves ($\deg(i) = 1$), while others become hubs with degree 3, 4, or higher.

6.4.2 Dualizing Degree Constraints

To eliminate these degree violations without sacrificing the polynomial tractability of 1-tree computations, Michael Held and Richard Karp [HK70; HK71] proposed dualizing the n degree constraints (6.29).

Let $\boldsymbol{\pi} = (\pi_0, \pi_1, \ldots, \pi_{n-1}) \in \mathbb{R}^n$ be a vector of Lagrange multipliers, where each π_i is associated with the degree constraint of vertex $i \in V$. Because degree constraints are *equalities* ($\deg(i) = 2$), the multipliers π_i are *unconstrained in sign*: $\pi_i \in \mathbb{R}$.

Formally, the primal TSP is stated as:

$$z^* = \min_{\mathbf{x}} \quad \sum_{e \in E} w_e x_e \tag{6.30}$$

$$\text{subject to} \quad \sum_{e \in \delta(i)} x_e = 2 \quad \forall i \in V, \tag{6.31}$$

$$\mathbf{x} \text{ is the incidence vector of a 1-tree.} \tag{6.32}$$

Dualizing the degree equations (6.31) yields the Lagrangian dual function:

$$\mathcal{L}(\boldsymbol{\pi}) = \min_{\mathbf{x}\in 1\text{-Tree}} \left\{ \sum_{e\in E} w_e x_e + \sum_{i\in V} \pi_i \left(2 - \sum_{e\in\delta(i)} x_e \right) \right\}. \tag{6.33}$$

We now rewrite the dualized penalty term. Since each edge $e = \{i, j\}$ is incident to exactly two endpoints i and j, summing over all vertices distributes the multiplier contributions across edges:

$$\sum_{i\in V} \pi_i \sum_{e\in\delta(i)} x_e = \sum_{\{i,j\}\in E} (\pi_i + \pi_j) x_{ij}. \tag{6.34}$$

Substituting this identity into (6.33) produces:

$$\begin{aligned} \mathcal{L}(\boldsymbol{\pi}) &= \min_{\mathbf{x}\in 1\text{-Tree}} \left\{ \sum_{\{i,j\}\in E} w_{ij} x_{ij} - \sum_{\{i,j\}\in E} (\pi_i + \pi_j) x_{ij} + 2 \sum_{i\in V} \pi_i \right\} \\ &= \min_{\mathbf{x}\in 1\text{-Tree}} \left\{ \sum_{\{i,j\}\in E} (w_{ij} - \pi_i - \pi_j) x_{ij} \right\} + 2 \sum_{i\in V} \pi_i. \end{aligned} \tag{6.35}$$

6.4.3 Modified Edge Costs and Multipliers

Formula (6.35) reveals the brilliance of the Held–Karp relaxation. For any fixed multiplier vector $\boldsymbol{\pi}$, the Lagrangian subproblem is simply a *Minimum 1-Tree Problem* evaluated with *modified edge weights*:

$$w'_{ij} = w_{ij} - \pi_i - \pi_j \quad \forall \{i, j\} \in E, \tag{6.36}$$

to which the constant offset $2\sum_{i\in V} \pi_i$ is added.

The mechanical intuition is immediately apparent:

- If vertex i has too high a degree ($\deg(i) > 2$) in the current 1-tree, it is an over-utilized hub. To discourage edges incident to i, we decrease π_i. Because $w'_{ij} = w_{ij} - \pi_i - \pi_j$, decreasing π_i *increases* the cost of all edges incident to i, making them less attractive in subsequent MST selections!
- Conversely, if vertex i has too low a degree ($\deg(i) = 1$), it is an under-utilized leaf. Increasing π_i *decreases* the modified cost of incident edges, making them cheaper and encouraging the 1-tree to incorporate them.

6.4.4 Multiplier Invariance and Optimality

A remarkable mathematical property of the modified weights (6.36) is that they preserve the exact relative costs of all valid Hamiltonian tours.

Theorem 6.4 *Tour Length Invariance under Held–Karp Multipliers*

Let $H \subset E$ be any Hamiltonian cycle in G. For any multiplier vector $\boldsymbol{\pi} \in \mathbb{R}^n$, the total modified weight of H plus the dual offset equals its exact original weight:

$$\sum_{\{i,j\}\in H} w'_{ij} + 2 \sum_{i\in V} \pi_i = \sum_{\{i,j\}\in H} w_{ij}. \tag{6.37}$$

Furthermore, if the multipliers satisfy $\sum_{i \in V} \pi_i = 0$, then:

$$\sum_{\{i,j\} \in H} w'_{ij} = \sum_{\{i,j\} \in H} w_{ij}. \tag{6.38}$$

Proof. Because H is a Hamiltonian cycle, every vertex $i \in V$ has degree exactly 2 ($\deg_H(i) = 2$). Summing the modified edge weights over H:

$$\begin{aligned} \sum_{\{i,j\} \in H} w'_{ij} &= \sum_{\{i,j\} \in H} (w_{ij} - \pi_i - \pi_j) \\ &= \sum_{\{i,j\} \in H} w_{ij} - \sum_{i \in V} \deg_H(i) \pi_i \\ &= \sum_{\{i,j\} \in H} w_{ij} - 2 \sum_{i \in V} \pi_i. \end{aligned} \tag{6.39}$$

Adding $2 \sum_{i \in V} \pi_i$ to both sides cancels the multiplier terms completely, establishing the theorem. When $\sum_{i \in V} \pi_i = 0$, the offset is zero, so the modified tour cost equals the original tour cost identically. □

Theorem 6.5 *Held–Karp Optimality Certificate*

Suppose that for some multiplier vector $\boldsymbol{\pi} \in \mathbb{R}^n$, the minimum 1-tree T^* computed under modified weights w' has degree $\deg_{T^*}(i) = 2$ for all $i \in V$. Then T^* is an optimal TSP tour, and $\mathcal{L}(\boldsymbol{\pi}) = z^*$.

Proof. If $\deg_{T^*}(i) = 2$ for all $i \in V$, then T^* is a connected 2-regular graph on n vertices, which is by definition a Hamiltonian cycle. By Theorem 6.4, the cost of T^* in the original graph is:

$$w(T^*) = \sum_{e \in T^*} w'_e + 2 \sum_{i \in V} \pi_i = \mathcal{L}(\boldsymbol{\pi}). \tag{6.40}$$

Since T^* is a valid TSP tour, its cost satisfies $w(T^*) \geq z^*$. On the other hand, by weak duality (Theorem 6.1), $\mathcal{L}(\boldsymbol{\pi}) \leq z^*$. Combining these inequalities yields:

$$z^* \leq w(T^*) = \mathcal{L}(\boldsymbol{\pi}) \leq z^* \implies \mathcal{L}(\boldsymbol{\pi}) = z^* = w(T^*). \tag{6.41}$$

Thus, T^* is provably optimal. □

6.4.5 A Worked Example on 4 Vertices

Figure 6.3 visualizes how Lagrangian multipliers tighten the 1-tree bound on a complete 4-vertex graph K_4 with $V = \{0, 1, 2, 3\}$. The original edge weights are:

$$w_{01} = 7, \quad w_{02} = 4, \quad w_{03} = 9, \quad w_{12} = 5, \quad w_{13} = 8, \quad w_{23} = 6.$$

Example 6.1 *Tightening the 1-Tree Bound via Multipliers*

We trace two iterations of the Held–Karp bounding procedure:

1. **Iteration 0 ($\boldsymbol{\pi} = \mathbf{0}$)**:

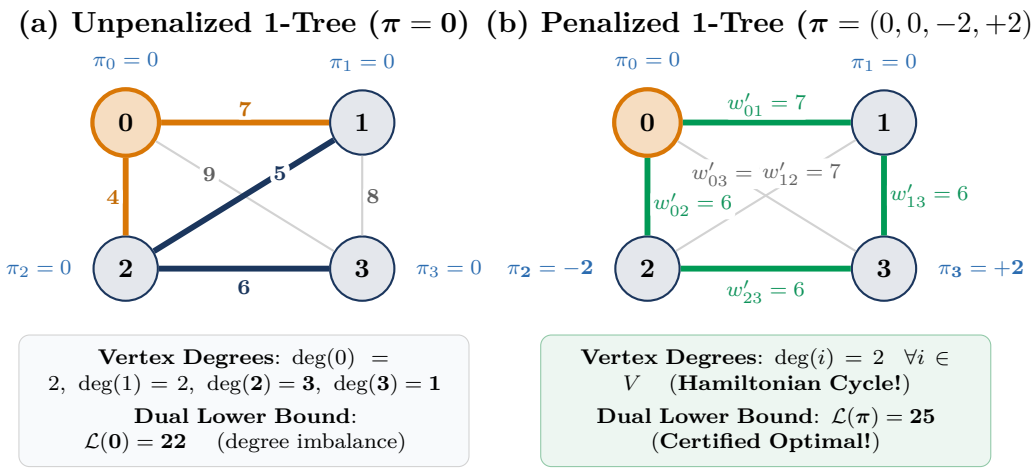


Figure 6.3: Lagrangian relaxation and multiplier transformation for a 4-vertex TSP. Panel (a): unpenalized 1-tree ($\boldsymbol{\pi} = \mathbf{0}$) with degree imbalance at vertices 2 and 3 ($\mathcal{L} = 22$). Panel (b): after introducing multipliers $\boldsymbol{\pi} = (0, 0, -2, +2)$, the minimum 1-tree under modified weights $w'_{ij} = w_{ij} - \pi_i - \pi_j$ becomes 2-regular, yielding a certified optimal Hamiltonian cycle of cost $\mathcal{L} = 25$.

- Two cheapest edges incident to root vertex 0: $\{0, 2\}$ ($w = 4$) and $\{0, 1\}$ ($w = 7$).
- Minimum Spanning Tree on $V \setminus \{0\} = \{1, 2, 3\}$: edges $\{1, 2\}$ ($w = 5$) and $\{2, 3\}$ ($w = 6$) are chosen (edge $\{1, 3\}$ with $w = 8$ is omitted).
- The unpenalized 1-tree contains edges $\{\{0, 2\}, \{0, 1\}, \{1, 2\}, \{2, 3\}\}$ with cost:

$$\mathcal{L}(\mathbf{0}) = 4 + 7 + 5 + 6 = 22.$$

- Vertex degrees in this 1-tree: $\deg(0) = 2$, $\deg(1) = 2$, $\deg(2) = 3$, $\deg(3) = 1$.
- Degree violations: vertex 2 has an excess of edges ($\deg(2) - 2 = +1$), while vertex 3 has a deficit ($\deg(3) - 2 = -1$).

2. **Iteration 1 ($\boldsymbol{\pi} = (0, 0, -2, +2)$)**: Applying a step size $\mu = 2$ in the subgradient direction $g_i = 2 - \deg(i)$ yields:

$$\pi_0 = 0, \quad \pi_1 = 0, \quad \pi_2 = 0 + 2(2 - 3) = -2, \quad \pi_3 = 0 + 2(2 - 1) = +2.$$

Notice that $\sum_{i=0}^{3} \pi_i = 0 + 0 - 2 + 2 = 0$. The modified edge weights $w'_{ij} = w_{ij} - \pi_i - \pi_j$ become:

$$\begin{aligned} w'_{01} &= 7 - 0 - 0 = 7, & w'_{02} &= 4 - 0 - (-2) = 6, \\ w'_{03} &= 9 - 0 - 2 = 7, & w'_{12} &= 5 - 0 - (-2) = 7, \\ w'_{13} &= 8 - 0 - 2 = 6, & w'_{23} &= 6 - (-2) - 2 = 6. \end{aligned}$$

We recompute the minimum 1-tree under w':

- Two cheapest edges from vertex 0: $\{0, 2\}$ ($w' = 6$) and $\{0, 1\}$ ($w' = 7$) [or $\{0, 3\}$ ($w' = 7$)].
- MST on $\{1, 2, 3\}$: edges $\{1, 3\}$ ($w' = 6$) and $\{2, 3\}$ ($w' = 6$).

- Total modified cost: $6 + 7 + 6 + 6 + 2\sum_i \pi_i = 25 + 2(0) = 25$.

The lower bound jumped from 22 to 25! Furthermore, the resulting 1-tree has edges $\{\{0,1\},\{1,3\},\{2,3\},\{0,2\}\}$, which forms a simple cycle of length 4: $0-1-3-2-0$. By Theorem 6.5, this tour is certified optimal with original length $7+8+6+4=25$.

6.5 Subgradient Optimization for Held–Karp

To maximize $\mathcal{L}(\boldsymbol{\pi})$ across all multiplier vectors $\boldsymbol{\pi} \in \mathbb{R}^n$, we deploy subgradient optimization.

6.5.1 Subgradients and Zero-Sum Property

By Theorem 6.3, for any current multiplier vector $\boldsymbol{\pi}$, the subgradient $\mathbf{g} \in \mathbb{R}^n$ of $\mathcal{L}(\boldsymbol{\pi})$ is the degree residual:

$$g_i = 2 - \deg_T(i) \quad \forall i \in V, \tag{6.42}$$

where $\deg_T(i)$ is the degree of vertex i in the minimum 1-tree T computed under modified weights $w'_{ij} = w_{ij} - \pi_i - \pi_j$.

Lemma 6.6 *Conservation of Multiplier Zero-Sum*

In any 1-tree T on n vertices, the sum of all vertex degrees is identically equal to $2n$. Consequently:

$$\sum_{i \in V} g_i = \sum_{i \in V} (2 - \deg_T(i)) = 2n - \sum_{i \in V} \deg_T(i) = 2n - 2n = 0. \tag{6.43}$$

Therefore, if the initial multiplier vector satisfies $\sum_{i \in V} \pi_i^{(0)} = 0$, then under the subgradient update rule:

$$\boldsymbol{\pi}^{(k+1)} = \boldsymbol{\pi}^{(k)} + \mu_k \mathbf{g}^{(k)}, \tag{6.44}$$

the zero-sum property is strictly preserved for all iterations $k \geq 0$:

$$\sum_{i \in V} \pi_i^{(k+1)} = \sum_{i \in V} \pi_i^{(k)} + \mu_k \sum_{i \in V} g_i^{(k)} = 0 + \mu_k(0) = 0. \tag{6.45}$$

Theorem 6.6 is remarkable: it guarantees that by simply initializing $\boldsymbol{\pi}^{(0)} = \mathbf{0}$, the multipliers remain zero-sum throughout all iterations automatically. No projection or normalization step is ever required!

6.5.2 Adaptive Step Size Control

In practice, subgradient convergence depends critically on the step size μ_k. While the theoretical harmonic step $\mu_k = \mu_0/(k+1)$ guarantees asymptotic convergence, its practical progress is often slow. For the Held–Karp bound, Polyak's adaptive step size rule [Pol87], as refined by Held and Karp [HK71] and Valenzuela and Jones [VJ97], provides vastly superior numerical performance:

$$\mu_k = \frac{\alpha_k(\mathrm{UB} - \mathcal{L}(\boldsymbol{\pi}^{(k)}))}{\sum_{i \in V}(2 - \deg_T(i))^2}, \tag{6.46}$$

where:

- UB is a known primal upper bound (the length of any valid TSP tour, obtained quickly via a heuristic such as nearest neighbor or 2-opt; see Chapter 8). If no upper bound is available, one can replace $(\mathrm{UB} - \mathcal{L}_k)$ by $\mathcal{L}_k$ as in Valenzuela and Jones [VJ97].
- $\sum_{i \in V} (2 - \deg_T(i))^2 = \|\mathbf{g}^{(k)}\|_2^2$ is the squared Euclidean norm of the subgradient vector.
- $\alpha_k \in (0, 2]$ is a scalar parameter initialized to $\alpha_0 = 2.0$ (or 1.0). If the lower bound fails to strictly improve over a predetermined window of iterations (e.g., 5 or 10 iterations), α is decayed: $\alpha_{k+1} \leftarrow 0.9 \cdot \alpha_k$ or $\alpha_{k+1} \leftarrow 0.5 \cdot \alpha_k$.

Figure 6.4 illustrates the convergence trajectory of subgradient optimization. Notice how the instantaneous bound $\mathcal{L}_k = \mathcal{L}(\boldsymbol{\pi}^{(k)})$ oscillates non-monotonically, while the best lower bound $\underline{\mathcal{L}}_k = \max_{j \le k} \mathcal{L}_j$ monotonically ascends in discrete jumps toward the optimal Held–Karp bound $\mathcal{L}^*$.

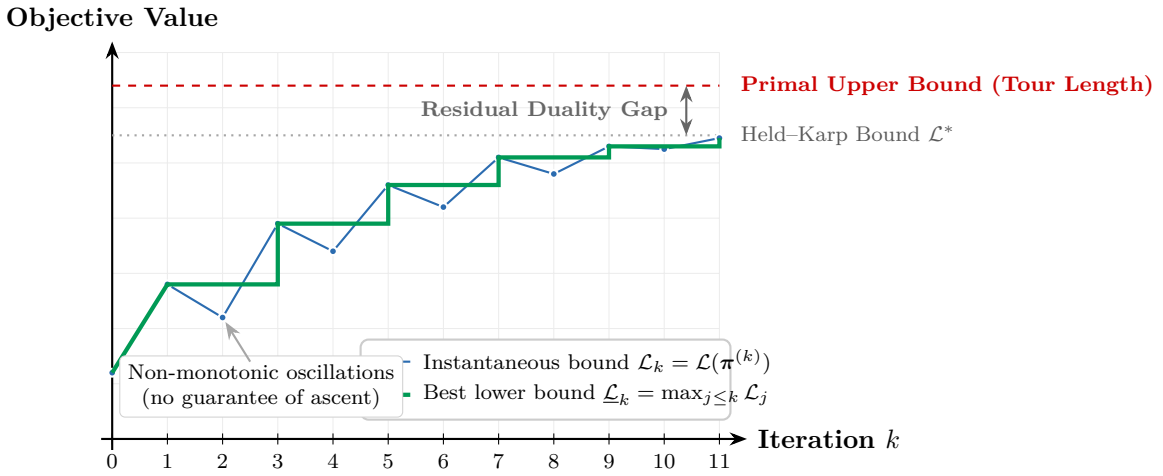


Figure 6.4: Subgradient optimization convergence for the Held–Karp TSP lower bound. The instantaneous dual value $\mathcal{L}_k$ oscillates non-monotonically due to slope discontinuities at vertices of the 1-tree polyhedron. The running best bound $\underline{\mathcal{L}}_k$ ascends monotonically toward $\mathcal{L}^*$, closing the gap against the primal upper bound.

6.5.3 The Held–Karp Subgradient Procedure

The subgradient optimization framework for the Held–Karp 1-tree relaxation proceeds iteratively. Starting from the unpenalized baseline ($\boldsymbol{\pi}^{(0)} = \mathbf{0}$), it repeatedly evaluates the minimum 1-tree under penalized costs and steers the multipliers along the subgradient direction.

The complete algorithmic procedure unfolds in the following systematic steps:

1. **Initialization**:
   - Set initial multiplier vector $\boldsymbol{\pi}^{(0)} = (0, 0, \ldots, 0) \in \mathbb{R}^n$.
   - Initialize the best recorded lower bound $\underline{\mathcal{L}} \leftarrow -\infty$.
   - Set an initial step-size scalar $\alpha_0 \in (0, 2]$ (typically $\alpha_0 = 2.0$), a decay factor $\gamma \in (0, 1)$ (e.g., $\gamma = 0.7$ or 0.5), and a non-improvement window parameter p (e.g., $p = 5$ or 10).
   - Compute or retrieve a heuristic upper bound UB (such as via nearest-neighbor followed by 2-opt local search; see Chapter 8).

2. **Edge Weight Transformation**: At iteration $k \geq 0$, modify the edge costs using the current multiplier vector $\boldsymbol{\pi}^{(k)}$:

$$w'_{ij} = w_{ij} - \pi_i^{(k)} - \pi_j^{(k)} \quad \forall \{i, j\} \in E. \tag{6.47}$$

3. **Minimum 1-Tree Subproblem Resolution**:
   - Select designated root vertex $r \in V$ (conventionally $r = 0$).
   - Compute a Minimum Spanning Tree T_{MST} on the induced subgraph $G[V \setminus \{0\}]$ under modified weights w' using Kruskal's algorithm with a disjoint-set (Union-Find) data structure (as detailed in Appendix A) or Prim's algorithm in $\mathcal{O}(n^2)$ time.
   - Identify the two distinct edges $e_1 = \{0, u\}$ and $e_2 = \{0, v\}$ $(u \neq v)$ incident to root vertex 0 that minimize w'.
   - Form the minimum 1-tree $T^{(k)} = T_{\text{MST}} \cup \{e_1, e_2\}$.
4. **Dual Lower Bound Evaluation**: Evaluate the Lagrangian dual function:

$$\mathcal{L}(\boldsymbol{\pi}^{(k)}) = \sum_{e \in T^{(k)}} w'_e + 2 \sum_{i \in V} \pi_i^{(k)}. \tag{6.48}$$

   Update the running best lower bound:

$$\underline{\mathcal{L}} \leftarrow \max\left(\underline{\mathcal{L}},\ \mathcal{L}(\boldsymbol{\pi}^{(k)})\right). \tag{6.49}$$

5. **Optimality Test**: Compute the degree $\deg_{T^{(k)}}(i)$ of every vertex $i \in V$ in $T^{(k)}$.
   - If $\deg_{T^{(k)}}(i) = 2$ for all $i \in V$, the 1-tree $T^{(k)}$ is connected and 2-regular, hence a valid Hamiltonian cycle! By Theorem 6.5, $T^{(k)}$ is an optimal TSP tour, and the algorithm halts immediately with an optimality certificate ($\mathcal{L}(\boldsymbol{\pi}^{(k)}) = z^*$).
6. **Subgradient Computation and Multiplier Update**: Evaluate the subgradient vector $\mathbf{g}^{(k)} \in \mathbb{Z}^n$:

$$g_i^{(k)} = 2 - \deg_{T^{(k)}}(i) \quad \forall i \in V. \tag{6.50}$$

   If $\|\mathbf{g}^{(k)}\|_2 = 0$, all degrees are satisfied; terminate. Otherwise, compute the Polyak step size:

$$\mu_k = \frac{\alpha_k \left(\text{UB} - \mathcal{L}(\boldsymbol{\pi}^{(k)})\right)}{\sum_{i \in V} \left(2 - \deg_{T^{(k)}}(i)\right)^2}. \tag{6.51}$$

   Update the multipliers:

$$\pi_i^{(k+1)} = \pi_i^{(k)} + \mu_k \left(2 - \deg_{T^{(k)}}(i)\right) \quad \forall i \in V. \tag{6.52}$$

7. **Zero-Sum Multiplier Centering**: While $\sum_{i \in V} g_i^{(k)} = 0$ theoretically guarantees that zero-sum multipliers remain zero-sum (Theorem 6.6), floating-point rounding errors can induce numerical drift over hundreds of iterations. In practice, one recenters the multipliers after each update:

$$\bar{\pi} = \frac{1}{n} \sum_{i \in V} \pi_i^{(k+1)}, \qquad \pi_i^{(k+1)} \leftarrow \pi_i^{(k+1)} - \bar{\pi} \quad \forall i \in V. \tag{6.53}$$

8. **Step Size Decay and Termination**: If the best lower bound $\underline{\mathcal{L}}$ fails to strictly improve over p consecutive iterations, reduce the step parameter:

$$\alpha_{k+1} \leftarrow \gamma \cdot \alpha_k. \tag{6.54}$$

Terminate when the maximum iteration limit $K_{\max}$ is reached, when $\alpha_k < \epsilon$, or when $\lceil \underline{\mathcal{L}} \rceil = \mathrm{UB}$ for integer edge weights.

Computational Complexity of the Subgradient Step. Each iteration requires:

- Computing modified edge weights: $\mathcal{O}(n^2)$ arithmetic operations.
- Minimum Spanning Tree on $n-1$ vertices: $\mathcal{O}(n^2)$ using Prim's algorithm or $\mathcal{O}(n^2 \log n)$ via Kruskal's algorithm on complete graphs.
- Finding the two cheapest incident edges to vertex 0: $\mathcal{O}(n)$ comparisons.
- Degree evaluation and subgradient update: $\mathcal{O}(n)$ operations.

Consequently, each subgradient iteration runs in $\mathcal{O}(n^2)$ time. In practice, 50 to 200 iterations suffice to capture over 99% of the asymptotic Held–Karp bound, executing in mere milliseconds even for graphs with hundreds of vertices.

6.6 Polyhedral Strength and Held–Karp

Why is the Held–Karp bound so celebrated in combinatorial optimization? To understand its power, we must examine the classical integer programming formulation of the TSP and its polyhedral relaxation.

6.6.1 The DFJ Formulation of the TSP

In their landmark 1954 paper, George Dantzig, Ray Fulkerson, and Selmer Johnson [DFJ54] introduced the first integer linear programming formulation of the TSP. Consider an undirected complete graph $G = (V, E)$ with edge weights $w_e \geq 0$. For each edge $e \in E$, let $x_e \in \{0, 1\}$ be a binary decision variable indicating whether edge e is traversed in the tour:

$$x_e = \begin{cases} 1 & \text{if edge } e \text{ belongs to the tour,} \\ 0 & \text{otherwise.} \end{cases} \tag{6.55}$$

Degree Constraints (2-Factor)

Because every vertex in a valid Hamiltonian cycle must have exactly one incoming edge and one outgoing edge, the degree of every vertex in the tour subgraph must equal 2:

$$\sum_{e \in \delta(i)} x_e = 2 \quad \forall i \in V, \tag{6.56}$$

where $\delta(i) = \{e \in E \mid i \in e\}$ denotes the set of edges incident to vertex i.

However, the degree constraints (6.56) alone are *insufficient* to define a Hamiltonian cycle. Any collection of mutually disjoint cycles that collectively visit every vertex in V (known as a *2-factor* or *subtour*) satisfies (6.56). For instance, in a 6-vertex graph with $V = \{0, 1, 2, 3, 4, 5\}$, two disconnected triangles $(0, 1, 2)$ and $(3, 4, 5)$ satisfy $\deg(i) = 2$ for all $i \in V$, yet fail to form a connected Hamiltonian tour.

Subtour Elimination Constraints (SECs)

To eliminate disconnected subtours, Dantzig, Fulkerson, and Johnson introduced constraints requiring that every proper, non-empty vertex subset $S \subset V$ cannot contain an isolated cycle.

For any vertex subset $S \subset V$, let $E(S) = \{\{u, v\} \in E \mid u, v \in S\}$ denote the set of edges whose both endpoints lie entirely within S. If the tour were to form an isolated cycle on the vertices of S, it would select exactly $|S|$ edges from $E(S)$. To forbid such a cycle, the number of selected edges within S cannot exceed $|S| - 1$:

$$\sum_{e \in E(S)} x_e \leq |S| - 1 \quad \forall S \subset V,\ 2 \leq |S| \leq n - 2. \tag{6.57}$$

Equivalently, let $\delta(S) = \{\{u, v\} \in E \mid u \in S,\ v \in V \setminus S\}$ denote the cut separating S from the rest of the graph. Any connected Hamiltonian cycle that enters S must eventually exit S, meaning it must cross the cut $\delta(S)$ at least twice:

$$\sum_{e \in \delta(S)} x_e \geq 2 \quad \forall S \subset V,\ 2 \leq |S| \leq n - 2. \tag{6.58}$$

Proposition 6.7 *Equivalence of SEC Formulations*

Under the degree constraints $\sum_{e \in \delta(i)} x_e = 2$ for all $i \in V$, the subset edge-capacity constraint (6.57) and the cut constraint (6.58) are mathematically equivalent for every vertex subset $S \subset V$.

Proof. Summing the degree equations over all vertices $i \in S$:

$$\sum_{i \in S} \sum_{e \in \delta(i)} x_e = 2|S|. \tag{6.59}$$

In this summation, every internal edge $e \in E(S)$ has both endpoints in S and is counted exactly twice. Every cut edge $e \in \delta(S)$ has exactly one endpoint in S and is counted once. Therefore:

$$2 \sum_{e \in E(S)} x_e + \sum_{e \in \delta(S)} x_e = 2|S| \iff \sum_{e \in \delta(S)} x_e = 2|S| - 2 \sum_{e \in E(S)} x_e. \tag{6.60}$$

Substituting this identity demonstrates that:

$$\sum_{e \in E(S)} x_e \leq |S| - 1 \iff 2|S| - 2 \sum_{e \in E(S)} x_e \geq 2 \iff \sum_{e \in \delta(S)} x_e \geq 2. \tag{6.61}$$

□

The complete Dantzig–Fulkerson–Johnson (DFJ) integer linear program is:

$$\min_{\boldsymbol{x}} \quad \sum_{e \in E} w_e x_e \tag{6.62}$$

$$\text{subject to} \quad \sum_{e \in \delta(i)} x_e = 2 \quad \forall i \in V, \tag{6.63}$$

$$\sum_{e \in E(S)} x_e \leq |S| - 1 \quad \forall S \subset V,\ 2 \leq |S| \leq n - 2, \tag{6.64}$$

$$x_e \in \{0, 1\} \quad \forall e \in E. \tag{6.65}$$

Notice that the number of subtour elimination constraints (6.64) is $2^{n-1} - n - 1$, which grows exponentially with n. For $n = 50$, there are more than 10^{14} constraints, precluding any static enumeration in an LP solver!

6.6.2 Subtour LP and Held–Karp Equivalence

The linear programming relaxation of the DFJ model, denoted *DFJ-LP*, is obtained by relaxing the integrality restriction $x_e \in \{0,1\}$ to $0 \le x_e \le 1$:

$$z_{\text{DFJ-LP}} = \min_{\boldsymbol{x}} \quad \sum_{e \in E} w_e x_e \tag{6.66}$$

$$\text{subject to} \quad \sum_{e \in \delta(i)} x_e = 2 \quad \forall i \in V, \tag{6.67}$$

$$\sum_{e \in E(S)} x_e \le |S| - 1 \quad \forall S \subset V,\ 2 \le |S| \le n-2, \tag{6.68}$$

$$0 \le x_e \le 1 \quad \forall e \in E. \tag{6.69}$$

In two seminal papers, Michael Held and Richard Karp [HK70; HK71] established the theoretical equivalence between the Lagrangian dual of the 1-tree relaxation and this exponential LP relaxation.

Theorem 6.8 *Held–Karp Equivalence Theorem*

The maximum value of the Lagrangian dual function $\mathcal{L}(\boldsymbol{\pi})$ over all node multiplier vectors $\boldsymbol{\pi} \in \mathbb{R}^n$ equals the optimal objective value of the linear programming relaxation of the DFJ subtour elimination formulation:

$$\max_{\boldsymbol{\pi} \in \mathbb{R}^n} \mathcal{L}(\boldsymbol{\pi}) = z_{\text{DFJ-LP}}. \tag{6.70}$$

Polyhedral Intuition and Outline. By Geoffrion's foundational theorem on Lagrangian duality [Geo74], the optimal Lagrangian dual value for an integer program with dualized linear constraints $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and base combinatorial set X is equal to the optimal value of the linear program that optimizes over the convex hull of X:

$$\max_{\boldsymbol{\lambda}} \mathcal{L}(\boldsymbol{\lambda}) = \min \left\{ \boldsymbol{c}^{\top} \boldsymbol{x} \mid \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \in \operatorname{conv}(X) \right\}. \tag{6.71}$$

In the Held–Karp relaxation, X is the set of all 1-trees on G. A 1-tree consists of a spanning tree on $V \setminus \{0\}$ combined with two edges incident to root vertex 0. By Jack Edmonds' polyhedral characterization of the spanning tree polytope [Edm71], the convex hull $\operatorname{conv}(X)$ is completely described by the non-negativity bounds, the total edge-count equality $\sum_{e \in E} x_e = n$, and the tree-subtour inequalities $\sum_{e \in E(S)} x_e \le |S| - 1$ for all $S \subseteq V \setminus \{0\}$.

When we constrain $\boldsymbol{x} \in \operatorname{conv}(X)$ by the dualized degree equations $\sum_{e \in \delta(i)} x_e = 2$ for all $i \in V$, the resulting polytope is precisely the DFJ-LP feasible region defined by (6.67)–(6.69). Hence, the dual optimum $\max_{\boldsymbol{\pi}} \mathcal{L}(\boldsymbol{\pi})$ coincides with $z_{\text{DFJ-LP}}$. □

The practical significance of Theorem 6.8 is profound:

- **Linear Programming Approach**: Computing $z_{\text{DFJ-LP}}$ directly with an LP solver requires delayed constraint generation (cutting planes): one begins with the degree equations, solves the LP, solves a minimum s-t cut problem (via maximum flow) to detect violated subtour constraints, adds them dynamically to the simplex tableau, and re-optimizes until no subtours are violated.
- **Lagrangian Relaxation Approach**: Subgradient optimization converges to the *exact same polyhedral bound* $z_{\text{DFJ-LP}}$ without any LP solver or cutting planes, relying solely on $\mathcal{O}(n^2)$ Minimum Spanning Tree computations!

6.6.3 Empirical Strength of the Bound

Beyond its polyhedral equivalence, the true justification for the prominence of the Held–Karp bound is its extraordinary tightness on practical and benchmark instances.

In their original computational investigations, Held and Karp [HK70; HK71] made the remarkable empirical observation that on instances up to 64 cities, the gap between the optimal tour length z^* and the Held–Karp bound $\mathcal{L}^*$ was frequently less than 1%, and often closed completely to 0.0% at the root node.

Subsequent extensive computational studies over thousands of instances—most notably the definitive studies by Johnson and McGeoch [JM97], Valenzuela and Jones [VJ97], and Applegate, Bixby, Chvátal, and Cook [App+06]—have confirmed this behavior:

- **Uniform Random Euclidean Instances**: For points distributed uniformly at random in the unit square, the expected optimality gap of the Held–Karp lower bound against the true optimal tour is empirically between **0.7% and 0.9%**, across instance sizes ranging from $n = 100$ to $n = 100,000$ [JM97; VJ97].
- **TSPLIB Benchmark Library**: On the standard TSPLIB library of geographic and printed circuit board instances, the Held–Karp bound is almost always within **1.0% to 1.5%** of the certified optimum [App+06].
- **Comparison with Unpenalized Bounds**: In stark contrast, the static unpenalized 1-tree ($\boldsymbol{\pi} = \mathbf{0}$) typically exhibits an optimality gap of **15% to 25%**, while the naive 2-cheapest incident edges bound typically yields gaps of **30% to 45%**.

The Cornerstone of Exact TSP Solvers. This near-zero optimality gap explains why the Held–Karp bound is the cornerstone of state-of-the-art exact TSP solvers, such as the renowned *Concorde* solver [App+06]. In a Branch and Bound or Branch and Cut framework, reducing the root gap from 20% to less than 1% reduces the size of the search tree from hundreds of millions of nodes to a manageable handful of branches, making exact optimization tractable for instances with thousands of vertices.

6.7 Chapter Summary

Key Takeaways: Lagrangian Relaxation.

- **Dualizing Complicating Constraints**: Lagrangian relaxation extracts the easy combinatorial core of an NP-hard problem (X) and dualizes the complicating constraints into the objective function via penalty multipliers ($\boldsymbol{\lambda}$ or $\boldsymbol{\pi}$).
- **Guaranteed Lower Bounds**: By weak duality, evaluating the relaxed subproblem under any valid multiplier vector yields a rigorous lower bound on the primal minimization problem.
- **Subgradient Optimization**: The dual function $\mathcal{L}$ is concave and piecewise linear. Subgradients are obtained for free from constraint violations ($\mathbf{Ax} - \mathbf{b}$). Updating multipliers along the subgradient direction drives them toward the optimal dual bound.
- **Zero-Sum Multiplier Conservation**: In the TSP, because every 1-tree has exactly n edges, the sum of degree violations $\sum_{i \in V}(2 - \deg(i)) = 0$ is conserved at every iteration. This guarantees that initial zero-sum multipliers remain zero-sum throughout the search.
- **Polyhedral Strength**: The Held–Karp bound matches the linear programming relaxation of the exponential DFJ subtour elimination formulation, yet requires only simple minimum spanning tree calculations.

Table 6.2 summarizes the structural components of Lagrangian relaxation across the two problem classes studied in this chapter.

Table 6.2: Systematic comparison of Lagrangian relaxation formulations for RCSPP and the TSP.

Component	RCSPP	Traveling Salesperson Problem
Primal Objective	Minimize path cost $\mathbf{c}^\top \mathbf{x}$	Minimize tour weight $\mathbf{w}^\top \mathbf{x}$
Easy Core Subproblem	Directed s-t paths (Dijkstra)	Minimum 1-Trees (Kruskal/Prim)
Complicating Constraints	Resource budget $\sum_e t_e x_e \leq T$	Degree regularity $\deg(i) = 2\ \forall i \in V$
Multipliers	Scalar $\lambda \geq 0$ (inequality)	Vector $\boldsymbol{\pi} \in \mathbb{R}^n$, $\sum_i \pi_i = 0$ (equality)
Modified Edge Weights	$c'_{ij} = c_{ij} + \lambda t_{ij}$	$w'_{ij} = w_{ij} - \pi_i - \pi_j$
Subgradient Vector	$g = \sum_e t_e x_e - T$	$g_i = 2 - \deg_T(i) \quad \forall i \in V$
Polyhedral Bound	Equals LP relaxation of RCSPP	Equals DFJ Subtour Elimination LP

6.8 Exercises

Exercise 6.1 *Lagrangian Relaxation of the 0–1 Knapsack Problem*

Consider the 0–1 Knapsack Problem:

$$\max_{\mathbf{x}} \quad \sum_{i=0}^{n-1} v_i x_i \quad \text{subject to} \quad \sum_{i=0}^{n-1} w_i x_i \leq W, \quad x_i \in \{0,1\}\ \forall i \in \{0,\dots,n-1\},$$

where $v_i > 0$ and $w_i > 0$.

1. Formulate the Lagrangian relaxation obtained by dualizing the capacity constraint with multiplier $\lambda \geq 0$.
2. Show that for a fixed λ, the objective decomposes across items into n independent single-variable subproblems:

$$\mathcal{L}(\lambda) = \lambda W + \sum_{i=0}^{n-1} \max_{x_i \in \{0,1\}} (v_i - \lambda w_i) x_i.$$

   State the optimal decision rule for $x_i(\lambda)$ in terms of the ratio v_i/w_i and the multiplier λ.
3. **Guiding the Dual Equivalence Proof**: Prove that the optimal Lagrangian dual bound $\min_{\lambda \geq 0} \mathcal{L}(\lambda)$ equals the continuous LP relaxation bound z_{LP} by following these steps:
   (a) Assume items are indexed in non-increasing ratio order: $v_0/w_0 \geq v_1/w_1 \geq \cdots \geq v_{n-1}/w_{n-1}$. Let s denote the critical (split) item in the greedy continuous knapsack relaxation:

$$\sum_{i=0}^{s-1} w_i \leq W < \sum_{i=0}^{s} w_i.$$

   (b) Choose the specific candidate multiplier $\lambda^* = v_s/w_s$. Evaluate $\mathcal{L}(\lambda^*)$ explicitly using the decision rule from Part 2.
   (c) Show algebraically that $\mathcal{L}(\lambda^*)$ is identically equal to the continuous knapsack upper bound:

$$\mathcal{L}(\lambda^*) = \sum_{i=0}^{s-1} v_i + \frac{W - \sum_{i=0}^{s-1} w_i}{w_s} v_s = z_{\text{LP}}.$$

(d) Conclude by weak duality ($\mathcal{L}(\lambda) \geq z_{\text{LP}}$ for all $\lambda \geq 0$ due to the integrality property of $\mathcal{X} = \{0,1\}^n$) that $\min_{\lambda \geq 0} \mathcal{L}(\lambda) = z_{\text{LP}}$.

Exercise 6.2 *Step-by-Step Held–Karp Calculation*

Consider a complete TSP graph with 5 vertices $V = \{0,1,2,3,4\}$ and the symmetric distance matrix:

$$\boldsymbol{D} = \begin{pmatrix} 0 & 9 & 3 & 8 & 11 \\ 9 & 0 & 7 & 12 & 6 \\ 3 & 7 & 0 & 5 & 10 \\ 8 & 12 & 5 & 0 & 4 \\ 11 & 6 & 10 & 4 & 0 \end{pmatrix}.$$

1. Compute the minimum 1-tree rooted at vertex 0 with initial multipliers $\boldsymbol{\pi}^{(0)} = \mathbf{0}$. Report the selected edges, the vertex degrees, and the initial lower bound $\mathcal{L}(\mathbf{0})$.
2. Identify the degree violations and compute the subgradient vector $\mathbf{g}^{(0)} \in \mathbb{R}^5$, where $g_i^{(0)} = 2 - \deg_{T_0}(i)$. Verify that $\sum_{i=0}^{4} g_i^{(0)} = 0$.
3. Using a simple fixed step size $\theta_0 = 1.0$, compute the updated multiplier vector:

$$\boldsymbol{\pi}^{(1)} = \boldsymbol{\pi}^{(0)} + \theta_0 \, \mathbf{g}^{(0)}.$$

   Verify that the zero-sum condition $\sum_{i=0}^{4} \pi_i^{(1)} = 0$ is preserved.
4. Compute the modified distance matrix $D'_{ij} = D_{ij} - \pi_i^{(1)} - \pi_j^{(1)}$ under $\boldsymbol{\pi}^{(1)}$ and evaluate the updated 1-tree lower bound:

$$\mathcal{L}(\boldsymbol{\pi}^{(1)}) = c(T_{\boldsymbol{\pi}^{(1)}}) + 2\sum_{i=0}^{4} \pi_i^{(1)}.$$

   Show that the lower bound strictly improves from 26 to 27.

Exercise 6.3 *Root Independence of the 1-Tree Relaxation*

In our presentation of the Held–Karp bound, vertex 0 was arbitrarily designated as the special root vertex for the 1-tree computation. Let $r \in V$ be any choice of root vertex, and let $\mathcal{L}_r(\boldsymbol{\pi})$ denote the corresponding Lagrangian dual function.

1. Prove that for any Hamiltonian cycle $H \subset E$, H is a 1-tree with respect to root r for *every* $r \in V$.
2. Prove that the optimal Held–Karp dual value is invariant to the choice of root vertex:

$$\max_{\boldsymbol{\pi} \in \mathbb{R}^n} \mathcal{L}_r(\boldsymbol{\pi}) = \max_{\boldsymbol{\pi} \in \mathbb{R}^n} \mathcal{L}_{r'}(\boldsymbol{\pi}) \quad \forall r, r' \in V.$$

3. In a practical finite-iteration subgradient search, does the choice of root vertex affect the rate of convergence? Discuss why practitioners sometimes cycle through different root vertices during search.

Exercise 6.4 *Held–Karp Lagrangian Heuristic*

Subgradient optimization generates a sequence of 1-trees $T_0, T_1, \ldots, T_k$. Although most of these 1-trees are not Hamiltonian cycles, they often share many edges with optimal or near-optimal TSP tours.

1. Design a greedy tour construction heuristic (a *Lagrangian heuristic*) that takes a 1-tree T_k and converts it into a feasible Hamiltonian cycle by removing redundant edges and reconnecting disconnected components.
2. Explain how running this heuristic periodically during subgradient optimization helps dynamically update the upper bound UB in Polyak's step size formula (6.46).

CHAPTER 7

Network Flows

Network flow theory occupies a foundational position in combinatorial optimization and operational research. It bridges the structural elegance of graph theory with the polyhedral rigor of linear programming duality. Many practical decision problems—ranging from freight logistics and telecommunication routing to project scheduling, bipartite matching, and domain filtering in constraint satisfaction—can be cast as finding optimal commodity flows through capacitated networks.

In this chapter, we develop the theoretical foundations and algorithmic engines of network flow optimization. We begin by formalizing the dual pair formed by the *Maximum Flow Problem* and the *Minimum Cut Problem*, establishing their equivalence through linear programming duality and the combinatorial *Max-Flow Min-Cut Theorem*. We study the classical augmenting path method of Ford and Fulkerson, uncover the pathological behavior of unguided path choices, and analyze polynomial-time path selection strategies: capacity scaling (Δ-scaling) and the shortest augmenting path rule of Edmonds and Karp. We provide clean, idiomatic Java implementations of the underlying data structures and algorithms, explore practical modeling transformations (multiple sources and sinks, vertex capacities, lower bounds, and bipartite matching), and introduce the *Minimum-Cost Flow Problem* alongside the Successive Shortest Path algorithm.

7.1 Foundations and Problem Formulations

7.1.1 Historical Context: Network Interdiction

The mathematical formulation of network flows originated during the early Cold War. In a 1955 classified RAND Corporation memorandum, Theodore E. Harris and Frank S. Ross evaluated the capacity of the Soviet railway network connecting the Soviet Union to its satellite nations in Eastern Europe [HR55]. Their goal was twofold: from a logistical perspective, to estimate the maximum tonnage of supplies that could be transported from origins to destinations; and from an interdiction perspective, to identify the critical rail segments whose destruction would disrupt the transit network at minimum cost.

Harris and Ross recognized that the maximum throughput of the rail network was inextricably bound to the capacity of its most restrictive bottleneck. Lester R. Ford and Delbert R. Fulkerson formalized this observation mathematically in 1956, proving the celebrated *Max-Flow Min-Cut Theorem* [FF56; FF62]. In 1999, the Pentagon declassified the original Harris–Ross memorandum, revealing that modern network flow theory arose directly from this duality between flow throughput and network cut capacity.

7.1.2 The Minimum Cut Problem

We represent a capacitated network as a directed graph $G = (V, E)$, where $V = \{0, 1, \ldots, n-1\}$ is a finite set of n vertices and $E \subseteq V \times V$ is a set of m directed edges. Two distinguished vertices are specified: a *source* $s \in V$ and a *sink* (or target) $t \in V$ with $s \neq t$. Each directed edge $(u, v) \in E$ is endowed with a non-negative capacity $c(u, v) \geq 0$, representing the maximum rate at which commodity can traverse the edge from u to v. When $(u, v) \notin E$, we adopt the convention $c(u, v) = 0$.

Definition 7.1 *s-t Cut and Cut Capacity*

An *s-t cut* (or simply a *cut*) in a network $G = (V, E)$ is a partition of the vertex set V into two disjoint subsets A and $B = V \setminus A$ such that $s \in A$ and $t \in B$.
The *capacity* of the cut (A, B), denoted $\mathrm{cap}(A, B)$, is the sum of the capacities of all directed edges originating in A and terminating in B:

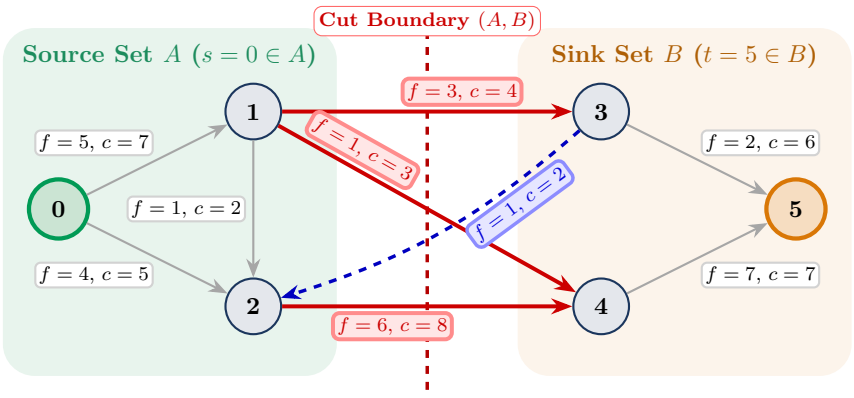
$$\mathrm{cap}(A, B) = \sum_{u \in A} \sum_{v \in B:\ (u,v) \in E} c(u, v). \tag{7.1}$$

A fundamental structural property of cut capacity is its *asymmetry*: edges originating in B and terminating in A do *not* contribute to $\mathrm{cap}(A, B)$. This reflects the physical principle that reverse edges cannot transport commodity forward from the source set A to the destination set B (Figure 7.1).

Definition 7.2 *Minimum s-t Cut Problem*

Given a capacitated network $G = (V, E)$ with source s, sink t, and non-negative edge capacities $c\colon E \to \mathbb{R}_{\geq 0}$, the *Minimum s-t Cut Problem* is to find an s-t cut (A, B) of minimum capacity:

$$\min_{(A,B)} \{\mathrm{cap}(A, B) \mid s \in A,\ t \in B,\ A \cup B = V,\ A \cap B = \emptyset\}. \tag{7.2}$$



Cut Capacity: $\mathrm{cap}(A, B) = c(1,3) + c(1,4) + c(2,4) = 4 + 3 + 8 = \mathbf{15}$ (*forward arcs only*)
Net Flow across Cut: $\mathrm{netflow}(A, B) = [f(1,3) + f(1,4) + f(2,4)] - f(3,2) = (3 + 1 + 6) - 1 = \mathbf{9} = v(f)$

Figure 7.1: An s-t cut (A, B) separating source $s = 0$ from sink $t = 5$. Only forward directed arcs crossing from A to B (solid red) contribute to the cut capacity $\mathrm{cap}(A, B) = 4+3+8 = 15$. The reverse arc $(3, 2)$ from B to A (dashed blue) carries flow back into A and is excluded from $\mathrm{cap}(A, B)$, but subtracts from the net flow across the cut.

7.1.3 The Maximum Flow Problem

We now formalize the dual primal concept: routing commodity through the network.

Definition 7.3 *s-t Flow and Flow Conservation*

An *s-t flow* (or simply a *flow*) is a function $f\colon E \to \mathbb{R}_{\geq 0}$ that assigns a non-negative scalar value $f(u,v)$ to each directed edge $(u,v) \in E$ satisfying two conditions:

1. **Capacity Constraints**: The flow along each edge cannot exceed its capacity:
$$0 \leq f(u,v) \leq c(u,v), \quad \forall (u,v) \in E. \tag{7.3}$$
2. **Flow Conservation (Local Equilibrium)**: At every intermediate vertex $v \in V \setminus \{s,t\}$, the total inflow equals the total outflow:
$$\sum_{u\colon (u,v)\in E} f(u,v) = \sum_{w\colon (v,w)\in E} f(v,w), \quad \forall v \in V \setminus \{s,t\}. \tag{7.4}$$

The *value* of the flow f, denoted $v(f)$, is the net commodity leaving the source s:
$$v(f) = \sum_{v\colon (s,v)\in E} f(s,v) - \sum_{u\colon (u,s)\in E} f(u,s). \tag{7.5}$$

Without loss of generality, we may assume that no directed edges enter s and no directed edges leave t. Under this standard assumption, the flow value simplifies to the total outflow from s, which by global conservation matches the total inflow into t:
$$v(f) = \sum_{v\colon (s,v)\in E} f(s,v) = \sum_{u\colon (u,t)\in E} f(u,t). \tag{7.6}$$

Definition 7.4 *Maximum s-t Flow Problem*

Given a capacitated network $G = (V,E)$ with source s, sink t, and capacities c, the *Maximum s-t Flow Problem* is to find a feasible flow f that maximizes the flow value $v(f)$:
$$\max_{f\colon E\to\mathbb{R}_{\geq 0}} \{v(f) \mid \text{constraints (7.3) and (7.4) hold}\}. \tag{7.7}$$

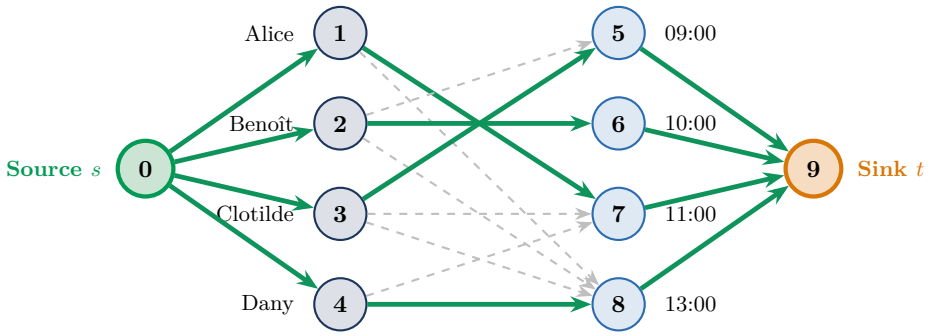
7.1.4 Case Study: Seminar Scheduling

To illustrate how discrete assignment problems naturally reduce to maximum flow formulations, consider an academic scheduling instance. Four speakers—Alice, Benoît, Clotilde, and Dany—must each deliver a one-hour presentation in a single lecture room. Four available time slots are proposed: 09:00, 10:00, 11:00, and 13:00. Each speaker indicates their availability:

- Alice: 11:00, 13:00.
- Benoît: 09:00, 10:00, 13:00.
- Clotilde: 09:00, 11:00, 13:00.
- Dany: 11:00, 13:00.

Can all four speakers be assigned to distinct, compatible time slots? We model this as a maximum flow problem in a bipartite network $G = (V,E)$ with $n = 10$ vertices indexed from 0 to 9 (Figure 7.2):

1. Source $s = 0$ connects to each speaker vertex $u \in \{1, 2, 3, 4\}$ with capacity $c(0, u) = 1$.
2. Each speaker connects to their available slot vertices $w \in \{5, 6, 7, 8\}$ with capacity $c(u, w) = 1$.
3. Each time slot connects to the sink $t = 9$ with capacity $c(w, 9) = 1$.



All capacities $c(e) = 1$. Solid green arcs indicate the maximum flow $v(f^*) = 4$ (matching).
Schedule: Clotilde → 09:00, Benoît → 10:00, Alice → 11:00, Dany → 13:00.

Figure 7.2: The seminar scheduling problem modeled as a maximum flow problem. Unit capacities ensure that each speaker delivers at most one lecture and each time slot hosts at most one presentation. The maximum flow value $v(f^*) = 4$ proves that a complete feasible schedule exists: Clotilde → 09:00, Benoît → 10:00, Alice → 11:00, and Dany → 13:00.

Because every edge capacity is an integer ($c_e = 1$), the Integrality Theorem (Theorem 7.7) guarantees that an optimal flow is integral: $f(u, w) \in \{0, 1\}$. An edge with $f(u, w) = 1$ corresponds to assigning speaker u to time slot w. If the maximum flow achieves $v(f^*) = 4$, every speaker is scheduled; if $v(f^*) < 4$, the minimum cut identifies the bottleneck preventing a complete schedule.

7.2 Linear Programming and Duality

7.2.1 Primal and Dual Formulations

Because the objective function and all constraints of the maximum flow problem are linear, it can be cast directly as a continuous Linear Program (LP). Let variable $f_{u,v} \geq 0$ denote the flow on edge $(u, v) \in E$.

$$\begin{aligned} \text{maximize} \quad & \sum_{v\colon (s,v)\in E} f_{s,v} \\ \text{subject to} \quad & \sum_{u\colon (u,v)\in E} f_{u,v} - \sum_{w\colon (v,w)\in E} f_{v,w} = 0, \quad \forall v \in V \setminus \{s, t\}, \\ & f_{u,v} \leq c_{u,v}, \quad \forall (u, v) \in E, \\ & f_{u,v} \geq 0, \quad \forall (u, v) \in E. \end{aligned} \tag{7.8}$$

To derive the dual linear program, we assign dual variables to each constraint of the primal problem:

- Dual potential $p_v \in \mathbb{R}$ associated with the flow conservation equation at vertex $v \in V \setminus \{s, t\}$. For convenience of notation, we fix $p_s = 1$ and $p_t = 0$.

- Dual variable $z_{u,v} \geq 0$ associated with the capacity constraint $f_{u,v} \leq c_{u,v}$ for each edge $(u, v) \in E$.

Formulating the Lagrangian of the primal maximization problem and applying standard LP duality principles yields the dual linear program:

$$\begin{aligned} \text{minimize} \quad & \sum_{(u,v)\in E} c_{u,v}\, z_{u,v} \\ \text{subject to} \quad & z_{u,v} - p_u + p_v \geq 0, \quad \forall (u,v) \in E, \\ & p_s = 1, \quad p_t = 0, \\ & z_{u,v} \geq 0, \quad \forall (u,v) \in E. \end{aligned} \tag{7.9}$$

The constraint $z_{u,v} \geq p_u - p_v$ combined with $z_{u,v} \geq 0$ implies that in any optimal dual solution, $z_{u,v} = \max\{0, p_u - p_v\}$.

Proposition 7.1 *Dual Problem as Minimum Cut Relaxation*

If the potential variables p_v are restricted to binary values $p_v \in \{0, 1\}$, the dual problem (7.9) is an exact formulation of the Minimum s-t Cut Problem.

Proof. Let $A = \{v \in V \mid p_v = 1\}$ and $B = \{v \in V \mid p_v = 0\}$. Since $p_s = 1$ and $p_t = 0$, $s \in A$ and $t \in B$, so (A, B) is a valid s-t cut. For any edge $(u, v) \in E$:

$$z_{u,v} = \max\{0, p_u - p_v\} = \begin{cases} 1 & \text{if } u \in A \text{ and } v \in B, \\ 0 & \text{otherwise.} \end{cases} \tag{7.10}$$

Thus $z_{u,v} = 1$ if and only if edge (u, v) originates in A and terminates in B. The dual objective function evaluates to:

$$\sum_{(u,v)\in E} c_{u,v} z_{u,v} = \sum_{u\in A} \sum_{v\in B:\ (u,v)\in E} c_{u,v} = \text{cap}(A, B). \tag{7.11}$$

Minimizing this objective over binary choices $p_v \in \{0, 1\}$ is therefore identical to finding an s-t cut of minimum capacity. □

7.2.2 Total Unimodularity and Integrality

The dual formulation (7.9) is a continuous linear relaxation over $p_v \in [0, 1]$. Why does solving this continuous relaxation guarantee binary cut indicators $p_v \in \{0, 1\}$, and why does the primal LP (7.8) yield integer flows whenever capacities $c_{u,v}$ are integers?

The answer lies in the algebraic concept of *Total Unimodularity* (TUM) introduced in Chapter 5. Recall that a matrix $\mathbf{A}$ is totally unimodular if the determinant of every square submatrix belongs to $\{-1, 0, 1\}$.

Theorem 7.2 *Integrality of Network Matrices*

Let $\mathbf{M} \in \{-1, 0, 1\}^{|V|\times|E|}$ be the vertex-edge incidence matrix of a directed graph

$G = (V, E)$, defined by:

$$M_{v,e} = \begin{cases} +1 & \text{if edge } e \text{ leaves vertex } v, \\ -1 & \text{if edge } e \text{ enters vertex } v, \\ 0 & \text{otherwise.} \end{cases} \tag{7.12}$$

Then $\mathbf{M}$ is totally unimodular.

Proof. We prove that every $k \times k$ submatrix $\mathbf{B}$ of $\mathbf{M}$ satisfies $\det(\mathbf{B}) \in \{-1, 0, 1\}$ by induction on k.

- **Base case ($k = 1$)**: Every entry of $\mathbf{M}$ is $+1$, -1, or 0. Thus $\det(\mathbf{B}) \in \{-1, 0, 1\}$.
- **Inductive step**: Assume the property holds for all submatrices of size $(k-1) \times (k-1)$. Consider a $k \times k$ submatrix $\mathbf{B}$. Each column of $\mathbf{B}$ corresponds to a directed edge in G and can contain at most one $+1$ and at most one -1:
  1. If any column of $\mathbf{B}$ contains only zeros, then $\det(\mathbf{B}) = 0$.
  2. If any column contains exactly one non-zero entry (± 1), expand $\det(\mathbf{B})$ along that column. By the Laplace expansion formula, $\det(\mathbf{B}) = \pm 1 \cdot \det(\mathbf{B}')$, where $\mathbf{B}'$ is a $(k-1) \times (k-1)$ submatrix. By the inductive hypothesis, $\det(\mathbf{B}') \in \{-1, 0, 1\}$, so $\det(\mathbf{B}) \in \{-1, 0, 1\}$.
  3. If every column contains exactly two non-zero entries (one $+1$ and one -1), then summing all row vectors of $\mathbf{B}$ yields the zero vector $\mathbf{0}^\top$. Hence the rows of $\mathbf{B}$ are linearly dependent, and $\det(\mathbf{B}) = 0$.

In all cases, $\det(\mathbf{B}) \in \{-1, 0, 1\}$, establishing total unimodularity. □

By the Hoffman–Kruskal theorem (Chapter 5), optimizing a linear objective over a polyhedron defined by a totally unimodular constraint matrix and integer bounds guarantees that every extreme point is integral. Consequently, solving the maximum flow linear program via the Simplex algorithm naturally yields an optimal integer flow and an optimal binary cut whenever all capacities are integers.

Simplex versus Graph Algorithms. Although the Simplex algorithm solves maximum flow problems with guaranteed integrality, the worst-case time complexity of the Simplex method on general polyhedra is exponential ($\mathcal{O}(2^n)$ pivots on Klee–Minty configurations; see Chapter 4). Specialized combinatorial graph algorithms exploit the underlying topology of the residual network directly, achieving strongly polynomial running times such as $\mathcal{O}(|V| \cdot |E|^2)$ without performing matrix inversions or simplex pivots.

7.3 The Ford-Fulkerson Method

7.3.1 Residual Graphs and Augmentation

A natural greedy heuristic for finding a maximum flow would be to repeatedly find a directed path from s to t in G that has spare capacity, push as much flow as possible along this path, and repeat until all paths are saturated. However, this naive greedy approach can become permanently trapped in a sub-optimal flow assignment, because it provides no mechanism to undo an earlier sub-optimal routing choice.

To overcome this fundamental limitation, Lester Ford and Delbert Fulkerson introduced the concept of an *augmenting path* with forward and backward edges [FF56]. A forward

edge allows increasing flow on an unsaturated edge, while a backward edge allows *decreasing* flow on an edge carrying positive flow. Decreasing flow on an edge (v, u) is mathematically equivalent to *redirecting* commodity along an alternate path, effectively undoing an earlier suboptimal routing decision.

This mechanism is encapsulated by the *residual network*:

Definition 7.5 *Residual Graph and Residual Capacity*

Let f be a feasible flow in a capacitated network $G = (V, E)$ with capacities c. The *residual graph* $G_f = (V, E_f)$ has the same vertex set V as G. For each pair of vertices $u, v \in V$, the residual capacity $c_f(u, v)$ is defined as:

$$c_f(u,v) = \begin{cases} c(u,v) - f(u,v) & \text{if } (u,v) \in E \text{ (forward edge)}, \\ f(v,u) & \text{if } (v,u) \in E \text{ (backward edge)}, \\ 0 & \text{otherwise.} \end{cases} \tag{7.13}$$

The residual edge set E_f consists of all ordered pairs (u, v) with strictly positive residual capacity:

$$E_f = \{(u,v) \in V \times V \mid c_f(u,v) > 0\}. \tag{7.14}$$

An edge $(u, v) \in E$ generates:

- A **forward residual edge** $(u, v) \in E_f$ with capacity $c_f(u, v) = c(u, v) - f(u, v) > 0$ whenever (u, v) is not saturated.
- A **backward residual edge** $(v, u) \in E_f$ with capacity $c_f(v, u) = f(u, v) > 0$ whenever (u, v) carries positive flow.

Figure 7.3 illustrates the exact correspondence between a flow assignment in G and its residual network G_f.

Definition 7.6 *Augmenting Path and Bottleneck Capacity*

An *augmenting path* P is a simple directed path from source s to sink t in the residual graph G_f.
The *bottleneck capacity* (or residual capacity) of P, denoted $\delta(P)$, is the minimum residual capacity among all edges along the path:

$$\delta(P) = \min_{(u,v) \in P} c_f(u,v) > 0. \tag{7.15}$$

Lemma 7.3 *Flow Augmentation Lemma*

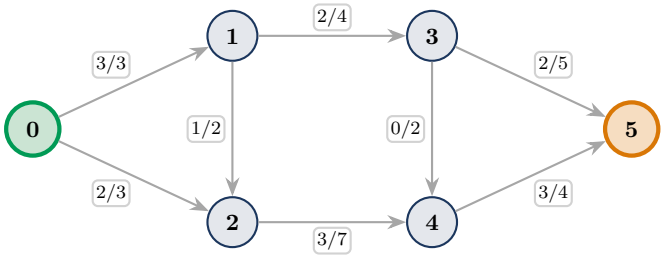
Let f be a feasible flow in G, and let P be an augmenting path in G_f with bottleneck capacity $\delta = \delta(P)$. Define an updated flow function $f' \colon E \to \mathbb{R}_{\geq 0}$ by:

$$f'(u,v) = \begin{cases} f(u,v) + \delta & \text{if } (u,v) \text{ is a forward edge on } P, \\ f(u,v) - \delta & \text{if } (v,u) \text{ is a backward edge on } P, \\ f(u,v) & \text{if neither } (u,v) \text{ nor } (v,u) \text{ appears on } P. \end{cases} \tag{7.16}$$

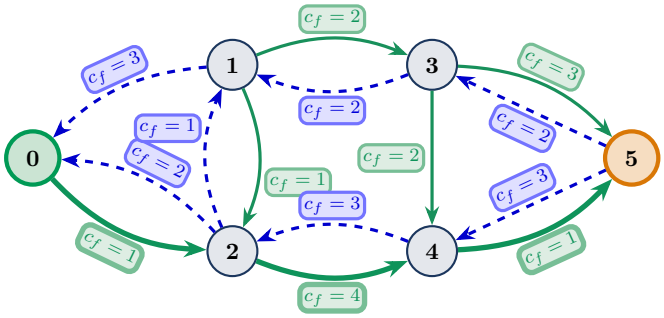
Then f' is a feasible flow in G, and its value is strictly increased by δ:

$$v(f') = v(f) + \delta. \tag{7.17}$$

(a) Flow Network G with Flow Assignment $f(e)/c(e)$, Current Value $v(f) = 5$



(b) Residual Network G_f (Solid green: Forward $c_f = c - f$; Dashed blue: Backward $c_f = f$)



Augmenting Path in G_f (bold green): $P = (0 \to 2 \to 4 \to 5)$, bottleneck $\delta = \min\{1, 4, 1\} = \mathbf{1}$
Pushing $\delta = 1$ along P yields updated flow $v(f') = 5 + 1 = \mathbf{6}$.

Figure 7.3: A flow assignment f in G and its corresponding residual network G_f. Saturated edges in G (such as $(0, 1)$ with $f = 3, c = 3$) have zero forward residual capacity and appear only as backward edges in G_f. Partially saturated edges (such as $(0, 2)$ with $f = 2, c = 3$) generate both a forward residual arc with $c_f(0, 2) = 1$ and a backward residual arc with $c_f(2, 0) = 2$.

Proof. We verify that f' satisfies capacity constraints and conservation laws:

1. **Capacity constraints**: For forward edges on P, $f'(u,v) = f(u,v) + \delta \leq f(u,v) + c_f(u,v) = f(u,v) + (c(u,v) - f(u,v)) = c(u,v)$. Non-negativity holds since $\delta > 0$. For backward edges on P, $f'(u,v) = f(u,v) - \delta \geq f(u,v) - c_f(v,u) = f(u,v) - f(u,v) = 0$.
2. **Flow conservation**: Consider any intermediate vertex $v \in V \setminus \{s,t\}$. If path P does not visit v, inflow and outflow remain unchanged. If P traverses v, it enters v along one residual edge and leaves v along another. There are four cases depending on whether incoming/outgoing residual edges correspond to forward or backward edges in G; in all four cases, the net change in total inflow to v exactly balances the net change in total outflow from v.
3. **Flow value**: Path P originates at s and terminates at t. The net flow leaving s increases by exactly δ, so $v(f') = v(f) + \delta$.

□

7.3.2 The Max-Flow Min-Cut Theorem

To establish the optimality conditions of the Ford–Fulkerson method, we first analyze the relationship between the flow through a network and the capacity of an arbitrary cut.

Definition 7.7 *Net Flow Across a Cut*

Let f be a flow in G, and let (A,B) be an s-t cut. The *net flow across the cut* (A,B), denoted $\text{netflow}(A,B)$, is the total flow on edges from A to B minus the total flow on edges from B to A:

$$\text{netflow}(A,B) = \sum_{u \in A} \sum_{v \in B \colon (u,v) \in E} f(u,v) - \sum_{u \in B} \sum_{v \in A \colon (u,v) \in E} f(u,v). \tag{7.18}$$

Lemma 7.4 *Flow-Value Lemma*

Let f be any feasible flow in G, and let (A,B) be any s-t cut. The net flow across (A,B) equals the value of the flow:

$$\text{netflow}(A,B) = v(f). \tag{7.19}$$

Proof. We sum the flow conservation equations over all vertices $u \in A$. For $u = s$, the net outflow is $v(f)$. For every other vertex $u \in A \setminus \{s\}$, flow conservation guarantees that net outflow is 0. Therefore:

$$v(f) = \sum_{u \in A} \left(\sum_{w \colon (u,w) \in E} f(u,w) - \sum_{w \colon (w,u) \in E} f(w,u) \right). \tag{7.20}$$

Any edge (u,w) where both $u \in A$ and $w \in A$ appears twice in this double summation: once with a positive sign (as an outflow from u) and once with a negative sign (as an inflow to w). These internal terms cancel out completely. The only surviving terms are edges crossing the cut boundary:

$$v(f) = \sum_{u \in A} \sum_{v \in B \colon (u,v) \in E} f(u,v) - \sum_{u \in B} \sum_{v \in A \colon (u,v) \in E} f(u,v) = \text{netflow}(A,B). \tag{7.21}$$

□

Lemma 7.5 *Weak Duality of Network Flows*

For any feasible flow f and any s-t cut (A, B):

$$v(f) \leq \text{cap}(A, B). \tag{7.22}$$

Proof. By the Flow-Value Lemma (Theorem 7.4) and the non-negativity of flows:

$$v(f) = \text{netflow}(A, B) = \sum_{u \in A, v \in B} f(u, v) - \sum_{u \in B, v \in A} f(u, v) \tag{7.23}$$

$$\leq \sum_{u \in A, v \in B} f(u, v) \leq \sum_{u \in A, v \in B} c(u, v) = \text{cap}(A, B). \tag{7.24}$$

□

Theorem 7.5 immediately implies that the capacity of *any* cut provides an upper bound on the value of *every* feasible flow. Consequently:

$$\max_f v(f) \leq \min_{(A,B)} \text{cap}(A, B). \tag{7.25}$$

We are now ready to state and prove the central theorem of network flow theory:

Theorem 7.6 *Max-Flow Min-Cut Theorem*

Let f be a feasible flow in a capacitated network $G = (V, E)$ with source s and sink t. The following three statements are equivalent:

(i) There exists an s-t cut (A, B) such that $\text{cap}(A, B) = v(f)$.

(ii) f is a maximum flow.

(iii) There is no augmenting path from s to t in the residual graph G_f.

Proof. We establish equivalence through the circular chain of implications (i) $\implies$ (ii) $\implies$ (iii) $\implies$ (i):

- **(i) $\implies$ (ii)**: Suppose there exists a cut (A, B) with $\text{cap}(A, B) = v(f)$. By Weak Duality (Theorem 7.5), for any other feasible flow f', $v(f') \leq \text{cap}(A, B) = v(f)$. Thus no flow can achieve a value strictly greater than $v(f)$, meaning f is a maximum flow.
- **(ii) $\implies$ (iii)**: Suppose, for the sake of contradiction, that f is a maximum flow but there exists an augmenting path P from s to t in G_f. By Theorem 7.3, pushing the bottleneck capacity $\delta(P) > 0$ produces a feasible flow f' with $v(f') = v(f) + \delta(P) > v(f)$. This contradicts the maximality of f.
- **(iii) $\implies$ (i)**: Suppose there is no augmenting path from s to t in G_f. We construct an explicit cut (A, B) as follows: define A to be the set of all vertices reachable from s via directed paths in the residual graph G_f, and let $B = V \setminus A$.

  1. Clearly $s \in A$ (a path of zero length). Since no augmenting path exists from s to t in G_f, $t \notin A$, which implies $t \in B$. Hence (A, B) is a valid s-t cut.
  2. For every edge $(u, v) \in E$ with $u \in A$ and $v \in B$, the edge (u, v) cannot belong to E_f (otherwise v would be reachable from s in G_f, placing $v \in A$). Therefore, $c_f(u, v) = 0$, which implies $f(u, v) = c(u, v)$ (all forward edges are fully saturated).

3. For every edge $(v, u) \in E$ with $v \in B$ and $u \in A$, the backward residual edge (u, v) cannot belong to E_f (otherwise v would be reachable from $u \in A$). Therefore, $c_f(u, v) = f(v, u) = 0$ (all backward edges carry zero flow).

Evaluating the net flow across this cut (A, B):

$$\text{netflow}(A, B) = \sum_{u \in A, v \in B} f(u, v) - \sum_{v \in B, u \in A} f(v, u) \tag{7.26}$$

$$= \sum_{u \in A, v \in B} c(u, v) - 0 = \text{cap}(A, B). \tag{7.27}$$

By the Flow-Value Lemma, $v(f) = \text{netflow}(A, B) = \text{cap}(A, B)$. This completes the proof.

□

Theorem 7.7 *Integrality Theorem*

If all edge capacities $c(u, v)$ in a network $G = (V, E)$ are integers, then there exists an optimal maximum flow f^* such that $f^*(u, v) \in \mathbb{Z}_{\geq 0}$ for every edge $(u, v) \in E$. Furthermore, the Ford–Fulkerson algorithm initialized with zero flow maintains integer flow values at every augmentation and terminates with an integral maximum flow.

Proof. We proceed by induction on the number of augmentations. The initial flow $f \equiv 0$ is trivially integer-valued. In the residual graph G_f, all forward capacities $c(u, v) - f(u, v)$ and backward capacities $f(v, u)$ are differences of integers, and therefore integers. The bottleneck capacity $\delta(P) = \min_{e \in P} c_f(e)$ is the minimum over a finite set of positive integers, so $\delta(P) \geq 1$ is an integer. Augmenting the flow along P adds or subtracts $\delta(P)$, preserving integrality. Since each augmentation increases the flow value by at least 1, the algorithm must terminate in a finite number of steps on any finite-capacity network. □

Extracting the Minimum Cut. Theorem 7.6 provides a direct, constructive method to compute a minimum cut once a maximum flow f^* is obtained: perform a Breadth-First Search (BFS) or Depth-First Search (DFS) starting from source s in the residual graph G_{f^*}. The set of all vertices marked as reachable from s constitutes set A; the set of unreachable vertices constitutes set $B = V \setminus A$. The cut (A, B) is guaranteed to be a minimum cut.

7.4 Path Selection and Complexity

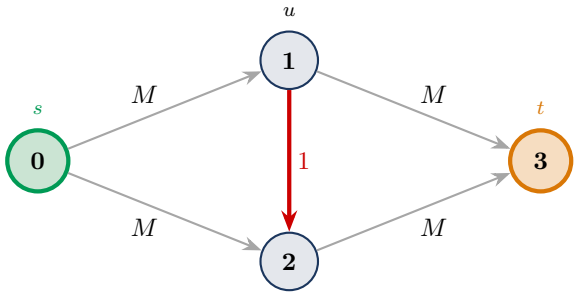
7.4.1 Arbitrary Paths and Pseudo-Polynomiality

The classical Ford–Fulkerson method does not specify how augmenting paths should be chosen in G_f. If augmenting paths are chosen arbitrarily (e.g., using an arbitrary Depth-First Search traversal), the number of augmentations can depend directly on the numerical magnitudes of the edge capacities, leading to pseudo-polynomial behavior.

Consider the pathological 4-node network illustrated in Figure 7.4. The network has vertices $V = \{0, 1, 2, 3\}$ with source $s = 0$ and sink $t = 3$. Edges $(0, 1)$, $(0, 2)$, $(1, 3)$, and $(2, 3)$ have a large capacity $M \in \mathbb{Z}^+$, while the central cross-edge $(1, 2)$ has unit capacity 1.

Suppose the DFS search alternates between two augmenting paths:

1. **Odd iterations**: Path $P_1 = (0 \to 1 \to 2 \to 3)$ with bottleneck $\delta = \min\{M, 1, M\} = 1$. This pushes 1 unit of flow, saturating $(1, 2)$ and creating reverse residual edge $(2, 1)$ with capacity 1.



Worst-Case Alternating Augmentations ($M = 10^6$):
Iteration 1: Path $0 \to 1 \to 2 \to 3$, Bottleneck $\delta = 1 \implies$ Flow pushes 1 unit across $(1, 2)$.
Iteration 2: Residual arc $(2, 1)$ has capacity 1. Path $0 \to 2 \to 1 \to 3$, Bottleneck $\delta = 1$.
... Alternating sequence requires $2M = 2 \times 10^6$ augmentations!
(*Edmonds–Karp or Capacity Scaling terminates in 2 augmentations.*)

Figure 7.4: The pathological 4-node diamond network. If an unguided DFS search repeatedly selects the augmenting path using the central cross-arc $(1, 2)$ and its reverse residual arc $(2, 1)$, each iteration pushes only $\delta = 1$ unit of flow. The algorithm requires $2M$ augmentations to reach the maximum flow value of $2M$.

2. **Even iterations**: Path $P_2 = (0 \to 2 \to 1 \to 3)$ with bottleneck $\delta = \min\{M, 1, M\} = 1$. This pushes 1 unit of flow, canceling the flow on $(1, 2)$ and recreating forward residual edge $(1, 2)$ with capacity 1.

Each augmentation increases the flow value by only 1. To reach the optimal maximum flow $v(f^*) = 2M$, the algorithm executes exactly $2M$ augmentations! If $M = 10^9$, the algorithm performs 2×10^9 graph searches. Because the input size required to encode M is $\mathcal{O}(\log M)$ bits, the complexity is exponential in the input size, rendering unguided Ford–Fulkerson a *pseudo-polynomial* algorithm:

$$\text{Running Time (Generic Ford–Fulkerson)} = \mathcal{O}(|E| \cdot v(f^*)) = \mathcal{O}(|V| \cdot |E| \cdot U), \tag{7.28}$$

where $U = \max_{e \in E} c(e)$ is the maximum edge capacity. Even worse, if capacities are allowed to be irrational numbers, an unguided Ford–Fulkerson algorithm can fail to terminate entirely and can converge to a value strictly smaller than the true maximum flow.

7.4.2 The Capacity Scaling Algorithm

To eliminate pseudo-polynomiality, we need path selection strategies that prioritize large augmentations. The *Capacity Scaling Algorithm* (or Δ-scaling) implements this intuition: it systematically searches for augmenting paths with bottleneck capacities of at least Δ, where Δ is a power-of-two threshold that is halved in successive scaling phases.

Algorithm 6: Capacity Scaling Max-Flow Algorithm

Input: Network $G = (V, E)$, source s, sink t, capacities c
Output: Maximum flow f^*

```
Initialize f(u, v) ← 0 for all (u, v) ∈ E;
U ← max_{e∈E} c(e);
Δ ← 2^⌊log₂ U⌋;
while Δ ≥ 1 do
    while there exists an augmenting path P in G_f with c_f(e) ≥ Δ for all e ∈ P do
        δ ← min_{e∈P} c_f(e);
        Augment flow along P by δ via (7.16);
    Δ ← Δ/2;
return f;
```

Let $G_f(\Delta)$ denote the subgraph of the residual network consisting solely of edges with residual capacity at least Δ: $E_f(\Delta) = \{e \in E_f \mid c_f(e) \geq \Delta\}$. A Δ-scaling phase repeatedly discovers augmenting paths in $G_f(\Delta)$ using standard BFS or DFS.

Lemma 7.8 *Residual Cut Bound in Scaling Phases*

At the end of a Δ-scaling phase, the difference between the optimal maximum flow value $v(f^*)$ and the current flow value $v(f)$ satisfies:

$$v(f^*) - v(f) \leq |E| \cdot \Delta. \tag{7.29}$$

Proof. At the termination of the Δ-scaling phase, there is no directed path from s to t in $G_f(\Delta)$. Let S be the set of vertices reachable from s in $G_f(\Delta)$, and let $T = V \setminus S$. Since $t \notin S$, (S, T) is an s-t cut.

By definition of S, no edge in G_f originating in S and terminating in T can have residual capacity $\geq \Delta$. Hence, for every directed edge $(u, v) \in E$ with $u \in S, v \in T$, the residual capacity satisfies $c_f(u, v) < \Delta$. The total residual capacity of the cut (S, T) in G_f is bounded by:

$$c_f(S, T) = \sum_{u \in S, v \in T} c_f(u, v) \leq |E| \cdot \Delta. \tag{7.30}$$

By the Flow-Value Lemma, the additional flow that can be routed from s to t in G_f cannot exceed the residual capacity of any cut, so $v(f^*) - v(f) \leq c_f(S, T) \leq |E| \cdot \Delta$. □

Theorem 7.9 *Complexity of Capacity Scaling*

The Capacity Scaling Algorithm performs at most $2|E|$ augmentations per scaling phase. Its total running time is $\mathcal{O}(|E|^2 \log U)$, which is weakly polynomial in the size of the network.

Proof. In the next phase, the threshold is $\Delta' = \Delta/2$, and each augmentation carries a bottleneck capacity of at least $\Delta' = \Delta/2$. Since the total remaining deficit between $v(f^*)$ and $v(f)$ at the start of this new phase was at most $|E| \cdot \Delta$, the number of augmentations performed in this phase cannot exceed:

$$\frac{|E| \cdot \Delta}{\Delta/2} = 2|E|. \tag{7.31}$$

The initial threshold is $\Delta \leq U$, and the algorithm terminates when $\Delta < 1$. The number of scaling phases is therefore $1 + \lfloor \log_2 U \rfloor = \mathcal{O}(\log U)$. In each phase, finding an augmenting path in $G_f(\Delta)$ via BFS or DFS takes $\mathcal{O}(|E|)$ time. The overall time complexity is:

$$\mathcal{O}(\log U) \times 2|E| \times \mathcal{O}(|E|) = \mathcal{O}(|E|^2 \log U). \tag{7.32}$$

□

7.4.3 Shortest Augmenting Path: Edmonds–Karp

In 1972, Jack Edmonds and Richard M. Karp discovered that choosing augmenting paths with the *minimum number of edges* guarantees a *strongly polynomial* running time, completely independent of edge capacities [EK72]. (This result was independently discovered by Efim A. Dinitz in 1970 [Din70].)

The Edmonds–Karp algorithm simply implements the Ford–Fulkerson method where each augmenting path is found via a **Breadth-First Search** (BFS) in G_f, treating each residual edge as having unit length 1.

Lemma 7.10 *Monotonicity of Residual Distances*

Let $d_{G_f}(s, v)$ denote the shortest path distance (measured in number of edges) from source s to vertex v in the residual graph G_f. For every vertex $v \in V$, the distance $d_{G_f}(s, v)$ increases monotonically after each flow augmentation:

$$d_{G_{f'}}(s, v) \geq d_{G_f}(s, v), \quad \forall v \in V, \tag{7.33}$$

where f' is the flow obtained after augmenting along a shortest augmenting path P.

Proof. Suppose, for the sake of contradiction, that there exists a non-empty set of vertices whose shortest distance from s decreases after augmenting along P. Let v be a vertex in this set that minimizes $d_{G_{f'}}(s, v)$, and let P' be a shortest path from s to v in $G_{f'}$. Let u be the predecessor of v on P', so that $(u, v) \in E_{f'}$ and:

$$d_{G_{f'}}(s, v) = d_{G_{f'}}(s, u) + 1. \tag{7.34}$$

Because $d_{G_{f'}}(s, u) < d_{G_{f'}}(s, v)$, by our minimal choice of v, the distance to u did not decrease:

$$d_{G_{f'}}(s, u) \geq d_{G_f}(s, u). \tag{7.35}$$

Now consider the status of edge (u, v) in G_f before the augmentation:

1. **Case 1:** $(u, v) \in E_f$. Then $d_{G_f}(s, v) \leq d_{G_f}(s, u) + 1$. Combining with (7.34) and (7.35) gives $d_{G_f}(s, v) \leq d_{G_{f'}}(s, u) + 1 = d_{G_{f'}}(s, v)$, contradicting the assumption that the distance to v decreased.
2. **Case 2:** $(u, v) \notin E_f$. Since $(u, v) \in E_{f'}$, the augmentation along P must have created (u, v) as a reverse residual edge. This implies that path P must have traversed (v, u) in the forward direction. Since P was a shortest path in G_f:

   $$d_{G_f}(s, u) = d_{G_f}(s, v) + 1. \tag{7.36}$$

   Substituting into (7.34) and (7.35):

   $$d_{G_{f'}}(s, v) = d_{G_{f'}}(s, u) + 1 \geq d_{G_f}(s, u) + 1 = d_{G_f}(s, v) + 2. \tag{7.37}$$

   This contradicts $d_{G_{f'}}(s, v) < d_{G_f}(s, v)$.

In both cases, we reach a contradiction. Hence $d_{G_f}(s, v)$ never decreases. □

Theorem 7.11 *Edmonds–Karp Bound on Augmentations*

The Edmonds–Karp algorithm performs at most $\frac{1}{2}|V||E|$ augmentations. Since each BFS takes $\mathcal{O}(|E|)$ time, the total running time is $\mathcal{O}(|V| \cdot |E|^2)$, which is strongly polynomial.

Proof. An edge (u, v) on an augmenting path P is called *critical* if it achieves the bottleneck capacity: $c_f(u, v) = \delta(P)$. Every augmentation saturates at least one critical edge, causing (u, v) to disappear from the residual network.

For (u, v) to become critical again in a subsequent augmentation, flow must first be pushed in the opposite direction along (v, u), which requires (v, u) to appear on some intermediate augmenting path. Let f be the flow when (u, v) was first critical, and let f' be the flow when (v, u) was subsequently augmented. By Theorem 7.10:

$$d_{G_f}(s, v) = d_{G_f}(s, u) + 1, \tag{7.38}$$

$$d_{G_{f'}}(s, u) = d_{G_{f'}}(s, v) + 1 \geq d_{G_f}(s, v) + 1 = d_{G_f}(s, u) + 2. \tag{7.39}$$

Thus, between two consecutive occurrences of (u, v) being critical, the distance from s to u must increase by at least 2. Because the maximum shortest-path distance in G_f from s to any vertex cannot exceed $|V| - 1$ without creating a cycle, edge (u, v) can become critical at most:

$$\frac{|V| - 1}{2} \leq \frac{|V|}{2} \text{ times.} \tag{7.40}$$

Since each of the $2|E|$ directed edges (forward and backward) can be critical at most $|V|/2$ times, and every augmentation saturates at least one edge, the total number of augmentations is at most:

$$2|E| \times \frac{|V|}{4} = \frac{1}{2}|V||E|. \tag{7.41}$$

Each augmenting path is discovered in $\mathcal{O}(|E|)$ time via BFS, yielding an overall running time of $\mathcal{O}(|V| \cdot |E|^2)$. □

7.4.4 Maximum Capacity Augmenting Paths

Another intuitive strategy is the *Maximum Capacity Augmenting Path* (or *fattest path*) rule: at each iteration, choose the path that maximizes the bottleneck capacity $\delta(P) = \min_{e \in P} c_f(e)$.

This path can be discovered using a priority queue in $\mathcal{O}(|E| \log |V|)$ time by modifying Dijkstra's algorithm (replacing the additive path length with the min-operator). A standard volume-doubling argument shows that the number of augmentations is at most $\mathcal{O}(|E| \log U)$, yielding a total running time of $\mathcal{O}(|E|^2 \log |V| \log U)$.

7.4.5 Comparison of Algorithmic Variants

Table 7.1 summarizes the theoretical characteristics of maximum flow algorithms.

7.5 Object-Oriented Implementation

7.5.1 The FlowEdge Data Structure

In modern algorithm engineering, the residual network G_f is not constructed as an explicit separate graph. Instead, each directed edge (u, v) stores both its capacity $c(u, v)$ and its

Algorithm	Path Selection Strategy	Time Complexity	Bound Type
Generic Ford–Fulkerson	Arbitrary path (DFS)	$\mathcal{O}(\lvert V\rvert \cdot \lvert E\rvert \cdot U)$	Pseudo-polynomial
Capacity Scaling	Bottleneck capacity $\geq \Delta$	$\mathcal{O}(\lvert E\rvert^2 \log U)$	Weakly polynomial
Fattest Path	Maximum bottleneck (Dijkstra)	$\mathcal{O}(\lvert E\rvert^2 \log \lvert V\rvert \log U)$	Weakly polynomial
Edmonds–Karp	Shortest path in edges (BFS)	$\mathcal{O}(\lvert V\rvert \cdot \lvert E\rvert^2)$	Strongly polynomial
Dinitz Algorithm	Blocking flows in layered networks	$\mathcal{O}(\lvert V\rvert^2 \cdot \lvert E\rvert)$	Strongly polynomial
Goldberg–Tarjan	Push-relabel with FIFO queue	$\mathcal{O}(\lvert V\rvert^3)$	Strongly polynomial

Table 7.1: Comparison of maximum flow algorithmic paradigms and their worst-case complexities.

current flow $f(u, v)$. Methods determine residual capacities and augment flows symmetrically from the perspective of either endpoint.

Listing 7.1 presents the `FlowEdge` class.

```
public class FlowEdge {
    private final int v;            // from vertex
    private final int w;            // to vertex
    private final double capacity; // forward capacity
    private double flow;            // current flow

    public FlowEdge(int v, int w, double capacity) {
        if (v < 0 || w < 0) {
            throw new IllegalArgumentException("Negative vertex");
        }
        if (capacity < 0.0) {
            throw new IllegalArgumentException("Negative capacity");
        }
        this.v = v;
        this.w = w;
        this.capacity = capacity;
        this.flow = 0.0;
    }

    public int from() { return v; }
    public int to() { return w; }
    public double capacity() { return capacity; }
    public double flow() { return flow; }

    public int other(int vertex) {
        if (vertex == v) return w;
        if (vertex == w) return v;
        throw new IllegalArgumentException("Invalid endpoint");
    }

    public double residualCapacityTo(int vertex) {
        if (vertex == v) return flow;              // backward edge
        if (vertex == w) return capacity - flow;  // forward edge
        throw new IllegalArgumentException("Invalid endpoint");
    }

    public void addResidualFlowTo(int vertex, double delta) {
```

```
        if (delta < 0.0) {
            throw new IllegalArgumentException("Negative delta");
        }
        if (vertex == v) flow -= delta;           // backward edge
        else if (vertex == w) flow += delta;      // forward edge
        else throw new IllegalArgumentException("Invalid endpoint");
    }
}
```

Listing 7.1: The capacitated edge data structure `FlowEdge.java`.

7.5.2 The FlowNetwork Representation

The `FlowNetwork` class maintains an adjacency list for each vertex. Crucially, when an edge $e = (v, w)$ is added, reference e is inserted into *both* `adj[v]` and `adj[w]`. When traversing from v, e represents a forward edge to w; when traversing from w, e represents a backward edge to v. This allows the residual network to be queried implicitly in $\mathcal{O}(1)$ time per edge.

```
import java.util.LinkedList;
import java.util.Queue;

public class FlowNetwork {
    private final int V;
    private int E;
    private Queue<FlowEdge>[] adj;

    @SuppressWarnings("unchecked")
    public FlowNetwork(int V) {
        if (V < 0) throw new IllegalArgumentException("Negative V");
        this.V = V;
        this.E = 0;
        this.adj = (Queue<FlowEdge>[]) new LinkedList[V];
        for (int v = 0; v < V; v++) {
            adj[v] = new LinkedList<FlowEdge>();
        }
    }

    public int V() { return V; }
    public int E() { return E; }

    public void addEdge(FlowEdge e) {
        int v = e.from();
        int w = e.to();
        adj[v].add(e); // forward reference
        adj[w].add(e); // backward reference
        E++;
    }

    public Iterable<FlowEdge> adj(int v) {
        return adj[v];
    }
}
```

Listing 7.2: Adjacency-list network representation `FlowNetwork.java`.

7.5.3 The Ford-Fulkerson Solver

Listing 7.3 implements the Edmonds–Karp algorithm. Method `hasAugmentingPath` performs a Breadth-First Search using a `Queue<Integer>`. It tracks visited vertices via `marked[]` and

remembers the discovering edge in `edgeTo[]`. Once an augmenting path is confirmed, the constructor computes its bottleneck capacity, pushes flow along the path, and updates the total flow value.

```
import java.util.LinkedList;
import java.util.Queue;

public class FordFulkerson {
    private final int V;
    private boolean[] marked;
    private FlowEdge[] edgeTo;
    private double value;

    public FordFulkerson(FlowNetwork G, int s, int t) {
        this.V = G.V();
        if (s == t || s < 0 || s >= V || t < 0 || t >= V) {
            throw new IllegalArgumentException("Invalid s or t");
        }
        this.value = 0.0;

        while (hasAugmentingPath(G, s, t)) {
            // Compute bottleneck capacity along path
            double bottle = Double.POSITIVE_INFINITY;
            for (int v = t; v != s; v = edgeTo[v].other(v)) {
                bottle = Math.min(bottle,
                    edgeTo[v].residualCapacityTo(v));
            }

            // Augment residual flow along path
            for (int v = t; v != s; v = edgeTo[v].other(v)) {
                edgeTo[v].addResidualFlowTo(v, bottle);
            }

            value += bottle;
        }
    }

    private boolean hasAugmentingPath(FlowNetwork G, int s, int t) {
        edgeTo = new FlowEdge[V];
        marked = new boolean[V];

        Queue<Integer> queue = new LinkedList<Integer>();
        queue.add(s);
        marked[s] = true;

        while (!queue.isEmpty() && !marked[t]) {
            int v = queue.remove();
            for (FlowEdge e : G.adj(v)) {
                int w = e.other(v);
                if (e.residualCapacityTo(w) > 0.0 && !marked[w]) {
                    edgeTo[w] = e;
                    marked[w] = true;
                    queue.add(w);
                }
            }
        }
        return marked[t];
    }

    public double value() { return value; }

```

```
58      public boolean inCut(int v) {
59          if (v < 0 || v >= V) {
60              throw new IllegalArgumentException("Vertex out of bounds");
61          }
62          return marked[v];
63      }
64  }
```

Listing 7.3: Edmonds–Karp maximum flow solver `FordFulkerson.java`.

7.6 Modeling Reductions and Extensions

The standard maximum flow problem requires a single source s, a single sink t, and simple upper bounds $0 \le f(e) \le c(e)$ on directed edges. In practical applications, problem descriptions frequently introduce complicating side constraints: multiple suppliers, node capacity limits, lower bounds, or bipartite matching structures. We now demonstrate how these variants can be transformed into standard single-source, single-sink maximum flow instances.

7.6.1 Multiple Sources and Multiple Sinks

Suppose a network has a set of k source vertices $S = \{s_0, s_1, \dots, s_{k-1}\}$ with supplies $S_i \ge 0$, and a set of ℓ sink vertices $T = \{t_0, t_1, \dots, t_{\ell-1}\}$ with demands $D_j \ge 0$.

Proposition 7.12 *Supersource and Supersink Transformation*

A multi-source, multi-sink flow problem is equivalent to a standard single-source, single-sink maximum flow problem on an augmented network $G' = (V \cup \{s^*, t^*\}, E')$ where:

1. A new *supersource* s^* is added, with a directed edge (s^*, s_i) of capacity $c(s^*, s_i) = S_i$ to each original source $s_i \in S$.
2. A new *supersink* t^* is added, with a directed edge (t_j, t^*) of capacity $c(t_j, t^*) = D_j$ from each original sink $t_j \in T$.

Any feasible flow from s^* to t^* in G' satisfies original conservation laws at all internal vertices, and the capacity constraints on edges from s^* and into t^* guarantee that individual supplies and demands are respected.

7.6.2 Vertex Capacities and Node Splitting

In telecommunication routing or railway yards, vertices themselves often represent physical switches or junction stations with bounded throughput capacities $c_V(v) < \infty$. The flow passing through vertex v must satisfy:

$$\sum_{u\colon (u,v)\in E} f(u,v) \le c_V(v). \tag{7.42}$$

Proposition 7.13 *Node Splitting Technique*

A network $G = (V, E)$ with vertex capacities $c_V \colon V \to \mathbb{R}_{\ge 0}$ is transformed into an equivalent network $G' = (V', E')$ with edge capacities only by splitting each vertex $v \in V$ into two vertices v_{in} and v_{out}:

1. For each $v \in V$, create an internal directed edge $(v_{\text{in}}, v_{\text{out}})$ with capacity $c'(v_{\text{in}}, v_{\text{out}}) = c_V(v)$.
2. For each original directed edge $(u, v) \in E$, replace it with directed edge $(u_{\text{out}}, v_{\text{in}})$ with capacity $c'(u_{\text{out}}, v_{\text{in}}) = c(u, v)$.
3. Set source $s' = s_{\text{out}}$ and sink $t' = t_{\text{in}}$.

Flow conservation across $(v_{\text{in}}, v_{\text{out}})$ enforces the vertex throughput constraint (7.42) exactly.

7.6.3 Lower Bounds and Edge Demands

In many distribution systems, contracts impose minimum delivery quantities: each edge $e \in E$ has both a lower bound $d(e) \geq 0$ and an upper bound $c(e) \geq d(e)$, requiring:

$$d(u, v) \leq f(u, v) \leq c(u, v), \quad \forall (u, v) \in E. \tag{7.43}$$

Unlike standard flow problems where $f \equiv 0$ is always feasible, determining whether a feasible flow exists under positive lower bounds is non-trivial.

Proposition 7.14 *Feasible Flow with Demands Reduction*

Let $G = (V, E)$ have edge lower bounds $d(e)$ and capacities $c(e)$. Define the vertex demand imbalance $b(v)$ as the net minimum flow required to leave vertex v:

$$b(v) = \sum_{w\colon (v,w)\in E} d(v, w) - \sum_{u\colon (u,v)\in E} d(u, v), \quad \forall v \in V. \tag{7.44}$$

Construct an auxiliary network $G' = (V \cup \{s', t'\}, E')$:

1. Add feedback edge (t, s) with capacity $c'(t, s) = \infty$ and lower bound 0 (converting s-t flow into a circulation).
2. For each original edge $(u, v) \in E$, set shifted capacity $c'(u, v) = c(u, v) - d(u, v) \geq 0$.
3. For each vertex $v \in V$:
   - If $b(v) < 0$ (flow surplus), add edge (s', v) with capacity $c'(s', v) = -b(v)$.
   - If $b(v) > 0$ (flow deficit), add edge (v, t') with capacity $c'(v, t') = b(v)$.

Then G admits a feasible flow satisfying (7.43) if and only if the maximum flow in G' from s' to t' saturates all edges leaving s'. If feasible, the original flow is recovered by setting $f(u, v) = f'(u, v) + d(u, v)$.

7.6.4 Bipartite Matching and Domain Filtering

As introduced in Section 7.1.4, the *Maximum Bipartite Matching Problem* on an undirected graph $G = (U \cup V, E)$ seeks a subset of edges $M \subseteq E$ of maximum cardinality such that no two edges in M share a common endpoint. By orienting all edges from U to V with capacity 1, connecting supersource s to each $u \in U$ with capacity 1, and connecting each $v \in V$ to supersink t with capacity 1, the maximum flow value equals $|M^*|$.

Beyond computing a single maximum matching, network flows play a central role in *Constraint Programming* (CP; see Chapter 9). In 1994, Jean-Charles Régin introduced

a filtering algorithm for the global constraint AllDifferent$(x_0, \ldots, x_{n-1})$ that achieves Generalized Arc Consistency (GAC) in $\mathcal{O}(n^{2.5})$ time [Rég94].

Régin's algorithm operates on the bipartite value graph between variables and domain values:

1. Compute a maximum matching M in the bipartite graph via maximum flow. If $|M| < n$, the constraint is unsatisfiable.
2. Construct the residual graph G_M oriented with respect to matching M (matching edges oriented from V to U, non-matching edges from U to V).
3. Compute the *Strongly Connected Components* (SCCs) of G_M in linear time $\mathcal{O}(n+m)$ using Tarjan's algorithm [Tar75].
4. An edge (u, v) can belong to *some* maximum matching if and only if it belongs to M, or its endpoints reside in the same SCC of G_M, or it belongs to an alternating path starting from an unmatched vertex.
5. Any edge (u, v) violating this condition can never participate in any valid assignment and is pruned from the domain of variable u.

7.7 Minimum-Cost Flow

7.7.1 Mathematical Formulation

The Maximum Flow Problem maximizes the quantity of commodity routed through a network. In many industrial applications, transporting commodity incurs monetary, physical, or environmental costs. The *Minimum-Cost Flow Problem* (MCF) seeks to route a required target flow volume $B \in \mathbb{R}_{\geq 0}$ from source s to sink t at minimum total cost.

Let $a(u, v)$ denote the cost per unit of flow routed along edge $(u, v) \in E$. The problem is formulated as:

$$\begin{aligned} \text{minimize} \quad & \sum_{(u,v)\in E} a(u,v)\, f(u,v) \\ \text{subject to} \quad & \sum_{w\colon (v,w)\in E} f(v,w) - \sum_{u\colon (u,v)\in E} f(u,v) = \begin{cases} B & \text{if } v = s, \\ -B & \text{if } v = t, \\ 0 & \forall v \in V \setminus \{s,t\}, \end{cases} \\ & 0 \leq f(u,v) \leq c(u,v), \quad \forall (u,v) \in E. \end{aligned} \tag{7.45}$$

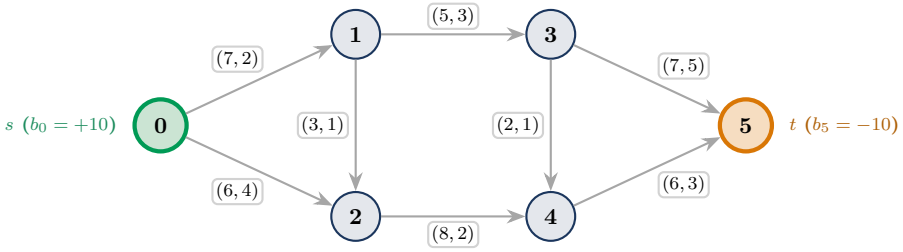
7.7.2 Residual Costs and Negative Cycles

When routing flow through a residual network G_f, each residual edge is assigned a modified unit cost $a_f(u, v)$:

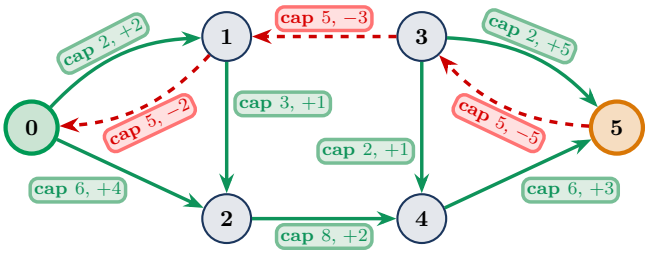
- For a forward residual edge $(u, v) \in E_f$, pushing flow increases the objective by $a(u, v)$:
$$a_f(u, v) = a(u, v). \tag{7.46}$$
- For a backward residual edge $(v, u) \in E_f$, pushing flow *cancels* flow previously sent along (u, v), refunding its unit cost:
$$a_f(v, u) = -a(u, v). \tag{7.47}$$

The presence of negative edge costs in G_f (Figure 7.5) leads to Klein's cycle-canceling optimality criterion [Kle67]:

(a) Minimum-Cost Network: Labels $(c_e, a_e) = (\textbf{capacity}, \textbf{unit cost})$, Target Flow $B = 10$



(b) Residual Network G_f after routing 5 units along $0 \to 1 \to 3 \to 5$ (unit path cost: $2 + 3 + 5 = 10$)



Arc Legend: → **Forward arc**: $c_f = c - f$, cost $+a$ ⇢ **Backward arc**: $c_f = f$, cost $-a$ (*refund*)
Successive Shortest Path: Bellman–Ford finds minimum-cost augmenting paths in G_f with negative costs.

Figure 7.5: The Minimum-Cost Flow Problem. (a) A network with capacities and unit costs (c_e, a_e) where $B = 10$ units of flow must be routed from $s = 0$ to $t = 5$. (b) Residual network G_f after routing 5 units along path $0 \to 1 \to 3 \to 5$. Backward residual arcs emerge with negative unit costs $a_f(v, u) = -a(u, v)$, representing the cost refund gained by canceling previously routed flow.

Theorem 7.15 *Negative Cycle Optimality Condition*

A feasible flow f of value B is a minimum-cost flow if and only if its residual graph G_f contains no directed cycle of negative total cost.

Proof. If G_f contains a directed cycle C with $\sum_{e \in C} a_f(e) < 0$, pushing $\delta = \min_{e \in C} c_f(e) > 0$ units around C preserves all conservation equations and flow value B while reducing the total cost by $\delta \cdot |\sum_{e \in C} a_f(e)|$. Hence f cannot be optimal. Conversely, if f' is an optimal flow of value B and f is non-optimal, the difference $f' - f$ decomposes into a set of cycles in G_f, at least one of which must have negative cost. □

7.7.3 The Successive Shortest Path Algorithm

Rather than canceling cycles from an existing feasible flow of value B, the *Successive Shortest Path* (SSP) algorithm starts with an initial flow of zero ($f \equiv 0$) and iteratively routes flow along shortest paths in G_f from s to t until the target demand B is met.

Algorithm 7: Successive Shortest Path Algorithm

Input: Network $G = (V, E)$, source s, sink t, capacities c, unit costs a, target flow B
Output: Minimum-cost flow f^*

```
Initialize f(u, v) ← 0 for all (u, v) ∈ E;
while v(f) < B do
    Find a shortest path P from s to t in G_f with respect to edge costs a_f;
    if no directed s-t path exists in G_f then
        return Target flow B is infeasible;
    δ ← min {B − v(f), min_{e∈P} c_f(e)};
    Augment flow along P by δ via (7.16);
return f;
```

Because backward edges in G_f have negative costs ($-a(u, v)$), Dijkstra's algorithm cannot be applied directly without modification. Instead, the *Moore–Bellman–Ford algorithm* is employed to find shortest paths in $\mathcal{O}(|V| \cdot |E|)$ time per augmentation.

Theorem 7.16 *Complexity of Successive Shortest Path*

If edge capacities and target demand B are integers, the Successive Shortest Path algorithm performs at most B unit augmentations, terminating in $\mathcal{O}(B \cdot |V| \cdot |E|)$ time.

Node Potentials and Dijkstra Acceleration. To avoid executing the $\mathcal{O}(|V||E|)$ Moore–Bellman–Ford routine at every augmentation, modern solvers maintain dual vertex potentials $\pi : V \to \mathbb{R}$. The *reduced cost* of edge (u, v) is defined as:

$$a^{\pi}(u, v) = a(u, v) + \pi(u) - \pi(v). \tag{7.48}$$

If potentials are updated at each iteration using shortest-path distances, reduced costs satisfy $a_f^{\pi}(u, v) \geq 0$ for all $e \in E_f$, enabling Dijkstra's algorithm with Fibonacci heaps to run in $\mathcal{O}(|E| + |V| \log |V|)$ time per augmentation [AMO93]. Furthermore, by applying capacity scaling to the target demands, polynomial-time bounds are achieved.

7.8 Chapter Summary and Exercises

7.8.1 Key Conceptual Takeaways

Table 7.2 synthesizes the core mathematical principles and algorithmic foundations established in this chapter.

Concept	Key Result and Algorithmic Significance
s-t Cut Capacity	Asymmetric definition: only forward edges $A \to B$ count towards capacity $\text{cap}(A, B)$.
Flow Conservation	Inflow equals outflow at all intermediate nodes; net flow across any cut equals $v(f)$.
Max-Flow Min-Cut	Equivalence: $\max v(f) = \min \text{cap}(A, B)$; dual of Max-Flow is the Min-Cut continuous LP.
Total Unimodularity	Vertex-edge incidence matrix is TUM; continuous Simplex guarantees integer flows.
Residual Graph G_f	Encodes available forward capacity $(c - f)$ and reversible backward flow (f).
Pathological DFS	Arbitrary path choices cause pseudo-polynomial running time $\mathcal{O}(\|V\|\|E\|U)$.
Capacity Scaling	Halving Δ filters small bottlenecks, achieving weakly polynomial $\mathcal{O}(\|E\|^2 \log U)$ time.
Edmonds–Karp	BFS shortest augmenting paths guarantee strongly polynomial $\mathcal{O}(\|V\| \cdot \|E\|^2)$ time.
Minimum-Cost Flow	Solved by successive shortest paths; backward edges refund cost $(a_f = -a)$.

Table 7.2: Foundational concepts in network flow optimization.

7.8.2 Exercises

Exercise 7.1 *Multi-Source Multi-Sink LP Formulation*

Consider a distribution network $G = (V, E)$ with two production plants s_0, s_1 with maximum supply capacities 40 and 50 units, and three retail distribution centers t_0, t_1, t_2 with demands 25, 35, and 30 units.

1. Formulate the maximum flow problem on this network as a continuous Linear Program in standard canonical form.
2. Construct the equivalent single-source single-sink network G' using supersource s^* and supersink t^*, specifying all node indices and edge capacities.
3. Formulate the dual linear program and state the complementary slackness conditions governing optimal flows and shadow prices.

Exercise 7.2 *Step-by-Step Ford-Fulkerson Trace*

Consider a network with vertices $V = \{0, 1, 2, 3, 4, 5\}$, source $s = 0$, sink $t = 5$, and directed capacities:

$$c(0,1) = 10, \quad c(0,2) = 10, \quad c(1,2) = 2, \quad c(1,3) = 4, \quad c(1,4) = 8,$$

$$c(2,4) = 9, \quad c(3,5) = 10, \quad c(4,3) = 6, \quad c(4,5) = 10.$$

1. Apply the Edmonds–Karp algorithm (BFS path selection) starting from zero flow. At each iteration, specify:
   - The augmenting path discovered in G_f.

- The bottleneck capacity δ.
- The updated flow assignment across all edges.

2. Draw the final residual graph G_{f^*} upon termination.
3. Identify the set of vertices A reachable from $s = 0$ in G_{f^*}, and compute the capacity of the resulting cut $(A, V \setminus A)$.
4. Verify that the maximum flow value equals the minimum cut capacity.

Exercise 7.3 *Capacity Scaling Execution Trace*

On the network from Exercise 7.2:

1. Determine the initial scaling parameter $\Delta = 2^{\lfloor \log_2 U \rfloor}$.
2. Trace the execution of the Capacity Scaling Algorithm (Algorithm 6) through each phase ($\Delta = 8, 4, 2, 1$).
3. Count the total number of augmenting paths discovered. Compare this count with the number of augmentations performed by the Edmonds–Karp algorithm.

Exercise 7.4 *Exact Node Throughput Demands*

Suppose each vertex $v \in V \setminus \{s, t\}$ in a network has an exact prescribed transit throughput $D(v) > 0$, requiring:

$$\sum_{u\colon (u,v)\in E} f(u, v) = \sum_{w\colon (v,w)\in E} f(v, w) = D(v).$$

1. Apply the node splitting transformation (Theorem 7.13) to convert this problem into a network with edge bounds.
2. Explain how to verify whether a feasible flow satisfying all node demands exists using a single maximum flow computation.

Exercise 7.5 *Circulations with Lower Bounds*

Consider a directed graph $G = (V, E)$ where each edge e has a lower bound $d(e) = 2$ and an upper capacity $c(e) = 6$.

1. Show that setting all flows to zero ($f \equiv 0$) is infeasible.
2. Apply the reduction in Theorem 7.14 to construct the auxiliary network G'.
3. Prove that if the maximum flow in G' saturates all edges leaving auxiliary source s', then setting $f(e) = f'(e) + d(e)$ yields a feasible circulation in G.

Exercise 7.6 *Project Selection and Maximum Weight Closure*

A company considers a set of potential projects $P = \{0, 1, \ldots, n-1\}$. Each project $i \in P$ has a net financial yield $p_i \in \mathbb{R}$ (positive for profitable projects, negative for costly prerequisite tasks). Prerequisites are modeled as a directed acyclic graph (DAG): directed edge (i, j) signifies that project i cannot be undertaken unless project j is also completed. A subset $S \subseteq P$ is *closed* if for every $i \in S$ and every $(i, j) \in E$, $j \in S$. The

goal is to find a closed subset $S^* \subseteq P$ maximizing total profit $\sum_{i \in S} p_i$.

1. Construct a network $G' = (P \cup \{s, t\}, E')$:
   - For each project i with $p_i > 0$, add edge (s, i) with capacity $c(s, i) = p_i$.
   - For each project j with $p_j < 0$, add edge (j, t) with capacity $c(j, t) = -p_j$.
   - For each dependency $(i, j) \in E$, add edge (i, j) with capacity $c(i, j) = \infty$.
2. Prove that any finite-capacity s-t cut (A, B) in G' corresponds to a valid closed subset $S = A \setminus \{s\}$.
3. Prove that minimizing cut capacity $\text{cap}(A, B)$ is mathematically equivalent to maximizing the total profit $\sum_{i \in S} p_i$.

Exercise 7.7 *Successive Shortest Path for Minimum-Cost Flow*

Consider a network with vertices $V = \{0, 1, 2, 3\}$, source $s = 0$, sink $t = 3$, and target demand $B = 4$. Directed edges have capacity and unit cost pairs (c_e, a_e):

$$e_1 = (0, 1) \colon (3, 2), \quad e_2 = (0, 2) \colon (3, 5), \quad e_3 = (1, 2) \colon (2, 1),$$

$$e_4 = (1, 3) \colon (2, 6), \quad e_5 = (2, 3) \colon (3, 2).$$

1. Trace the Successive Shortest Path algorithm (Algorithm 7) step-by-step from zero flow until $B = 4$ units are delivered.
2. At each step, identify the shortest path in G_f, verify that no negative cost cycles are created, and compute the cumulative routing cost.
3. Verify that the final flow assignment minimizes the total cost function (7.45).

CHAPTER 8

Local Search and Metaheuristics

This chapter is under preparation and will arrive later.

CHAPTER 9

Constraint Programming

This chapter is under preparation and will arrive later.

CHAPTER 10

Benchmarking Optimization Algorithms

This chapter is under preparation and will arrive later.

APPENDIX A

Kruskal's Algorithm and Disjoint Sets

The *Minimum Spanning Tree* (MST) problem is one of the most fundamental and historically significant problems in combinatorial optimization. First formulated by Otakar Borůvka in 1926 to optimize the electrification network of Southern Moravia, the MST problem asks for a tree of minimum total edge weight connecting all vertices of a weighted undirected graph.

In this textbook, minimum spanning trees play a critical role as combinatorial relaxations for **NP**-hard problems, most notably in establishing the 1-tree lower bounds for the Traveling Salesperson Problem (Chapter 3) and in providing the baseline subproblems for Lagrangian relaxation (Chapter 6). Among the classical algorithms for the MST problem, *Kruskal's algorithm*, introduced by Joseph Kruskal in 1956 [Kru56], is particularly versatile because its greedy edge selection directly accommodates pre-committed (mandatory) edges and forbidden (excluded) edges.

This appendix provides a self-contained, mathematically rigorous treatment of Kruskal's algorithm, the foundational Cut Property, the Disjoint-Set (Union-Find) data structure, and an idiomatic Java implementation.

A.1 The Minimum Spanning Tree Problem

Let $G = (V, E)$ be a connected, undirected graph with vertex set $V = \{0, 1, \ldots, n-1\}$, edge set E, and a weight function $w : E \to \mathbb{R}$ assigning a real weight $w(e)$ to each edge $e = \{u, v\} \in E$. We denote $n = |V|$ and $m = |E|$.

Definition A.1 *Spanning Tree*

A subgraph $T = (V, E_T)$ of G is a *spanning tree* of G if:

1. T contains all vertices of G (it is *spanning*).
2. T is connected.
3. T contains no simple cycles (it is *acyclic*).

Equivalently, T is a connected subgraph spanning all n vertices with exactly $n-1$ edges: $|E_T| = n - 1$.

The total weight of a spanning tree $T = (V, E_T)$ is the sum of the weights of its constituent edges:

$$w(T) = \sum_{e \in E_T} w(e). \tag{A.1}$$

Definition A.2 *Minimum Spanning Tree (MST)*

A spanning tree T^* of G is a *Minimum Spanning Tree* (MST) if its total weight is minimal among all spanning trees of G:

$$w(T^*) = \min\{w(T) : T \text{ is a spanning tree of } G\}. \tag{A.2}$$

If all edge weights in G are distinct, the minimum spanning tree is unique. If some edges have identical weights, multiple minimum spanning trees may exist with the same minimum total weight.

A.2 The Cut Property and Greedy Optimality

The design of efficient algorithms for the MST problem relies on a fundamental combinatorial property of graphs known as the *Cut Property*.

Definition A.3 *Cut and Crossing Edges*

Let $G = (V, E)$ be a graph.

1. A *cut* $(S, V \setminus S)$ is a partition of the vertex set V into two non-empty, disjoint subsets S and $V \setminus S$.
2. An edge $e = \{u, v\} \in E$ is said to *cross* the cut $(S, V \setminus S)$ if one endpoint lies in S and the other lies in $V \setminus S$ (without loss of generality, $u \in S$ and $v \in V \setminus S$). We denote the set of crossing edges by $\delta(S)$.
3. A set of edges $A \subseteq E$ *respects* the cut $(S, V \setminus S)$ if no edge in A crosses the cut: $A \cap \delta(S) = \emptyset$.
4. An edge $e^* \in \delta(S)$ is a *light edge* crossing the cut $(S, V \setminus S)$ if its weight is minimum among all edges crossing the cut:

$$w(e^*) = \min_{e \in \delta(S)} w(e). \tag{A.3}$$

Theorem A.1 *The Cut Property*

Let $G = (V, E)$ be a connected, weighted undirected graph. Let $A \subset E$ be a subset of edges that is contained in some Minimum Spanning Tree of G. Let $(S, V \setminus S)$ be any cut of G that respects A. If e^* is a light edge crossing $(S, V \setminus S)$, then:

$$A \cup \{e^*\} \text{ is also contained in some Minimum Spanning Tree of } G. \tag{A.4}$$

Proof. Let $T = (V, E_T)$ be an MST of G that contains A ($A \subseteq E_T$). If $e^* \in E_T$, then $A \cup \{e^*\} \subseteq E_T$, and the claim holds immediately.

Now assume $e^* \notin E_T$. Because T is a spanning tree, adding $e^* = \{u, v\}$ to T creates a unique simple cycle C in $T \cup \{e^*\}$. Since e^* crosses the cut $(S, V \setminus S)$, its endpoints u and v lie on opposite sides of the cut ($u \in S$ and $v \in V \setminus S$). Therefore, tracing cycle C from u to v along the path in T requires crossing the cut at least once in the opposite direction. Thus, there exists at least one edge $e' = \{x, y\} \in E_T$ on cycle C that also crosses the cut: $e' \in E_T \cap \delta(S)$ with $e' \neq e^*$.

Because the cut $(S, V \setminus S)$ respects A, no edge of A crosses the cut. Consequently, e' cannot belong to A ($e' \notin A$). We construct a new spanning subgraph:

$$T' = (E_T \setminus \{e'\}) \cup \{e^*\}. \tag{A.5}$$

Removing e' breaks the cycle C, leaving T' acyclic and connected with exactly $n-1$ edges; thus, T' is a valid spanning tree of G. Furthermore, since $e' \notin A$, we have $A \subseteq E_T \setminus \{e'\}$, which implies $A \cup \{e^*\} \subseteq E_{T'}$.

The weight of the new tree is:

$$w(T') = w(T) - w(e') + w(e^*). \tag{A.6}$$

Since e^* is a light edge crossing the cut $(S, V \setminus S)$ and e' also crosses this cut, we have $w(e^*) \leq w(e')$, which yields:

$$w(T') \leq w(T). \tag{A.7}$$

Because T is an MST, $w(T)$ is minimal, forcing $w(T') = w(T)$. Hence, T' is also an MST of G. Since $A \cup \{e^*\} \subseteq E_{T'}$, the edge e^* can be safely added to A. □

The Cut Property establishes that greedy, locally optimal edge choices on crossing cuts never compromise global optimality.

A.3 Kruskal's Algorithm and Invariants

Kruskal's algorithm [Kru56] maintains an acyclic forest $F = (V, E_F)$ initially consisting of n isolated vertices ($E_F = \emptyset$). The algorithm examines the edges of G in non-decreasing order of weight and greedily adds an edge to E_F if and only if it connects two previously disconnected components.

Algorithm 8: Kruskal's Minimum Spanning Tree Algorithm.

Input: Connected weighted graph $G = (V, E, w)$ with $V = \{0, 1, \ldots, n-1\}$
Output: Minimum Spanning Tree edges E_F

$E_F \leftarrow \emptyset$;
// Initialize empty forest
Sort E in non-decreasing order of weight: $w(e_0) \leq w(e_1) \leq \cdots \leq w(e_{m-1})$;
Initialize Disjoint-Set $\mathcal{D}$ with n singleton sets $\{0\}, \{1\}, \ldots, \{n-1\}$;
for $k \leftarrow 0$ **to** $m-1$ **do**
    $e_k = \{u, v\} \leftarrow E[k]$;
    if $\mathcal{D}.find(u) \neq \mathcal{D}.find(v)$ **then**
        $E_F \leftarrow E_F \cup \{e_k\}$;
        // Add safe edge to forest
        $\mathcal{D}$.union(u, v);
        // Merge the two connected components
        if $|E_F| == n-1$ **then**
            break;
            // Tree is complete
return E_F

Theorem A.2 *Correctness of Kruskal's Algorithm*

Kruskal's algorithm terminates with a Minimum Spanning Tree of G.

Proof. We proceed by induction on the size of the edge set E_F.

- **Base Case**: Before the loop begins, $E_F = \emptyset$, which is trivially a subset of any MST.
- **Inductive Step**: Suppose E_F is a subset of some MST. When Kruskal's algorithm selects edge $e = \{u, v\}$ to add to E_F, vertices u and v belong to distinct connected components in the forest (V, E_F). Let $S \subset V$ be the set of vertices in the connected component containing u. Because $v \notin S$, edge e crosses the cut $(S, V \setminus S)$.

  By construction, no edge currently in E_F crosses $(S, V \setminus S)$, because S is an entire connected component of (V, E_F). Thus, the cut $(S, V \setminus S)$ respects E_F. Furthermore, since edges are considered in non-decreasing order of weight, any edge crossing $(S, V \setminus S)$ that was examined previously must have formed a cycle within S or within $V \setminus S$, meaning it did not cross the cut. Therefore, e is a light edge crossing $(S, V \setminus S)$. By the Cut Property (Theorem A.1), $E_F \cup \{e\}$ is also contained in some MST.

When $|E_F| = n-1$, the forest is connected and acyclic, and hence constitutes a valid MST. □

A.4 The Disjoint-Set Data Structure

At line 6 of Algorithm 8, Kruskal's algorithm must determine whether the two endpoints u and v of candidate edge $e = \{u, v\}$ belong to the same connected component. Performing a depth-first or breadth-first search at every edge would require $\mathcal{O}(n)$ time per edge, resulting in an impractical $\mathcal{O}(m \cdot n)$ total running time.

The *Disjoint-Set* (or *Union-Find*) data structure maintains a collection of disjoint dynamic sets $\mathcal{S} = \{S_0, S_1, \dots\}$ partitioning the universe $V = \{0, 1, \dots, n-1\}$. Each set is identified by a unique *representative* element (the root of a directed tree). The data structure supports three fundamental operations:

- `makeSet`(i): Creates a new set containing the single element i, which serves as its own representative.
- `find`(i): Returns the canonical representative of the set containing element i.
- `union`(i, j): Merges the set containing i with the set containing j into a single set.

A.4.1 Union by Rank and Path Compression

A naive parent-pointer tree implementation where `parent[i]` points to i's parent can degrade into a linear chain of length n, yielding an undesirable $\mathcal{O}(n)$ time per `find`. To guarantee near-constant amortized performance, two complementary optimizations are essential:

1. **Union by Rank**: Maintain an integer array `rank[i]`, which represents an upper bound on the height of the subtree rooted at i. When merging two sets with roots r_u and r_v:
   - If `rank`$[r_u] <$ `rank`$[r_v]$, attach r_u as a child of r_v (the rank of r_v does not change).
   - If `rank`$[r_u] >$ `rank`$[r_v]$, attach r_v as a child of r_u.
   - If `rank`$[r_u] ==$ `rank`$[r_v]$, arbitrarily make r_v a child of r_u and increment `rank`$[r_u]$ by 1.

   Union by rank ensures that the height of any tree on k elements never exceeds $\lfloor \log_2 k \rfloor$.
2. **Path Compression**: During a call to `find`(i), make every node along the path from i to the root point directly to the root. While path compression does not change the tree during the traversal to the root, it flattens the tree during the recursion unwinding, so subsequent lookups along that branch take $\mathcal{O}(1)$ time.

Theorem A.3 *Complexity of Disjoint-Set with Both Heuristics*

A sequence of m operations on a Disjoint-Set data structure of n elements utilizing both union by rank and path compression executes in:

$$\mathcal{O}(m \cdot \alpha(n)) \text{ time}, \tag{A.8}$$

where $\alpha(n)$ denotes the extremely slow-growing inverse Ackermann function [Tar75].

The value of $\alpha(n)$ satisfies $\alpha(n) \leq 4$ for all $n \leq 10^{80}$ (a number vastly exceeding the estimated number of atoms in the observable universe). For all practical computational purposes, $\alpha(n)$ is a small constant, rendering each Union-Find operation effectively $\mathcal{O}(1)$ amortized.

A.5 Java Implementation

Listing A.1 provides a clean, production-grade implementation of the `UnionFind` class with path compression and union by rank.

```
public class UnionFind {
    private final int[] parent;
    private final int[] rank;
    private int count; // Number of disjoint sets

    /**
     * Initializes Union-Find on {0, 1, ..., n-1}.
     * @param n Universe size.
     */
    public UnionFind(int n) {
        this.parent = new int[n];
        this.rank = new int[n];
        this.count = n;
        for (int i = 0; i < n; i++) {
            parent[i] = i;
            rank[i] = 0;
        }
    }

    /**
     * Finds canonical root of element i
     * using path compression.
     */
    public int find(int i) {
        if (parent[i] != i) {
            parent[i] = find(parent[i]); // Path compression
        }
        return parent[i];
    }

    /**
     * Merges sets containing i and j using union by rank.
     * @return true if merged; false if already connected.
     */
    public boolean union(int i, int j) {
        int rootI = find(i);
        int rootJ = find(j);
        if (rootI == rootJ) {
```

```
            // Already in same component (cycle detected)
            return false;
        }
        // Union by rank: attach smaller under larger
        if (rank[rootI] < rank[rootJ]) {
            parent[rootI] = rootJ;
        } else if (rank[rootI] > rank[rootJ]) {
            parent[rootJ] = rootI;
        } else {
            parent[rootJ] = rootI;
            rank[rootI]++;
        }
        count--;
        return true;
    }

    /** Returns true if i and j belong to the same set. */
    public boolean connected(int i, int j) {
        return find(i) == find(j);
    }

    /** Returns current number of disjoint components. */
    public int getCount() {
        return count;
    }
}
```

Listing A.1: The Disjoint-Set (Union-Find) data structure in Java.

Listing A.2 implements Kruskal's algorithm using the `UnionFind` structure. Crucially, we also provide the method `computeConditionedMST`, which pre-inserts mandatory edges E_{fixed} before processing free edges, directly supporting the TSP Branch and Bound bounding procedure introduced in Section 3.3.3.

```
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;

public class KruskalMST {

    public static class Edge implements Comparable<Edge> {
        public final int u;
        public final int v;
        public final double weight;

        public Edge(int u, int v, double weight) {
            this.u = u;
            this.v = v;
            this.weight = weight;
        }

        @Override
        public int compareTo(Edge other) {
            return Double.compare(
                    this.weight, other.weight);
        }
    }

    /**
     * Standard Kruskal MST computation on undirected graph.
     * @param n Number of vertices {0, 1, ..., n-1}.
```

```
28       * @param edges Available candidate edges.
29       * @return List of n - 1 MST edges, or empty list.
30       */
31      public static List<Edge> computeMST(
32              int n, List<Edge> edges) {
33          List<Edge> sortedEdges = new ArrayList<>(edges);
34          Collections.sort(sortedEdges);
35
36          UnionFind uf = new UnionFind(n);
37          List<Edge> mst = new ArrayList<>();
38
39          for (Edge e : sortedEdges) {
40              if (uf.union(e.u, e.v)) {
41                  mst.add(e);
42                  if (mst.size() == n - 1) {
43                      break;
44                  }
45              }
46          }
47          return mst;
48      }
49
50      /**
51       * Computes a Conditioned MST with mandatory edges.
52       * Used for bounding in TSP Branch and Bound.
53       *
54       * @param n Number of vertices {0, 1, ..., n-1}.
55       * @param freeEdges Undecided edges for selection.
56       * @param mandatoryEdges Pre-committed edges.
57       * @return List of n - 1 edges, or null if infeasible.
58       */
59      public static List<Edge> computeConditionedMST(
60              int n,
61              List<Edge> freeEdges,
62              List<Edge> mandatoryEdges) {
63          UnionFind uf = new UnionFind(n);
64          List<Edge> mst = new ArrayList<>();
65
66          // Phase 1: Pre-insert mandatory edges
67          for (Edge e : mandatoryEdges) {
68              if (!uf.union(e.u, e.v)) {
69                  // Premature cycle: infeasible!
70                  return null;
71              }
72              mst.add(e);
73          }
74
75          // Phase 2: Greedily complete the spanning forest
76          List<Edge> sortedFree = new ArrayList<>(freeEdges);
77          Collections.sort(sortedFree);
78
79          for (Edge e : sortedFree) {
80              if (uf.union(e.u, e.v)) {
81                  mst.add(e);
82                  if (mst.size() == n - 1) {
83                      break;
84                  }
85              }
86          }
87
88          // Must connect all n vertices with n - 1 edges
```

```
89            return (mst.size() == n - 1) ? mst : null;
90        }
91    }
```

Listing A.2: Kruskal's algorithm and conditioned MST computation in Java.

A.6 Complexity and Comparison with Prim

Theorem A.4 *Asymptotic Complexity of Kruskal's Algorithm*

For a graph $G = (V, E)$ with $n = |V|$ vertices and $m = |E|$ edges:

1. **Sorting Time**: Sorting the m edges by weight requires $\mathcal{O}(m \log m)$ time. Since $m \leq n^2$, we have $\log m \leq 2 \log n$, so sorting takes $\mathcal{O}(m \log n)$ time.
2. **Union-Find Operations**: The algorithm performs n initializations (`makeSet`), at most $2m$ calls to `find`, and exactly $n - 1$ calls to `union`. By Theorem A.3, these execute in $\mathcal{O}(m \cdot \alpha(n))$ time.
3. **Total Time Complexity**: $\mathcal{O}(m \log n + m \cdot \alpha(n)) = \mathcal{O}(m \log n)$.
4. **Space Complexity**: $\mathcal{O}(n + m)$ to store the edge list and the `parent` and `rank` arrays.

A.6.1 Kruskal vs. Prim: Trade-Offs

While Kruskal's algorithm grows a global spanning forest across the entire graph, *Prim's algorithm* grows a single tree outwardly from an arbitrary seed vertex, always adding the cheapest edge connecting the growing tree to a remaining unvisited vertex. Table A.1 compares their computational trade-offs.

Table A.1: Comparison of Minimum Spanning Tree Algorithms.

Algorithm	Data Structure	Time Complexity	Best Suited For
Kruskal	Array + Union-Find	$\mathcal{O}(m \log n)$	Sparse graphs, conditioned MSTs (E_{fixed})
Prim	Binary Heap	$\mathcal{O}(m \log n)$	Sparse to moderately dense graphs
Prim	Fibonacci Heap	$\mathcal{O}(m + n \log n)$	Theoretical optimum for dense graphs
Prim	Adjacency Matrix	$\mathcal{O}(n^2)$	Complete/dense graphs ($m = \Theta(n^2)$)

Why Kruskal is Indispensable in Branch and Bound. For unconstrained complete graphs where $m = \frac{n(n-1)}{2}$, Prim's algorithm with an adjacency matrix runs in $\mathcal{O}(n^2)$ time, which is asymptotically faster than Kruskal's $\mathcal{O}(n^2 \log n)$ edge-sorting overhead.

However, within a Branch and Bound search tree for the TSP (Chapter 3):

- **Handling Pre-Committed Decisions**: Kruskal's algorithm integrates fixed edges E_{fixed} immediately: by pre-merging all mandatory edges in the Union-Find structure, any cycle formed by E_{fixed} on fewer than n vertices immediately reveals an invalid premature subtour in $\mathcal{O}(1)$ amortized time.
- **Re-using Sorted Edges**: The candidate edges E can be sorted once at the root node in $\mathcal{O}(m \log n)$ time. At child nodes, Kruskal can simply iterate over the already-sorted edge list, skipping edges marked in E_{excluded}, which reduces the per-node cost to near-linear $\mathcal{O}(m \cdot \alpha(n))$ time!

This structural synergy makes Kruskal's algorithm with Disjoint-Set Union one of the most effective building blocks in discrete optimization.

References

[AMO93] Ravindra K Ahuja, Thomas L Magnanti, and James B Orlin. *Network Flows: Theory, Algorithms, and Applications*. Prentice Hall, 1993 (cit. on pp. 120, 161).

[App+06] David L Applegate et al. *The Traveling Salesman Problem: A Computational Study*. Princeton University Press, 2006 (cit. on pp. 3, 135).

[Bel57] Richard Bellman. *Dynamic Programming*. Princeton University Press, 1957 (cit. on p. 11).

[BT97] Dimitris Bertsimas and John N Tsitsiklis. *Introduction to Linear Optimization*. Athena Scientific, 1997 (cit. on p. 68).

[Bla77] Robert G Bland. "New Finite Pivoting Rules for the Simplex Method". In: *Mathematics of Operations Research* 2.1 (1977), pp. 34–44 (cit. on p. 95).

[DFJ54] George Dantzig, Ray Fulkerson, and Selmer Johnson. "Solution of a Large-Scale Traveling-Salesman Problem". In: *Journal of the Operations Research Society of America* 2.4 (1954), pp. 393–410 (cit. on pp. 40, 132).

[Dan57] George B Dantzig. "Discrete-Variable Extremum Problems". In: *Operations Research* 5.2 (1957), pp. 266–277 (cit. on pp. 31, 35).

[Dan63] George B Dantzig. *Linear Programming and Extensions*. Princeton University Press, 1963 (cit. on p. 65).

[Din70] Efim A Dinitz. "Algorithm for Solution of a Problem of Maximum Flow in a Network with Power Estimation". In: *Soviet Mathematics Doklady* 11 (1970), pp. 1277–1280 (cit. on p. 152).

[Edm71] Jack Edmonds. "Matroids and the Greedy Algorithm". In: *Mathematical Programming* 1.1 (1971), pp. 127–136 (cit. on p. 134).

[EK72] Jack Edmonds and Richard M Karp. "Theoretical Improvements in Algorithmic Efficiency for Network Flow Problems". In: *Journal of the ACM* 19.2 (1972), pp. 248–264 (cit. on p. 152).

[Eve63] Hugh Everett. "Generalized Lagrange Multiplier Method for Solving Problems of Optimum Allocation of Resources". In: *Operations Research* 11.3 (1963), pp. 399–417 (cit. on p. 117).

[Fis81] Marshall L Fisher. "The Lagrangian Relaxation Method for Solving Integer Programming Problems". In: *Management Science* 27.1 (1981), pp. 1–18 (cit. on p. 117).

[FF56] Lester R Ford and Delbert R Fulkerson. "Maximal Flow through a Network". In: *Canadian Journal of Mathematics* 8 (1956), pp. 399–404 (cit. on pp. 139, 144).

[FF62] Lester R Ford and Delbert R Fulkerson. *Flows in Networks*. Princeton University Press, 1962 (cit. on p. 139).

[GJ79] Michael R Garey and David S Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W. H. Freeman and Company, 1979 (cit. on pp. 1, 11).

[Geo74] Arthur M Geoffrion. "Lagrangian Relaxation for Integer Programming". In: *Mathematical Programming Study* 2 (1974), pp. 82–114 (cit. on pp. 117, 134).

[Gom58] Ralph E Gomory. "Outline of an Algorithm for Integer Solutions to Linear Programs". In: *Bulletin of the American Mathematical Society* 64.5 (1958), pp. 275–278 (cit. on p. 111).

[HR55] Theodore E Harris and Frank S Ross. *Fundamentals of a Method for Evaluating Rail Net Capacities*. Tech. rep. RM-1573. Santa Monica, CA: The RAND Corporation, 1955 (cit. on p. 139).

[HK70] Michael Held and Richard M Karp. "The Traveling-Salesman Problem and Minimum Spanning Trees". In: *Operations Research* 18.6 (1970), pp. 1138–1162 (cit. on pp. 45, 117, 125, 134, 135).

[HK71] Michael Held and Richard M Karp. "The Traveling-Salesman Problem and Minimum Spanning Trees: Part II". In: *Mathematical Programming* 1.1 (1971), pp. 6–25 (cit. on pp. 117, 125, 129, 134, 135).

[JM97] David S Johnson and Lyle A McGeoch. "The Traveling Salesman Problem: A Case Study in Local Optimization". In: *Local Search in Combinatorial Optimization*. Ed. by Emile Aarts and Jan Karel Lenstra. John Wiley & Sons, 1997, pp. 215–310 (cit. on p. 135).

[JV87] Roy Jonker and Anton Volgenant. "A Shortest Augmenting Path Algorithm for Dense and Sparse Linear Assignment Problems". In: *Computing* 38.4 (1987), pp. 325–340 (cit. on p. 42).

[Kar84] Narendra Karmarkar. "A New Polynomial-Time Algorithm for Linear Programming". In: *Combinatorica* 4.4 (1984), pp. 373–395 (cit. on p. 95).

[Kar72] Richard M Karp. "Reducibility among Combinatorial Problems". In: *Complexity of Computer Computations*. Springer, 1972, pp. 85–103 (cit. on p. 12).

[Kha79] Leonid G Khachiyan. "A Polynomial Algorithm in Linear Programming". In: *Soviet Mathematics Doklady* 20 (1979), pp. 191–194 (cit. on p. 95).

[KM72] Victor Klee and George J Minty. "How Good is the Simplex Algorithm?" In: *Inequalities III*. Academic Press, 1972, pp. 159–175 (cit. on p. 95).

[Kle67] Morton Klein. "A Primal Method for Minimal Cost Flows with Applications to the Assignment and Transportation Problems". In: *Management Science* 14.3 (1967), pp. 205–220 (cit. on p. 159).

[Kru56] Joseph B Kruskal. "On the Shortest Spanning Subtree of a Graph and the Traveling Salesman Problem". In: *Proceedings of the American Mathematical Society* 7.1 (1956), pp. 48–50 (cit. on pp. 171, 173).

[Kuh55] Harold W Kuhn. "The Hungarian Method for the Assignment Problem". In: *Naval Research Logistics Quarterly* 2.1-2 (1955), pp. 83–97 (cit. on p. 42).

[LD60] Ailsa H Land and Alison G Doig. "An Automatic Method of Solving Discrete Programming Problems". In: *Econometrica* 28.3 (1960), pp. 497–520 (cit. on p. 31).

[Lem54] Carlton E Lemke. "The Dual Method of Solving the Linear Programming Problem". In: *Naval Research Logistics Quarterly* 1.1 (1954), pp. 36–47 (cit. on p. 108).

[Lit+63] John DC Little et al. “An Algorithm for the Traveling Salesman Problem”. In: *Operations Research* 11.6 (1963), pp. 972–989 (cit. on p. 31).

[Pol87] Boris T Polyak. *Introduction to Optimization*. Optimization Software, New York, 1987 (cit. on p. 129).

[Rég94] Jean-Charles Régin. “A Filtering Algorithm for Constraints of Difference in CSPs”. In: *AAAI* 94 (1994), pp. 362–367 (cit. on p. 159).

[SW11] Robert Sedgewick and Kevin Wayne. *Algorithms*. 4th. Addison-Wesley Professional, 2011 (cit. on pp. 10, 66).

[Sho85] Naum Zuselevich Shor. *Minimization Methods for Non-Differentiable Functions*. Springer-Verlag, 1985 (cit. on pp. 123, 124).

[Sti45] George J Stigler. “The Cost of Subsistence”. In: *Journal of Farm Economics* 27.2 (1945), pp. 303–314 (cit. on p. 96).

[Tar75] Robert Endre Tarjan. “Efficiency of a Good But Not Linear Set Union Algorithm”. In: *Journal of the ACM* 22.2 (1975), pp. 215–225 (cit. on pp. 159, 175).

[VJ97] Christine L Valenzuela and Antonia J Jones. “Estimating the Held-Karp Lower Bound for the Geometric TSP”. In: *European Journal of Operational Research* 102.1 (1997), pp. 157–175 (cit. on pp. 129, 130, 135).

[Wol20] Laurence A Wolsey. *Integer Programming*. 2nd. John Wiley & Sons, 2020 (cit. on p. 56).